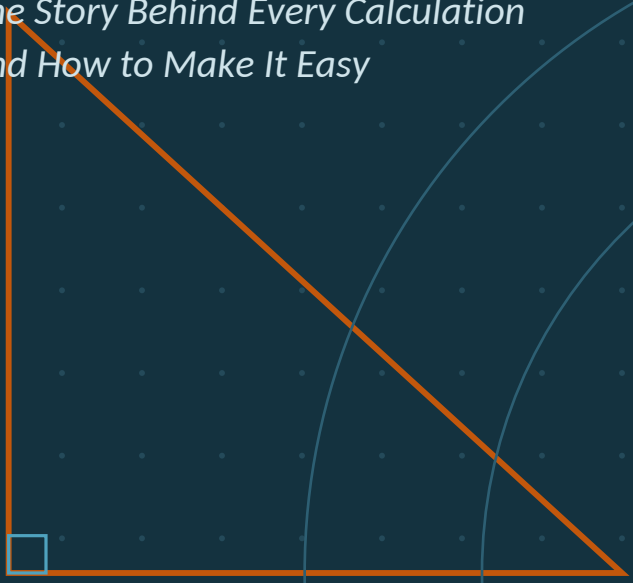
A GUIDE TO MATHEMATICAL THINKING

Think Before the Formula

The Story Behind Every Calculation and How to Make It Easy



SANTIANO

Santiano Books · First Edition · 2026

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This book is an independently written educational work intended for students and general readers. It explores the reasoning behind mathematical calculations, the practical situations that make them useful, and the habits that help learners approach problems with understanding rather than memorisation alone.

The views and interpretations expressed in this book are those of the author.

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Thank God for the opportunity to learn, create, and contribute to the preservation of knowledge.

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DEDICATION

This book is dedicated to every student who has ever looked at a calculation and wondered, "How am I supposed to know which formula to use?"

It is also dedicated to the great thinkers whose work has helped humanity understand numbers, shapes, money, nature, and the world around us.

Their discoveries remind us that the things we learn today were developed, improved, and passed down through the work of many people.

May we learn to understand their work and develop the courage to solve the problems that come before us.

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Preface

Why Do We Learn Calculations?

Imagine that you are sitting in a classroom.

Your teacher writes a question on the board and asks you to solve it.

You look at the question.

You look at your notebook.

Then you begin to wonder:

“Which formula should I use?”

You search your memory. You try to remember the formula your teacher explained yesterday. You may even remember the answer to a similar question.

But what happens when the question changes?

What happens when the numbers are different or the problem is written in a way you have never seen before?

You may become confused, even though you studied the topic.

Now, let us look at the same situation differently.

Imagine that you are no longer sitting in a classroom. You are a builder working on a real building. You need to know how long a ladder must be to reach a particular height.

You cannot simply guess.

You need to understand the situation, identify the measurements, and find a reliable way to calculate the answer.

Suddenly, the calculation has a purpose.

You are no longer solving the question only because it appears in an examination. You are solving it because someone needs an answer.

This is the idea behind this book.

Many students begin learning a calculation by asking which formula they should use. There is nothing wrong with learning formulas. Formulas are important tools.

However, we should also ask another question:

What problem does this formula help us solve?

When we understand the problem, we begin to understand why the formula is useful.

Think about an accountant. An accountant does not calculate profit merely because a textbook contains a profit formula. The accountant needs to understand whether a business is making money.

Think about a builder. A builder does not measure a wall merely to practise multiplication. The builder needs measurements to plan and complete a structure.

Think about an economist. An economist does not calculate costs merely to fill a page with numbers. The economist wants to understand how people use limited resources.

In each situation, the calculation serves a purpose.

This book will help you enter those situations with your imagination.

When we study accounting, you will become the person responsible for understanding the financial records of a business.

When we study geometry, you will become the person measuring a building.

When we study algebra, you will become the person searching for an unknown quantity.

When we study statistics, you will become the person examining information to discover what it tells you.

You will not only read about the work of great thinkers. You will also practise the reasoning that makes their ideas useful.

Of course, we cannot assume that we would have discovered every famous mathematical idea if we had lived in the same period as its original contributors.

Their work often required years of thought, and many discoveries depended on the contributions of earlier thinkers.

But we can place ourselves in similar practical situations, ask meaningful questions, and attempt to understand the reasoning behind the solutions.

That is how we turn history into a learning experience.

How to Use This Book

Whenever you meet a new calculation, follow these steps:

1. **Understand the situation.** What is happening, and what needs to be found?
2. **Investigate the idea.** Why is this kind of calculation useful? What is known about its history?
3. **Think like the person facing the problem.** Imagine that the answer matters to you.
4. **Identify the information.** What do you know, and what do you still need to find?
5. **Understand the formula.** Learn what each part means and why the formula applies.
6. **Calculate carefully.** Follow each step and understand the reason for it.
7. **Check your answer.** Ask whether the result is correct and reasonable.
8. **Apply what you learned.** Try another problem with different numbers or a different situation.

You do not need to rush through these steps.

Sometimes the easiest way to solve a difficult question is to slow down long enough to understand it.

This book will also show you how to estimate answers, recognize patterns, and use shortcuts when they are appropriate. You will learn that making calculations easier does not mean avoiding the reasoning. It means understanding the reasoning well enough to work more confidently.

One final thing is important.

Not every formula was invented to solve one specific practical problem. Some mathematical ideas began as abstract investigations and later became useful in

science, engineering, economics, and other fields. Whenever we discuss history, we must respect the evidence and avoid inventing stories about how discoveries happened.

Our goal is to understand both the history and the mathematics.

By the time you finish this book, I want you to approach a calculation differently.

Instead of asking only, “Which formula should I use?” I want you to begin asking:

“What am I trying to understand? Why does this method work? How can I use it to solve the problem before me?”

That is the habit we will develop together.

Welcome to *Think Before the Formula*.

— **Santiano**

CHAPTER 1

Before You Ask for the Formula

01

1.1 Introduction: The Question Every Student Asks

Imagine that you are sitting in a classroom, and your mathematics teacher writes a question on the board.

A ladder is placed against a wall. The foot of the ladder is 6 metres from the wall, and the top reaches a height of 8 metres. Calculate the length of the ladder.

You look at the question and begin to think.

Which formula should I use?

Have I seen this question before?

Is it Pythagoras' theorem, or is there another formula?

You open your notebook and search for a formula that looks familiar. After finding it, you substitute the numbers and calculate the answer.

Perhaps you get it right. Perhaps you make a mistake.

But let us pause for a moment.

Have you ever asked yourself why this question exists in the first place?

Why would anyone need to calculate the length of a ladder? Why would someone need to know the height of a building? What kind of person faces this problem outside a classroom?

These questions may appear simple, but they can change the way you learn mathematics.

Now imagine that you are not a student sitting in an examination hall. You are a construction worker who needs to reach a point high on a building. You have a ladder, but you need to understand the distance it must cover.

The problem is no longer just a question on paper.

It is a practical situation that requires a solution.

You must understand the measurements, identify the relationship between them, choose an appropriate mathematical method, and calculate carefully.

This is where our journey begins.

In this chapter, we will learn how to approach a calculation before reaching for a formula. We will examine why understanding the problem matters, how to think like the person facing it, and how to make difficult calculations easier by breaking them into understandable steps.

1.2 Two Students, One Calculation

Let us meet two students: David and Sarah.

Their teacher gives them the same question.

A rectangular room is 10 metres long and 6 metres wide. Calculate its area.

David remembers a formula from his notebook:

$$A = l \times w$$

He writes:

$$A = 10 \times 6$$

$$A = 60\,\mathrm{m}^2$$

He has obtained the correct answer.

Sarah also knows the formula, but before calculating, she asks herself what the area represents.

She imagines standing inside the room. The floor has a length and a width. If the owner wants to cover the floor with tiles, the owner needs to know how much surface must be covered.

Sarah understands that both dimensions matter. If the room becomes longer while its width stays the same, the floor area increases. If its width increases while its length stays the same, the area also increases.

She now uses the formula:

$$\text{Area} = \text{Length} \times \text{Width}$$

Substituting the measurements:

$$A = 10 \times 6$$

Therefore:

$$A = 60\,\text{m}^2$$

Sarah understands that the answer means the floor covers 60 square metres.

Both students answered correctly. So what is the difference?

David knows how to perform the calculation. Sarah knows how to perform it and understands what it represents.

This difference becomes important when the question changes.

Suppose the teacher asks:

A rectangular room has an area of 60 square metres and a width of 6 metres. Find its length.

David may become confused because the question is written differently.

Sarah remembers the relationship:

$$A = l \times w$$

She knows that the area is 60 square metres and the width is 6 metres.

To find the missing length, she divides the area by the width:

$$\text{Length} = \text{Area} \div \text{Width}$$

$$l = 60 \div 6$$

$$l = 10 \text{ metres}$$

Sarah can solve the new question because she understands the relationship between the quantities.

The lesson: Memorizing a formula helps you remember a method. Understanding the meaning of a formula helps you decide how to use it.

Neither memorization nor practice is useless. Both are important. But when you understand what a calculation means, you are better prepared to solve unfamiliar problems.

1.3 Mathematics Did Not Begin in Your Textbook

Imagine a world in which people could not measure land accurately, keep track of goods, compare prices, or calculate the materials needed for a building.

How would a farmer divide a field between family members?

How would a trader determine the value of goods exchanged?

How would a builder estimate the amount of material needed for a structure?

How would people organize records of what they owned or owed?

These are examples of situations in which numerical reasoning can be useful.

Mathematics has a long history. Its development did not happen in one place or at one moment. People across different societies developed counting methods, measurement systems, numerical records, and mathematical techniques over many generations.

Ancient Egyptian and Babylonian records, for example, contain evidence of mathematical work involving measurement, practical calculations, and administration. Other societies developed their own methods and contributed ideas that later became part of the wider history of mathematics.

Over time, mathematical thinking also developed beyond immediate practical needs. People began investigating patterns, relationships, logical arguments, and abstract questions. Some of these ideas later found important applications in science, engineering, economics, and technology.

This history teaches us something important.

When we encounter a mathematical topic today, we see a finished system of knowledge. We may see definitions, formulas, examples, and exercises arranged neatly on a page.

But the development of mathematics involved questions, observations, mistakes, arguments, improvements, and discoveries.

The final formula does not always show us the full journey.

That is why you should investigate a topic rather than simply memorizing the rules presented to you.

The Question to Ask

Whenever you begin a new topic, ask:

What kind of problem does this mathematical idea help us understand or solve?

For example:

- Fractions help us represent parts of a whole and quantities that are not necessarily whole numbers.
- Percentages help us express a quantity as a proportion out of one hundred.

- Algebra allows us to represent unknown quantities and relationships using symbols.
- Geometry helps us study shapes, sizes, distances, and spatial relationships.
- Statistics helps us organize and interpret data.
- Accounting calculations help people record and analyze financial information.

These explanations describe the uses of the topics. To learn their history more deeply, you can investigate when particular methods appeared, which people contributed to them, and how they developed.

You do not need to become a professional historian before solving a question. You simply need to develop the habit of asking why the topic matters.

1.4 Enter the Situation Before You Calculate

One of the most useful habits you can develop is to imagine yourself as the person who needs the answer.

Let us try three examples.

Situation A: You Are a Shopkeeper

You buy goods for ₦40,000 and sell them for ₦55,000. You want to know how much more money you received from sales than you spent buying the goods.

Before calculating, identify the two amounts and decide what their difference represents.

Situation B: You Are a Builder

You need to understand the relationship between a ladder, a vertical wall, and the ground. You know two sides of the right-angled triangle formed by these objects.

Before calculating, draw the situation and label the known measurements.

Situation C: You Are a Teacher

You have a list of examination scores and want one number that represents their arithmetic average.

Before calculating, ask what information should be combined and why the number of scores matters.

Notice that each situation begins with a need.

The shopkeeper wants to understand money received and money spent. The builder needs to understand a physical arrangement. The teacher wants to summarize a set of observations.

The calculations differ because the problems differ.

This is why it is not enough to look at numbers and immediately choose an operation. You must understand what the numbers represent and how they relate to the question.

1.5 The First Practical Calculation: Think Like a Shopkeeper

Let us now solve the shopkeeper's problem completely.

Imagine that you own a small shop.

At the beginning of the day, you purchase goods for ₦40,000. Later, you sell those goods for ₦55,000.

You want to determine the difference between the selling revenue and the purchase cost.

Step 1: Understand the Question

You are not being asked to add the two amounts.

You are being asked how much greater the selling revenue is than the purchase cost.

Step 2: Identify the Information

$$\text{Purchase cost} = ₦40{,}000$$

$$\text{Sales revenue} = ₦55{,}000$$

Step 3: Choose the Mathematical Operation

Subtraction is appropriate because we want to find the difference between the two amounts.

$$\text{Difference} = \text{Sales Revenue} - \text{Purchase Cost}$$

Step 4: Substitute the Values

$$\text{Difference} = ₦55{,}000 - ₦40{,}000$$

Subtract:

$$\text{Difference} = ₦15{,}000$$

Therefore, the difference is ₦15,000.

Step 5: Interpret the Answer

The sales revenue exceeds the purchase cost by ₦15,000.

However, we must be careful about what we call this amount.

If the shopkeeper also paid for transportation, electricity, wages, rent, and other expenses, those costs would reduce the final profit.

For example, suppose the additional expenses were ₦5,000.

Then:

$$\text{Profit} = \text{Sales Revenue} - \text{Purchase Cost} - \text{Additional Expenses}$$
$$\text{Profit} = ₦55{,}000 - ₦40{,}000 - ₦5{,}000$$
$$\text{Profit} = ₦10{,}000$$

This second calculation gives the profit after accounting for the additional expenses specified in the example.

Why This Calculation Is Useful

The shopkeeper does not calculate merely to obtain ₦15,000 or ₦10,000. The purpose is to understand the financial result of the business activity.

The numbers help the shopkeeper make decisions, such as whether the selling price covers the costs and whether the business is operating as expected.

This is the habit we want to develop: understand the decision behind the calculation.

1.6 The Second Practical Calculation: Think Like a Builder

Now imagine that you are helping a builder understand how long a ladder must be to reach a point on a wall.

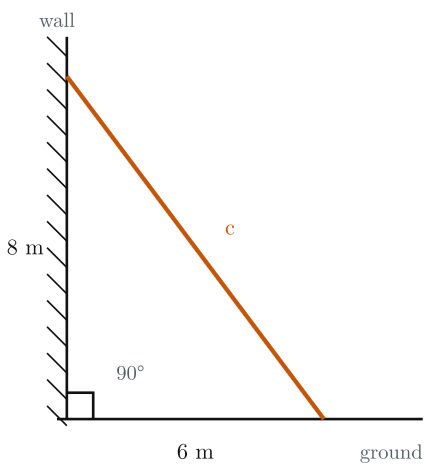
The ladder's foot is 6 metres from the wall, and the required height is 8 metres.

Assume that the wall and the ground meet at a right angle.

The ladder, wall, and ground form a right-angled triangle.

FIGURE 1

Ladder, Wall, and Ground



A vertical wall meets the ground at 90°. The ladder runs from 6 m along the ground to 8 m up the wall; its length c is the unknown.

Step 1: Understand the Question

We know the horizontal distance and the vertical height.

We need to find the diagonal length of the ladder.

Step 2: Identify the Mathematical Relationship

For a right-angled triangle, Pythagoras' theorem states:

$$a^2 + b^2 = c^2$$

Here, a and b are the two shorter sides, and c is the hypotenuse, the side opposite the right angle.

In our example:

$a = 6$, $b = 8$

The ladder is the hypotenuse.

Step 3: Substitute the Measurements

$$6^2 + 8^2 = c^2$$

The small raised number 2 means that the number is multiplied by itself.

Therefore:

$$6^2 = 6 \times 6 = 36$$

And:

$$8^2 = 8 \times 8 = 64$$

Our equation becomes:

$$36 + 64 = c^2$$

Step 4: Add the Numbers

$$100 = c^2$$

This tells us that the square of the ladder's length is 100.

But we need the length itself.

Step 5: Find the Square Root

The square root of a number is a value that, when multiplied by itself, gives the original number.

Since:

$$10 \times 10 = 100$$

we obtain:

$$c = \sqrt{100}$$
$$c = 10 \text{ metres}$$

We use the positive square root because a physical length is positive.

Step 6: Check the Answer

Substitute the result into the original relationship:

$$6^2 + 8^2 = 10^2$$
$$36 + 64 = 100$$
$$100 = 100$$

The equation is satisfied.

Under the assumptions of this geometric model, the ladder is 10 metres long.

Why Understanding Made the Calculation Easier

If you had memorized only the formula, you might have wondered which number belonged in each position.

But when you understood the situation, you could identify the two known sides and the unknown diagonal.

The diagram helped you see the relationship. The formula expressed that relationship. The calculation then followed from it.

This does not mean that understanding the history of Pythagoras' theorem alone will teach you every detail of the calculation. You still need to learn the theorem and practise using it.

The advantage is that you now know what you are calculating and why the method applies.

Practical note: A geometric result does not, by itself, establish that a ladder is safe to use. Real construction work requires appropriate equipment, stable placement, and proper safety procedures.

1.7 How to Read a Calculation Question

Many students struggle because they try to solve a question before understanding its wording.

Let us use a simple example.

A trader buys a bag of rice for ₦72,000 and sells it for ₦85,000. Calculate the difference between the selling price and the purchase price.

A student might immediately search for a formula.

But we can approach the question more carefully.

First, Identify the Action

The trader buys something and sells it.

The question asks for the difference between the two prices.

Second, Identify the Known Quantities

$$\text{Purchase price} = ₦72{,}000$$

$$\text{Selling price} = ₦85{,}000$$

Third, Identify the Unknown Quantity

We need the difference between the two prices.

Fourth, Select the Operation

Because we want the difference, we subtract the purchase price from the selling price.

$$₦85{,}000 - ₦72{,}000 = ₦13{,}000$$

Fifth, Explain the Answer

The selling price is ₦13,000 higher than the purchase price.

Notice that we did not need a complicated formula.

The important work was understanding what the question meant.

Now let us change the question.

What percentage of the purchase price is the difference?

The numbers remain the same, but the question has changed.

First, calculate the difference:

$$₦85{,}000 - ₦72{,}000 = ₦13{,}000$$

Next, divide the difference by the purchase price:

$$13{,}000 \div 72{,}000$$

Then multiply by 100 to express the ratio as a percentage:

$$(13{,}000 \div 72{,}000) \times 100$$

$$\approx 18.06\%$$

Therefore, the difference is approximately 18.06% of the purchase price.

Why did we divide by ₦72,000 rather than ₦85,000?

Because the question asks us to compare the difference with the purchase price. The reference quantity is the purchase price.

If the question instead asked for the difference as a percentage of the selling price, the denominator would be ₦85,000.

This is an important lesson:

> **The same numbers can require different calculations because the question asks for a different relationship.**

1.8 The Seven Questions to Ask Before Calculating

Whenever you meet a new calculation, pause and ask yourself these seven questions.

Question 1: What is happening?

Understand the situation in ordinary language.

Question 2: What am I being asked to find?

Identify the exact unknown quantity.

Question 3: What information do I already have?

List the known values and their units.

Question 4: What does each number represent?

A number might represent money, distance, time, a proportion, or something else. Its meaning affects the calculation.

Question 5: What relationship connects the known values to the unknown?

This is where you identify the mathematical principle or formula.

Question 6: Why does this method apply?

Check that the conditions of the method match the problem.

For example, Pythagoras' theorem applies to right-angled triangles, not to every triangle without distinction.

Question 7: How will I know whether my answer makes sense?

Check the arithmetic, units, assumptions, and meaning of the result.

These questions will not remove every difficulty. Some advanced problems require several methods and many steps. But they give you a reliable starting point.

1.9 How to Research a Mathematical Topic

The central idea of this book is that you should investigate a topic before treating its formula as something to memorize blindly.

But what does that mean in practice?

Suppose your next lesson is about percentages.

Instead of beginning only with the formula, you might investigate questions such as:

- What does the word *percent* mean?
- Why is a percentage expressed out of 100?
- How are percentages useful in trade, finance, measurement, and statistics?
- How do percentages relate to fractions and decimals?
- What is the difference between a percentage increase and a percentage-point increase?

Suppose you are learning Pythagoras' theorem.

You might investigate:

- What is a right-angled triangle?
- What is the hypotenuse?
- What relationship does the theorem describe?
- What historical evidence do we have about the theorem and the development of its mathematical ideas?
- Where is the theorem useful in surveying, construction, and measurement?

Suppose you are learning accounting.

You might investigate:

- Why do businesses keep financial records?
- What is the difference between revenue and profit?
- Why must expenses be recorded?
- What questions do financial statements help answer?

- Why must calculations be checked carefully?

A Simple Research Method

Use this four-part approach.

1. **Find the purpose.** Explain what problem the topic helps address.
2. **Check the history.** Use reliable books, educational sources, and historical records where available. Distinguish established facts from

uncertain claims.

3. **Study the relationship.** Learn why the formula or method works.
4. **Create a practical example.** Invent or investigate a realistic situation where the calculation would be useful.

For a short lesson, this research might take only a few minutes. For a research-based book, it may require much more careful reading and verification.

The point is not to delay every calculation unnecessarily. It is to build understanding that makes later calculations easier to recognize and apply.

1.10 Common Mistakes That Make Calculations Harder

Some mistakes arise not because a student cannot perform arithmetic, but because the student has not understood the problem properly.

Mistake 1: Choosing a Formula Too Early

A student sees numbers and immediately selects a formula without checking whether it applies.

How to improve: Explain the problem in your own words before selecting the formula.

Mistake 2: Confusing the Meaning of the Numbers

A student divides by the wrong quantity when calculating a percentage.

How to improve: Identify the reference quantity. Ask, “What is this amount being compared with?”

Mistake 3: Skipping Steps

A student tries to perform several operations mentally and loses track of the calculation.

How to improve: Write each important step clearly, especially when learning a new method.

Mistake 4: Ignoring Units

A student writes an answer of 60 without saying whether it represents metres, square metres, naira, or another quantity.

How to improve: Carry units through the calculation and include the appropriate unit in the final answer.

For example, length is measured in metres, while area is measured in square metres.

Mistake 5: Refusing to Check the Answer

A student obtains a number and immediately moves to the next question.

How to improve: Substitute the result into the original relationship, repeat the calculation by another valid method, or estimate the expected size of the answer.

Mistake 6: Believing That Difficulty Means Inability

A student encounters an unfamiliar question and assumes they cannot solve it.

How to improve: Break the problem into smaller questions. What is known? What is unknown? What relationship connects them? What is the first step?

You do not have to understand the entire problem instantly to make progress.

1.11 How to Make Calculations Look Easy

There is a difference between making a calculation look short and making it genuinely easier to understand.

A short calculation can hide several important steps. A longer explanation can make the reasoning clearer.

The goal is to understand a method well enough that you can eventually use it efficiently.

Here are five habits that help.

Habit 1: Draw a Picture When the Problem Is Physical

For geometry, construction, movement, and measurement, a diagram can help you identify relationships that are difficult to see from words alone.

Habit 2: Separate the Information from the Question

Write down what is known and what you need to find. This reduces confusion.

Habit 3: Learn Why Operations Work

Do not memorize only that you should subtract, divide, or take a square root. Understand what each operation is doing in the problem.

Habit 4: Estimate Before Calculating

If a shopkeeper receives ₦85,000 and spends ₦72,000 on the goods, you should expect the difference to be somewhere around ₦10,000, not ₦100,000.

An estimate can help you notice unreasonable results.

Habit 5: Practise with Different Situations

Solve several questions involving the same relationship, but use different numbers and contexts.

For example, Pythagoras' theorem can be applied to a ladder, a diagonal across a rectangular floor, or a distance calculated from perpendicular movements. The physical stories differ, but the underlying mathematical relationship can remain the same.

The more situations you understand, the easier it becomes to recognize when a method is useful.

1.12 Practical Workshop: You Are the Problem-Solver

Now it is your turn.

Do not rush to search for formulas. Read each situation and identify the problem before calculating.

Practical 1: The Shopkeeper's Decision

You purchase goods for ₦120,000 and sell them for ₦150,000. Other business expenses total ₦12,000.

Answer these questions:

1. What is the business trying to determine?
2. What is the difference between sales revenue and purchase cost?
3. What is the profit after the stated additional expenses?
4. Why would it be misleading to ignore the additional expenses when calculating profit?
5. What other information might a business owner need before deciding whether the business is performing well?

Practical 2: The Builder's Measurement

A right-angled triangle has two shorter sides measuring 9 metres and 12 metres. Find the hypotenuse.

Before calculating:

1. Explain why a triangle is relevant to the problem.
2. State the condition that makes Pythagoras' theorem applicable.
3. Write the theorem.
4. Substitute the values.
5. Show how you calculate each square.
6. Find the hypotenuse.
7. Verify your answer.

Practical 3: The Teacher's Records

A teacher records five examination scores:

10, 12, 14, 16, 18.

The teacher wants the arithmetic mean.

First ask what an average is intended to summarize.

Then add the five scores:

$$10 + 12 + 14 + 16 + 18 = 70$$

There are five scores, so divide their total by five:

$$\text{Mean} = 70 \div 5$$

$$\text{Mean} = 14$$

Now answer these questions:

1. Why did we add the scores?
2. Why did we divide by five?
3. Does a mean of 14 mean that every student scored 14?
4. What additional information might the teacher need to understand the class's performance?

Practical 4: Investigate Before You Calculate

Choose one of these topics:

- Fractions
- Percentages
- Algebra
- Pythagoras' theorem
- Profit and loss
- Statistics

Write a short investigation containing:

1. What the topic means.
2. What kind of problem it helps solve.

3. What you can establish about its history.
4. One practical situation in which it is useful.
5. One worked calculation, with an explanation of every important step.

The aim is to connect the topic's meaning, history, and practical application.

1.13 Chapter Review

Before moving to the next chapter, make sure you can explain these ideas in your own words.

1. Why is understanding a problem important before selecting a formula?
2. How can a practical situation help you understand a calculation?
3. What is the difference between memorizing a formula and understanding its meaning?
4. Why should historical claims about mathematical discoveries be checked?
5. How can diagrams, estimation, and step-by-step working make calculations easier?
6. Why should you interpret and check your final answer?

Remember that the aim is not to avoid formulas. It is to learn to use them intelligently.

1.14 Conclusion: Become the Person Who Needs the Answer

Imagine returning to the classroom tomorrow.

Your teacher introduces a new calculation topic. This time, before asking which formula you should memorize, you ask what kind of problem the topic helps solve.

You investigate the purpose of the method. You study the relationship it describes. You imagine yourself facing the practical problem. Then you learn the formula, work through the steps, and check the answer.

You may still make mistakes. You may need to ask questions or practise more examples. That is a normal part of learning.

But you will have something more useful than the ability to repeat a procedure: you will be developing the ability to reason through a problem.

The great thinkers whose work we study did not all approach mathematics in the same way. They lived in different periods, investigated different questions, and made different contributions. Their work reminds us that mathematical knowledge develops through human effort, thought, and discovery.

We cannot know which ideas we might have discovered if we had lived in their circumstances. But we can learn to ask better questions and understand the methods they left behind.

The next time you see a calculation, do not be in a hurry to reach for the formula.

Look at the problem.

Understand the situation.

Ask what you need to find.

Then begin.

> **A formula becomes much more useful when you understand the problem it is meant to solve.**

That is the first lesson of *Think Before the Formula*.

CHAPTER 2

Why Calculations Were Created

02

2.1 Introduction: Mathematics Began with Human Problems

Imagine living thousands of years ago.

There are no calculators, no computers, no smartphones, and no modern textbooks. You are a farmer who needs to divide a harvest among your family. You are a trader who wants to exchange goods with another trader. You are a builder who needs to construct a house that will not collapse.

You face problems that require answers.

How much grain do you have? How many animals belong to you? How much land do you need? How can you divide your harvest fairly? How will you know whether the goods you received are equal in value to the goods you gave away?

You cannot simply ignore these questions. Your survival, business, family, and community may depend on finding the right answers.

This is where the story of calculation begins.

Mathematics did not begin as a collection of formulas printed in a textbook. It developed over a long period as human beings found ways to count, measure, compare, divide, predict, and solve practical problems.

People needed to understand the world around them. Mathematics became one of the tools they developed to do so.

In this chapter, we will explore why calculations became necessary, how different human activities encouraged mathematical development, and how understanding the original problem behind a calculation can make it easier to learn today.

The central lesson is simple:

> **To understand a calculation, first understand the problem that made the calculation necessary.**

2.2 The First Problem: How Do You Count What You Have?

Imagine that you are a farmer living in an early agricultural community.

You own a number of goats. Every morning, you take them out to graze. In the evening, you bring them back.

At first, you may recognize each animal individually. But as your herd grows, remembering every animal becomes more difficult.

How can you know whether all your goats have returned?

You need a reliable method of counting.

One possible method is to use physical objects. For every goat that leaves, you place one small stone in a container. When the goats return, you match each returning animal with one stone.

If every stone has a matching goat, your count agrees with the number of animals that returned. If stones remain after all the goats have returned, you have a reason to investigate whether an animal is missing.

This example illustrates an important idea: people needed ways to represent quantities even before modern written arithmetic became widespread.

Physical objects, marks, and other counting methods could help people keep track of what they owned or exchanged.

As societies developed, counting became increasingly important for agriculture, trade, taxation, construction, and administration.

Why Is This Important Today?

Consider a shopkeeper who wants to know how many bags of rice remain in a store.

The shopkeeper begins with 80 bags and sells 23.

How many bags remain?

We can represent the situation as:

$$\text{Number remaining} = \text{Initial number} - \text{Number sold}$$

$$\text{Number remaining} = 80 - 23$$

$$\text{Number remaining} = 57 \text{ bags}$$

Notice what happened.

We did not begin by asking, “Which formula should I memorize?”

We began with a real situation:

- The shopkeeper had 80 bags.
- Some bags were sold.
- The number of bags decreased.
- We therefore subtracted the number sold from the original quantity.

The calculation became understandable because we understood the event.

Whenever you encounter a mathematical problem, try to imagine the actual situation before working with the numbers.

2.3 The Second Problem: How Do You Divide Things Fairly?

Suppose a farmer has 12 oranges and wants to share them equally among 4 children.

How many oranges should each child receive?

You could distribute the oranges one at a time:

- Give one orange to each child.
- Repeat the process.
- Continue until all the oranges have been shared.

Each child receives 3 oranges.

Mathematics expresses this process more briefly:

$$12 \div 4 = 3$$

Division is useful because it helps us determine how a quantity can be shared into equal groups.

But why do we divide instead of add or multiply?

Because the problem asks for the number of oranges each child receives when 12 oranges are shared equally among 4 children.

The meaning of the problem determines the operation.

Another Example: Sharing Money

A group of 5 students contributes money to buy materials for a class project. Together, they contribute ₦10,000. They decide to share the cost equally.

Each student's contribution is:

$$₦10{,}000 \div 5 = ₦2{,}000$$

Each student contributes ₦2,000.

Now imagine a different situation.

Five students each contribute ₦2,000. How much money do they have altogether?

This time, we multiply:

$$5 \times ₦2{,}000 = ₦10{,}000$$

The numbers are similar, but the questions are different.

In the first situation, we know the total and want to find each person's share. We divide.

In the second situation, we know the number of people and each person's contribution. We multiply.

> **The operation is chosen because of the relationship between the quantities, not merely because certain numbers appear in the question.**

2.4 The Third Problem: How Do You Measure What You Cannot Count Easily?

Counting works well for separate objects such as books, chairs, or oranges.

But what if you want to know the length of a road, the area of a field, or the amount of water in a container?

You cannot solve every measurement problem simply by counting individual objects.

You need units of measurement.

A unit provides a standard quantity against which another quantity can be compared.

For example:

- A metre is a unit of length.
- A square metre is a unit of area.
- A kilogram is a unit of mass.
- A litre is a unit of volume.

- A second is a unit of time.

Measurement allows people to describe quantities consistently.

Imagine that two builders are discussing the length of a wooden beam.

One says the beam is 5 metres long. The other says it is 500 centimetres long.

Are they describing different lengths?

No.

Since 1 metre equals 100 centimetres:

$$5 \text{ metres} = 5 \times 100 \text{ centimetres}$$

$$5 \text{ metres} = 500 \text{ centimetres}$$

The two measurements describe the same length using different units.

This is why understanding units matters. A numerical answer without an appropriate unit may be incomplete or misleading.

A Practical Example: Measuring a Room

Suppose a rectangular room is 6 metres long and 4 metres wide.

You want to cover the floor with tiles.

The first question is not simply, “Which formula do I use?”

Ask yourself:

What information do I have?

I know the length and width of the floor.

What do I want to find?

I want the amount of floor area that needs to be covered.

What mathematical relationship connects these quantities?

For a rectangle, the area is the length multiplied by the width.

Therefore:

$$\text{Area} = \text{Length} \times \text{Width}$$

$$\text{Area} = 6 \times 4$$

$$\text{Area} = 24 \text{ square metres}$$

The floor has an area of 24 m^2.

Why do we use square metres rather than metres?

Because we are measuring an area, which covers a two-dimensional surface. The unit reflects the quantity being measured.

This is an example of how practical needs encourage mathematical ideas and how understanding the meaning of a measurement helps us choose the correct calculation.

2.5 The Fourth Problem: How Do You Trade Without Confusion?

Trade has been one of the important reasons for the development and use of arithmetic.

When people exchange goods, they need ways to compare quantities and values.

Imagine that a trader exchanges 3 baskets of yams for 6 baskets of oranges.

The trader may want to know how many baskets of oranges are received for each basket of yams.

The rate is:

$$6 \div 3 = 2$$

The trader receives 2 baskets of oranges for every basket of yams exchanged.

This is an example of a rate: a comparison between two quantities.

Rates appear in many situations today:

- kilometres travelled per hour;
- price per kilogram;
- pages read per day;
- wages earned per hour;
- litres of fuel used per kilometre.

The same underlying idea appears in different forms.

Example: Comparing Prices

A shop sells a 2-kilogram bag of rice for ₦6,000.

Another shop sells a 5-kilogram bag for ₦14,000.

Which bag offers the lower price per kilogram?

For the first bag:

$$₦6{,}000 \div 2 = ₦3{,}000 \text{ per kilogram}$$

For the second bag:

$$₦14{,}000 \div 5 = ₦2{,}800 \text{ per kilogram}$$

The second bag has the lower price per kilogram.

However, the buyer should also consider other factors, such as quality, transport costs, available money, and whether the larger quantity will be used before it spoils.

The calculation helps answer a specific question. It does not automatically answer every question involved in making the decision.

This distinction is important.

Mathematics provides evidence for decisions, but understanding the wider situation helps us use that evidence wisely.

2.6 The Fifth Problem: How Do You Plan for the Future?

Human beings do not only need to understand what is happening now. They also need to make plans.

Farmers want to estimate harvests. Traders want to plan purchases. Builders want to estimate materials. Governments want to prepare budgets. Families want to manage expenses.

These activities require calculations involving quantities that may change.

Imagine a farmer who plants 100 seeds. Suppose, for a simple planning exercise, the farmer expects 80% of them to grow successfully.

How many plants does the farmer expect?

First, remember that 80% means 80 out of every 100.

Therefore:

$$80\% = \frac{80}{100} = 0.8$$

Expected number of plants:

$$0.8 \times 100 = 80$$

The farmer expects 80 plants.

Notice that the answer is an expectation based on an assumed success rate, not a guarantee that exactly 80 seeds will grow.

Real situations contain uncertainty. Weather, soil conditions, pests, and other factors may affect the result.

Mathematics helps people estimate and plan, but the reliability of a calculation depends partly on the quality of the information used.

Why Percentages Became Useful

Percentages allow us to express a proportion out of 100.

They make it easier to compare quantities of different sizes.

For example, suppose one student answers 8 questions correctly out of 10, while another answers 36 questions correctly out of 40.

The first student's percentage is:

$$(8 \div 10) \times 100\% = 80\%$$

The second student's percentage is:

$$(36 \div 40) \times 100\% = 90\%$$

The second student has the higher percentage of correct answers.

Without converting the results into a common form, comparing 8 out of 10 with 36 out of 40 may be less immediate.

Percentages provide a shared scale for comparison.

They are now widely used in education, business, economics, science, statistics, and everyday financial decisions.

2.7 The Historical Development of Mathematics

The history of mathematics is not the story of one person inventing every calculation at once.

It is a long story involving many societies, generations of practical experience, written records, and intellectual discoveries.

Different civilizations developed mathematical methods to address their needs. Later scholars learned from earlier traditions, adapted existing ideas, and developed new ones.

Understanding this history helps us see mathematics as a human achieve-ment that grew over time.

2.7.1 Ancient Mesopotamia

Mesopotamia was a historical region associated with the land between the Tigris and Euphrates rivers, largely in present-day Iraq and surrounding areas.

Ancient Mesopotamian societies used mathematics in administration, trade, land measurement, astronomy, and other activities.

Some Babylonian mathematical records were written on clay tablets. These records show sophisticated methods of calculation, including work with multiplication, division, fractions, and problems involving unknown quantities.

Babylonian mathematics is also associated with a base-60 number system.

This historical system helps explain why the number 60 remains important in how we measure time and angles.

For example:

- 1 minute contains 60 seconds.
- 1 hour contains 60 minutes.
- A full circle contains 360 degrees.

The modern use of these units developed through historical traditions rather than being chosen randomly in every case.

The important lesson is that the mathematical systems people use today have histories. Some conventions have survived because they became useful, widely adopted, or deeply established in measurement systems.

2.7.2 Ancient Egypt

Ancient Egyptian mathematics was used in activities such as accounting, surveying, construction, and the administration of resources.

Egyptian mathematical documents contain problems involving arithmetic, fractions, and practical measurements.

The annual flooding of the Nile affected agricultural land, making land measurement and the administration of property important concerns.

Builders and administrators also needed ways to calculate quantities of materials and manage resources.

It would be inaccurate to imagine that every ancient calculation was performed exactly as it would be taught in a modern classroom. Mathematical notation and methods differed across periods and cultures.

Nevertheless, surviving Egyptian records demonstrate that people developed organized techniques for solving numerical problems long before modern schools existed.

When you study fractions or area today, you are learning mathematical ideas that have long histories, even though the notation and methods used in classrooms have changed.

2.7.3 Ancient Greece

Ancient Greek mathematicians made important contributions to mathematical reasoning, geometry, and the development of formal proofs.

A practical problem might ask how to find the length of a side of a triangle. A mathematical investigation could go further and ask why a relationship between the sides must be true.

This difference is important.

Calculation produces a result. Mathematical reasoning seeks to explain why a method works and under what conditions it can be trusted.

Euclid, who lived around the third century BCE, organized a large body of mathematical knowledge in his influential work known as *Elements*. It presented definitions, propositions, and proofs in a systematic structure.

The Pythagorean theorem is associated with the ancient Greek mathematical tradition, although knowledge of related numerical relationships existed in other cultures before or around the period of Greek mathematical development.

The theorem states that in a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

In symbols:

$$a^2 + b^2 = c^2$$

Here, a and b represent the shorter sides of the right-angled triangle, while c represents the hypotenuse, the side opposite the right angle.

This relationship is useful in construction, surveying, navigation, and many other fields.

Its importance goes beyond memorizing the equation. It gives us a reliable relationship that can be used to find an unknown side when the other two sides are known.

2.7.4 Indian Mathematics

Mathematical traditions in India made major contributions to arithmetic, algebra, and the development and use of the decimal place-value system with zero.

A place-value system allows the position of a digit to determine its value.

Consider the number 555.

The first 5 represents 500.

The second 5 represents 50.

The third 5 represents 5.

Therefore:

$$555 = 500 + 50 + 5$$

The digits are the same, but their positions give them different values.

The development and transmission of the decimal place-value system were historical processes involving scholars and communities across regions. Indian mathematical traditions played a central role in the system's development, and knowledge of it later spread through other parts of the world.

The use of zero as a number and as a place-value marker made the system especially powerful.

Consider the difference between 5, 50, and 500.

Zero helps distinguish these quantities clearly.

Modern arithmetic, calculators, computers, financial systems, and scientific calculations depend heavily on place-value notation.

A symbol that may seem ordinary in a textbook is part of a major historical development.

2.7.5 The Development of Algebra

Algebra provides methods for representing relationships and solving problems involving unknown quantities.

Imagine that a trader has some bags of rice. The trader purchases 5 additional bags and now has 17 bags.

How many bags did the trader originally have?

We could solve the problem by reasoning backward:

$$17 - 5 = 12$$

The trader originally had 12 bags.

Algebra represents the unknown quantity with a symbol.

Let x represent the original number of bags.

Then:

$$x + 5 = 17$$

Subtract 5 from both sides:

$$x = 17 - 5$$

Therefore:

$$x = 12$$

The symbol x does not make the problem more mysterious. It gives a name to the quantity we do not yet know.

Algebra developed through the contributions of many cultures and scholars. Its history includes mathematical traditions in ancient civilizations, important work by Greek and Indian mathematicians, and major develop-ments in the medieval Islamic world and later European mathematics.

The word *algebra* comes through Arabic from a term used in the title of a mathematical work by the ninth-century scholar al-Khwarizmi.

Over time, algebra developed into a general language for describing mathematical relationships.

It allows us to solve not only one problem about bags of rice but also thousands of different problems that share the same structure.

2.8 From Practical Problems to Mathematical Rules

Let us examine how a practical problem can develop into a general mathematical rule.

Imagine that you are helping a farmer calculate the total number of oranges in several equal-

sized baskets.

Each basket contains 8 oranges.

If there is 1 basket, the total is:

$$1 \times 8 = 8$$

If there are 2 baskets:

$$2 \times 8 = 16$$

If there are 3 baskets:

$$3 \times 8 = 24$$

If there are 10 baskets:

$$10 \times 8 = 80$$

Instead of solving every possible situation separately, we can express the relationship generally.

Let n represent the number of baskets.

The total number of oranges is:

$$\text{Total oranges} = 8n$$

This expression works for any non-negative whole number of baskets.

The practical problem has revealed a pattern. The pattern can then be expressed mathematically.

This is one of the most powerful features of mathematics: a rule discovered from one type of situation can often be applied to many similar situations.

However, we must always pay attention to the conditions under which a rule applies.

If the baskets contain different numbers of oranges, the expression 8n may no longer give the correct total.

A useful mathematical rule is not merely a short expression. It is an expression that correctly represents a relationship under specified conditions.

2.9 Why Memorizing Formulas Is Not Enough

Imagine two students preparing for an examination.

Both students memorize the formula for the area of a rectangle:

$$\text{Area} = \text{Length} \times \text{Width}$$

The first student remembers the formula but does not understand what it means.

The second student understands that area measures the amount of surface covered by the rectangle and that multiplying the length by the width gives the area when both measurements are expressed in compatible units.

Now consider the following question:

A rectangular floor is 8 metres long and has an area of 40 square metres. Find its width.

The first student may hesitate because the question does not directly ask for the area.

The second student recognizes that the relationship is:

$$\text{Area} = \text{Length} \times \text{Width}$$

Substitute the known quantities:

$$40 = 8 \times \text{Width}$$

Divide both sides by 8:

$$\text{Width} = 40 \div 8$$
$$\text{Width} = 5 \text{ metres}$$

The second student understands that the same relationship can be rearranged to find a missing quantity.

This is the advantage of understanding a formula rather than merely memorizing it.

A Formula Is a Relationship

A formula describes a relationship between quantities.

For example:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

This relationship can help us calculate distance when speed and time are known.

It can also be rearranged to calculate speed:

$$\text{Speed} = \text{Distance} \div \text{Time}$$

Or time:

$$\text{Time} = \text{Distance} \div \text{Speed}$$

These forms express the same underlying relationship, assuming the quantities are measured consistently and the speed is constant over the journey.

Understanding the relationship helps you decide which form is appropriate for the question.

When you learn a formula, ask:

1. What does each symbol represent?
2. What relationship does the formula express?
3. Why are these quantities multiplied, divided, added, or subtracted?
4. What information must be known before the formula can be used?
5. What assumptions must be true?
6. What units should the answer have?
7. How can I check whether the answer makes sense?

These questions help transform a formula from something you memorize into something you understand.

2.10 How Different Professions Use Calculations

Mathematics becomes easier to appreciate when we see how different people use it to solve real problems.

Let us examine a few professions.

2.10.1 The Accountant

An accountant works with financial information.

Suppose a business earns ₦250,000 in revenue during a particular period and has expenses of ₦180,000.

The difference is:

$$\text{Profit} = \text{Revenue} - \text{Expenses}$$
$$\text{Profit} = ₦250{,}000 - ₦180{,}000$$
$$\text{Profit} = ₦70{,}000$$

The accountant is not subtracting simply because subtraction is the topic of a lesson. The accountant is comparing money earned with money spent to determine the difference.

Real accounting involves more detailed rules concerning when revenue and expenses are recognized, as well as assets, liabilities, taxes, and other matters. Nevertheless, the example demonstrates the basic relationship.

2.10.2 The Builder

A builder needs to estimate quantities of materials, measure lengths, calculate areas, and check dimensions.

Suppose a rectangular wall is 5 metres wide and 3 metres high.

Its surface area is:

$$\text{Area} = 5 \times 3$$

$$\text{Area} = 15\,\text{m}^2$$

This measurement may help estimate the amount of paint required.

However, the builder must also consider the coverage of the paint, the number of coats, and the areas occupied by windows and doors.

The first calculation gives the wall's gross area. It does not automatically give the exact amount of paint required.

Understanding the real problem tells the builder which additional information is necessary.

2.10.3 The Economist

An economist studies how people, businesses, and societies make decisions about resources.

Suppose the price of a product increases from ₦2,000 to ₦2,500.

The increase is:

$$₦2{,}500 - ₦2{,}000 = ₦500$$

To calculate the percentage increase, divide the increase by the original price and multiply by 100%.

$$\text{Percentage increase} = (\text{Increase} \div \text{Original price}) \times 100\%$$
$$\text{Percentage increase} = (500 \div 2{,}000) \times 100\%$$
$$\text{Percentage increase} = 25\%$$

The price increased by 25%.

Notice why we divide by the original price. We want to measure the increase relative to the starting value.

If we divided by the new price instead, we would answer a different question.

The meaning of the calculation determines the correct reference quantity.

2.10.4 The Scientist

Scientists use mathematics to measure observations, compare results, test explanations, and construct models.

Suppose a scientist measures the temperature of a substance every minute during an experiment.

The recorded values may help reveal whether the temperature is rising, falling, or remaining approximately constant.

The scientist may use averages, graphs, rates of change, and statistical methods to interpret the measurements.

But the calculations do not replace careful observation or a well-designed experiment.

If the measurements are inaccurate, the conclusions may also be unreliable.

Mathematics helps scientists reason from evidence, while scientific methods help them decide whether the evidence is trustworthy.

2.10.5 The Teacher

A teacher may use calculations to summarize student performance.

Suppose five students receive the following scores:

10, 12, 14, 16, 18

The mean score is calculated by adding the scores and dividing by the number of students.

$$\text{Mean} = (10 + 12 + 14 + 16 + 18) \div 5$$
$$\text{Mean} = 70 \div 5$$
$$\text{Mean} = 14$$

The mean score is 14.

This gives the teacher one useful summary of the group. However, the mean alone does not reveal every student's performance or explain why students achieved those scores.

Different mathematical summaries answer different questions.

The skill is not only knowing how to calculate a mean. It is understanding what the mean tells us and what it does not tell us.

2.11 The Importance of Asking the Right Question

Sometimes the biggest difficulty in a mathematical problem is not the calculation itself. It is identifying what the problem is asking.

Imagine a trader who purchases goods for ₦50,000 and sells them for ₦65,000.

The trader asks, "How much did I gain?"

The difference is:

$$₦65{,}000 - ₦50{,}000 = ₦15{,}000$$

If the trader has no other relevant costs, the difference represents the amount gained over the purchase cost.

But suppose the trader also paid ₦4,000 for transportation and ₦2,000 for other business expenses.

The amount remaining after these costs is:

$$₦15{,}000 - ₦4{,}000 - ₦2{,}000 = ₦9{,}000$$

The result changes because the question becomes more detailed.

The original calculation compared the selling price with the purchase cost. The second calculation also considered additional expenses.

Now suppose the trader asks, “What percentage profit did I make?”

We must clarify which cost forms the basis of the percentage.

Using the purchase cost alone:

$$\text{Percentage profit} = (₦15{,}000 \div ₦50{,}000) \times 100\%$$
$$\text{Percentage profit} = 30\%$$

If the additional expenses are included in the total cost:

$$\text{Total cost} = ₦50{,}000 + ₦4{,}000 + ₦2{,}000$$
$$\text{Total cost} = ₦56{,}000$$
$$\text{Profit after these expenses} = ₦65{,}000 - ₦56{,}000$$
$$\text{Profit after these expenses} = ₦9{,}000$$
$$\text{Percentage profit on total cost} = (₦9{,}000 \div ₦56{,}000) \times 100\%$$
$$\text{Percentage profit on total cost} \approx 16.07\%$$

Both percentages arise from different definitions of cost and profit.

This example teaches a vital lesson: a correct calculation can still give a misleading answer if the question is not clearly defined.

Before calculating, identify exactly what you are trying to measure.

2.12 A Simple Method for Understanding Any Calculation

You can use the following method whenever you face a mathematical problem.

Step 1: Read the Situation

Read the problem carefully. Do not rush to the numbers.

Ask yourself what is happening.

Is something being shared? Is a quantity increasing? Is a distance being measured? Are two values being compared?

Step 2: Identify What You Know

Write down the information provided.

For example:

A car travels at 60 kilometres per hour for 3 hours.

Known information:

$$\text{Speed} = 60\,\text{km/h}$$
$$\text{Time} = 3\ \text{hours}$$

Step 3: Identify What You Need

What is the question asking you to find?

In this case, we want the distance travelled.

Step 4: Find the Relationship

$$\text{Distance} = \text{Speed} \times \text{Time}$$

This relationship is appropriate because the car's speed is assumed to remain constant during the three hours.

Step 5: Substitute the Values

$$\text{Distance} = 60 \times 3$$
$$\text{Distance} = 180 \text{ kilometres}$$

Step 6: Check the Units

Speed is measured in kilometres per hour, while time is measured in hours.

Multiplying them gives kilometres per hour multiplied by hours, leaving kilometres.

The units agree with the quantity we are trying to find.

Step 7: Ask Whether the Answer Makes Sense

If the car travels 60 kilometres every hour for 3 hours, a total distance of 180 kilometres is reasonable under the constant-speed assumption.

This final check can help you identify mistakes before you accept an answer.

2.13 Making Calculations Easier Through Estimation

You do not always need to begin with an exact calculation.

Sometimes, estimating the answer first makes the exact calculation easier to check.

Suppose you want to calculate:

$$198 \times 51$$

You could multiply the numbers directly.

But first, estimate:

198 is close to 200.

51 is close to 50.

Therefore:

198×51 is approximately 200×50.

$$200 \times 50 = 10{,}000$$

We expect the exact answer to be close to 10,000.

Now calculate accurately:

$$\begin{aligned} 198 \times 51 &= 198 \times (50 + 1) \\ &= (198 \times 50) + (198 \times 1) \\ &= 9{,}900 + 198 \\ &= 10{,}098 \end{aligned}$$

The exact answer is 10,098, which is close to our estimate.

The estimate provides a useful check. If your exact answer had been 1,098 or 100,980, you would have reason to examine your work.

Why Estimation Matters

Estimation is useful because it helps you:

- recognize answers that are far too large or too small;
- check mental calculations;
- make quick decisions in everyday situations;
- identify possible input or arithmetic errors;
- understand the approximate size of a quantity before finding its exact value.

Estimation is not a replacement for exact calculation when precision is required. It is a supporting skill that helps you reason more effectively.

2.14 Practical Workshop: Discover the Problem Behind the Calculation

The following exercises are designed to help you think before using a formula.

Do not immediately search for a formula. Begin by explaining the situation in your own words.

Exercise 1: The Farmer

A farmer collects 96 eggs and packs them equally into 8 containers.

Answer the following questions:

1. What is the farmer trying to do?
2. Which mathematical operation is appropriate?
3. Why is that operation appropriate?
4. How many eggs go into each container?

Exercise 2: The Trader

A trader buys a bag of rice for ₦48,000 and sells it for ₦60,000. The trader also spends ₦3,000 on transportation.

Calculate:

1. The difference between the selling price and purchase price.
2. The amount remaining after transportation expenses.
3. The amount remaining as a percentage of the purchase price.

Explain why the second and third answers require you to consider information beyond the original purchase and selling prices.

Exercise 3: The Builder

A rectangular room is 7 metres long and 5 metres wide.

Calculate:

1. The floor area.
2. The total area of the floor if a second identical room is added.
3. Why the answer to the first question is expressed in square metres rather than metres.

Exercise 4: The Student

A student scores 15, 18, 12, 20, and 15 in five tests.

Calculate the mean score.

Then answer:

1. What does the mean tell you?
2. Does the mean show the student's score in every individual test?
3. Why might a teacher need additional information to understand the student's performance fully?

Exercise 5: The Journey

A bus travels at a constant speed of 70 kilometres per hour for 4 hours.

Calculate the distance travelled.

Then explain:

1. What each number represents.
2. Why multiplication is appropriate.
3. What assumption is being made about the bus's speed.

Exercise 6: Research a Mathematical Idea

Choose one of the following topics:

- Fractions
- Percentages
- Geometry
- Algebra
- Calculus
- Statistics

Research its historical development using reliable sources.

Write a short account answering these questions:

1. What kinds of problems has the topic helped people solve?
2. Which societies or mathematicians made important contributions to its development?
3. How has the topic changed over time?
4. Where is it used today?
5. What part of its history surprised you most?

When researching history, distinguish between a well-supported historical fact and a simplified story that may leave out important details. Mathematics often develops through contributions from many people and cultures rather than through one isolated discovery.

2.15 Chapter Review

Before moving to the next chapter, make sure you understand these key ideas.

1. **Mathematics developed to help people solve problems.**

Counting, measuring, sharing, trading, building, and planning all encouraged the development and use of mathematical methods.

2. **Mathematical operations have meanings.**

Addition combines quantities, subtraction finds differences, multiplication can represent repeated equal groups, and division can represent sharing or determining how many groups fit into a quantity.

3. **Measurement requires appropriate units.**

A length, area, mass, volume, and time are different types of quantities. Their units must match what is being measured.

4. **Mathematical history involves many societies.**

Mesopotamian, Egyptian, Greek, Indian, Islamic-world, and other mathematical traditions contributed to the development and transmission of mathematical knowledge.

5. **Formulas represent relationships.**

A formula becomes easier to understand when you know what its symbols represent, why the quantities are connected, and when the relationship applies.

6. **The right question matters.**

A calculation must answer the question actually being asked. Extra costs, assumptions, and definitions may change the result.

7. **Estimation helps you check answers.**

An approximate answer gives you a useful way to judge whether an exact answer is reasonable.

2.16 Conclusion: Mathematics Is a Human Tool

The next time you open a mathematics textbook, remember that every calculation has a meaning.

A subtraction problem may represent a shopkeeper calculating what remains after expenses. A percentage may help a trader compare prices. An equation may represent an unknown quantity. A geometrical relationship may help a builder determine a length. A statistical calculation may help a teacher summarize examination results.

The symbols and formulas are not the whole story.

Behind them are quantities, relationships, decisions, and human problems that require solutions.

When you understand the situation, the calculation often becomes more natural. When you understand why a method works, you can adapt it to unfamiliar problems. When you learn to question your assumptions, you become less likely to accept an answer simply because it looks mathematical.

This is the habit we are developing throughout this book.

Do not begin by asking only, “Which formula should I use?”

Begin by asking:

“What is happening, what do I need to find, and why would this calculation help me find it?”

Once you can answer those questions, you are ready to calculate with understanding.

CHAPTER 3

Numbers, Fractions and Percentages

03

3.1 Introduction: The Numbers Have a Story

Imagine that you are a trader in a busy market.

In the morning, you purchase 20 bags of rice. By the afternoon, you have sold 8 bags. Before closing your shop, you want to know how many bags remain.

You calculate:

$$20 - 8 = 12$$

Twelve bags remain.

Now imagine that you want to divide the remaining bags equally among three customers. You calculate:

$$12 \div 3 = 4$$

Each customer receives four bags.

Later, another trader tells you that the price of rice has increased by 15%. You need to understand what that percentage means before deciding whether the new price is reasonable.

All these situations involve numbers, but they do not all ask the same question.

One asks how many objects remain. Another asks how a quantity can be shared. A third asks how one value compares with another.

This is why understanding numbers involves more than learning to perform arithmetic.

You must understand what numbers represent, how they relate to one another, and why certain operations are appropriate for particular situations. In this chapter, we will explore numbers, fractions, decimals, ratios, and

percentages through practical problems. We will also examine how these mathematical ideas help people make decisions in business, education, construction, economics, and everyday life.

Our goal is not simply to calculate correctly.

Our goal is to understand what the calculation means.

3.2 Understanding Numbers: What Are We Actually Counting?

Suppose you enter a classroom and see 35 students.

The number 35 tells you how many students are present.

Now suppose a student scores 35 marks in a test. The same number represents a score.

If a trader sells a product for ₦35, the number represents an amount of money.

If a journey takes 35 minutes, the number represents a duration.

The numeral is the same, but its meaning depends on the quantity it describes.

This distinction matters because numbers do not always represent the same kind of thing.

Consider the following examples:

- 12 books represent a count of objects.

- 12 metres represent a length.
- ₦12,000 represents money.
- 12 seconds represent time.
- 12°C represents temperature.

Although each example contains the number 12, you cannot treat all the quantities as interchangeable.

For instance, adding 12 metres to 12 seconds does not produce a meaningful physical measurement.

Before calculating, identify the quantities involved.

Ask yourself:

1. What does each number represent?
2. What unit is attached to it?
3. Can the quantities be combined or compared meaningfully?
4. What does the question require me to find?

These questions may seem simple, but they help prevent many mathematical mistakes.

3.3 The Different Kinds of Numbers

Mathematicians classify numbers into groups because different kinds of numbers are useful for different purposes.

Understanding these groups helps you recognize the numbers you are working with.

3.3.1 Natural Numbers

Natural numbers are used for counting.

They are commonly written as:

1, 2, 3, 4, 5, 6, ...

Some mathematical conventions include zero among the natural numbers, while others begin with 1. When the distinction matters, the convention should be stated.

Consider a shopkeeper counting bags of flour.

There may be:

1 bag, 2 bags, 3 bags, and so on.

Natural numbers are useful for describing these whole-number counts.

However, they are not sufficient for every situation.

What happens if you have no bags of flour? What if you owe someone money? What if you divide one cake into four equal pieces?

These situations require other kinds of numbers or representations.

3.3.2 Whole Numbers

Whole numbers include zero and the positive counting numbers:

0, 1, 2, 3, 4, 5, ...

Suppose a shopkeeper has sold every item in a box.

The number of items remaining is zero.

Zero is useful because it represents the absence of a quantity being counted.

It does not mean that nothing can be described mathematically. Zero has an important role in arithmetic, algebra, measurement, and computing.

For example:

$$8 - 8 = 0$$

The answer tells us that the two equal quantities cancel when one is subtracted from the other.

3.3.3 Integers

Integers include negative whole numbers, zero, and positive whole numbers.

They can be represented as:

..., -3, -2, -1, 0, 1, 2, 3, ...

Integers are useful when quantities can lie above or below a reference point.

For example, suppose the temperature is measured relative to zero degrees Celsius.

A temperature of 5°C is five degrees above zero.

A temperature of -5°C is five degrees below zero.

Integers can also represent changes in a bank balance, elevation relative to a reference level, or gains and losses in a simplified accounting model.

Imagine that your account balance is ₦2,000, but you owe ₦3,000.

If we represent the debt as negative relative to zero, the net position is:

$$₦2{,}000 - ₦3{,}000 = -₦1{,}000$$

The negative result indicates that the amount owed exceeds the money available by ₦1,000.

The negative sign is not decoration. It communicates the direction or nature of the quantity relative to the chosen reference point.

3.3.4 Rational Numbers

A rational number is a number that can be expressed as a fraction of two integers, provided the denominator is not zero.

The general form is:

a/b, where a and b are integers and $b \neq 0$.

Examples include:

1/2, 3/4, $-5/2$, 7, and 0.

A whole number can also be expressed as a fraction:

$$7 = \frac{7}{1}$$

A terminating decimal such as 0.25 is rational because:

$$0.25 = \frac{25}{100} = \frac{1}{4}$$

A repeating decimal such as 0.333... is rational because:

$$0.333\ldots = \frac{1}{3}$$

Rational numbers are especially useful when quantities are divided into parts or compared through ratios.

3.3.5 Irrational Numbers

Irrational numbers cannot be expressed exactly as a fraction of two integers.

Their decimal expansions continue indefinitely without repeating in a fixed pattern.

Two familiar examples are:

π and $\sqrt{2}$.

The number π is important in calculations involving circles. The number $\sqrt{2}$ appears in geometry, including the calculation of the diagonal of a square with side length 1 unit.

These numbers may seem unusual, but they arise naturally when we study shapes and mathematical relationships.

For practical calculations, we often use approximations.

For example:

$$\pi \approx 3.14159$$

The symbol $\approx$ means “approximately equal to.”

The value 3.14159 is not the exact value of π. It is a useful approximation.

This distinction matters whenever precision is important.

3.3.6 Real Numbers

Real numbers include both rational and irrational numbers.

They can be represented on the number line.

Examples include:

-4, 0, 2.5, $3/4$, $\sqrt{2}$, and π.

Real numbers are widely used in measurement, geometry, algebra, calculus, and scientific modelling.

For now, the most important point is that different problems require different kinds of numbers.

Counting objects may involve whole numbers. Measuring lengths may involve decimals. Describing debt may involve negative numbers. Calculating the diagonal of a square may introduce an irrational number.

Understanding the situation helps you understand which numbers are appropriate.

3.4 Why Negative Numbers Are Useful

Negative numbers can be confusing when students first encounter them because it may seem strange to have a quantity that is less than zero.

The confusion often disappears when we connect negative numbers to a reference point.

Imagine a building with a ground floor labelled 0.

Floors above the ground floor are labelled:

1, 2, 3, 4, ...

Basement floors below the ground floor are labelled:

-1, -2, -3, ...

The negative numbers indicate positions below the reference level.

The same idea can be used to describe temperature relative to zero, changes in a financial balance, or movement in the opposite direction along a line.

Example 1: A Change in Money

A person has ₦5,000 and spends ₦7,000.

The change in the person's money is:

$$₦5{,}000 - ₦7{,}000 = -₦2{,}000$$

In a simplified model, the result indicates a shortfall of ₦2,000 relative to the original amount available.

The calculation makes sense because we are comparing two quantities with the same unit.

Example 2: Movement Along a Road

Suppose a person walks 8 metres east and then walks 3 metres west.

If east is defined as the positive direction, the movement can be represented as:

$$+8 + (-3) = +5$$

The person's net displacement is 5 metres east of the starting point.

Notice the difference between total distance and displacement.

Total distance travelled:

$$8 + 3 = 11 \text{ metres}$$

Net displacement:

$$8 - 3 = 5 \text{ metres east}$$

The two calculations answer different questions.

Distance tells us how much ground was covered. Displacement tells us the net change in position.

This is another example of why understanding the question is essential before choosing an operation.

3.5 Fractions: Understanding Parts of a Whole

Imagine that you buy a pizza and divide it into four equal pieces.

You eat one piece.

What fraction of the pizza have you eaten?

You have eaten:

$$\frac{1}{4}$$

The number above the line is called the numerator. It tells us how many parts are being considered.

The number below the line is called the denominator. It tells us how many equal parts make up the whole.

Therefore, in the fraction 1/4:

- Numerator = 1
- Denominator = 4

The fraction represents one part out of four equal parts.

But why do we need fractions?

Because many real-life quantities cannot be described adequately using whole numbers alone.

You may own half a loaf of bread, spend three-quarters of your money, use two-thirds of a bottle of paint, or travel one-and-a-half kilometres.

Fractions allow us to describe these parts precisely.

Example 1: Sharing a Cake

A cake is divided equally among 6 students.

One student receives:

1/6 of the cake.

If two students each receive one piece, they receive:

$$\frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

This can be simplified to:

$$\frac{2}{6} = \frac{1}{3}$$

Therefore, two of the six equal pieces make up one-third of the cake.

The fraction tells us how much of the original whole is being considered.

Example 2: A Tank of Water

Suppose a water tank is three-quarters full.

The fraction is:

$$\frac{3}{4}$$

This means that the tank contains three of the four equal portions that would make up its full capacity.

If the tank can hold 200 litres, how much water does it contain?

First, divide the full capacity by 4:

$$200 \div 4 = 50 \text{ litres}$$

Each quarter holds 50 litres.

Then multiply by 3:

$$50 \times 3 = 150 \text{ litres}$$

The tank contains 150 litres of water.

We could also calculate:

$$\frac{3}{4} \times 200 = 150 \text{ litres}$$

The formula becomes understandable when we see that we are finding three equal quarters of the total capacity.

3.6 Equivalent Fractions: Different Forms, Same Value

Suppose one student eats half a cake.

Another student eats two-quarters of an identical cake.

Have they eaten the same fraction of their cakes?

Yes.

Because:

$$\frac{1}{2} = \frac{2}{4}$$

Both fractions represent the same proportion of the whole.

Equivalent fractions are fractions that have different numerators and denominators but represent the same value.

To create an equivalent fraction, multiply or divide the numerator and denominator by the same non-zero number.

For example:

$$\frac{2}{3} = \frac{2 \times 2}{3 \times 2}$$

$$\frac{2}{3} = \frac{4}{6}$$

Similarly:

$$\frac{6}{8} = \frac{6 \div 2}{8 \div 2}$$

$$\frac{6}{8} = \frac{3}{4}$$

The value has not changed. Only the representation has changed.

Why Are Equivalent Fractions Useful?

Suppose you want to add:

$$\frac{1}{2} + \frac{1}{4}$$

You cannot simply add the numerators and denominators to obtain 2/6. That would be incorrect because the parts represented by the two denominators are not the same size.

Imagine a cake divided into two equal pieces. Each piece is half the cake.

Now imagine the same cake divided into four equal pieces. Each piece is one-quarter of the cake.

A half is equal to two quarters.

Therefore:

$$\frac{1}{2} = \frac{2}{4}$$

Now the fractions have the same denominator:

$$\frac{1}{2} + \frac{1}{4} = \frac{2}{4} + \frac{1}{4}$$

$$= \frac{3}{4}$$

The answer is three-quarters.

We use equivalent fractions because addition and subtraction of fractions require us to combine quantities expressed in compatible-sized parts.

3.7 Adding and Subtracting Fractions

Let us explore this carefully.

Example 1: Adding Fractions with the Same Denominator

Suppose you drink 2/7 of a bottle of juice in the morning and another 3/7 in the afternoon.

How much of the bottle have you consumed?

The denominator is the same in both fractions.

Therefore:

$$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$$

You have consumed five-sevenths of the bottle.

Why do we add only the numerators?

Because both fractions count the same kind of part: sevenths.

You have 2 sevenths and then another 3 sevenths, giving 5 sevenths.

Example 2: Subtracting Fractions with the Same Denominator

A container is 7/8 full. You use 3/8 of its capacity.

How much remains?

$$\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$$

Simplify:

$$\frac{4}{8} = \frac{1}{2}$$

Half the container's capacity remains.

Example 3: Adding Fractions with Different Denominators

Calculate:

$$\frac{1}{3} + \frac{1}{6}$$

The denominators are different, so first express the fractions using a common denominator.

Since $\frac{1}{3} = 2/6$:

$$\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6}$$

$$= \frac{3}{6}$$

Simplify:

$$\frac{3}{6} = \frac{1}{2}$$

Therefore:

$$\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

Example 4: Subtracting Fractions with Different Denominators

Calculate:

$$\frac{3}{4} - \frac{1}{6}$$

A common denominator for 4 and 6 is 12.

Convert each fraction:

$$\frac{3}{4} = \frac{9}{12}$$
$$\frac{1}{6} = \frac{2}{12}$$

Subtract:

$$\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$$

The answer is 7/12.

The Principle Behind the Method

When fractions have different denominators, we convert them into equivalent fractions with a common denominator.

This allows us to combine or compare parts of equal size.

Instead of memorizing a mysterious procedure, remember the meaning of the fractions.

You cannot directly combine three quarters and two sixths by simply adding their numerators and denominators because quarters and sixths represent differently sized parts.

A common denominator gives both fractions a shared unit.

3.8 Multiplying Fractions

Multiplication of fractions is useful when we need to find a fraction of another quantity.

Suppose a farmer owns 24 hectares of land and uses three-quarters of the land to grow maize.

How many hectares are used for maize?

We calculate:

$$\frac{3}{4} \times 24$$

First, divide 24 by 4:

$$24 \div 4 = 6$$

Then multiply by 3:

$$6 \times 3 = 18$$

The farmer uses 18 hectares for maize.

The calculation works because we are finding three equal quarters of 24 hectares.

Example 2: Finding a Fraction of a Fraction

Suppose a container is 3/4 full, and you use 1/2 of the water currently in it.

What fraction of the container's full capacity have you used?

You are finding half of three-quarters.

Therefore:

$$\frac{1}{2} \times \frac{3}{4} = \frac{1 \times 3}{2 \times 4}$$
$$= \frac{3}{8}$$

You have used 3/8 of the container's full capacity.

The word "of" often indicates multiplication when dealing with fractions and percentages, although the full context should always be checked.

The General Rule

To multiply fractions, multiply the numerators together and multiply the denominators together.

$\frac{a}{b} \times c/d = \frac{ac}{bd}$

provided b and d are not zero.

For example:

$$\frac{2}{5} \times \frac{3}{7} = \frac{6}{35}$$

The result is 6/35.

In practical problems, interpret the answer in relation to the original quantities.

3.9 Dividing Fractions

Division asks us to determine how many times one quantity fits into another, or to determine a quantity per unit.

Suppose you have 3/4 of a metre of ribbon.

You want to cut it into pieces that are each 1/8 of a metre long.

How many pieces can you make?

We calculate:

$$\frac{3}{4} \div \frac{1}{8}$$

Convert 3/4 into eighths:

$$\frac{3}{4} = \frac{6}{8}$$

Now ask how many pieces of size 1/8 fit into 6/8.

There are 6 pieces.

Therefore:

$$\frac{3}{4} \div \frac{1}{8} = 6$$

The calculation tells us that six pieces can be made, assuming there is no material lost during cutting.

The General Rule

To divide by a fraction, multiply by its reciprocal.

$\frac{a}{b} \div c/d = a/b \times d/\mathrm{c}$

where the denominators are non-zero and the divisor c/d is not zero.

For example:

$$\frac{2}{3} \div \frac{4}{5}$$
$$= \frac{2}{3} \times \frac{5}{4}$$
$$= \frac{10}{12}$$
$$= \frac{5}{6}$$

The answer is 5/6.

Why does this work?

Dividing by a quantity is equivalent to multiplying by the amount of that quantity contained in one unit, expressed through its reciprocal.

You do not need to master the full theoretical explanation immediately, but you should understand the practical meaning: fraction division often asks how many pieces of a particular size fit into another quantity.

3.10 Decimals: Another Way to Represent Fractions

Fractions are not the only way to represent parts of a whole.

Decimals use place value and a decimal point to represent numbers that include fractional parts.

Consider:

0.5

This means five-tenths:

$$0.5 = \frac{5}{10} = \frac{1}{2}$$

Similarly:

$$0.25 = \frac{25}{100} = \frac{1}{4}$$

And:

$$0.75 = \frac{75}{100} = \frac{3}{4}$$

Decimals are especially common in money, measurement, scientific calculations, and calculators.

Example 1: Money

Suppose a product costs ₦2,500.50.

The decimal part represents fifty kobo in the currency's decimal subdivision.

The notation allows the amount to be written compactly.

Example 2: Length

A wooden board is 2.75 metres long.

This can be expressed as:

$$2.75\,\text{m} = 2\,\text{m} + 0.75\,\text{m}$$

Since $0.75 = 3/4$:

$$2.75\,\text{m} = 2\tfrac{3}{4}\,\text{m}$$

Both forms represent the same length.

Example 3: Converting a Fraction to a Decimal

Convert 3/8 to a decimal.

Divide 3 by 8:

$$3 \div 8 = 0.375$$

Therefore:

$$\frac{3}{8} = 0.375$$

Example 4: Converting a Decimal to a Fraction

Convert 0.6 to a fraction in its simplest form.

Write 0.6 as six-tenths:

$$0.6 = \frac{6}{10}$$

Divide the numerator and denominator by 2:

$$\frac{6}{10} = \frac{3}{5}$$

Therefore:

$$0.6 = \frac{3}{5}$$

Decimals and fractions are different representations of numerical values. Choosing the most convenient representation can make calculations easier.

3.11 Why Place Value Matters

Consider the numbers:

7, 70, 700, and 7,000.

Each number contains the digit 7, but the position of the digit changes its value.

In 7, the digit represents seven units.

In 70, it represents seven tens.

In 700, it represents seven hundreds.

In 7,000, it represents seven thousands.

The decimal system is a base-10 place-value system.

Each position to the left represents ten times the value of the position immediately to its right.

For example:

$$4{,}582 = 4{,}000 + 500 + 80 + 2$$

This can also be written as:

$$4 \times 1{,}000 + 5 \times 100 + 8 \times 10 + 2 \times 1$$

The same principle continues to the right of the decimal point.

For example:

$$3.47 = 3 + \frac{4}{10} + \frac{7}{100}$$

The 4 represents four-tenths, while the 7 represents seven-hundredths.

This explains why 3.47 is different from 3.407.

In 3.47, the 4 represents four-tenths.

In 3.407, the 4 represents four-tenths and the 7 represents seven-thousandths, with zero tenths? More precisely, the digits represent:

$$3.407 = 3 + \frac{4}{10} + \frac{0}{100} + \frac{7}{1000}$$

The two decimals have different values because their digits occupy different places.

Place value is essential for understanding arithmetic, decimals, rounding, percentages, and scientific notation.

3.12 Ratios: Comparing Quantities

A ratio compares two quantities by division.

Suppose a class contains 12 boys and 18 girls.

The ratio of boys to girls is:

12:18

Both numbers can be divided by 6:

$$12{:}18 = 2{:}3$$

The simplified ratio is 2:3.

This means that for every 2 boys, there are 3 girls in the stated comparison.

The ratio does not mean that the class contains only 5 students. It describes the relative sizes of the two groups.

The actual class contains:

$12 + 18 = 30$ students.

Example 1: Mixing a Drink

Suppose a drink is prepared using 2 cups of concentrate for every 5 cups of water.

The ratio of concentrate to water is:

2:5

If you want to prepare a larger quantity while keeping the same proportions, multiply both parts of the ratio by the same number.

For three times the original quantity:

$$2 \times 3 = 6 \text{ cups of concentrate}$$

$$5 \times 3 = 15 \text{ cups of water}$$

The new ratio is:

$$6{:}15 = 2{:}5$$

The proportions remain the same.

Example 2: A Map

A map uses a scale of 1:100,000.

This means that one unit of length on the map represents 100,000 of the same units on the ground.

If 1 centimetre on the map represents 100,000 centimetres in reality, then:

$$100{,}000\ \text{cm} = 1{,}000\,\text{m} = 1\ \text{km}$$

Therefore, 1 centimetre on the map represents 1 kilometre on the ground.

If two places are 4 centimetres apart on the map, their represented straight-line distance is:

$$4 \times 1\ \text{km} = 4\ \text{km}$$

The actual route between the places may be longer because roads do not necessarily follow a straight line.

This illustrates an important point: a ratio or scale represents a particular relationship, and the result must be interpreted according to what that relationship measures.

3.13 Percentages: Comparing Values Out of One Hundred

The word percentage refers to a quantity expressed per hundred.

The symbol is %.

Therefore:

$$1\% = \frac{1}{100}$$

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$50\% = \frac{50}{100} = \frac{1}{2}$$

$$75\% = \frac{75}{100} = \frac{3}{4}$$

$$100\% = \frac{100}{100} = 1$$

A percentage expresses a proportion relative to a reference quantity.

This is why percentages are useful for comparing test results, price changes, profits, discounts, interest rates, and many other quantities.

Example 1: Examination Results

A student answers 36 questions correctly out of 40.

What percentage of the questions did the student answer correctly?

First, divide the number of correct answers by the total number of questions:

$$36 \div 40 = 0.9$$

Multiply by 100%:

$$0.9 \times 100\% = 90\%$$

The student answered 90% of the questions correctly.

The denominator is 40 because that is the total number of questions.

Example 2: A Discount

A bag costs ₦20,000, and a shop offers a 10% discount.

How much is the discount?

Convert 10% to a decimal:

$$10\% = \frac{10}{100} = 0.10$$

Multiply by the original price:

$$0.10 \times ₦20{,}000 = ₦2{,}000$$

The discount is ₦2,000.

The new price is:

$$₦20{,}000 - ₦2{,}000 = ₦18{,}000$$

The customer pays ₦18,000, assuming there are no additional charges.

Example 3: Finding a Percentage of a Quantity

A school has 800 students. If 35% participate in a particular activity, how many students participate?

Convert the percentage to a decimal:

$$35\% = 0.35$$

Multiply by the total:

$$0.35 \times 800 = 280$$

Therefore, 280 students participate.

The method works because we are finding 35 hundredths of the total number of students.

3.14 The Three Main Types of Percentage Problems

Many percentage questions can be classified into three groups.

Understanding which type you are facing helps you choose the right calculation.

Type 1: Finding the Percentage

You know the part and the whole, and you want to know what percentage the part represents.

Example:

A student scores 45 marks out of 60.

$$\text{Percentage} = (\text{Part} \div \text{Whole}) \times 100\%$$

$$\text{Percentage} = (45 \div 60) \times 100\%$$

$$\text{Percentage} = 75\%$$

The student scores 75%.

Type 2: Finding the Part

You know the whole and the percentage, and you want to calculate the corresponding part.

Example:

A shop offers a 20% discount on an item priced at ₦15,000.

$$\text{Discount} = 20\% \times ₦15{,}000$$

$$\text{Discount} = 0.20 \times ₦15{,}000$$

$$\text{Discount} = ₦3{,}000$$

The discount is ₦3,000.

Type 3: Finding the Whole

You know a part and the percentage it represents, and you want to find the original whole.

Example:

A student scores 30 marks, representing 75% of the total possible marks.

What is the total possible score?

Let the total score be x.

Then:

$$75\% \text{ of } x = 30$$

$$0.75x = 30$$

Divide both sides by 0.75:

$$x = 30 \div 0.75$$

$$x = 40$$

The total possible score is 40 marks.

The key is to identify which quantity is missing.

Do not memorize a single percentage procedure and apply it to every question without checking what is known and what must be found.

3.15 Percentage Increase and Decrease

Percentages are frequently used to describe changes in quantities.

Examples include:

- an increase in the price of a product;
- a reduction in transport fares;
- growth in business revenue;
- a decrease in the number of customers;
- a rise in a student's examination score.

The percentage change is calculated relative to the original value.

Example 1: Price Increase

A product's price rises from ₦4,000 to ₦5,000.

Step 1: Calculate the increase.

$$\text{Increase} = ₦5{,}000 - ₦4{,}000$$

$$\text{Increase} = ₦1{,}000$$

Step 2: Divide the increase by the original price.

$$₦1{,}000 \div ₦4{,}000 = 0.25$$

Step 3: Convert to a percentage.

$$0.25 \times 100\% = 25\%$$

The price increased by 25%.

Example 2: Price Decrease

A product's price falls from ₦8,000 to ₦6,000.

Step 1: Calculate the decrease.

$$\text{Decrease} = ₦8{,}000 - ₦6{,}000$$

$$\text{Decrease} = ₦2{,}000$$

Step 2: Divide by the original price.

$$₦2{,}000 \div ₦8{,}000 = 0.25$$

Step 3: Convert to a percentage.

$$0.25 \times 100\% = 25\%$$

The price decreased by 25%.

Why Must We Use the Original Value?

The original value provides the reference point for measuring the change.

A change of ₦1,000 means something different when the original price is ₦2,000 than when the original price is ₦100,000.

For the first price:

$$(₦1{,}000 \div ₦2{,}000) \times 100\% = 50\%$$

For the second price:

$$(₦1{,}000 \div ₦100{,}000) \times 100\% = 1\%$$

The absolute change is the same, but the percentage change is different.

The percentage communicates the size of the change relative to the starting value.

3.16 A Common Mistake: Equal Percentage Changes Do Not Always Cancel

Suppose a product costs ₦10,000.

Its price increases by 20%.

New price:

$$\begin{aligned} &₦10{,}000 + (0.20 \times ₦10{,}000) \\ &= ₦10{,}000 + ₦2{,}000 \\ &= ₦12{,}000 \end{aligned}$$

Later, the new price decreases by 20%.

The decrease is calculated from ₦12,000, not from the original ₦10,000.

Decrease:

$$0.20 \times ₦12{,}000 = ₦2{,}400$$

Final price:

$$₦12{,}000 - ₦2{,}400 = ₦9{,}600$$

The final price is ₦9,600.

It is not ₦10,000.

Why?

The first percentage increase was calculated using ₦10,000 as the reference value. The second percentage decrease was calculated using ₦12,000 as the reference value.

The reference quantities were different.

This is why a 20% increase followed by a 20% decrease does not generally return a quantity to its original value.

The result depends on the base to which each percentage is applied.

3.17 Ratios and Percentages in Economics and Business

Ratios and percentages are particularly important when comparing financial information.

Imagine that two businesses report the following figures.

Business A earns ₦100,000 in revenue and records ₦20,000 in profit.

Business B earns ₦500,000 in revenue and records ₦50,000 in profit.

Which business earns more profit in total?

Business A earns ₦20,000.

Business B earns ₦50,000.

Business B earns more profit in absolute terms.

But what percentage of revenue is represented by profit in each business?

For Business A:

$$\begin{aligned}\text{Profit margin} &= (\text{Profit} \div \text{Revenue}) \times 100\% \\ &= (₦20{,}000 \div ₦100{,}000) \times 100\% \\ &= 20\%\end{aligned}$$

For Business B:

$$\begin{aligned}\text{Profit margin} &= (₦50{,}000 \div ₦500{,}000) \times 100\% \\ &= 10\%\end{aligned}$$

Business A has a higher profit margin, while Business B has a higher total profit.

These are different measures.

A business with a higher profit margin is not automatically more successful in every respect. Other factors, such as the scale of operations, risk, growth, debt, and future costs, may also matter.

Mathematics helps separate questions that may sound similar but require different answers.

3.18 Making Fraction and Percentage Calculations Easier

You can simplify many calculations by choosing a convenient method.

Method 1: Simplify Before Multiplying

Calculate:

$$\frac{3}{4} \times 20$$

Instead of performing unnecessary steps, divide 20 by 4 first:

$$20 \div 4 = 5$$

Then multiply by 3:

$$5 \times 3 = 15$$

Therefore:

$$\frac{3}{4} \times 20 = 15$$

Method 2: Use Familiar Percentages

Some percentages have simple fractional equivalents.

For example:

$$50\% = \frac{1}{2}$$

$$25\% = \frac{1}{4}$$

$$75\% = \frac{3}{4}$$

$$10\% = \frac{1}{10}$$

If you want to calculate 25% of ₦12,000, find one-quarter:

$$₦12{,}000 \div 4 = ₦3{,}000$$

Therefore, 25% of ₦12,000 is ₦3,000.

This may be faster than converting the percentage into a decimal and multiplying.

Method 3: Find 10% First

Suppose you want to calculate 30% of ₦7,000.

First, find 10%:

$$₦7{,}000 \div 10 = ₦700$$

Then multiply by 3:

$$₦700 \times 3 = ₦2{,}100$$

Therefore:

$$30\% \text{ of } ₦7{,}000 = ₦2{,}100$$

This works because 30% is three times 10%.

Method 4: Break Numbers into Convenient Parts

Calculate:

15% of ₦8,000

You can break 15% into $10\% + 5\%$.

$$10\% \text{ of } ₦8{,}000 = ₦800$$
$$5\% \text{ of } ₦8{,}000 = \text{half of } ₦800 = ₦400$$

Add the results:

$$₦800 + ₦400 = ₦1{,}200$$

Therefore, 15% of ₦8,000 is ₦1,200.

Method 5: Estimate Before Calculating

Suppose you need to calculate 19% of ₦4,950.

Before calculating exactly, estimate using 20% of ₦5,000.

$$20\% \text{ of } ₦5{,}000 = ₦1{,}000$$

The exact answer should be slightly less than ₦1,000.

Now calculate:

$$\begin{aligned} 19\% \text{ of } ₦4{,}950 &= 0.19 \times ₦4{,}950 \\ &= ₦940.50 \end{aligned}$$

The result is reasonably close to the estimate.

Estimation helps you recognize whether the answer is plausible.

3.19 Choosing the Best Representation

A quantity can sometimes be represented as a fraction, decimal, or percentage.

For example:

$$\frac{1}{2} = 0.5 = 50\%$$

These forms have the same value.

Which should you use?

It depends on the problem.

If you are dividing a cake into equal pieces, the fraction 1/2 may be the clearest representation.

If you are entering a value into a calculator, 0.5 may be convenient.

If you are comparing examination results, 50% may be easier to interpret.

The same principle applies to other quantities.

A fraction may be convenient for exact proportional reasoning. A decimal may be useful for measurement or computation. A percentage may be ideal for comparing a part with a whole.

The strongest mathematical thinker is not someone who uses one representation for everything. It is someone who understands the available representations and chooses the one that best communicates the answer.

3.20 Practical Workshop

Work through the following questions carefully.

For every problem, identify what is known, what must be found, and why your chosen operation is appropriate.

Exercise 1: Classifying Numbers

Identify the most suitable number category or categories for each number.

1. 7
2. 0
3. -9
4. $3/5$
5. 0.25
6. $\sqrt{2}$
7. π
8. 4.75

Remember that some numbers belong to more than one category.

Exercise 2: Fractions

A farmer has 48 oranges.

1. One-quarter of the oranges are given to a neighbour. How many oranges are given away?

2. One-third of the original 48 oranges are sold. How many are sold?
3. What fraction of the original oranges remains if 12 oranges are removed?
4. Explain the difference between finding one-quarter of 48 and dividing 48 by 4.

Exercise 3: Fraction Operations

Calculate and simplify where possible.

1. $\frac{2}{7} + 3/7$
2. $\frac{5}{6} - 1/3$
3. $\frac{3}{4} \times 8/9$
4. $\frac{2}{5} \div 4/15$
5. $\frac{1}{2} + 2/3$

For each question, explain why the method works.

Exercise 4: Decimals

1. Convert $3/5$ to a decimal.
2. Convert 0.875 to a fraction in its simplest form.
3. Calculate $12.5 + 3.75$.
4. Calculate $20 - 7.65$.
5. Explain why 4.08 is greater than 4.008.

Exercise 5: Ratios

A class contains 15 boys and 25 girls.

1. Write the ratio of boys to girls in its simplest form.
2. Write the ratio of girls to boys in its simplest form.
3. Calculate the total number of students.
4. What fraction of the class consists of girls?
5. Express the proportion of girls as a percentage.

Exercise 6: Percentages

1. Calculate 15% of ₦20,000.
2. A student scores 42 out of 50. Find the percentage score.
3. A product priced at ₦30,000 receives a 12% discount. Find the discount and the new price.
4. A price rises from ₦5,000 to ₦6,500. Calculate the percentage increase.
5. A price falls from ₦12,000 to ₦9,000. Calculate the percentage decrease.
6. A student scores 36 marks, representing 80% of the total possible score. Find the total possible score.

Exercise 7: The Business Decision

A trader buys 40 bags of rice for ₦1,600,000 in total.

The trader sells all the bags for ₦1,840,000 and spends ₦80,000 on transportation.

Calculate:

1. The total purchase cost.
2. The difference between the selling price and purchase cost.
3. The amount remaining after transportation expenses.
4. The remaining amount as a percentage of the purchase cost.
5. Why the percentage in question 4 is not necessarily the same as a profit margin calculated using sales revenue as the denominator.

Exercise 8: Create Your Own Problem

Choose a real-life situation involving money, food, distance, time, school results, or business.

Write a short mathematical problem about it.

Then solve it using the following structure:

The situation: Explain what is happening.

Known information: List the quantities given.

What I need to find: State the question clearly.

The mathematical relationship: Explain why the chosen operation is appropriate.

The calculation: Show each step.

The meaning of the answer: Interpret the result in the context of the problem.

The check: Explain how you know the answer is reasonable.

3.21 Chapter Summary

In this chapter, we have explored several mathematical ideas and their practical meanings.

Numbers help us represent quantities, positions, measurements, and relationships.

Natural numbers and whole numbers are useful for counting. Integers help represent quantities relative to a reference point. Rational numbers include fractions and terminating or repeating decimals, while irrational numbers include values such as π and $\sqrt{2}$.

Fractions represent parts of a whole or ratios between quantities. Equivalent fractions allow us to express the same value in different forms. Addition and subtraction require compatible fractional units, multiplication can find a fraction of a quantity, and division can determine how many pieces of a given size fit into another quantity.

Decimals provide another way to represent fractional values. Place value explains why the position of a digit determines its contribution to a number.

Ratios compare quantities, while percentages express proportions relative to one hundred. Percentage calculations are useful for discounts, examination scores, price changes, profit comparisons, and many other situations.

Above all, the chapter has emphasized one principle:

> **The correct mathematical method begins with understanding the relationship between the quantities involved.**

Do not treat fractions, decimals, ratios, and percentages as unrelated procedures. They are different ways of representing and comparing quantities.

3.22 Conclusion: Learn to See the Meaning Behind the Numbers

Imagine two students looking at the same problem.

One sees only numbers and searches through memory for a formula.

The other sees a situation, identifies the quantities, understands their relationships, and then selects a mathematical method.

Both students may eventually reach the same answer. But the second student is better prepared for unfamiliar problems because the reasoning does not depend entirely on memorization.

That is the habit you should continue developing.

When you see a fraction, ask what whole is being divided and what the parts represent.

When you see a percentage, ask what quantity provides the reference point.

When you see a ratio, ask which quantities are being compared.

When you see a negative number, ask what reference point gives it meaning.

When you see a decimal, pay attention to place value and units.

Mathematics becomes more manageable when you stop treating numbers as isolated symbols and begin seeing them as representations of real quantities and relationships.

In the next chapter, we will enter the world of builders, shapes, measurements, and the Pythagorean theorem. We will discover how geometry can help us solve practical problems by understanding the structure of the objects around us.

CHAPTER 4

Think Like a Builder — Geometry and Pythagoras

04

4.1 Introduction: Before the Building, There Is a Problem

Imagine that you are a builder.

A customer asks you to construct a house. Before you can begin, you must answer several questions.

How long should the walls be? How wide should each room be? How much land will the building occupy? How many tiles will cover the floor? How much paint will the walls require? How can you ensure that the roof is properly supported?

These are not merely mathematical questions. They are practical problems involving materials, space, money, safety, and design.

A builder cannot simply choose numbers at random. Each measurement must correspond to something real.

If a room is too small, it may not serve its intended purpose. If a measurement is incorrect, materials may be wasted. If structural calculations are wrong, the consequences can be serious.

This is why builders use geometry.

Geometry is the branch of mathematics concerned with shapes, sizes, positions, distances, angles, and spatial relationships.

It helps us understand how objects are arranged, how large surfaces are, how much space objects occupy, and how different measurements are connected.

But geometry becomes easier when we stop treating its formulas as mysterious rules.

Instead of beginning with the question, “Which formula should I memorize?”, begin with these questions:

- What am I building or measuring?
- What information do I already have?
- What quantity do I need to find?
- How are the measurements related?
- What calculation represents that relationship?

In this chapter, we will think like builders and discover how geometry helps solve real problems.

We will explore lengths, perimeter, area, volume, angles, triangles, and the Pythagorean theorem.

Most importantly, we will learn to understand the problem before choosing the formula.

4.2 Understanding Geometry Through Real Objects

Look around you.

A door is approximately rectangular. A football may be represented by a sphere. A water tank may be cylindrical. A roof may contain triangular sections. A floor is often rectangular or square.

Geometry gives us mathematical ways to describe these shapes.

Consider a rectangular classroom.

It has a length, a width, and a height.

These measurements describe different dimensions of the same room.

The length and width can help us calculate the floor area. The length, width, and height can help us calculate the room's volume.

Although all three measurements describe the room, they answer different questions.

If you want to buy tiles for the floor, you need the floor area.

If you want to estimate the amount of air inside the room, you need its volume.

If you want to place a fence around a rectangular plot, you need its perimeter.

The first lesson of geometry is therefore simple:

> **Identify what you are measuring before choosing a mathematical method.**

Let us examine these ideas more carefully.

4.3 Length: Measuring How Far Something Extends

Length measures the distance between two points or the extent of an object in a particular direction.

We measure length using units such as millimetres, centimetres, metres, and kilometres.

For example:

- The thickness of a sheet of paper may be measured in millimetres.
- The length of a book may be measured in centimetres.
- The length of a classroom may be measured in metres.
- The distance between two towns may be measured in kilometres.

The choice of unit depends on the size of the object or distance.

Imagine that a carpenter needs a wooden board measuring 2 metres.

The carpenter must determine the correct length before cutting the board.

If the board is cut too short, it may not fit the intended space. If it is cut too long, additional work may be required.

The measurement is therefore not an abstract number. It guides a physical action.

Example 1: Converting Units

A wooden board is 3.5 metres long.

How many centimetres long is it?

We know that:

$$1 \text{ metre} = 100 \text{ centimetres}$$

Therefore:

$$3.5 \text{ metres} = 3.5 \times 100 \text{ centimetres}$$

$$= 350 \text{ centimetres}$$

The board is 350 cm long.

Example 2: Comparing Lengths

A pipe is 240 cm long. Another pipe is 1.8 m long.

Which pipe is longer?

Convert both lengths to the same unit.

First pipe:

240 cm

Second pipe:

$$1.8 \times 100 = 180 \text{ cm}$$

Now compare:

$240 \text{ cm} > 180 \text{ cm}$

The first pipe is longer.

The important step was converting both measurements to the same unit before comparing them.

4.4 Perimeter: How Much Boundary Must You Cover?

Imagine that you own a rectangular plot of land.

You want to build a fence around its boundary.

You need to know the total distance around the plot.

This distance is called the perimeter.

The perimeter of a shape is the total length of its outer boundary.

For a rectangle, the opposite sides are equal.

If the length is l and the width is w, the perimeter is:

$$P = l + w + l + w$$

Combining equal terms gives:

$$P = 2l + 2w$$

Therefore:

$$\boxed{P = 2(l + w)}$$

This formula is not an arbitrary rule. It represents the addition of the four sides of a rectangle.

Example 1: Fencing a Garden

A rectangular garden is 12 metres long and 7 metres wide.

How much fencing is required to go around its boundary?

The garden has:

- Two sides of length 12 m.
- Two sides of length 7 m.

Add all four sides:

$$P = 12 + 7 + 12 + 7$$
$$P = 38 \text{ m}$$

Alternatively:

$$P = 2(12 + 7)$$
$$P = 2(19)$$
$$P = 38 \text{ m}$$

The perimeter is 38 metres.

If the fencing is sold in fixed lengths, the builder may need to purchase more than exactly 38 metres to allow for gates, overlaps, cutting, and installation requirements.

The calculation provides the boundary length, while the actual purchasing decision depends on the construction details.

Example 2: Finding a Missing Length

A rectangular plot has a perimeter of 50 metres and a width of 10 metres.

Find its length.

We know:

$$P = 2(l + w)$$

Substitute the known values:

$$50 = 2(l + 10)$$

Divide both sides by 2:

$$25 = l + 10$$

Subtract 10 from both sides:

$$l = 25 - 10$$

$$\boxed{l = 15 \text{ m}}$$

The length is 15 metres.

Notice that we did not need a completely different formula. We used the same relationship and rearranged it to find the unknown quantity.

This is an important mathematical habit.

A formula describes a relationship. When the known and unknown quantities change, the relationship can often be rearranged to answer a different question.

4.5 Area: How Much Surface Must You Cover?

Now imagine that you want to tile the floor of a rectangular room.

Knowing the perimeter does not tell you how much of the floor must be covered.

You need to know the area.

Area measures the amount of two-dimensional surface occupied by a shape.

For a rectangle:

$$\boxed{A = l \times w}$$

where:

- A represents area;
- l represents length;
- w represents width.

Why do we multiply the length by the width?

Imagine dividing a rectangular floor into equal squares, each measuring 1 metre by 1 metre.

The number of squares that fit across the floor depends on its width. The number of rows depends on its length.

Multiplying the number of squares across by the number of rows gives the total number of square metres.

This is the idea behind the area formula.

Example 1: Tiling a Room

A rectangular room is 8 metres long and 5 metres wide.

Find its floor area.

$$A = l \times w$$
$$A = 8 \times 5$$
$$\boxed{A = 40 \text{ m}^2}$$

The floor area is 40 square metres.

If each tile covers 0.25 square metres, how many tiles would be needed in an ideal calculation with no cutting or wastage?

Divide the floor area by the area of one tile:

$$\text{Number of tiles} = \frac{40}{0.25}$$
$$\boxed{160 \text{ tiles}}$$

In practice, additional tiles may be required because some tiles must be cut, and some may break during installation.

The mathematical calculation gives the basic quantity. The construction plan determines the appropriate allowance.

Example 2: Comparing Two Rooms

Room A is 6 m long and 4 m wide.

Room B is 8 m long and 3 m wide.

Find the area of each room.

Room A:

$$A = 6 \times 4 = 24 \text{ m}^2$$

Room B:

$$A = 8 \times 3 = 24 \text{ m}^2$$

Both rooms have the same floor area.

However, their shapes are different.

Room A is shorter and wider, while Room B is longer and narrower.

This reveals an important distinction: two shapes can have the same area without having the same dimensions or perimeter.

If you were arranging furniture, designing a hallway, or planning a classroom, the dimensions might matter as much as the total area.

A mathematical result is useful, but you must understand what it tells you and what it leaves unanswered.

4.6 Why Area Is Measured in Square Units

Suppose a room is 6 metres long and 4 metres wide.

When we calculate:

$$6 \text{ m} \times 4 \text{ m}$$

the units multiply as well as the numbers:

$$\text{m} \times \text{m} = \text{m}^2$$

Therefore:

$$6 \text{ m} \times 4 \text{ m} = 24 \text{ m}^2$$

The answer is expressed in square metres.

A square metre represents the area of a square whose sides are each 1 metre long.

Similarly:

- cm^2 means square centimetres.
- m^2 means square metres.
- km^2 means square kilometres.

The squared unit is not simply a writing convention. It reflects the fact that area involves two dimensions.

A Common Mistake

A student calculates the area of a room as 30 metres.

The numerical calculation may be correct, but the unit is wrong.

If the room is 6 m long and 5 m wide, its area is:

$$6 \times 5 = 30 \text{ m}^2$$

The correct answer is 30 square metres, not 30 metres.

Metres measure length. Square metres measure area.

Always check whether the unit matches the quantity being calculated.

4.7 The Area of a Triangle

Builders and designers often work with triangular shapes.

A triangular roof section, a triangular support, or a triangular piece of land may require an area calculation.

The area of a triangle is:

$$\boxed{A = \frac{1}{2}bh}$$

where:

- b is the base;
- h is the perpendicular height.

Why do we multiply the base by the height and then divide by 2?

Imagine a rectangle with a base of 8 cm and a height of 6 cm.

Its area is:

$$8 \times 6 = 48 \text{ cm}^2$$

Now draw a diagonal from one corner of the rectangle to the opposite corner.

The diagonal divides the rectangle into two equal triangles.

Each triangle has half the rectangle's area:

$$48 \div 2 = 24 \text{ cm}^2$$

This explains why the area formula for a triangle includes one-half.

The formula works for any triangle when the height is perpendicular to the chosen base, even if the triangle is not part of a rectangle drawn in that exact way.

Example 1: Finding a Triangular Area

A triangular garden has a base of 10 metres and a perpendicular height of 6 metres.

Calculate its area.

$$A = \frac{1}{2}bh$$

Substitute:

$$A = \frac{1}{2} \times 10 \times 6$$

First multiply:

$$10 \times 6 = 60$$

Then divide by 2:

$$60 \div 2 = 30$$

Therefore:

$$\boxed{A = 30 \text{ m}^2}$$

Example 2: Finding the Height

A triangle has an area of 42 cm^2 and a base of 12 cm.

Find its perpendicular height.

Start with:

$$A = \frac{1}{2}bh$$

Substitute the known values:

$$42 = \frac{1}{2} \times 12 \times h$$

Simplify:

$$42 = 6h$$

Divide both sides by 6:

$$h = 42 \div 6$$

$$\boxed{h = 7 \text{ cm}}$$

The height is 7 cm.

This example shows how understanding the relationship makes it possible to find an unknown measurement.

4.8 Circles: Measuring a Different Kind of Shape

Not every object is rectangular or triangular.

Wheels, pipes, plates, circular gardens, and many machine parts involve circles.

Two important measurements of a circle are the radius and the diameter.

The radius is the distance from the centre of a circle to its boundary.

The diameter is the distance across the circle through its centre.

Therefore:

$$\boxed{d = 2r}$$

where d is the diameter and r is the radius.

If a circular water tank has a radius of 2 metres, its diameter is:

$$d = 2 \times 2 = 4 \text{ m}$$

The diameter is 4 metres.

4.8.1 Circumference

The circumference is the distance around a circle.

It plays a role similar to the perimeter of a polygon.

The formula is:

$$\boxed{C = 2\pi r}$$

or:

$$\boxed{C = \pi d}$$

The symbol π, called pi, represents a mathematical constant approximately equal to 3.14159.

Why does pi appear in the formula?

For every circle, the ratio of its circumference to its diameter is the same constant:

$$\pi = \frac{C}{d}$$

Rearranging gives:

$$C = \pi d$$

Since $d = 2r$, we can also write:

$$C = 2\pi r$$

Example 1: Finding the Circumference

A circular garden has a diameter of 14 metres.

Using $\pi = \frac{22}{7}$, calculate its circumference.

$$C = \pi d$$
$$C = \frac{22}{7} \times 14$$
$$C = 22 \times 2$$
$$\boxed{C = 44 \text{ m}}$$

The circumference is 44 metres using the stated approximation for pi.

If a fence must be placed around the garden, this calculation provides the basic boundary length.

4.8.2 Area of a Circle

The area of a circle is:

$$\boxed{A = \pi r^2}$$

The radius is squared because the area is a two-dimensional quantity, and the geometry of the circle connects its area to the square of its radius.

The complete derivation can be studied through more advanced geometry and calculus. For now, focus on understanding what the formula measures and how to use it correctly.

Example 2: Finding the Area

A circular table has a radius of 7 cm.

Using $\pi = \frac{22}{7}$, calculate its area.

$$A = \pi r^2$$

Substitute:

$$A = \frac{22}{7} \times 7^2$$
$$A = \frac{22}{7} \times 49$$
$$A = 22 \times 7$$
$$\boxed{A = 154 \text{ cm}^2}$$

The area of the table's circular surface is 154 cm^2.

Remember the distinction:

- Circumference measures the distance around the circle.
- Area measures the surface inside the circle.

A question about fencing requires circumference. A question about covering the top of a table requires area.

Choosing the right formula begins with identifying the quantity you need.

4.9 Volume: How Much Space Can an Object Hold?

Imagine that you are building a water tank.

You know its length and width, but you also need to know its height.

Why?

Because a tank's capacity depends on all three dimensions.

Volume measures the amount of three-dimensional space occupied by an object or enclosed within a container.

For a rectangular prism, also called a cuboid:

$$\boxed{V = l \times w \times h}$$

where:

- l is length;
- w is width;
- h is height.

Why do we multiply all three dimensions?

Imagine filling a rectangular container with equal-sized cubes.

The length determines how many cubes fit along one direction.

The width determines how many fit across another direction.

The height determines how many layers can be stacked.

Multiplying the three quantities gives the total number of cubic units.

Example 1: A Water Tank

A rectangular tank is 4 m long, 3 m wide, and 2 m high.

Find its internal volume, assuming these are the internal dimensions.

$$V = lwh$$
$$V = 4 \times 3 \times 2$$
$$\boxed{V = 24 \text{ m}^3}$$

The internal volume is 24 cubic metres.

Since one cubic metre equals 1,000 litres:

$$24 \text{ m}^3 = 24{,}000 \text{ litres}$$

The tank's geometric capacity is 24,000 litres, assuming it can be filled to the full internal volume.

In practice, usable capacity may be lower because of the design, required free space, outlets, and other features.

Example 2: Finding a Missing Height

A rectangular container has a volume of 60 m^3. Its length is 5 m and its width is 4 m.

Find its height.

Start with:

$$V = lwh$$

Substitute:

$$60 = 5 \times 4 \times h$$
$$60 = 20h$$

Divide both sides by 20:

$$h = 60 \div 20$$
$$\boxed{h = 3 \text{ m}}$$

The height is 3 metres.

Once again, the same relationship can be rearranged to find an unknown measurement.

4.10 Understanding Units of Volume

Volume uses cubic units because it measures three dimensions.

Examples include:

- cm^3, or cubic centimetres;
- m^3, or cubic metres;
- km^3, or cubic kilometres.

For example:

$$2\ \text{m} \times 3\ \text{m} \times 4\ \text{m} = 24\ \text{m}^3$$

The units multiply as follows:

$$\text{m} \times \text{m} \times \text{m} = \text{m}^3$$

This is why the answer is expressed in cubic metres rather than square metres.

Area involves two dimensions. Volume involves three.

Converting Cubic Metres to Litres

One cubic metre equals 1,000 litres.

Therefore:

$$1\ \text{m}^3 = 1{,}000\ \text{L}$$

And:

$$3.5\ \text{m}^3 = 3{,}500\ \text{L}$$

This conversion is useful when working with tanks, water supply, and other liquid-storage problems.

However, remember that the capacity of an actual container depends on its internal dimensions, not necessarily its external dimensions.

4.11 Angles: Understanding Direction and Shape

Imagine that a carpenter needs to cut two pieces of wood so that they meet at a particular angle.

Or imagine a builder constructing a roof with two sloping sides.

The lengths of the materials alone may not fully describe the required arrangement. The angle between them also matters.

An angle measures the amount of rotation or separation between two rays that share a common endpoint.

Angles are commonly measured in degrees.

A full turn is:

$$360^\circ$$

A half-turn is:

$$180^\circ$$

A quarter-turn is:

$$90^\circ$$

A right angle measures exactly 90°.

A straight angle measures 180°.

An acute angle is greater than 0° but less than 90°.

An obtuse angle is greater than 90° but less than 180°.

Why Do Angles Matter?

Angles help describe the orientation of objects.

They are important in:

- building roofs and stairs;
- designing roads and bridges;
- surveying land;
- constructing machines;
- navigation;
- drawing and technical design.

Suppose two angles lie on a straight line and form a pair of adjacent angles.

Together, they measure 180°.

If one angle measures 65°, the other is:

$$180^\circ - 65^\circ = 115^\circ$$

The second angle measures 115°.

The calculation works because the two angles form a straight angle.

It would not be correct to apply this rule to any two angles simply because they appear near one another in a drawing. The geometric relationship must actually be present.

This is another important lesson: formulas and rules apply under specific conditions.

4.12 Triangles and the Importance of the Right Angle

A triangle is a shape with three sides and three interior angles.

The interior angles of any ordinary Euclidean triangle add up to:

$$180^\circ$$

For example, suppose a triangle has angles of 50° and 60°.

The third angle is:

$$180^\circ - 50^\circ - 60^\circ = 70^\circ$$

The third angle measures 70°.

Triangles are especially important in construction because triangular arrangements can provide useful geometric relationships and structural stability when designed appropriately.

One particular triangle has a special role in mathematics: the right-angled triangle.

A right-angled triangle contains one angle measuring 90°.

The side opposite the right angle is called the hypotenuse.

The hypotenuse is always the longest side of a right-angled triangle.

The relationship between the three sides is expressed by the Pythagorean theorem.

Before learning the theorem, let us understand the problem that makes it useful.

4.13 The Pythagorean Theorem: A Builder's Problem

Imagine that you are working on a building.

A ladder must reach a point on a vertical wall. The foot of the ladder stands on level ground a certain distance away from the wall.

You know the height the ladder must reach and the horizontal distance from the wall.

But you need to know how long the ladder is.

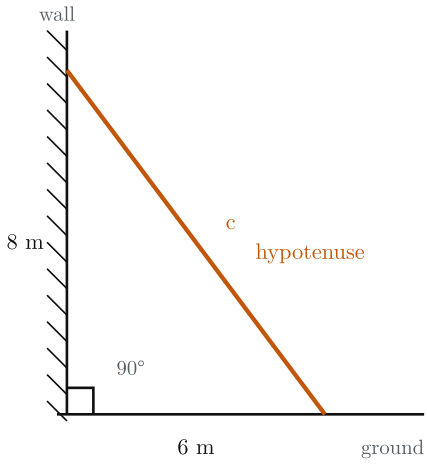
The situation forms a right-angled triangle:

- The wall represents the vertical side.
- The ground represents the horizontal side.
- The ladder represents the hypotenuse.

The wall and the ground meet at a right angle, assuming the wall is vertical and the ground is level.

FIGURE 2

Ladder Against a Wall



The same ladder as a right-angled triangle: height 8 m, base 6 m, ladder c as the hypotenuse.

How can we calculate the ladder's length?

The Pythagorean theorem provides the relationship:

$$\boxed{a^2 + b^2 = c^2}$$

where:

- a is one shorter side;
- b is the other shorter side;
- c is the hypotenuse.

The theorem applies to right-angled triangles.

It tells us that the square of the hypotenuse equals the sum of the squares of the other two sides.

Let us calculate the ladder's length.

The vertical height is 8 m.

The horizontal distance is 6 m.

The ladder length is c.

Using the theorem:

$$a^2 + b^2 = c^2$$

Substitute:

$$6^2 + 8^2 = c^2$$

Calculate the squares:

$$36 + 64 = c^2$$

Add:

$$100 = c^2$$

Take the positive square root:

$$c = \sqrt{100}$$

Therefore:

$$\boxed{c = 10 \text{ m}}$$

The ladder is 10 metres long.

Why Do We Use the Pythagorean Theorem?

We use it because the wall, ground, and ladder form a right-angled triangle, and the theorem connects the lengths of the three sides of such a triangle.

We are not using the theorem merely because the problem contains three measurements.

If the angle between the wall and ground were not $90°$, the ordinary Pythagorean relationship would not apply directly.

Understanding the shape and its conditions tells us whether the theorem is appropriate.

An Important Safety Note

The calculation establishes the geometric length required for the stated dimensions. It does not establish that a ladder is safe to use.

Real ladder safety depends on factors such as the ladder's design, condition, angle, footing, load rating, and the manufacturer's instructions. A geometric answer must not replace appropriate safety procedures.

4.14 A Second Example of Pythagoras

Suppose a right-angled triangle has shorter sides measuring 5 cm and 12 cm.

Find the hypotenuse.

Start with:

$$a^2 + b^2 = c^2$$

Substitute:

$$5^2 + 12^2 = c^2$$

Calculate:

$$25 + 144 = c^2$$
$$169 = c^2$$

Take the positive square root:

$$c = \sqrt{169}$$

$$\boxed{c = 13 \text{ cm}}$$

The hypotenuse measures 13 cm.

The side lengths 5, 12, and 13 form a well-known Pythagorean triple.

A Pythagorean triple consists of three positive integers that satisfy:

$$a^2 + b^2 = c^2$$

Other examples include:

- 3, 4, 5;
- 8, 15, 17;
- 7, 24, 25.

These triples can make certain calculations quicker because the square roots are whole numbers.

However, do not assume that every right-angled triangle has integer side lengths.

The theorem works for all right-angled triangles, including those with decimal or irrational side lengths.

4.15 Finding a Missing Shorter Side

Sometimes the hypotenuse is known, but one of the shorter sides is missing.

Suppose a right-angled triangle has a hypotenuse of 13 m and one shorter side of 5 m.

Find the other shorter side.

Let the unknown side be b.

Start with:

$$a^2 + b^2 = c^2$$

Substitute:

$$5^2 + b^2 = 13^2$$

Calculate the squares:

$$25 + b^2 = 169$$

Subtract 25 from both sides:

$$b^2 = 169 - 25$$
$$b^2 = 144$$

Take the positive square root:

$$b = \sqrt{144}$$
$$\boxed{b = 12 \text{ m}}$$

The missing side is 12 m.

We subtract the known squared side from the squared hypotenuse because the theorem states that the two shorter-side squares add up to the hypotenuse's square.

The rearrangement follows directly from the relationship.

4.16 When the Answer Is Not a Whole Number

Consider a right-angled triangle with shorter sides of 4 cm and 7 cm.

Find the hypotenuse.

Using Pythagoras:

$$4^2 + 7^2 = c^2$$
$$16 + 49 = c^2$$
$$65 = c^2$$

Therefore:

$$c = \sqrt{65}$$

The exact answer is:

$$\boxed{c = \sqrt{65}\ \text{cm}}$$

Using a calculator:

$$c \approx 8.06\ \text{cm}$$

Both forms are useful.

The exact form preserves the mathematical value without rounding. The approximate form may be more practical when a physical measurement is required.

If the answer is rounded, state the degree of precision used.

For example, 8.06 cm is the approximate value to two decimal places.

Do not treat a rounded answer as perfectly exact.

4.17 How to Recognize a Pythagoras Problem

Students sometimes memorize the Pythagorean theorem but struggle to recognize when it should be used.

Look for these features:

1. The problem involves a right-angled triangle.
2. Two side lengths are known, or one side and the hypotenuse are known.
3. The unknown is another side of that triangle.
4. The right angle is established by the information given or by the geometry of the situation.

If these conditions are satisfied, the theorem may be appropriate.

However, a diagram alone is not always sufficient to prove that an angle is exactly $90°$. In formal mathematics, use the information supplied in the question or a clearly established geometric relationship.

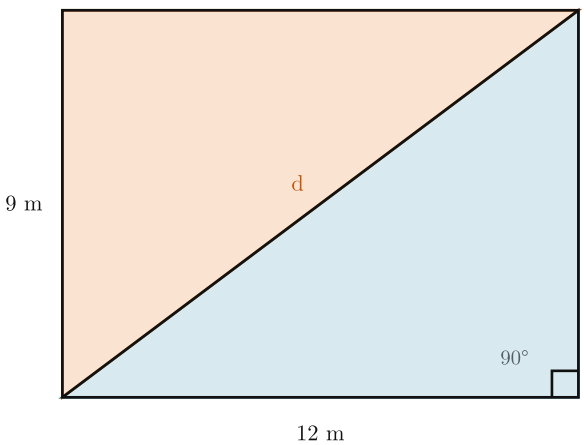
Example: A Rectangular Field

A rectangular field is 9 m long and 12 m wide.

Find the length of the diagonal from one corner to the opposite corner.

FIGURE 3

Diagonal of a Rectangular Field



The diagonal d splits the field into two right-angled triangles, so 92 + 122 = d2.

The diagonal forms the hypotenuse of a right-angled triangle.

Therefore:

$$9^2 + 12^2 = d^2$$

$$81 + 144 = d^2$$

$$225 = d^2$$

$$d = \sqrt{225}$$

$$\boxed{d = 15 \text{ m}}$$

The diagonal measures 15 metres.

This calculation may be useful when checking diagonal measurements during layout. In practical construction, the measurements and tolerances must be chosen appropriately for the project.

4.18 The Difference Between Perimeter, Area, and Volume

These three concepts are easy to confuse because they often involve the same shape.

Let us compare them using a rectangular room.

Suppose the room is 8 m long, 5 m wide, and 3 m high.

Perimeter of the Floor

The perimeter describes the boundary around the rectangular floor.

$$P = 2(l + w)$$
$$P = 2(8 + 5)$$
$$\boxed{P = 26 \text{ m}}$$

This could help estimate the distance around the floor's boundary.

Area of the Floor

The area describes the floor surface.

$$A = lw$$
$$A = 8 \times 5$$
$$\boxed{A = 40 \text{ m}^2}$$

This could help estimate the floor covering required.

Volume of the Room

The volume describes the three-dimensional space inside the room.

$$V = lwh$$
$$V = 8 \times 5 \times 3$$
$$\boxed{V = 120 \text{ m}^3}$$

This measures the room's internal geometric volume, assuming the dimensions describe a rectangular interior.

Notice the difference in the units:

- Perimeter: metres.
- Area: square metres.
- Volume: cubic metres.

The units help you recognize which quantity has been calculated.

A useful question to ask yourself is:

Am I measuring a boundary, a surface, or a three-dimensional space?

Your answer determines which concept you need.

4.19 Geometry in Everyday Life

You do not need to be a professional builder to use geometry.

It appears in ordinary decisions and activities.

4.19.1 Buying Floor Tiles

You measure the length and width of a room, calculate the floor area, and compare it with the area covered by each tile.

The calculation helps estimate the number of tiles needed.

4.19.2 Painting a Wall

You calculate the wall's area and consider how much surface a given quantity of paint can cover.

You may need to subtract the areas of doors and windows.

You should also account for the number of coats and the paint manufacturer's coverage guidance.

4.19.3 Planning a Garden

You calculate perimeter when planning a fence and area when estimating the amount of ground available for planting.

4.19.4 Constructing Furniture

A carpenter measures lengths, widths, and heights to ensure that the pieces fit together.

Angles may also be important when cutting or assembling the pieces.

4.19.5 Designing Packaging

A business that packages products may calculate the volume of a box to determine whether an item will fit inside.

The external dimensions may also matter when estimating storage or transport space.

4.19.6 Reading Maps

Scale and distance calculations help estimate distances between locations.

If the map scale is known, a measured distance on the map can be converted into a represented real-world distance.

The actual route may differ from the straight-line distance.

4.19.7 Sports

Sports fields and courts involve lengths, widths, boundaries, angles, and areas.

Geometry helps describe these spaces and compare measurements.

In all these examples, the mathematical operation becomes easier to choose once the practical purpose is clear.

4.20 How to Make Geometry Easier

Geometry may appear difficult when many formulas are presented together. The following method helps organize your thinking.

Step 1: Identify the Object

Is the problem about a rectangle, triangle, circle, cube, cylinder, or another shape?

Do not assume the shape based only on how a drawing looks. Use the information provided.

Step 2: Identify the Quantity

Are you finding:

- a side length?
- a perimeter?
- an area?
- a volume?
- an angle?
- a diagonal?

Different questions may involve the same shape but require different calculations.

Step 3: Write Down the Known Measurements

For example:

$$\text{Length} = 10\,\text{m}$$

$$\text{Width} = 6\,\text{m}$$

$$\text{Unknown} = \text{Area}$$

Step 4: Choose the Relationship

For a rectangle:

$$A = lw$$

For a triangle:

$$A = \frac{1}{2}bh$$

For a rectangular prism:

$$V = lwh$$

For a right-angled triangle:

$$a^2 + b^2 = c^2$$

Choose the relationship that matches the shape and the quantity you need to find.

Step 5: Substitute Carefully

Write the known values in the appropriate positions.

For a rectangle with length 10 m and width 6 m:

$$A = 10 \times 6$$
$$A = 60 \text{ m}^2$$

Step 6: Check the Units

Does your answer have the correct type of unit?

If you calculate an area, the unit should be squared.

If you calculate a volume, the unit should be cubed.

If you calculate a length, the unit should be a length unit.

Step 7: Check Whether the Answer Makes Sense

If a rectangular room is 10 m long and 6 m wide, an area of 60 m^2 is reasonable.

An answer of 600 m^2 would be worth checking because it is much larger than expected from the given dimensions.

Estimation and common sense are valuable partners in geometry.

4.21 Common Mistakes in Geometry

Mistake 1: Confusing Area with Perimeter

A student calculates the perimeter of a rectangular floor when the question asks how many tiles are required.

The perimeter measures the boundary. The area measures the surface.

The correct quantity depends on the task.

Mistake 2: Forgetting Units

Writing 40 as the answer to an area question is incomplete when the appropriate unit is square metres.

Always include the correct unit.

Mistake 3: Using the Wrong Height in a Triangle

The height in the triangle area formula must be perpendicular to the chosen base.

A sloping side is not automatically the height.

Mistake 4: Using Pythagoras for Every Triangle

The Pythagorean theorem applies to right-angled triangles.

It cannot be applied directly to every triangle without additional information or a different mathematical method.

Mistake 5: Forgetting to Square the Side Lengths

For a right-angled triangle with sides 3 and 4, the calculation is:

$$3^2 + 4^2 = 9 + 16 = 25$$

It is not:

$$3 + 4 = 7$$

The theorem involves the squares of the side lengths.

Mistake 6: Forgetting the Square Root

If:

$$c^2 = 100$$

then the positive side length is:

$$c = \sqrt{100} = 10$$

The value 100 is the square of the side length, not the side length itself.

Mistake 7: Using External Dimensions for Internal Capacity

When calculating the usable capacity of a container, the internal dimensions are generally the relevant measurements.

Wall thickness and other structural features may make the internal volume smaller than the external geometric volume.

Mistake 8: Assuming the Formula Guarantees Practical Success

A calculation may determine a length, area, or volume correctly while leaving practical questions unanswered.

A construction project may also require engineering assessment, suitable materials, safe installation, and compliance with relevant standards.

Mathematics supports practical decisions, but it does not replace every other form of expertise.

4.22 Practical Workshop: Think Like a Builder

Attempt the following questions without looking at the worked examples.

For each problem, identify the shape, the known measurements, the unknown quantity, and the relationship you need.

Exercise 1: Fencing a Garden

A rectangular garden is 15 m long and 9 m wide.

1. Calculate its perimeter.
2. Calculate its area.
3. Explain why the answers have different units.
4. If fencing is sold in lengths of 3 m, how many lengths would cover the perimeter in an ideal calculation with no wastage?

Exercise 2: Tiling a Room

A rectangular room is 7 m long and 4 m wide.

Each tile covers 0.25 m^2.

1. Calculate the floor area.
2. Calculate the theoretical number of tiles required.
3. Explain why the actual number purchased may be greater.

Exercise 3: A Triangular Sign

A triangular sign has a base of 12 cm and a perpendicular height of 8 cm.

1. Calculate its area.
2. If its area is 60 cm^2 and its base is 15 cm, find its perpendicular height.
3. Explain why the height must be perpendicular to the base.

Exercise 4: A Circular Flower Bed

A circular flower bed has a radius of 3.5 m.

Using $\pi = \frac{22}{7}$:

1. Calculate its diameter.
2. Calculate its circumference.
3. Calculate its area.
4. Explain which measurement would be useful for fencing and which would be useful for estimating the surface to be covered with soil.

Exercise 5: A Water Tank

A rectangular tank is 5 m long, 2 m wide, and 3 m high.

1. Calculate its internal volume, assuming the measurements are internal dimensions.
2. Convert the volume into litres.
3. Explain why volume is expressed in cubic metres.

Exercise 6: Pythagoras and a Ladder

A ladder reaches a point 12 m above the ground. Its foot is 5 m from the wall.

Assume the wall is vertical, the ground is level, and the ladder forms a right-angled triangle with them.

1. Draw and label the triangle.
2. Identify the hypotenuse.
3. Calculate the ladder's geometric length using the Pythagorean theorem.
4. Explain why the theorem is appropriate.
5. State one practical safety consideration that the calculation does not answer.

Exercise 7: The Diagonal of a Room

A rectangular room measures 8 m by 15 m.

1. Draw the rectangle and its diagonal.
2. Explain how the diagonal creates a right-angled triangle.
3. Calculate the diagonal length.
4. Explain one practical situation in which this measurement may be useful.

Exercise 8: Find the Mistake

A student is asked to calculate the area of a rectangle measuring 9 m by 4 m.

The student writes:

$$A = 2(9 + 4) = 26 \text{ m}^2$$

Answer the following questions:

1. What did the student calculate?

2. Why is that not the correct method for finding area?
3. Calculate the correct area.
4. Explain the difference between the perimeter and area of the rectangle.

Exercise 9: Design Your Own Problem

Imagine that you are designing a small room, garden, shop, or water tank.

Create a realistic mathematical problem involving at least two measurements.

Then solve it using the following structure:

The situation: Describe what you are designing or measuring.

Known measurements: List the values and their units.

The question: State what you need to find.

The mathematical relationship: Write the formula and explain why it applies.

The calculation: Show each step.

The answer: Include the correct units.

The practical interpretation: Explain what the result means for your design.

The limitation: Identify at least one real-world factor that the calculation does not automatically account for.

4.23 Chapter Review

Before moving to the next chapter, make sure you understand the following principles.

1. **Geometry describes shapes and spatial relationships.**

It helps us measure lengths, boundaries, surfaces, angles, and three-dimensional space.

2. **Perimeter measures the boundary.**

For a rectangle:

$$P = 2(l + w)$$

3. **Area measures a two-dimensional surface.**

For a rectangle:

$$A = lw$$

For a triangle:

$$A = \frac{1}{2}bh$$

For a circle:

$$A = \pi r^2$$

4. **Circumference measures the boundary of a circle.**

$$C = 2\pi r = \pi d$$

5. **Volume measures three-dimensional space.**

For a rectangular prism:

$$V = lwh$$

6. **Units communicate the meaning of the answer.**

Lengths use units such as metres. Areas use square units. Volumes use cubic units.

7. **The Pythagorean theorem connects the sides of a right-angled**

triangle.

$$a^2 + b^2 = c^2$$

The hypotenuse is the side opposite the right angle.

8. **A formula must match the problem.**

Do not choose a formula simply because you recognize the shape or the numbers. Identify the quantity you need to find and the relationship that connects it to the known information.

9. **Practical calculations have limits.**

Correct geometry does not automatically account for material waste, structural safety, installation requirements, or every other real-world consideration.

4.24 Conclusion: Think Like the Person Who Must Use the Answer

Imagine that two people are asked to calculate the floor area of a room.

The first person memorizes the formula and writes:

$$A = l \times w$$

The second person understands that the question concerns the amount of floor surface to be covered. They identify the length and width, check the units, multiply the measurements, and interpret the result as square metres.

Both may obtain the same answer.

But the second person is better prepared to handle a different problem, such as finding the width when the area and length are known.

That is the advantage of understanding relationships rather than memorizing isolated rules.

Geometry becomes more meaningful when you think about the object, the purpose of the measurement, and the decision that depends on the answer.

A builder does not calculate area simply to produce a number. The builder may need to purchase tiles, estimate paint, plan a garden, or compare designs.

A person using the Pythagorean theorem is not merely manipulating squares and square roots. They are using a relationship that connects the sides of a right-angled triangle.

The formula is a tool. Understanding tells you when and how to use it.

As you continue studying mathematics, keep asking:

> **What real problem does this calculation solve, why does the method work, and what does the answer allow me to understand or do?**

In the next chapter, we will move from the building site to the world of business and accounting. We will examine how people calculate costs, revenue, profit, loss, discounts, and financial percentages—and why understanding the meaning of each financial quantity matters just as much as performing the arithmetic correctly.

CHAPTER 5

Think Like an Accountant

05

5.1 Introduction: Before You Calculate Profit

Imagine that you own a small shop.

You buy bags of rice, cartons of noodles, bottles of cooking oil, and other goods. Every day, customers come in and buy from you. Money enters your cash box, and you feel that your business is doing well.

At the end of the month, you count the money and discover that you have ₦250,000.

You might immediately think, "I have made ₦250,000!"

But is that really your profit?

What about the money you spent buying the goods? What about transportation, electricity, rent, packaging, and other expenses?

The money remaining in your cash box is not necessarily your profit.

To understand your actual profit, you must first understand what happened to your money.

This is where accounting begins.

Accounting is not simply about writing figures in a book or using a calculator. It is about understanding the financial story behind a business. It helps us identify where

money came from, where it went, what remains, and whether the business is making progress.

Before an accountant chooses a formula, the accountant asks questions:

- How much money entered the business?
- How much did the business spend?
- What goods were sold?
- What goods remain?
- Does the business owe anyone money?
- How much profit or loss was made?

These questions give meaning to the calculations that follow.

In this chapter, we will learn to think like an accountant before reaching for a formula.

5.2 Why Accounting Was Necessary

Imagine a trader who sells goods every day but keeps no records.

On Monday, the trader buys goods for ₦40,000 and sells them for ₦52,000.

On Tuesday, the trader buys more goods for ₦30,000 and sells them for ₦39,000.

On Wednesday, the trader spends ₦5,000 transporting goods from the market.

At the end of the week, the trader looks at the money available and tries to remember every transaction.

The trader may believe that the business is profitable, but memory alone cannot provide a reliable financial picture.

Perhaps some customers have not paid. Perhaps some goods remain unsold. Perhaps money was withdrawn for personal use. Without proper records, it becomes difficult to know the actual position of the business.

Accounting developed because people needed dependable ways to record and understand economic activities.

Long before modern computers, merchants and administrators recorded goods, debts, payments, taxes, and property. As trade expanded, keeping accurate records became increasingly important.

One major development was double-entry bookkeeping, which became widely known in European commerce during the Renaissance. In 1494, the Italian mathematician Luca Pacioli published a book that described the Venetian method of bookkeeping. He did not invent the entire system, but his work helped document and spread its principles.

Today, accounting supports shops, banks, schools, hospitals, factories, governments, and international companies.

The central problem remains similar: **How can we record financial**

events accurately and use those records to understand what is happening?

That is the question behind accounting calculations.

5.3 Enter the Situation Before Calculating

Let us return to our shopkeeper.

Suppose the shopkeeper sells goods worth ₦100,000 during one week.

The shopkeeper spent ₦65,000 purchasing the goods that were sold. The shopkeeper also paid ₦8,000 for transportation and ₦7,000 for other operating expenses.

How much profit was made?

A student might see the figures and immediately subtract two numbers without understanding what each represents.

An accountant first organizes the situation.

Money received from sales:

₦100,000

Cost of goods sold:

₦65,000

Other expenses:

$$₦8{,}000 + ₦7{,}000 = ₦15{,}000$$

The shopkeeper received ₦100,000 in sales, but the business also incurred costs.

The total costs relevant to this example are:

$$₦65{,}000 + ₦15{,}000 = ₦80{,}000$$

Therefore, the profit is:

$$₦100{,}000 - ₦80{,}000 = ₦20{,}000$$

The business made a profit of ₦20,000.

Notice what happened. We did not begin by searching for a formula. We first identified the money received and the costs associated with earning it.

Only then did subtraction become meaningful.

5.4 Revenue: Understanding the Money Earned from Sales

Revenue is the income a business earns from its ordinary activities, such as selling goods or providing services, before deducting its expenses.

Imagine that a student sells 20 notebooks for ₦500 each.

The total revenue is the price received for each notebook multiplied by the number sold.

$$\text{Revenue} = \text{Price per notebook} \times \text{Number sold}$$

$$\text{Revenue} = ₦500 \times 20$$

$$\text{Revenue} = ₦10{,}000$$

The student has earned ₦10,000 in sales revenue.

But the student has not necessarily made ₦10,000 in profit.

Perhaps the notebooks cost ₦350 each to purchase. Perhaps transportation and packaging also cost money.

Revenue tells us how much was earned from sales. It does not tell us how much the business kept after costs.

Example 1: Selling exercise books

A trader sells 40 exercise books at ₦700 each.

Step 1: Understand the situation.

The trader receives ₦700 for each book and sells 40 books.

Step 2: Calculate the total sales.

$$\text{Revenue} = ₦700 \times 40$$

$$\text{Revenue} = ₦28{,}000$$

Answer: The trader's revenue is ₦28,000.

Example 2: Different products

A small shop sells the following items in one day:

- 10 bottles of oil at ₦2,500 each.
- 15 packets of noodles at ₦600 each.

Revenue from oil:

$$10 \times ₦2{,}500 = ₦25{,}000$$

Revenue from noodles:

$$15 \times ₦600 = ₦9{,}000$$

Total revenue:

$$₦25{,}000 + ₦9{,}000 = ₦34{,}000$$

Answer: Total revenue is ₦34,000.

The important lesson is that when a business sells different products, calculate the revenue from each product and then add the results.

5.5 Cost: Understanding What the Business Gives Up

Revenue tells us what comes in from sales. Cost tells us what the business sacrifices to obtain goods, produce items, or provide services.

Suppose you purchase 50 calculators for ₦2,000 each.

The total purchase cost is:

$$50 \times ₦2{,}000 = ₦100{,}000$$

You have spent ₦100,000 purchasing the calculators.

However, a business may face several kinds of costs.

Cost of goods purchased

This is the amount paid to acquire goods for resale.

For example, a trader purchases 30 school bags at ₦4,000 each.

Total purchase cost:

$$30 \times ₦4{,}000 = ₦120{,}000$$

Operating expenses

These are expenses associated with running the business.

Examples include:

- Rent.
- Electricity.
- Transportation.
- Advertising.
- Packaging.
- Internet and communication.
- Staff salaries.

Suppose a shop spends the following amounts in one month:

Expense	Amount
Rent	₦30,000
Electricity	₦5,000
Transportation	₦8,000
Packaging	₦2,000
Total	**₦45,000**

To find the total, add every expense:

$$₦30{,}000 + ₦5{,}000 + ₦8{,}000 + ₦2{,}000 = ₦45{,}000$$

The business spent ₦45,000 on these operating expenses.

Why classification matters

Imagine that a shop purchases 100 bags for resale but sells only 70 during the month.

It would be misleading to automatically treat the purchase cost of all 100 bags as the cost of the 70 bags sold when calculating the cost of goods sold.

Some bags remain in inventory.

For a simple example, suppose each bag costs ₦3,000 and 70 bags are sold.

Cost of the bags sold:

$$70 \times ₦3{,}000 = ₦210{,}000$$

The remaining 30 bags have a purchase cost of:

$$30 \times ₦3{,}000 = ₦90{,}000$$

Those remaining bags are inventory, assuming they are still held for sale and have not lost value.

This distinction matters because accounting must describe the actual financial situation, not merely subtract every payment made during a period.

5.6 Profit: What Remains After Costs

Profit is the amount by which revenue exceeds the costs and expenses included in the calculation.

Let us examine a business that sells school supplies.

During one month:

- Sales revenue = ₦200,000.
- Cost of goods sold = ₦120,000.
- Operating expenses = ₦30,000.

First, calculate the amount remaining after the cost of goods sold.

$$₦200{,}000 - ₦120{,}000 = ₦80{,}000$$

This is called **gross profit** in this simplified example.

$$\text{Gross profit} = \text{Revenue} - \text{Cost of goods sold}$$

The business must still account for its operating expenses.

$$₦80{,}000 - ₦30{,}000 = ₦50{,}000$$

This gives an operating result of ₦50,000 before any other items that may apply, such as interest or taxes.

For this example, assuming there are no other relevant expenses or income items, the profit before tax is ₦50,000.

The important distinction is that gross profit and final profit are not always the same.

Example: A phone accessories shop

A trader sells phone accessories for ₦150,000 during a month.

The cost of the accessories sold is ₦90,000. The shop also incurs operating expenses of ₦25,000.

Step 1: Calculate gross profit.

$$₦150{,}000 - ₦90{,}000 = ₦60{,}000$$

Step 2: Deduct operating expenses.

$$₦60{,}000 - ₦25{,}000 = ₦35{,}000$$

Answer: The operating profit is ₦35,000, before any other applicable income or expenses.

The trader did not make ₦150,000 in profit. That amount represents revenue.

Understanding the difference between revenue and profit is one of the most important habits in financial thinking.

5.7 Loss: When Costs Exceed Revenue

A business does not always make a profit.

Suppose a trader sells goods for ₦80,000. The cost of the goods sold is ₦60,000, while operating expenses are ₦25,000.

First, calculate gross profit:

$$₦80{,}000 - ₦60{,}000 = ₦20{,}000$$

Then deduct operating expenses:

$$₦20{,}000 - ₦25{,}000 = -₦5{,}000$$

The result is negative ₦5,000.

This means the business made a loss of ₦5,000 under the conditions described.

Why did the business make a loss even though the sales were greater than the cost of the goods?

Because the total expenses were greater than the amount remaining after paying for the goods sold.

This illustrates an important principle: **Making sales does not auto-**

matically mean making a profit.

A business must understand both its revenue and its costs.

A situation to think about

Imagine two traders.

Trader A has sales of ₦300,000 and total relevant costs of ₦240,000.

Trader B has sales of ₦200,000 and total relevant costs of ₦150,000.

Trader A's profit:

$$₦300{,}000 - ₦240{,}000 = ₦60{,}000$$

Trader B's profit:

$$₦200{,}000 - ₦150{,}000 = ₦50{,}000$$

Trader A has the higher profit in this example.

But suppose Trader A had total relevant costs of ₦270,000 instead.

$$₦300{,}000 - ₦270{,}000 = ₦30{,}000$$

Trader B would then make more profit, despite having lower sales.

This shows why revenue alone cannot tell us which business is performing better.

5.8 Discounts: Understanding Reductions in Price

Imagine that a shop displays a school bag for ₦20,000 but announces a discount of 10%.

What does the discount mean?

It means that the customer will pay 10% less than the original price.

Before calculating, identify three things:

1. Original price = ₦20,000.
2. Discount rate = 10%.
3. The amount to find = the reduction in price.

To calculate 10% of ₦20,000, convert the percentage into a fraction:

$$10\% = \frac{10}{100} = 0.10$$

Then multiply:

$$₦20{,}000 \times 0.10 = ₦2{,}000$$

The discount is ₦2,000.

To find the new price, subtract the discount from the original price:

$$₦20{,}000 - ₦2{,}000 = ₦18{,}000$$

Answer: The customer pays ₦18,000.

A faster method

If the discount is 10%, the customer pays the remaining 90%.

$$100\% - 10\% = 90\%$$

Therefore:

$$₦20{,}000 \times 0.90 = ₦18{,}000$$

Both methods give the same answer.

The second method may be quicker when you already understand the relationship between a discount and the amount remaining.

Example: A 15% discount

A calculator costs ₦12,000. A shop offers a discount of 15%.

First, find the discount:

$$15\% \times ₦12{,}000$$

$$= 0.15 \times ₦12{,}000$$

$$= ₦1{,}800$$

Next, calculate the reduced price:

$$₦12{,}000 - ₦1{,}800 = ₦10{,}200$$

Answer: The customer pays ₦10,200.

Why businesses offer discounts

A discount may encourage customers to buy, help a shop clear old stock, or attract attention to a product.

However, a discount also reduces the selling price.

If a business offers discounts without understanding its costs, it may sell many goods but earn little profit or even make a loss.

This is why discount calculations are not merely school exercises. They help businesses make decisions.

5.9 Percentages: Measuring Profit Relative to Cost

Suppose a trader buys a school bag for ₦5,000 and sells it for ₦6,000.

The profit is:

$$₦6{,}000 - ₦5{,}000 = ₦1{,}000$$

The trader wants to know the profit as a percentage of the cost price.

We compare the profit with the original cost:

$$\text{Profit percentage on cost} = (\text{Profit} \div \text{Cost price}) \times 100\%$$

Substitute the values:

$$\begin{aligned} &= (₦1{,}000 \div ₦5{,}000) \times 100\% \\ &= 0.20 \times 100\% \\ &= 20\% \end{aligned}$$

The profit is 20% of the cost price.

Why do we divide by the cost price?

Because the question asks how large the profit is compared with what the trader originally paid.

The cost price is the reference amount.

If the question instead asks for profit as a percentage of sales, the reference amount changes.

Profit margin: A different calculation

Using the same example:

- Cost price = ₦5,000.
- Selling price = ₦6,000.
- Profit = ₦1,000.

Profit margin compares profit with revenue, not cost.

$$\begin{aligned}\text{Profit margin} &= (\text{Profit} \div \text{Sales revenue}) \times 100\% \\ &= (₦1{,}000 \div ₦6{,}000) \times 100\% \\ &\approx 16.67\%\end{aligned}$$

Therefore:

- Profit as a percentage of cost = 20%.
- Profit margin as a percentage of sales = approximately 16.67%.

These percentages differ because they use different reference amounts.

A common mistake is to treat them as the same calculation. Always read the question carefully and identify the amount against which the profit is being compared.

5.10 Markup: Deciding the Selling Price

Suppose a trader buys a wristwatch for ₦8,000 and wants to add a markup of 25% on the cost price.

A markup is the amount added to cost to determine a selling price.

First, calculate 25% of ₦8,000:

$$25\% \times ₦8{,}000 = 0.25 \times ₦8{,}000 = ₦2{,}000$$

The markup is ₦2,000.

Add the markup to the cost:

$$₦8{,}000 + ₦2{,}000 = ₦10{,}000$$

The selling price is ₦10,000.

Notice that a 25% markup on cost does not produce a 25% profit margin on sales.

The profit is ₦2,000, while the sales revenue is ₦10,000.

Profit margin:

$$(₦2{,}000 \div ₦10{,}000) \times 100\% = 20\%$$

Thus, a 25% markup on cost produces a 20% profit margin in this example, assuming the cost considered is the full relevant cost and no additional expenses affect the result.

This distinction is important for traders who need to set prices carefully.

5.11 Break-Even: When the Business Makes Neither Profit Nor Loss

Imagine that you start a small printing business.

You pay ₦20,000 for monthly expenses that do not change with the number of pages printed. These are called fixed costs in this simplified example.

Each printed page costs ₦10 in paper, ink, and other variable materials.

You charge customers ₦20 per page.

How many pages must you print and sell to cover all these costs?

Before calculating, understand the situation.

For every page sold:

- Revenue = ₦20.
- Variable cost = ₦10.
- Amount remaining to cover fixed costs = ₦10.

That ₦10 is the contribution per page.

The first pages sold help cover the ₦20,000 fixed costs. Once those costs have been covered, additional contribution can produce profit, assuming no other relevant costs or changes.

The break-even quantity is:

$$\text{Fixed costs} \div \text{Contribution per unit}$$
$$= ₦20{,}000 \div ₦10$$
$$= 2{,}000 \text{ pages}$$

The business must sell 2,000 pages to break even under these assumptions.

Let us verify the result.

Revenue from 2,000 pages:

$$2{,}000 \times ₦20 = ₦40{,}000$$

Variable costs:

$$2{,}000 \times ₦10 = ₦20{,}000$$

Add fixed costs:

$$₦20{,}000 + ₦20{,}000 = ₦40{,}000$$

Total costs equal revenue:

$$₦40{,}000 - ₦40{,}000 = ₦0$$

The business makes neither a profit nor a loss.

That is the meaning of break-even.

What happens after break-even?

Suppose the printing business sells 2,500 pages.

Revenue:

$$2{,}500 \times ₦20 = ₦50{,}000$$

Variable costs:

$$2{,}500 \times ₦10 = ₦25{,}000$$

Total costs:

$$₦25{,}000 + ₦20{,}000 = ₦45{,}000$$

Profit:

$$₦50{,}000 - ₦45{,}000 = ₦5{,}000$$

The business makes a profit of ₦5,000.

This example shows how accounting calculations help business owners answer practical questions before making decisions.

5.12 Cash and Profit Are Not Always the Same

This is one of the most important ideas in accounting.

Imagine that a business sells goods worth ₦100,000 to a customer. The customer promises to pay next month.

The sale may count as revenue under accrual accounting, even though the business has not yet received the cash.

Now imagine that the business receives a bank loan of ₦200,000.

The cash balance increases by ₦200,000, but receiving a loan does not ordinarily mean the business has earned ₦200,000 in profit. The business also has an obligation to repay the lender.

These examples show that cash movements and profit are different concepts.

Cash flow focuses on money entering and leaving the business.

Profit measures the financial result of income and expenses over a period.

A business may report a profit but still face difficulty paying its bills if customers have not paid on time. Conversely, a business may have plenty of cash because it borrowed money, even when its operations are not profitable.

That is why an accountant must understand what each transaction represents before recording or calculating anything.

5.13 Keeping Records: Let the Numbers Tell the Truth

Imagine that a trader makes five sales in one day but writes down only three.

At the end of the day, the records will not show the full sales activity.

Or imagine that the trader records all sales but forgets to record transportation expenses.

The calculated profit may appear higher than the actual result.

Accurate calculations depend on accurate information.

A simple daily record might look like this:

Transaction	Money received	Money paid
Sale of goods	₦25,000	—
Purchase of goods	—	₦12,000
Transportation	—	₦2,000
Another sale	₦18,000	—
Packaging	—	₦1,000
Total	**₦43,000**	**₦15,000**

The difference between money received and money paid is:

$$₦43{,}000 - ₦15{,}000 = ₦28{,}000$$

But should we immediately call ₦28,000 the business's profit?

Not necessarily.

This table records cash movements, not a complete profit calculation. Some purchases may be inventory that has not yet been sold. Some sales may relate to goods whose costs need to be identified. Other expenses, debts, and timing differences may also matter.

The table is useful for tracking cash, but a proper profit calculation requires classifying the transactions correctly.

The lesson is simple: **Good calculations cannot rescue bad records.**

5.14 How to Make Accounting Calculations Easier

Accounting can appear complicated when many transactions are mixed together. The solution is to organize the information before calculating.

Method 1: Separate the categories

Do not mix sales revenue, cost of goods sold, operating expenses, loans, and personal withdrawals as though they all mean the same thing.

Identify the category of each item first.

Method 2: Use a table

When a question contains many figures, place them in a table.

Item	Amount
Sales revenue	₦180,000
Cost of goods sold	₦100,000
Operating expenses	₦35,000

The table makes the financial story easier to follow.

Method 3: Calculate in stages

Instead of performing every operation at once, divide the problem into smaller steps.

For example:

1. Calculate revenue.
2. Calculate cost of goods sold.
3. Find gross profit.
4. Deduct operating expenses.
5. Consider any other relevant income or expenses.

This reduces confusion and makes mistakes easier to identify.

Method 4: Check the meaning of the answer

If a shop earns ₦100,000 in sales and incurs ₦120,000 in relevant costs, a positive profit would not make sense under those assumptions.

The result should be a loss of ₦20,000.

Understanding the situation helps you recognize unreasonable answers.

Method 5: Distinguish percentages carefully

Before calculating a percentage, ask:

"Percentage of what?"

- Profit percentage on cost compares profit with cost.
- Profit margin compares profit with sales revenue.
- Discount percentage compares the reduction with the original price.
- Expense percentage compares an expense with a chosen reference amount.

The reference amount determines the meaning of the percentage.

Method 6: Check the units and currency

Do not mix naira and kobo carelessly, and do not add unrelated quantities.

If one figure is recorded in thousands of naira and another is recorded in naira, convert them to the same unit before adding or subtracting.

Method 7: Use a calculator as a checking tool

A calculator is useful for arithmetic, especially when transactions are numerous.

However, it cannot decide whether a loan is revenue, whether an unsold item belongs in inventory, or whether a percentage should be based on cost or sales.

Those decisions require understanding.

The best approach is to think first, calculate second, and check the result afterward.

5.15 Common Accounting Mistakes

Mistake 1: Treating revenue as profit

A business that earns ₦500,000 in sales has not necessarily made ₦500,000 in profit.

Costs and expenses must be considered.

Mistake 2: Ignoring operating expenses

A trader may calculate the difference between sales and purchase costs but forget rent, transport, electricity, and other expenses.

This can overstate the business's profit.

Mistake 3: Confusing markup with profit margin

A markup based on cost and a margin based on sales use different reference amounts.

Always identify the percentage being requested.

Mistake 4: Assuming cash received is profit

A loan can increase cash without creating profit. A credit sale can generate revenue before the customer pays.

Understand the transaction before classifying it.

Mistake 5: Treating all purchases as goods already sold

Some purchased goods may remain in inventory. Their cost should not automatically be treated as the cost of goods sold for the period.

Mistake 6: Forgetting the time period

Revenue and expenses must be considered for the relevant accounting period.

A calculation for one week should not accidentally include an entire year's figures unless the question requires it.

Mistake 7: Using the wrong reference amount

When calculating a percentage, the denominator matters.

For example, a profit of ₦10,000 is 20% of ₦50,000, but only 10% of ₦100,000.

The same profit produces different percentages when the reference amount changes.

5.16 Practical Workshop: Run a Small Business on Paper

Imagine that you operate a small shop selling notebooks.

During one month, you record the following:

- You sell notebooks for a total of ₦120,000.
- The notebooks sold cost the business ₦72,000.
- Transportation costs ₦6,000.
- Packaging costs ₦2,000.
- Rent and other operating expenses total ₦10,000.

Answer the questions below.

Question 1: What is the total sales revenue?

The revenue is given directly as ₦120,000.

Question 2: What is the gross profit?

$$\begin{aligned}\text{Gross profit} &= \text{Sales revenue} - \text{Cost of goods sold} \\ &= ₦120{,}000 - ₦72{,}000 \\ &= ₦48{,}000\end{aligned}$$

Question 3: What are the total operating expenses?

Add the expenses:

$$₦6{,}000 + ₦2{,}000 + ₦10{,}000 = ₦18{,}000$$

Question 4: What is the operating profit?

$$\begin{aligned}\text{Operating profit} &= \text{Gross profit} - \text{Operating expenses} \\ &= ₦48{,}000 - ₦18{,}000 \\ &= ₦30{,}000\end{aligned}$$

Assuming no other relevant income or expenses, the business has an operating profit of ₦30,000.

Question 5: What is the profit percentage on the cost of goods sold?

Profit percentage on cost = $(\text{Profit} \div \text{Cost of goods sold}) \times 100\%$

$$= (₦30{,}000 \div ₦72{,}000) \times 100\%$$

$$\approx 41.67\%$$

This calculation compares the operating profit with the cost of the goods sold, as specified in the question. It should not be confused with the usual gross markup calculated using gross profit and the cost of goods sold.

Question 6: What is the operating profit margin?

$$\begin{aligned}\text{Operating profit margin} &= (\text{Operating profit} \div \text{Revenue}) \times 100\% \\ &= (₦30{,}000 \div ₦120{,}000) \times 100\% \\ &= 25\%\end{aligned}$$

The operating profit is 25% of sales revenue.

Question 7: What could happen if transportation costs increased?

If transportation costs increased while every other figure remained unchanged, operating expenses would rise and operating profit would fall.

For example, if transportation increased from ₦6,000 to ₦11,000, the additional cost would be ₦5,000.

The new operating profit would be:

$$₦30{,}000 - ₦5{,}000 = ₦25{,}000$$

This demonstrates how calculations can help a business owner understand the effect of changing costs.

5.17 Chapter Exercises

Try to solve the following questions before checking your work.

Part A: Understanding the situation

1. Explain the difference between revenue and profit.
2. Why can a business have high sales but still make a loss?
3. Explain why receiving a loan does not ordinarily count as earning profit.
4. Why must an accountant distinguish between goods purchased and goods sold?

Part B: Calculations

5. A trader sells 25 school bags at ₦3,200 each. Calculate the total revenue.
6. A shop earns ₦180,000 in sales revenue. The cost of goods sold is ₦110,000, and operating expenses are ₦25,000. Calculate the gross profit and operating profit.
7. A trader buys a calculator for ₦4,000 and sells it for ₦5,200. Calculate the profit and the profit percentage on cost.
8. A pair of shoes costs ₦18,000. The shop offers a discount of 20%. Calculate the discount amount and the final selling price.

9. A trader buys a bag for ₦6,000 and adds a markup of 30% on cost.

Calculate the markup and the selling price. Then calculate the profit margin, assuming no additional costs.

10. A business has fixed costs of ₦30,000. Each product sells for ₦500

and has a variable cost of ₦300. Calculate the contribution per product and the break-even quantity.

11. A business earns ₦250,000 in revenue and has a cost of goods sold

of ₦150,000. Its operating expenses are ₦40,000. Calculate the operating profit margin.

12. A business reports an operating profit of ₦45,000 on sales revenue of

₦300,000. Calculate its operating profit margin.

Part C: Think Like an Accountant

13. Two shops have the same revenue of ₦200,000.

Shop A has total relevant costs of ₦150,000. Shop B has total relevant costs of ₦175,000.

Which shop makes more profit, and by how much?

14. A trader receives ₦90,000 in cash during a week. The trader also

receives a loan of ₦50,000. Explain why the trader's total cash received should not automatically be described as revenue or profit.

15. A shopkeeper calculates profit by subtracting the cost of goods pur-

chased during a month from sales revenue, without checking whether some goods remain unsold. Explain the possible problem with this method.

16. Create a simple financial record for a business of your choice. Include

at least three sales, three expenses, and a calculation of the result. Clearly state any assumptions you make.

5.18 Chapter Review

In this chapter, we learned that accounting begins with understanding financial events.

We explored the difference between revenue, cost, gross profit, operating expenses, operating profit, and loss. We calculated discounts, markups, profit percentages, and profit margins. We also examined break-even analysis and learned why cash flow is not the same as profit.

Most importantly, we learned to classify financial information before calculating.

A formula can produce a numerical answer, but only an understanding of the situation can tell us whether we have selected the right figures and interpreted the answer correctly.

When you see a financial problem, ask:

- What money came into the business?
- What costs were associated with the goods or services?
- What expenses were incurred?
- Which period does the question cover?
- What exactly am I being asked to find?
- Which figures belong in the calculation?
- Does the answer make sense?

These questions transform accounting from a collection of formulas into a way of thinking.

5.19 Conclusion: The Accountant Sees the Story Behind the Money

A person may see ₦100,000 and think only about the amount.

An accountant asks where it came from, what it represents, which costs relate to it, and what it reveals about the business.

That difference is the foundation of sound financial reasoning.

The accountant does not simply ask, “Which formula should I use?”

The accountant first asks, “What happened?”

Once the financial story is understood, the appropriate calculation becomes easier to identify.

The same principle applies beyond business. Economists examine the choices people make with limited resources. They study how consumers respond to prices, how businesses make decisions, and how governments manage resources.

To understand those decisions, we must learn to look beyond the numbers and examine the human choices and economic forces behind them.

In the next chapter, we will move from the shopkeeper’s accounts to the wider economy.

We will learn to **think like an economist**.

CHAPTER 6

Think Like an Economist

06

6.1 Introduction: Why Does Everything Have a Price?

Imagine you have ₦5,000.

You want to buy a textbook, eat lunch, and save some money. Unfortunately, ₦5,000 may not be enough to do everything you want.

You must make a choice.

Perhaps you buy the textbook and eat a less expensive meal. Perhaps you eat lunch and postpone buying the textbook. Or perhaps you divide the money between several needs.

Whatever you choose, something else may have to wait.

Now imagine that millions of people face similar decisions every day. Families decide how to spend their income. Businesses decide what to produce. Workers decide which jobs to accept. Governments decide how to distribute public funds.

These decisions influence prices, production, employment, trade, and the overall economy.

This is where economics begins.

Economics studies how individuals, businesses, governments, and societies make choices about using limited resources to satisfy competing wants.

An economist does not simply look at a price and ask which formula to use. An economist first asks:

- Why does this product have this price?
- What happens when its price increases?
- Why do consumers choose one product over another?
- How does a business decide what to produce?
- What happens when resources are limited?
- Why do some choices create benefits while others create costs?

The calculations used in economics are tools for answering these questions.

To understand the mathematics, we must first understand the economic situation.

6.2 Scarcity: The Problem Behind Economics

Imagine a university has ₦10 million available for a new project.

The university wants to improve its library, purchase laboratory equipment, repair classrooms, and upgrade its internet facilities.

Unfortunately, the available money cannot fully fund every project.

The university must decide how to allocate its resources.

This problem is an example of **scarcity**.

Scarcity means that available resources are limited relative to people's wants.

Resources include:

- Money.
- Time.
- Labour.
- Land.
- Machinery.
- Raw materials.

- Knowledge and skills.

Scarcity does not necessarily mean that a person has absolutely nothing. It means that available resources are insufficient to satisfy every possible want simultaneously.

Even a wealthy person has limited time. A company with substantial capital cannot employ every worker, purchase every machine, or produce every product.

Scarcity forces people to choose.

Example: A student's time

Suppose you have four hours available after lectures.

You want to study mathematics, read economics, practise a skill, and rest.

If you spend all four hours studying mathematics, you will have less time for the other activities.

The question is not simply how many hours you have. It is how you should allocate those hours among competing uses.

Economists study these choices because they influence how resources are used.

Why this matters for calculation

Suppose a student has ₦6,000 and wants to buy notebooks costing ₦500 each.

How many notebooks can the student afford if all the money is spent on notebooks?

$$\text{Number of notebooks} = \text{Total money} \div \text{Price per notebook}$$

$$= ₦6{,}000 \div ₦500$$

$$= 12 \text{ notebooks}$$

The calculation tells us that the student can purchase 12 notebooks.

But the real economic question is broader: should the student spend all ₦6,000 on notebooks?

Perhaps some money is needed for transport, food, or other materials.

Mathematics tells us what is possible under the stated conditions. Economic reasoning helps us examine whether the choice is sensible.

6.3 Opportunity Cost: What Do You Give Up?

Imagine you have Saturday afternoon free.

You can either attend a paid training programme or work at a shop where you would earn ₦4,000.

You decide to attend the training.

You may have gained useful knowledge, but you gave up the opportunity to earn ₦4,000.

That forgone opportunity is part of the cost of your decision.

In economics, **opportunity cost** is the value of the next-best alternative that is sacrificed when a choice is made.

It is not necessarily the total value of everything you did not choose. It focuses on the best alternative you gave up.

Example 1: Studying or working

Suppose you have one free evening.

Your options are:

- Attend a training session.
- Work a shift that pays ₦5,000.
- Watch a film.

You choose the training session.

If working the shift was your next-best alternative, the opportunity cost of attending the training is the ₦5,000 you gave up, along with any other relevant non-monetary considerations.

The opportunity cost is not automatically the sum of the wages and the value of the film. Those are separate alternatives, and you cannot pursue all of them at the same time.

Example 2: Using business space

A shop owner owns a small building.

The owner can use it as a shop or rent it to another trader for ₦100,000 per month.

The owner decides to operate a business in the building.

Even if the owner does not pay rent to another person, using the building has an opportunity cost: the rental income that could have been earned.

This is an important idea because a decision may have a cost even when no cash payment is made.

Example 3: Choosing between two projects

A business has enough resources to complete only one of two projects.

- Project A is expected to generate ₦500,000 in net benefit.
- Project B is expected to generate ₦350,000 in net benefit.

The business chooses Project A.

If these estimates are comparable and Project B is the next-best alternative, the opportunity cost of choosing A is the ₦350,000 benefit forgone from B.

The business should not subtract ₦350,000 from ₦500,000 and call the result the opportunity cost. The opportunity cost is the value of the forgone alternative itself.

Why economists care about opportunity cost

Without this concept, people may believe that a decision is free simply because it requires no direct payment.

But every choice uses resources that could have been used elsewhere.

The central question is:

What is the best alternative I must give up to make this choice?

That question can improve decisions in education, business, government, and personal life.

6.4 Demand: Why Do People Buy Products?

Imagine that a shop sells bottled water for ₦200 per bottle.

Many students buy it during the day.

The following week, the price increases to ₦500. Some students continue buying it, but others decide to buy a cheaper drink or carry water from home.

What has changed?

The price has increased, and some consumers have reduced the quantity they are willing and able to buy.

Economists use the concept of **demand** to study this relationship.

Demand refers to the quantities of a product that consumers are willing and able to purchase at different prices during a given period, assuming other relevant factors remain unchanged.

The phrase *willing and able* is important.

A person may want a car but lack the income to buy it. Wanting the car alone does not establish effective market demand for that purchase.

The law of demand

The law of demand states that, other things being equal, the quantity demanded of a good generally increases when its price falls and decreases when its price rises.

This describes a common relationship, not a claim that every product behaves identically under every circumstance.

Example: Bottled water

Suppose the demand for bottled water at different prices is recorded as follows.

Price per bottle	Quantity demanded per day
₦100	100 bottles
₦200	80 bottles
₦300	60 bottles
₦400	40 bottles
₦500	20 bottles

Price per bottle	Quantity demanded per day
₦100	100 bottles
₦200	80 bottles
₦300	60 bottles
₦400	40 bottles
₦500	20 bottles

As the price increases, the quantity demanded decreases in this example.

The table describes a hypothetical market. It is not a claim about the actual demand for bottled water in a particular location.

Understanding the situation before drawing a graph

Imagine that we want to represent the information visually.

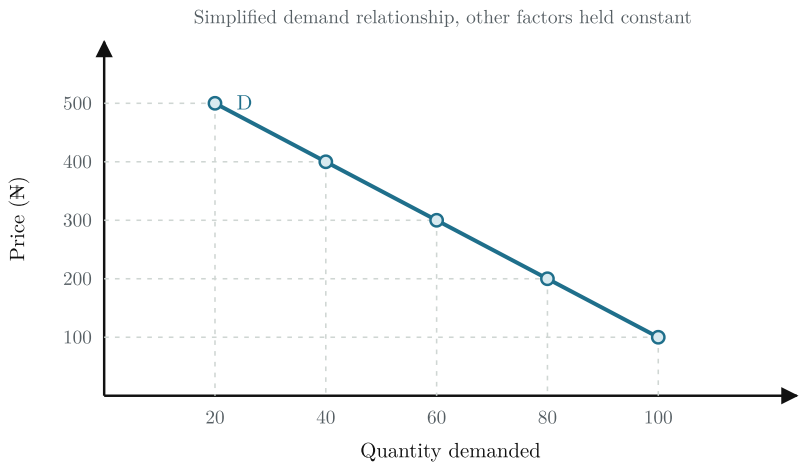
The horizontal axis shows quantity demanded.

The vertical axis shows price.

Each row of the table provides one point on the graph.

FIGURE 4

Demand Curve



Five hypothetical points from (100, 100) to (20, 500). As price rises, quantity demanded falls.

The downward slope represents the inverse relationship between price and quantity demanded in this example.

Why might quantity demanded fall when price rises?

Several explanations are possible.

> **First: Consumers may switch to alternatives.**

If bottled water becomes more expensive, some consumers may purchase another drink or use a refillable bottle.

> **Second: Higher prices reduce purchasing power.**

With the same income, a consumer may be unable to afford the same quantity at the higher price.

> **Third: Consumers may change their priorities.**

They may decide that the product is no longer worth the amount being charged.

These explanations help us understand the behaviour behind the demand curve.

A crucial distinction: Demand and quantity demanded

Suppose the price of bottled water increases, and people buy fewer bottles.

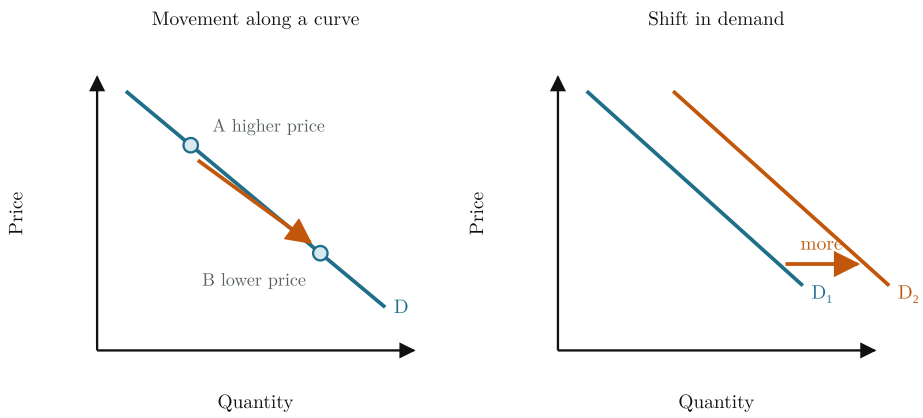
That is a change in **quantity demanded**, represented by movement along the same demand curve, assuming other factors remain unchanged.

Now imagine that a heatwave causes more people to want bottled water at every price.

Demand may increase. The entire demand curve shifts.

FIGURE 5

Movement Along a Demand Curve and a Shift in Demand



Left: a price change moves us from A to B on the same curve. Right: an increase in demand moves the whole curv from D1 to D2.

The distinction matters because a change in a product's own price normally changes quantity demanded, while factors such as income, tastes, population, and prices of related goods can shift demand.

6.5 Supply: Why Do Businesses Sell Products?

Now imagine that the price of bottled water rises from ₦200 to ₦500 per bottle.

The higher price may encourage some sellers to supply more bottled water because selling it becomes more attractive, assuming their costs and other conditions remain unchanged.

This is related to **supply**.

Supply refers to the quantities producers are willing and able to offer for sale at different prices during a given period.

The law of supply

The law of supply states that, other things being equal, the quantity supplied generally increases when the price rises and decreases when the price falls.

This relationship often occurs because higher prices can make additional production more profitable.

However, supply also depends on production costs, available equipment, labour, technology, weather, and other conditions.

Example: A small bakery

Suppose a bakery supplies loaves of bread at different prices.

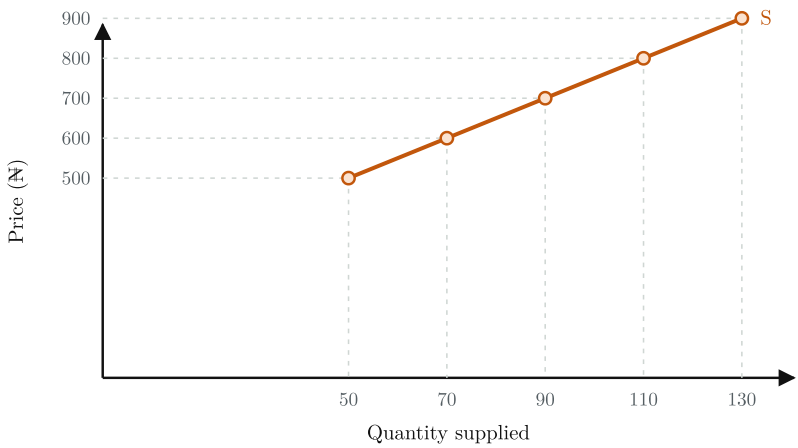
Selling price per loaf	Quantity supplied per day
₦500	50 loaves
₦600	70 loaves
₦700	90 loaves
₦800	110 loaves
₦900	130 loaves

Selling price per loaf	Quantity supplied per day
₦500	50 loaves
₦600	70 loaves
₦700	90 loaves
₦800	110 loaves
₦900	130 loaves

In this hypothetical example, the bakery is willing to supply more bread at higher prices.

FIGURE 6

Supply Curve



Points (50, 500) to (130, 900). Higher prices lead producers to supply more.

The upward slope represents the positive relationship between price and quantity supplied in the example.

Why does supply often increase with price?

A higher selling price may enable producers to cover additional production costs and earn more from supplying extra units.

For example, a bakery may find it worthwhile to hire additional workers or extend its operating hours if the selling price makes the extra production profitable.

But a higher price does not guarantee that every producer can increase supply immediately. The bakery may have limited ovens, scarce ingredients, or insufficient workers.

Economics studies both the incentive to produce and the limits on production.

6.6 Market Equilibrium: Where Demand Meets Supply

Imagine a market in which consumers want to buy bottled water and sellers want to sell it.

At some prices, consumers may want more bottles than sellers are willing to supply.

At other prices, sellers may offer more bottles than consumers want to buy.

Economists use the concept of **market equilibrium** to describe a situation in which quantity demanded equals quantity supplied.

Consider the following hypothetical information.

Price per bottle	Quantity demanded	Quantity supplied
₦100	100	20
₦200	80	40
₦300	60	60
₦400	40	80
₦500	20	100

At ₦300:

- Quantity demanded = 60 bottles.
- Quantity supplied = 60 bottles.

The two quantities are equal.

Therefore, ₦300 is the equilibrium price in this simplified table, and 60 bottles is the equilibrium quantity.

Why is this important?

At ₦100, consumers want 100 bottles, but sellers supply only 20.

The shortage is:

$100 - 20 = 80$ bottles.

At ₦500, sellers supply 100 bottles, but consumers want only 20.

The surplus is:

$100 - 20 = 80$ bottles.

At ₦300, the quantities match.

This is the equilibrium described by the table.

Shortage and surplus

A **shortage** occurs when quantity demanded exceeds quantity supplied at a given price.

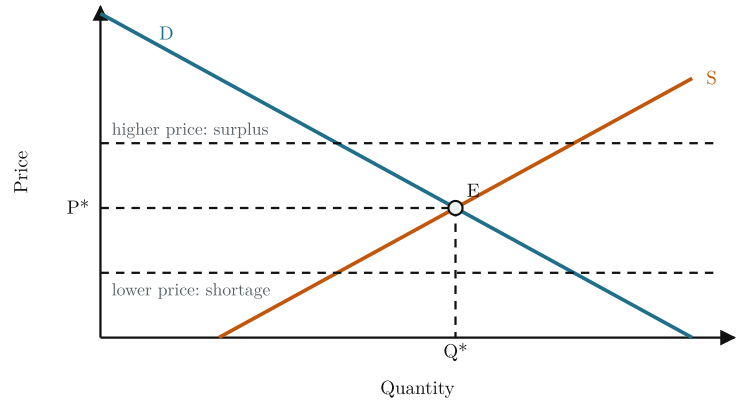
A **surplus** occurs when quantity supplied exceeds quantity demanded at a given price.

In a simplified competitive market, a shortage can put upward pressure on price, while a surplus can put downward pressure on price.

Real markets may behave differently because of price controls, contracts, market power, storage, expectations, and other factors.

FIGURE 7

Market Equilibrium



D and S cross at E. A price above equilibrium leaves a surplus, and a price below it leaves a shortage. Drawn with 100 − 2P and Q = 20 + 2P, so E is at P = 20, Q = 60.

Calculating equilibrium algebraically

Suppose the quantity demanded is described by:

$$Q_d = 100 - 2P$$

The quantity supplied is described by:

$$Q_s = 20 + 2P$$

Here:

- Qd means quantity demanded.
- Qs means quantity supplied.
- P means price.

At equilibrium:

$$Q_d = Q_s$$

Therefore:

$$100 - 2P = 20 + 2P$$

Move the constant terms to one side and the price terms to the other:

$$100 - 20 = 2P + 2P$$

$$80 = 4P$$

Divide both sides by 4:

$$P = 20$$

Substitute $P = 20$ into either equation.

Quantity demanded:

$$Q_d = 100 - 2(20)$$

$$Q_d = 100 - 40$$

$$Q_d = 60$$

Quantity supplied:

$$Q_s = 20 + 2(20)$$

$$Q_s = 20 + 40$$

$$Q_s = 60$$

Therefore, the equilibrium price is 20 monetary units, and the equilibrium quantity is 60 units.

The numbers in this algebraic model are hypothetical. The equations were chosen to illustrate how the idea of equilibrium becomes a mathematical problem.

Notice the order of reasoning: we first understood that equilibrium means equal quantities. Only then did we set the equations equal to each other.

6.7 Elasticity: How Strongly Do People Respond to Price Changes?

Imagine that the price of salt increases by 10%.

Many households may continue buying nearly the same quantity because salt is inexpensive relative to many other purchases and is commonly used in cooking.

Now imagine that the price of a particular luxury item increases by 10%.

Some consumers may postpone buying it or choose a substitute.

These are possible examples of different responses to price changes.

Economists measure the responsiveness of quantity demanded to price changes using **price elasticity of demand**.

In simple terms, elasticity asks:

When price changes, how strongly does quantity demanded respond?

The standard percentage formula is:

Price elasticity of demand = Percentage change in quantity demanded ÷ Percentage change in price

Because price and quantity demanded often move in opposite directions, the calculated value is usually negative under the standard convention. Economists frequently discuss the absolute value when classifying demand as elastic or inelastic.

Example 1: Calculating elasticity

Suppose the price of a product increases by 10%, and the quantity demanded decreases by 20%.

Using the standard percentage-change formula:

$$\text{Elasticity} = (-20\%) \div 10\%$$

$$\text{Elasticity} = -2$$

The absolute value is 2.

Since 2 is greater than 1, demand is elastic under this calculation.

This means the percentage change in quantity demanded is greater than the percentage change in price.

Example 2: Inelastic demand

Suppose the price increases by 10%, but quantity demanded decreases by only 4%.

$$\text{Elasticity} = (-4\%) \div 10\%$$

$$\text{Elasticity} = -0.4$$

The absolute value is 0.4.

Since 0.4 is less than 1, demand is inelastic under this calculation.

Quantity demanded changes by a smaller percentage than price.

Why do these calculations matter?

A business may want to understand how a price increase could affect sales revenue.

If demand is elastic, a price increase may reduce quantity demanded proportionately more than the price rises, potentially reducing revenue.

If demand is inelastic, a price increase may reduce quantity demanded proportionately less than the price rises, potentially increasing revenue.

These are useful general relationships, but actual outcomes depend on the market, the size of the price change, and other relevant factors.

A note about large price changes

For large changes between two observed prices and quantities, economists often use the midpoint method to calculate elasticity. It uses the average of the starting and ending values as the reference amount.

This helps avoid obtaining different percentage changes merely because the calculation starts from a different endpoint.

The important point is to understand what is being measured before choosing the calculation method.

6.8 Revenue, Cost and Profit in Economics

In the previous chapter, we examined accounting calculations.

Economists also study revenue, costs, and profit, but they may ask broader questions about how decisions should be made.

For example:

- Should a business produce more goods?
- Will a price increase improve revenue?
- Is a new machine worth purchasing?
- What is the cost of choosing one project over another?
- How do changes in production affect profit?

To answer these questions, we must understand how revenue and costs behave.

Total revenue

Total revenue is the amount a business receives from selling its output.

$$\text{Total revenue} = \text{Price per unit} \times \text{Quantity sold}$$

Suppose a business sells 100 notebooks at ₦800 each.

Total revenue:

$$100 \times ₦800 = ₦80{,}000$$

If it sells 150 notebooks at the same price:

$$150 \times ₦800 = ₦120{,}000$$

Revenue increases because more notebooks are sold at the same price.

However, a business cannot always sell additional units at the same price. Demand conditions may require a price reduction to attract more customers.

Total cost

Total cost includes the costs associated with producing a particular quantity of output.

For a simple model:

$$\text{Total cost} = \text{Fixed cost} + \text{Variable cost}$$

Suppose a bakery has fixed costs of ₦20,000 per day and variable costs of ₦300 for every loaf produced.

If the bakery produces 100 loaves:

$$\text{Variable cost} = 100 \times ₦300 = ₦30{,}000$$

$$\text{Total cost} = ₦20{,}000 + ₦30{,}000$$

$$\text{Total cost} = ₦50{,}000$$

If the bakery produces 150 loaves:

$$\text{Variable cost} = 150 \times ₦300 = ₦45{,}000$$

$$\text{Total cost} = ₦20{,}000 + ₦45{,}000$$

$$\text{Total cost} = ₦65{,}000$$

The variable cost increases as production increases, while the fixed cost remains unchanged under the assumptions of this example.

In real businesses, fixed costs may change when production expands enough to require additional premises, machinery, or management.

Profit

Economic profit in the simplest revenue-cost model is:

$$\text{Profit} = \text{Total revenue} - \text{Total cost}$$

If revenue exceeds cost, the difference is positive.

If cost exceeds revenue, the difference is negative.

For the bakery producing 100 loaves, suppose each loaf sells for ₦700.

Revenue:

$$100 \times ₦700 = ₦70{,}000$$

Total cost:

₦50,000

Profit in this simplified model:

$$₦70{,}000 - ₦50{,}000 = ₦20{,}000$$

Now suppose the bakery produces 150 loaves and sells every loaf at ₦700.

Revenue:

$$150 \times ₦700 = ₦105{,}000$$

Total cost:

₦65,000

Profit:

$$₦105{,}000 - ₦65{,}000 = ₦40{,}000$$

Under these assumptions, increasing production raises profit.

But this result depends on the assumption that all additional loaves can be sold at the same price and that the variable cost remains ₦300 per loaf. If those conditions change, the outcome may change too.

6.9 Inflation: Why Does Money Sometimes Buy Less?

Imagine that last year, ₦2,000 could buy a particular collection of goods.

This year, the same collection costs ₦2,400.

Your money has not changed in numerical value, but the goods have become more expensive.

If this pattern occurs across a broad range of goods and services in an economy, it may reflect inflation.

Inflation is a sustained increase in the general price level of goods and services over time.

It does not mean that every price rises by the same percentage. Some prices may rise sharply, others may rise slowly, and some may fall.

Economists measure inflation using price indexes that track changes in the prices of a selected basket of goods and services.

Calculating a price increase

Suppose a basket of goods costs ₦2,000 in one period and ₦2,400 in the next.

The increase in cost is:

$$₦2{,}400 - ₦2{,}000 = ₦400$$

The percentage increase is:

$$\begin{aligned}\text{Percentage increase} &= (\text{Increase} \div \text{Original value}) \times 100\% \\ &= (₦400 \div ₦2{,}000) \times 100\% \\ &= 0.2 \times 100\% \\ &= 20\%\end{aligned}$$

The basket costs 20% more than before.

This example illustrates a price change for one basket. It does not establish that the overall inflation rate in an economy is 20%, because an official inflation measure uses a broader set of prices and a defined methodology.

Why inflation matters

Inflation can affect:

- Household purchasing power.
- Business production costs.
- Savings and borrowing decisions.
- Wages and salaries.
- Government spending.
- The prices of imported goods.

Suppose your income remains unchanged while the prices of the goods you regularly purchase rise. You may be able to afford fewer of those goods.

That is why economists study not only how much money people earn but also what that money can buy.

Nominal income and real income

Nominal income is measured in money terms.

Real income adjusts nominal income for changes in the price level, providing a better indication of purchasing power.

For example, suppose a person's nominal income increases by 10%, while the relevant price index increases by 20%.

The person's income has increased in money terms, but purchasing power has not necessarily increased.

Using a simplified index calculation, let the initial price index be 100 and the later index be 120.

If initial income is ₦100,000, the later nominal income after a 10% increase is ₦110,000.

Expressed in the initial period's purchasing-power terms:

$$\text{Real income measure} = ₦110{,}000 \times (100 \div 120)$$

$$\approx ₦91{,}666.67$$

The initial income was ₦100,000 in the same reference-period terms.

Under these assumptions, the person's purchasing power has fallen, despite the increase in nominal income.

This calculation shows why an increase in salary does not automatically mean an improvement in living standards.

6.10 Gross Domestic Product: Measuring Economic Production

Imagine that a country produces food, clothing, furniture, transportation services, software, and many other goods and services during one year.

How can economists summarize the value of the country's production?

One major measure is **Gross Domestic Product**, commonly called GDP.

GDP measures the market value of final goods and services produced within a country's borders during a specified period.

The word *final* matters because counting the same product repeatedly at different stages of production can exaggerate total output.

A simple example

Imagine an economy produces only three final products in one year:

Final product	Quantity	Price per unit	Value
Rice bags	1,000	₦20,000	₦20,000,000
Chairs	500	₦15,000	₦7,500,000
Tables	200	₦40,000	₦8,000,000
Total			**₦35,500,000**

The total market value of the final products is:

$$₦20{,}000{,}000 + ₦7{,}500{,}000 + ₦8{,}000{,}000 = ₦35{,}500{,}000$$

Under the simplified assumption that these are all the final goods produced within the economy during the period, this gives a GDP figure of ₦35,500,000 for the example.

A real economy is much more complex. It produces a vast range of goods and services, and GDP accounting must follow defined rules about production, prices, imports, exports, and other items.

Why not count every sale?

Imagine that a farmer sells wheat to a flour mill. The mill processes the wheat and sells flour to a bakery. The bakery uses the flour to produce bread.

If we simply added the full value of the wheat, flour, and bread, we could count the value of the same underlying materials more than once.

GDP accounting avoids this kind of double counting by focusing on final output or, equivalently, by summing value added across production stages.

Value added is the value of a producer's output minus the value of intermediate goods and services used to produce it.

What GDP tells us—and what it does not

GDP helps economists measure the size and growth of economic production.

However, GDP alone does not tell us everything about people's well-being.

It does not directly show how income is distributed, whether people feel secure, how much unpaid household work is performed, or the full environmental consequences of production.

A country may have growing GDP while some households experience financial hardship.

Therefore, GDP is an important measure, but it should not be treated as a complete measure of human progress.

6.11 Economic Growth: Comparing Different Years

Suppose a country's real GDP increases from ₦100 billion to ₦110 billion.

What is the growth rate?

First, calculate the increase:

$$₦110 \text{ billion} - ₦100 \text{ billion} = ₦10 \text{ billion}$$

Next, divide the increase by the original value and multiply by 100%.

$$\begin{aligned}\text{Growth rate} &= (\text{Change in real GDP} \div \text{Initial real GDP}) \times 100\% \\ &= (₦10 \text{ billion} \div ₦100 \text{ billion}) \times 100\% \\ &= 10\%\end{aligned}$$

Real GDP grew by 10%.

Why use real GDP?

Nominal GDP measures output using the prices of the period being measured.

If prices rise, nominal GDP may increase even when the amount of goods and services produced has not increased.

Real GDP adjusts for price changes, helping economists compare production across time more meaningfully.

Example: Nominal growth versus real growth

Suppose the total money value of a country's production increases from ₦100 billion to ₦120 billion.

Nominal growth is:

$$[(₦120 \text{ billion} - ₦100 \text{ billion}) \div ₦100 \text{ billion}] \times 100\%$$

$$= 20\%$$

But suppose the general price level increased by 10% over the same period.

The economy's real growth would be lower than its nominal growth.

A precise real-growth calculation requires an appropriate price index or GDP deflator, rather than simply subtracting inflation from nominal growth in every situation.

The general lesson is that we must understand what a number measures before interpreting it.

A 20% increase in money value is not automatically a 20% increase in actual production.

6.12 Unemployment: More Than a Percentage

Imagine that a country has a population of 10 million people.

Not everyone in that population is part of the labour force. Some are children, some are retired, some are studying without working or seeking work, and others may not currently be available for employment.

Suppose the country has a labour force of 4 million people, including those who are employed and those who are unemployed under the applicable statistical definition.

If 400,000 people are unemployed, what is the unemployment rate?

$$\begin{aligned}\text{Unemployment rate} &= (\text{Number of unemployed people} \div \text{Labour force}) \times 100\% \\ &= (400{,}000 \div 4{,}000{,}000) \times 100\% \\ &= 0.1 \times 100\% \\ &= 10\%\end{aligned}$$

The unemployment rate is 10%.

Why is the denominator important?

A common mistake is to divide the number of unemployed people by the entire population.

In this example, dividing 400,000 by 10 million would give 4%.

But that is not the unemployment rate under the definition used here.

The unemployment rate uses the labour force as its denominator.

The denominator determines what the percentage means.

What does unemployment measure?

Unemployment statistics generally focus on people who are without work, available for work, and seeking work according to the relevant survey or official definition.

People who want employment but are not currently counted as unemployed under that definition may still face serious difficulties. Their situation may be reflected in other measures, such as labour underutilization or broader employment statistics.

Therefore, one percentage cannot describe every aspect of the labour market.

Why economists study unemployment

Unemployment can affect household income, living standards, production, government revenue, and social conditions.

Economists investigate why unemployment occurs, how it changes over time, and which policies may improve employment opportunities.

The calculation provides a summary measure, while further analysis helps explain the causes and consequences.

6.13 Economic Models: Simplifying Reality to Understand It

The economy contains millions of people, businesses, institutions, and decisions.

It would be difficult to understand every detail simultaneously.

Economists therefore use models.

An economic model is a simplified representation of an economic situation that helps explain relationships or examine possible outcomes.

A model does not need to include every detail of reality. It needs to represent the relevant features of the question being studied.

Example: A simple market model

Suppose we describe demand with:

$$Q_d = 100 - 2P$$

And supply with:

$$Q_s = 20 + 2P$$

These equations are not intended to describe every real market. They are simplified representations of how quantity demanded and quantity supplied might relate to price.

The model allows us to calculate equilibrium and examine what happens when conditions change.

What happens if demand increases?

Suppose demand changes to:

$$Q_d = 120 - 2P$$

Supply remains:

$$Q_s = 20 + 2P$$

At equilibrium:

$$120 - 2P = 20 + 2P$$

Subtract 20 from both sides:

$$100 - 2P = 2P$$

Add 2P to both sides:

$$100 = 4P$$
$$P = 25$$

Substitute into the supply equation:

$$Q_s = 20 + 2(25)$$
$$Q_s = 20 + 50$$
$$Q_s = 70$$

The new equilibrium price is 25 monetary units, and the new equilibrium quantity is 70 units.

Compare this with the earlier equilibrium:

- Earlier equilibrium price = 20.
- Earlier equilibrium quantity = 60.
- New equilibrium price = 25.
- New equilibrium quantity = 70.

In this simplified model, an increase in demand raises both equilibrium price and equilibrium quantity.

Why this conclusion depends on assumptions

The result assumes that the supply relationship remains unchanged and that the market behaves according to the model.

In a real market, producers may face capacity limits, government policies may affect prices, or other changes may occur at the same time.

The model helps isolate one relationship so that we can study it more clearly.

This is similar to a scientist using a simplified experiment: controlling certain factors makes it easier to understand the effect of another factor.

6.14 How to Make Economic Calculations Easier

Economic problems often look difficult because they combine several ideas.

The best approach is to separate the reasoning from the arithmetic.

Method 1: Identify the economic concept

Before calculating, determine whether the question concerns:

- Scarcity.
- Opportunity cost.
- Demand or supply.
- Equilibrium.
- Elasticity.
- Revenue or profit.
- Inflation.
- GDP.
- Economic growth.
- Unemployment.

The concept determines what the calculation should measure.

Method 2: Identify the relevant quantities

Suppose a question asks for the unemployment rate.

You need the number of unemployed people and the labour force.

You do not automatically need the entire population.

If a question asks for economic growth, you need the initial and final values of the relevant GDP measure.

Choosing the correct figures is as important as performing the calculation.

Method 3: Understand the denominator

Percentages compare one quantity with another.

Always ask:

“Compared with what?”

For example:

- Inflation measures a change in a price index.
- Unemployment rate compares unemployed people with the labour force.
- Economic growth compares a change in output with an initial output measure.
- Elasticity compares percentage changes in quantity and price.

Using the wrong denominator produces a different measure.

Method 4: Write down the known information

Suppose the question states:

- Initial GDP = ₦500 billion.
- Final GDP = ₦550 billion.

Write the information down before calculating.

Change in GDP:

$$₦550 \text{ billion} - ₦500 \text{ billion} = ₦50 \text{ billion}$$

Growth rate:

$$(₦50 \text{ billion} \div ₦500 \text{ billion}) \times 100\% = 10\%$$

Separating the steps makes the calculation easier to follow.

Method 5: Use a table for comparisons

When a problem contains several prices and quantities, a table helps organize the data.

It can also reveal relationships that are difficult to notice when figures are presented in a long paragraph.

Method 6: Check whether the result makes sense

If quantity demanded is calculated as negative in a situation where the model is intended to describe a normal positive quantity, check the equation and the assumptions.

If a country's GDP increases by ₦20 billion from an initial value of ₦100 billion, the growth rate should be 20%, not 2%.

Reasoning helps detect arithmetic and interpretation errors.

Method 7: Separate the model from reality

A model may assume that other factors remain unchanged.

Do not forget that assumption when interpreting the answer.

The result tells you what follows within the model, not necessarily everything that will happen in the real world.

6.15 Common Mistakes in Economic Thinking

Mistake 1: Thinking that a higher price always means higher profit

A higher price may increase revenue per unit, but it can also reduce the quantity sold.

The effect on total revenue and profit depends on demand, costs, and other conditions.

Mistake 2: Confusing demand with desire

A person may desire a product without having the willingness and ability to purchase it at the stated price.

Economics distinguishes ordinary wants from effective market demand.

Mistake 3: Ignoring opportunity cost

A choice may appear free when no cash changes hands, but it may still require giving up a valuable alternative.

Mistake 4: Assuming that GDP measures everything important

GDP measures economic production. It does not directly measure every aspect of well-being, equality, environmental quality, or happiness.

Mistake 5: Confusing nominal and real growth

An increase in the money value of production may partly reflect higher prices rather than an increase in actual output.

Mistake 6: Using the wrong denominator

An unemployment rate uses the labour force, not the total population, under the standard definition presented in this chapter.

Mistake 7: Treating a model as a perfect description of reality

Economic models simplify situations. Their conclusions depend on their assumptions and the relationships they include.

Mistake 8: Calculating before understanding

A correct calculation applied to the wrong concept can still produce a misleading answer.

Always identify the economic question before selecting the mathematical method.

6.16 Practical Workshop: Think Like an Economist

Consider the following situations.

Situation A: The university budget

A university has ₦20 million available for development.

It is considering:

- Improving the library.
- Purchasing laboratory equipment.
- Renovating classrooms.

The university cannot fully fund every project.

Questions:

1. What economic problem does the university face?
2. Why must the university make a choice?
3. Explain the opportunity cost of selecting one project over the next-best alternative.
4. What additional information would help the university make a better decision?

There is no single numerical answer because the problem requires identifying the relevant information and evaluating alternatives.

Situation B: A change in price

A shop sells 100 bottles of a drink per day at ₦200 each.

After the price rises to ₦250, it sells 80 bottles per day.

Question 1: What was the original revenue?

$$100 \times ₦200 = ₦20{,}000$$

Question 2: What is the new revenue?

$$80 \times ₦250 = ₦20{,}000$$

Question 3: Did revenue increase?

No. Revenue remained ₦20,000 per day.

The higher price was offset by the reduction in quantity sold.

Question 4: Can we conclude that profit remained unchanged?

No. We would need information about costs and other relevant factors.

This example shows why sales revenue and profit must be distinguished.

Situation C: Inflation and purchasing power

A collection of goods costs ₦10,000 in one year and ₦11,500 the following year.

Question 1: What is the increase in cost?

$$₦11{,}500 - ₦10{,}000 = ₦1{,}500$$

Question 2: What is the percentage increase?

$$(₦1{,}500 \div ₦10{,}000) \times 100\% = 15\%$$

The collection costs 15% more than before.

Question 3: If a person's income increases by only 5%, has the person's purchasing power necessarily increased?

No. If the relevant prices rise by 15% while income rises by only 5%, the person will generally be able to purchase less of that collection, assuming the other conditions remain comparable.

Situation D: Calculating economic growth

A country's real GDP rises from ₦800 billion to ₦880 billion.

Calculate the growth rate.

Change in GDP:

$$₦880 \text{ billion} - ₦800 \text{ billion} = ₦80 \text{ billion}$$

Growth rate:

$$(₦80 \text{ billion} \div ₦800 \text{ billion}) \times 100\% = 10\%$$

Answer: Real GDP grew by 10%.

Situation E: Market equilibrium

Suppose:

$$Q_d = 90 - P$$
$$Q_s = 30 + 2P$$

At equilibrium:

$$90 - P = 30 + 2P$$

Subtract 30:

$$60 - P = 2P$$

Add P:

$$60 = 3P$$
$$P = 20$$

Substitute into the demand equation:

$$Q_d = 90 - 20 = 70$$

Substitute into the supply equation:

$$Q_s = 30 + 2(20) = 70$$

Answer: The equilibrium price is 20 monetary units, and the equilibrium quantity is 70 units.

Situation F: Make your own economic model

Choose a product that people commonly buy, such as bread, notebooks, transport services, or bottled water.

Create a hypothetical table showing how the quantity demanded might change at five different prices.

Then answer:

1. What relationship do you observe?
2. What other factors could affect demand?
3. Would every consumer respond to price changes in the same way?
4. What information would you need to investigate the actual market?
5. How could a graph help explain your findings?

The purpose is not to guess what every real market will do. It is to learn how to build a clear explanation from stated assumptions and evidence.

6.17 Chapter Exercises

Attempt these questions before reviewing the chapter.

Part A: Understanding economic ideas

1. Define scarcity and explain why it exists even in wealthy societies.
2. What is opportunity cost? Give one example from a student's life.
3. Distinguish between demand and quantity demanded.
4. Explain the difference between a shortage and a surplus.
5. What is market equilibrium?
6. Why is GDP not a complete measure of human well-being?
7. Explain the difference between nominal income and real income.
8. Why do economists use models?

Part B: Calculations

9. A student has ₦15,000 and buys six textbooks at ₦2,000 each. How much money remains?
10. The price of a product rises from ₦400 to ₦500. Calculate the percentage increase.
11. A country's real GDP rises from ₦2 trillion to ₦2.3 trillion. Calculate the growth rate.
12. A country has 500,000 unemployed people and a labour force of 5 million. Calculate the unemployment rate.
13. A product's price increases by 8%, while its quantity demanded decreases by 16%. Calculate price elasticity of demand using the standard percentage-change method.
14. Suppose quantity demanded is:

$$Q_d = 120 - 2P$$

And quantity supplied is:

$$Q_s = 30 + P$$

Calculate the equilibrium price and quantity.

15. A business sells 200 products at ₦1,500 each. Its total cost is ₦240,000.

Calculate its revenue and profit under the stated assumptions.

Part C: Economic reasoning

16. A government spends more money on roads and less on public libraries.

Explain how opportunity cost applies to this decision.

17. A business raises the price of a product and discovers that its revenue

falls. Give a possible economic explanation.

18. A person's salary increases by 8%, but the prices of the goods they

regularly buy rise by 12%. Explain what may happen to purchasing power.

19. A market has a shortage at the current price. Under a simplified

competitive-market model, what pressure might this place on price? Explain why the outcome may differ in a market with price controls.

20. Design a simple economic model for a product of your choice. State

the assumptions, explain what the model is intended to show, and identify at least two limitations.

6.18 Chapter Review

In this chapter, we explored the thinking behind economic calculations.

We learned that scarcity forces people to make choices and that opportunity cost represents the value of the next-best alternative given up.

We examined demand, supply, market equilibrium, and elasticity. We also studied revenue, cost, profit, inflation, GDP, economic growth, and unemployment.

Throughout the chapter, we returned to one central principle:

> **The calculation should answer an economic question, not replace the need to understand it.**

When you see an economic problem, ask:

- What decision is being made?
- Which resources are limited?
- What alternative is being sacrificed?
- What relationship does the question describe?
- Which quantities are relevant?
- What does the percentage compare?
- What assumptions are being made?
- What can the result tell us, and what can it not tell us?

These questions help turn economic mathematics into a tool for understanding human decisions and the world around us.

6.19 Conclusion: From Money to Human Choices

The accountant examines the financial story behind a business.

The economist looks beyond an individual business to understand choices, incentives, markets, and the allocation of resources.

A rise in price is not merely a number. It may change what a family can afford, how much a business produces, and which alternatives consumers choose.

An increase in income is not automatically an increase in purchasing power. A larger economy is not automatically a fairer or happier society. A decision that produces one benefit may also involve a cost that is less obvious.

Economic thinking teaches us to examine these relationships carefully.

The next time you encounter an economic formula, do not begin by asking only how to use it.

Ask what human problem it was designed to explain.

Once you understand the problem, the mathematics becomes more meaningful.

In the next chapter, we will explore another powerful idea: using symbols to represent unknown quantities and relationships.

We will learn to **think like an algebraist**.

CHAPTER 7

Algebra — Finding What You Do Not Know

07

7.1 Introduction: The Mystery of the Missing Number

Imagine that you walk into a shop to buy a notebook and a pen.

The shopkeeper tells you that three notebooks and two pens cost ₦1,600 altogether.

You know the total price, but you do not know the individual prices of the notebooks and pens.

How can you find them?

You might begin by guessing. Perhaps each notebook costs ₦400 and each pen costs ₦200. That would give:

$$3(400) + 2(200) = 1600$$

In this case, the guess works.

But what if the prices are not so easy to guess? What if you are given several totals from different purchases? What if the unknown quantity is not the price of a product but the distance travelled, the number of workers required, or the amount of money someone has?

This is where algebra becomes useful.

Algebra allows us to represent unknown quantities using symbols and describe relationships between them using mathematical expressions and equations.

Instead of guessing a number, we can build a mathematical representation of the situation and solve it systematically.

But before we begin manipulating letters, let us understand the problem algebra was developed to help us solve.

7.2 Why Did Mathematics Need Algebra?

For centuries, people used arithmetic to count objects, measure land, calculate payments, and share resources.

Arithmetic works very well when the quantities are known.

For example:

$$25 + 15 = 40$$

If you know that you have 25 books and receive another 15, you can calculate the total.

But suppose you know that you have some books, receive 15 more, and end up with 40.

How many books did you have originally?

You could reason backward:

$$40 - 15 = 25$$

The unknown quantity can be found through arithmetic.

Now imagine a more complicated situation.

A trader buys several bags of rice at the same price per bag and pays a transportation fee. The trader knows the total amount spent but wants to determine the price of each bag.

As the number of unknowns and relationships increases, ordinary arithmetic becomes less convenient.

Algebra provides a systematic language for expressing these relationships.

A brief history of algebra

The development of algebra took place over many centuries and involved mathematicians from several civilizations.

Ancient Babylonian mathematical texts contain methods for solving problems that can be represented using equations. Ancient Egyptian mathematical texts also describe procedures for finding unknown quantities.

Greek mathematicians studied geometric relationships that later influenced the development of algebraic reasoning.

Indian mathematicians made important contributions to arithmetic and algebra, including work involving negative numbers and equations.

In the medieval Islamic world, the Persian mathematician Muhammad ibn Musa al-Khwarizmi wrote a major work on solving equations in the ninth century. His work helped systematize methods for solving linear and

quadratic equations. The word *algebra* comes from a term in the title of that work, commonly translated as *restoration* or *reduction*.

Over time, mathematicians developed more symbolic methods, making it easier to represent unknowns and general relationships with letters.

Modern algebra now supports science, engineering, economics, computing, statistics, and many other fields.

The important point is that algebra was not created merely to make students solve strange questions containing letters.

It developed because people needed systematic ways to describe and solve problems involving unknown quantities.

7.3 Enter the Situation Before Using a Letter

Imagine that you are helping a friend organize a small event.

You know that the venue costs ₦15,000. You also know that each participant's meal costs ₦2,000.

The total cost depends on the number of participants.

If 5 people attend, the cost is:

$$15000 + 5(2000) = 25000$$

If 10 people attend:

$$15000 + 10(2000) = 35000$$

Now suppose you want to calculate the cost for any number of participants, without repeating the calculation every time.

You can represent the number of participants with a letter.

Let:

$$n = \text{number of participants}$$

The total cost becomes:

$$C = 15000 + 2000n$$

Here, C represents the total cost.

This equation describes the relationship between the number of participants and the amount of money required.

If $n = 5$:

$$C = 15000 + 2000(5)$$
$$C = 25000$$

If $n = 10$:

$$C = 15000 + 2000(10)$$
$$C = 35000$$

The same equation works for different numbers of participants.

That is one of algebra's greatest strengths: **one mathematical relationship can describe many different situations.**

7.4 What Do Letters Mean in Algebra?

In algebra, letters represent quantities.

Depending on the situation, a letter may represent a number that is unknown, a number that can change, or a fixed but unspecified number.

For example:

$$x + 5 = 12$$

Here, x represents the unknown number we want to find.

Subtract 5 from both sides:

$$x = 12 - 5$$

Therefore:

$$x = 7$$

But consider:

$$y = 3x + 2$$

In this equation, x may take different values, and the value of y depends on x.

If $x = 1$:

$$y = 3(1) + 2 = 5$$

If $x = 4$:

$$y = 3(4) + 2 = 14$$

In this case, x and y can represent changing quantities.

Unknowns and variables

An **unknown** is a quantity whose value must be determined in a particular problem.

A **variable** is a quantity that can take different values within a mathematical relationship.

The same letter can serve either role, depending on the context.

For example, in:

$$x + 4 = 10$$

we usually seek one value of x.

In:

$$y = 2x$$

we may study how y changes as x changes.

Why use letters instead of blank spaces?

Suppose we write:

$$\square + 8 = 20$$

The blank represents an unknown number.

Algebra uses a letter instead:

$$x + 8 = 20$$

The letter makes it easier to write longer expressions, multiple equations, and general rules.

It also allows us to describe patterns without choosing a particular numerical value.

7.5 Expressions: Writing Mathematical Ideas

A mathematical expression is a combination of numbers, variables, and operations.

Examples include:

$$x + 5$$
$$3y$$
$$2a + 7$$
$$4m - 3n$$

An expression does not necessarily state that two quantities are equal.

For example:

$$3x + 5$$

means three times x, plus five.

If $x = 4$, we can calculate the value:

$$3(4) + 5 = 12 + 5 = 17$$

The expression has a value of 17 when $x = 4$.

Translating words into algebra

Algebra becomes much easier when you understand the language used to describe operations.

Words	Algebraic representation
A number plus 7	$x + 7$
A number minus 4	$x - 4$
Three times a number	$3x$
A number divided by 5	$\frac{x}{5}$
Twice a number plus 6	$2x + 6$
Five less than a number	$x - 5$
Five less than twice a number	$2x - 5$
The sum of a number and its square	$x + x^2$
The product of two numbers	xy
The difference between two numbers	$x - y$, if (x) is the first and (y) the second

The order of words matters.

For example, “five less than a number” means subtract five from the number:

$$x - 5$$

It does not mean:

$$5 - x$$

Similarly, “five less than twice a number” means:

$$2x - 5$$

The phrase “twice a number” must be evaluated before subtracting five.

Example 1: Translating a statement

Write “seven more than three times a number” algebraically.

Let the number be x.

Three times the number:

$$3x$$

Seven more than that:

$$3x + 7$$

Answer:

$$3x + 7$$

Example 2: Translating a more complicated statement

A number is multiplied by four, and six is then subtracted from the result.

Let the number be x.

Multiplying by four gives:

$$4x$$

Subtracting six gives:

$$4x - 6$$

Answer:

$$4x - 6$$

The goal is not to memorize every phrase. It is to understand the sequence of operations described by the words.

7.6 Simplifying Algebraic Expressions

Imagine that you sell notebooks and pens.

You earn x naira from one notebook and another x naira from a second notebook.

Your total earnings from the notebooks are:

$$x + x = 2x$$

Instead of writing $x+x$ repeatedly, algebra allows us to combine like terms.

Like terms have the same variable raised to the same power.

Examples of like terms include:

$$3x, \quad 5x, \quad -2x$$

They can be combined because they represent multiples of the same quantity.

For example:

$$3x + 5x = 8x$$

The coefficients are added:

$$3 + 5 = 8$$

The variable remains x.

Example 1: Combining like terms

Simplify:

$$4x + 7x$$

Add the coefficients:

$$4 + 7 = 11$$

Therefore:

$$4x + 7x = 11x$$

Example 2: Combining positive and negative terms

Simplify:

$$9x - 4x + 2x$$

Combine the coefficients:

$$9 - 4 + 2 = 7$$

Therefore:

$$9x - 4x + 2x = 7x$$

Example 3: Simplifying an expression with two variables

Simplify:

$$3a + 4b + 2a - b$$

Group the a-terms:

$$3a + 2a = 5a$$

Group the b-terms:

$$4b - b = 3b$$

Therefore:

$$3a + 4b + 2a - b = 5a + 3b$$

Why can we combine $3x$ and $5x$, but not $3x$ and $5y$?

Because $3x$ and $5x$ represent three and five copies of the same quantity.

But $3x$ and $5y$ may represent entirely different quantities.

Unless additional information relates x and y, they cannot be combined into one like term.

Therefore:

$$3x + 5y$$

is already simplified.

Example 4: Expressions with powers

Simplify:

$$4x^2 + 3x + 2x^2 + 5x$$

Combine the x^2-terms:

$$4x^2 + 2x^2 = 6x^2$$

Combine the x-terms:

$$3x + 5x = 8x$$

Therefore:

$$6x^2 + 8x$$

Notice that x^2 and x are not like terms. They represent different powers of the variable.

7.7 Multiplication and the Distributive Law

Suppose a student buys three packs of pens. Each pack costs $x + 200$ naira, where x represents the price of the pens themselves and ₦200 represents the packaging cost.

The total cost is:

$$3(x + 200)$$

This means three times the entire quantity inside the brackets.

We can calculate it by distributing the multiplication:

$$3(x + 200) = 3x + 600$$

This is called the distributive law.

In general:

$$a(b + c) = ab + ac$$

The number or expression outside the brackets multiplies every term inside the brackets.

Example 1: Expanding brackets

Expand:

$$4(x + 3)$$

Multiply 4 by each term:

$$4(x) + 4(3)$$

Therefore:

$$4x + 12$$

Example 2: A negative number outside the bracket

Expand:

$$-2(x + 5)$$

Multiply -2 by both terms:

$$-2(x) + (-2)(5)$$

Therefore:

$$-2x - 10$$

A negative multiplier changes the signs of both positive terms inside the bracket.

Example 3: Subtracting a bracket

Simplify:

$$8x - (3x + 4)$$

Subtracting the entire bracket means multiplying every term inside it by -1:

$$8x - 3x - 4$$

Combine like terms:

$$5x - 4$$

Example 4: Expanding two brackets

Expand:

$$(x + 2)(x + 3)$$

Multiply each term in the first bracket by each term in the second bracket:

$$x(x + 3) + 2(x + 3)$$

Expand:

$$x^2 + 3x + 2x + 6$$

Combine like terms:

$$x^2 + 5x + 6$$

Therefore:

$$(x+2)(x+3) = x^2 + 5x + 6$$

This method is useful in algebra, geometry, and many other areas of mathematics.

7.8 Equations: Finding the Missing Value

An equation states that two expressions are equal.

For example:

$$x + 6 = 14$$

This equation tells us that the unknown number plus six equals fourteen.

To find the unknown, we must preserve the equality while isolating x.

Subtract six from both sides:

$$x + 6 - 6 = 14 - 6$$

Therefore:

$$x = 8$$

Why subtract six?

Because addition by six is the operation applied to x, and subtraction by six reverses that operation.

We perform the same operation on both sides to preserve equality.

Think of an equation as a balanced scale

Imagine a balance scale with equal weights on both sides.

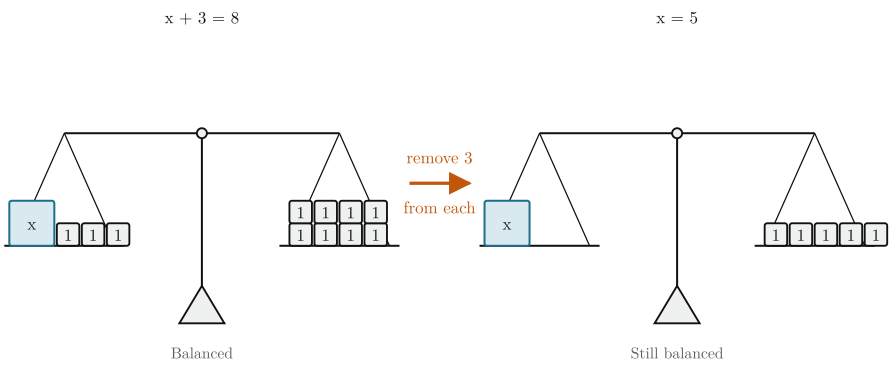
If you remove the same weight from both sides, the scale remains balanced.

If you add the same weight to both sides, it also remains balanced.

Equations follow the same principle: whatever operation you perform on one side must be performed on the other side if equality is to be preserved.

FIGURE 8

The Balanced Equation



Whatever is done to one pan must be done to the other. Removing three unit blocks from each side leaves x = 5.

Example 1: Addition

Solve:

$$x + 9 = 20$$

Subtract 9 from both sides:

$$x = 20 - 9$$

Therefore:

$$x = 11$$

Example 2: Subtraction

Solve:

$$x - 7 = 15$$

Add 7 to both sides:

$$x = 15 + 7$$

Therefore:

$$x = 22$$

Example 3: Multiplication

Solve:

$$5x = 35$$

Divide both sides by 5:

$$x = \frac{35}{5}$$

Therefore:

$$x = 7$$

Example 4: Division

Solve:

$$\frac{x}{4} = 9$$

Multiply both sides by 4:

$$x = 9(4)$$

Therefore:

$$x = 36$$

The principle is the same in every example: isolate the unknown while preserving equality.

7.9 Solving Equations with More Than One Operation

Some equations require several steps.

Consider:

$$3x + 5 = 20$$

The unknown is multiplied by three and then five is added.

To isolate x, reverse the operations in the opposite order.

First, subtract five from both sides:

$$3x = 20 - 5$$
$$3x = 15$$

Next, divide both sides by three:

$$x = \frac{15}{3}$$

Therefore:

$$x = 5$$

Example 1: A more complicated equation

Solve:

$$4x - 7 = 21$$

Add seven to both sides:

$$4x = 28$$

Divide both sides by four:

$$x = 7$$

Example 2: Unknowns on both sides

Solve:

$$5x + 3 = 2x + 18$$

We want to collect the terms containing x on one side and the constant numbers on the other.

Subtract $2x$ from both sides:

$$3x + 3 = 18$$

Subtract three from both sides:

$$3x = 15$$

Divide both sides by three:

$$x = 5$$

Example 3: Fractions in an equation

Solve:

$$\frac{x}{3} + 2 = 8$$

Subtract two:

$$\frac{x}{3} = 6$$

Multiply by three:

$$x = 18$$

Checking the answer

An important habit is to substitute the answer into the original equation.

For:

$$5x + 3 = 2x + 18$$

we found:

$$x = 5$$

Substitute 5 into the left-hand side:

$$5(5) + 3 = 25 + 3 = 28$$

Substitute 5 into the right-hand side:

$$2(5) + 18 = 10 + 18 = 28$$

Both sides equal 28.

Therefore, the solution is correct.

Checking an answer helps reveal arithmetic errors and mistakes made while rearranging an equation.

7.10 Turning Real-Life Situations into Equations

Algebra becomes much more meaningful when we use it to solve actual problems.

Example 1: Finding a person's age

A father is three times as old as his child. Their ages add up to 48 years.

How old is the child?

Step 1: Identify the unknown.

Let the child's age be x.

Step 2: Express the father's age.

The father is three times as old:

$$3x$$

Step 3: Write the relationship.

Their ages add up to 48:

$$x + 3x = 48$$

Step 4: Solve.

$$4x = 48$$
$$x = 12$$

The child is 12 years old.

The father's age is:

$$3(12) = 36$$

Check:

$$12 + 36 = 48$$

The result agrees with the information in the question.

Example 2: Buying notebooks

A student buys five identical notebooks and pays ₦3,500.

What is the price of one notebook?

Let the price of one notebook be x.

The total cost is:

$$5x = 3500$$

Divide both sides by five:

$$x = \frac{3500}{5}$$
$$x = 700$$

Each notebook costs ₦700.

Example 3: A taxi fare

A taxi service charges a fixed starting fee of ₦500 plus ₦200 for every kilometre travelled.

A passenger pays ₦2,100.

How far did the passenger travel under this pricing model?

Let the distance be d kilometres.

Total fare:

$$500 + 200d = 2100$$

Subtract 500:

$$200d = 1600$$

Divide by 200:

$$d = 8$$

The passenger travelled 8 kilometres.

Example 4: A business target

A small business has fixed expenses of ₦20,000 for an event. Each item it sells contributes ₦500 toward covering those expenses.

How many items must it sell to cover the ₦20,000?

Let the number of items be n.

$$500n = 20000$$

Divide both sides by 500:

$$n = 40$$

The business must sell 40 items to cover the fixed expenses under these assumptions.

This is a simplified break-even calculation. It assumes that each item contributes ₦500 toward the fixed costs and that there are no other relevant costs.

7.11 Simultaneous Equations: When One Equation Is Not Enough

Return to the notebook and pen problem from the beginning of the chapter.

Three notebooks and two pens cost ₦1,600.

That gives us:

$$3x + 2y = 1600$$

Here:

- x is the price of one notebook.
- y is the price of one pen.

Can we determine both prices from this equation alone?

No.

There are two unknowns but only one independent equation. Many combinations could produce the same total.

For example:

If $x = 400$ and $y = 200$:

$$3(400) + 2(200) = 1600$$

But if $x = 300$ and $y = 350$:

$$3(300) + 2(350) = 1600$$

Both combinations satisfy the equation.

We need additional information.

Suppose we are also told that two notebooks and three pens cost ₦1,650.

Now we have:

$$\begin{aligned} 3x + 2y &= 1600 \\ 2x + 3y &= 1650 \end{aligned}$$

These are simultaneous equations because both must be satisfied by the same values of x and y.

Solving by elimination

We have:

$$\begin{aligned} 3x + 2y &= 1600 \quad (1) \\ 2x + 3y &= 1650 \quad (2) \end{aligned}$$

We want to eliminate one unknown.

Multiply equation (1) by 3:

$$9x + 6y = 4800 \quad (3)$$

Multiply equation (2) by 2:

$$4x + 6y = 3300 \qquad (4)$$

Subtract equation (4) from equation (3):

$$(9x - 4x) + (6y - 6y) = 4800 - 3300$$
$$5x = 1500$$

Divide by five:

$$x = 300$$

The price of one notebook is ₦300.

Substitute $x = 300$ into equation (1):

$$3(300) + 2y = 1600$$
$$900 + 2y = 1600$$

Subtract 900:

$$2y = 700$$

Divide by two:

$$y = 350$$

The price of one pen is ₦350.

Check the answer

First purchase:

$$3(300) + 2(350) = 900 + 700 = 1600$$

Second purchase:

$$2(300) + 3(350) = 600 + 1050 = 1650$$

Both totals are correct.

Therefore:

- Notebook = ₦300.
- Pen = ₦350.

Why simultaneous equations matter

The method is useful when two or more unknown quantities are connected by several pieces of information.

Economists may use simultaneous equations to study markets. Engineers may use them to calculate unknown forces. Businesses may use them to compare products and costs.

The method is not about eliminating letters for its own sake. It is about combining pieces of information until the unknown quantities can be determined.

7.12 Inequalities: When the Answer Is a Range

Not every real-life problem asks for one exact value.

Sometimes we need to determine whether a quantity is greater than, less than, or equal to a limit.

Imagine that you have ₦10,000 for buying notebooks costing ₦500 each.

How many notebooks can you buy without exceeding your budget?

Let n represent the number of notebooks.

The total cost is:

$$500n$$

The cost must not exceed ₦10,000:

$$500n \leq 10000$$

Divide both sides by 500:

$$n \leq 20$$

You can buy at most 20 notebooks, assuming there are no other costs.

The symbol $\leq$ means "less than or equal to."

Common inequality symbols

Symbol	Meaning
$<$	Less than
$>$	Greater than
$\leq$	Less than or equal to
$\geq$	Greater than or equal to

Example 1: A spending limit

Solve:

$$3x + 4 < 19$$

Subtract four from both sides:

$$3x < 15$$

Divide by three:

$$x < 5$$

The solution is every value of x less than 5.

Example 2: A negative multiplier

Solve:

$$-2x > 8$$

Divide both sides by -2.

When multiplying or dividing an inequality by a negative number, the inequality sign must reverse:

$$x < -4$$

This is a crucial rule.

Why does the sign reverse?

Consider:

$$3 < 5$$

Multiply both sides by -1:

$$-3 > -5$$

The order reverses because -3 is greater than -5.

This explains why multiplying or dividing an inequality by a negative number reverses the sign.

7.13 Algebra and Patterns: Discovering General Rules

Imagine that you are arranging chairs in rows.

One row contains 5 chairs.

Two rows contain 10 chairs.

Three rows contain 15 chairs.

The total number of chairs depends on the number of rows.

If each row contains 5 chairs, then:

$$C = 5n$$

where:

- C is the total number of chairs.
- n is the number of rows.

This is a general rule.

Instead of calculating each arrangement separately, we can use the equation for any number of rows.

If $n = 20$:

$$C = 5(20) = 100$$

There are 100 chairs.

A pattern involving addition

Consider the sequence:

$$3,\ 6,\ 9,\ 12,\ 15, \ldots$$

Each term is a multiple of three.

The n-th term is:

$$T_n = 3n$$

For the seventh term:

$$T_7 = 3(7) = 21$$

The seventh term is 21.

A more complex pattern

Consider:

$$2,\ 5,\ 8,\ 11,\ 14, \ldots$$

The sequence increases by 3 each time.

The first term is 2, and each later term adds another 3.

Therefore:

$$T_n = 2 + 3(n - 1)$$

Simplifying:

$$T_n = 2 + 3n - 3$$
$$T_n = 3n - 1$$

For the tenth term:

$$T_{10} = 3(10) - 1 = 29$$

The tenth term is 29.

Algebra allows us to move from observing a pattern to describing a general relationship.

7.14 Functions: Understanding How One Quantity Depends on Another

A function is a relationship that assigns exactly one output to each permitted input.

Imagine that a printing shop charges ₦100 for every page printed.

The total cost depends on the number of pages.

If n is the number of pages, the cost is:

$$C(n) = 100n$$

This is a function.

The notation $C(n)$ means that the cost depends on the input n.

If 5 pages are printed:

$$C(5) = 100(5) = 500$$

If 20 pages are printed:

$$C(20) = 100(20) = 2000$$

The function allows us to calculate the cost for any permitted number of pages.

A function with a fixed charge

Suppose a service charges a fixed fee of ₦500 plus ₦100 per page.

The function becomes:

$$C(n) = 500 + 100n$$

For 10 pages:

$$C(10) = 500 + 100(10)$$
$$C(10) = 1500$$

The total cost is ₦1,500.

Why functions matter

Functions describe relationships in which one quantity depends on another.

They are used to represent:

- Total cost as a function of quantity.
- Distance as a function of time.
- Revenue as a function of units sold.
- Population as a function of time.

- Temperature changes during an experiment.
- The output of a computer program.

Functions provide the foundation for studying graphs, mathematical modelling, and calculus.

They help us ask not only what a quantity is, but also how it changes when another quantity changes.

7.15 How to Make Algebra Easier

Algebra becomes more manageable when we focus on the meaning of the symbols and operations.

Method 1: Define every variable

Before writing an equation, state what each letter represents.

For example:

Let x be the price of one notebook.

This prevents confusion when several unknown quantities appear in the same problem.

Method 2: Translate the situation carefully

If a problem says that the total cost is ₦5,000 plus ₦300 for each item, write:

$$C = 5000 + 300n$$

Do not rush into calculations before representing the situation correctly.

Method 3: Keep both sides balanced

When solving an equation, perform the same operation on both sides.

If you subtract five from the left, subtract five from the right.

This preserves equality.

Method 4: Reverse operations in the correct order

For:

$$3x + 7 = 25$$

First subtract seven:

$$3x = 18$$

Then divide by three:

$$x = 6$$

Reverse the operations applied to the unknown.

Method 5: Collect like terms

Simplify expressions by grouping terms with the same variables and powers.

For example:

$$4x + 2x + 3y - y = 6x + 2y$$

This makes the expression easier to understand.

Method 6: Check your solution

Substitute your answer into the original equation.

If both sides agree, the solution is consistent with the equation.

If they do not agree, revisit the steps.

Method 7: Use a table when there are many possibilities

If an unknown quantity can take several values, a table may help you test the relationship.

For example, for:

$$y = 2x + 3$$

x	y
0	3
1	5
2	7
3	9

The table shows how the output changes as the input changes.

Method 8: Do not memorize without understanding

Rules are useful, but understanding why they work makes them easier to remember and apply correctly.

When you see:

$$3(x+4)$$

ask what the expression means.

It means three copies of the entire quantity $x+4$.

Therefore:

$$3(x+4) = 3x+12$$

The meaning leads naturally to the rule.

7.16 Common Algebra Mistakes

Mistake 1: Treating addition and multiplication as the same

The expression:

$$3x$$

means three multiplied by x.

It does not mean $3+x$.

Mistake 2: Combining unlike terms

You can simplify:

$$3x + 5x = 8x$$

But you cannot simplify:

$$3x + 5y$$

to $8xy$ or $8x$.

The variables represent different quantities.

Mistake 3: Forgetting to distribute to every term

Incorrect:

$$3(x + 4) = 3x + 4$$

Correct:

$$3(x + 4) = 3x + 12$$

The multiplier applies to both terms inside the bracket.

Mistake 4: Changing only one side of an equation

If you subtract three from the left side of an equation, you must also subtract three from the right side.

Otherwise, equality is not preserved.

Mistake 5: Misreading word problems

“Three less than a number” means:

$$x - 3$$

"Three less than twice a number" means:

$$2x - 3$$

Translate the words carefully.

Mistake 6: Forgetting to reverse an inequality sign

When multiplying or dividing both sides of an inequality by a negative number, reverse the inequality sign.

Mistake 7: Accepting an answer without checking

If an equation produces $x = 6$, substitute 6 into the original equation.

Checking is especially useful when equations involve several operations or fractions.

Mistake 8: Assuming every problem has one solution

An equation may have one solution, several solutions, or no solution.

A system of equations may also be underdetermined if there is not enough independent information.

Before solving, understand what the problem tells you.

7.17 Practical Workshop: Become an Algebraist

Work through these situations.

Situation A: The missing number

A number is multiplied by four, and eight is added. The result is 32.

Let the unknown number be x.

Write the equation:

$$4x + 8 = 32$$

Subtract eight:

$$4x = 24$$

Divide by four:

$$x = 6$$

Answer: The number is 6.

Situation B: Buying school materials

You buy four notebooks and one pen for ₦3,100.

The pen costs ₦300.

Let x be the price of one notebook.

The equation is:

$$4x + 300 = 3100$$

Subtract 300:

$$4x = 2800$$

Divide by four:

$$x = 700$$

Answer: Each notebook costs ₦700.

Situation C: Planning an event

You have ₦25,000 for an event. The venue costs ₦5,000, and each participant requires ₦1,000 for refreshments.

How many participants can you accommodate if all the money is spent on these costs?

Let n be the number of participants.

$$5000 + 1000n = 25000$$

Subtract 5,000:

$$1000n = 20000$$

Divide by 1,000:

$$n = 20$$

Answer: You can accommodate 20 participants under the stated assumptions.

Situation D: Finding two unknown prices

Two notebooks and one pen cost ₦1,700.

One notebook and two pens cost ₦1,600.

Let:

- x = price of one notebook.
- y = price of one pen.

Write the equations:

$$2x + y = 1700$$
$$x + 2y = 1600$$

Multiply the second equation by two:

$$2x + 4y = 3200$$

Subtract the first equation:

$$3y = 1500$$

Therefore:

$$y = 500$$

Substitute into the first equation:

$$2x + 500 = 1700$$
$$2x = 1200$$
$$x = 600$$

Answer:

- Notebook = ₦600.
- Pen = ₦500.

Check:

$$2(600) + 500 = 1700$$
$$600 + 2(500) = 1600$$

Both equations are satisfied.

Situation E: Creating a rule

A delivery service charges ₦700 as a starting fee and ₦150 for each kilometre travelled.

Write a function for the total charge.

Let d represent the distance in kilometres.

$$C(d) = 700 + 150d$$

Calculate the charge for 12 kilometres:

$$C(12) = 700 + 150(12)$$
$$C(12) = 700 + 1800$$
$$C(12) = 2500$$

Answer: The charge is ₦2,500.

7.18 Chapter Exercises

Try these questions before checking your work.

Part A: Understanding algebra

1. Explain why algebra is useful when the value of a quantity is unknown.
2. What is the difference between an expression and an equation?
3. Explain why $3x + 5x = 8x$, but $3x + 5y$ cannot generally be simplified to one term.
4. What does it mean to preserve equality when solving an equation?
5. Explain why the sign of an inequality reverses when both sides are multiplied or divided by a negative number.

Part B: Simplifying expressions

6. Simplify:

$$7x + 4x - 3x$$

7. Simplify:

$$5a + 3b + 2a - 7b$$

8. Expand:

$$4(x + 6)$$

9. Expand:

$$3(2x - 5)$$

10. Expand and simplify:

$$(x + 4)(x + 2)$$

Part C: Solving equations

11. Solve:

$$x + 12 = 31$$

12. Solve:

$$5x = 65$$

13. Solve:

$$3x + 7 = 28$$

14. Solve:

$$4x - 9 = 23$$

15. Solve:

$$6x + 5 = 2x + 25$$

16. Solve:

$$\frac{x}{5} + 3 = 11$$

Part D: Real-life applications

17. A trader buys six identical bags for ₦18,000. Find the price of one bag.

18. A taxi charges ₦400 plus ₦250 per kilometre. Write an equation for the fare and calculate the fare for 10 kilometres.

19. The sum of two numbers is 42. One number is twice the other. Find both numbers.

20. Two notebooks and three pens cost ₦1,900. Three notebooks and one pen cost ₦1,800. Find the price of each item.

21. A student has ₦12,000 and wants to buy books costing ₦800 each. Write an inequality to show the maximum number of books the student can buy if all the money is spent only on books.

22. A business charges a fixed fee of ₦10,000 plus ₦500 for each item produced. Write a function for the total cost of producing n items.

23. A sequence begins:

$$4,\ 7,\ 10,\ 13,\ 16, \ldots$$

Find a formula for its n-th term.

24. Create a real-life problem involving an unknown quantity. Define your variable, write an equation, solve it, and verify your answer.

7.19 Chapter Review

In this chapter, we learned that algebra provides a way to represent unknown quantities and describe relationships between them.

We explored expressions, like terms, the distributive law, equations, simultaneous equations, inequalities, patterns, and functions.

We also saw how algebra can help us solve problems involving prices, ages, transport fares, business costs, and event planning.

Most importantly, we learned to think before manipulating symbols.

When you encounter an algebra problem, ask:

- What quantity is unknown?
- What does each variable represent?

- What relationship connects the quantities?
- How can I express that relationship mathematically?
- Which operation will isolate the unknown?
- Have I preserved equality?
- Does the answer satisfy the original problem?

These questions help you move from guessing to systematic reasoning.

7.20 Conclusion: Algebra Is a Language for Possibility

Imagine trying to describe every possible price, distance, cost, or quantity using only individual numerical examples.

You would need a separate calculation for every situation.

Algebra makes something more powerful possible. It allows us to describe relationships that remain useful even when the numbers change.

The equation:

$$C = 5000 + 300n$$

can represent the cost of producing different quantities of items.

The equation:

$$y = 2x + 3$$

can describe how one quantity depends on another.

A system of equations can help us determine several unknown quantities from connected pieces of information.

Algebra teaches us that an unknown is not necessarily a barrier to understanding. It can be represented, studied, and solved when sufficient information is available.

Once you understand the relationship behind the symbols, the letters become easier to manage.

In the next chapter, we will explore what happens when quantities change continuously.

We will move beyond finding unknown values and begin asking questions about speed, growth, slopes, and rates of change.

We will learn to **think like a calculus student**.

CHAPTER 8

Calculus — Understanding Change

08

8.1 Introduction: The World Does Not Stand Still

Imagine you are travelling in a car.

At the beginning of the journey, the car is moving slowly. A few seconds later, it is moving faster. When the driver approaches a busy junction, the car slows down. Eventually, it stops.

Now consider these questions:

- How fast is the car moving at one particular moment?
- How quickly is its speed increasing?
- How far will it travel in the next five seconds?
- If the car continues accelerating, how will its position change?

These questions are not simply about adding, subtracting, multiplying, or dividing numbers. They are about understanding **how quantities change**.

Calculus was developed to help mathematicians and scientists investigate problems involving continuous change, motion, growth, and accumulation.

The same ideas can help an economist understand how cost changes when production increases, how a business can maximise profit, or how the population of a country may grow over time.

But before learning any calculus formula, we must understand the problem that makes the formula necessary.

Suppose someone tells you that the derivative of x^2 is $2x$.

You could memorise this rule and use it in an examination. However, a deeper understanding begins when you ask:

1. What does x^2 represent?
2. What is changing?
3. What information do we want to discover?
4. Why is ordinary arithmetic not enough?
5. What does the answer $2x$ tell us?

These questions transform calculus from a collection of rules into a way of understanding the world.

That is the purpose of this chapter.

8.2 The Problem That Led to Calculus

Consider a ball rolling down a slope.

At different moments, the ball occupies different positions. Its position changes with time.

Suppose we record the ball's position:

Time (seconds)	Position (metres)
0	0
1	2
2	8
3	18
4	32

Look carefully at the table.

During the first second, the ball travels 2 metres.

During the second second, it travels 6 metres.

During the third second, it travels 10 metres.

During the fourth second, it travels 14 metres.

The distance travelled during each one-second interval is increasing.

This tells us that the ball is not moving at a constant speed. Its motion is changing.

Now imagine that you want to know the ball's speed exactly at $t = 2$ seconds.

The table tells us how far it travelled during an interval, but it does not directly tell us its exact speed at one instant.

We need a better mathematical method.

This is one of the central problems calculus was developed to investigate:

> **how can we measure change at an exact moment, and how can we calculate the total effect of change over an interval?**

Calculus provides two major tools for answering these questions:

- **Differentiation**, which helps us measure rates of change.
- **Integration**, which helps us calculate accumulation and totals.

Understanding the difference between these two ideas is the foundation of this chapter.

8.3 A Brief History of Calculus

Calculus did not appear suddenly in the mind of one person. Its development involved many centuries of mathematical investigation.

Ancient mathematical ideas

Ancient Greek mathematicians investigated problems involving areas and volumes.

Archimedes, who lived in the third century BCE, developed methods for estimating and determining areas and volumes. His method of exhaustion involved approximating a shape using simpler shapes and improving the approximation.

For example, imagine trying to determine the area inside a circle.

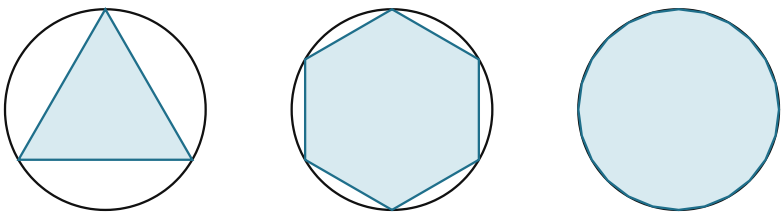
A circle is curved, so it is difficult to calculate its area by simply counting squares or using the formulas for rectangles and triangles.

However, you can place polygons inside the circle. As the number of sides increases, the polygon can approximate the circle more closely.

This approach anticipated some of the reasoning later used in integral calculus.

FIGURE 9

Approximating a Circle with Polygons



As the number of sides increases, the inscribed polygon follows the circular boundary more closely.

The development of calculus in the seventeenth century

During the seventeenth century, European mathematicians investigated motion, planetary orbits, tangents to curves, and areas beneath curved graphs.

Two particularly important figures were Isaac Newton and Gottfried Wilhelm Leibniz.

Newton developed methods of calculus in connection with motion and changing quantities. Leibniz independently developed a systematic form of calculus and introduced much of the notation that mathematicians still use.

For example, Leibniz's notation

$$\frac{dy}{dx}$$

is widely used to represent the derivative of y with respect to x.

Their work helped establish the relationship between differentiation and integration.

This relationship became known as the fundamental theorem of calculus.

The history of calculus demonstrates an important lesson: mathematics often develops because existing methods cannot fully answer the questions people want to investigate.

The problem comes first. The mathematical tool develops in response.

8.4 Understanding Change Before Using Calculus

Before introducing differentiation, let us examine an everyday situation.

Suppose a student walks 100 metres in 20 seconds.

The student's average speed is

$$\text{Average speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

Substituting the values,

$$\text{Average speed} = \frac{100}{20}$$

Therefore,

$$\boxed{\text{Average speed} = 5 \text{ m/s}}$$

This means the student travelled an average of 5 metres per second.

But there is an important distinction.

The student might have started slowly, walked faster in the middle, and slowed down near the end.

The average speed does not necessarily tell us the speed at every moment.

To understand the difference, consider a journey in which the position of an object is described by

$$s = t^2$$

where:

- s represents position in metres;
- t represents time in seconds.

At $t = 1$,

$$s = 1^2 = 1$$

At $t = 2$,

$$s = 2^2 = 4$$

The object moves from position 1 metre to position 4 metres during that interval.

Its average rate of change is

$$\frac{\text{Change in position}}{\text{Change in time}} = \frac{4-1}{2-1}$$

Therefore,

$$\boxed{3 \text{ m/s}}$$

Now calculate the average rate of change between $t = 2$ and $t = 3$.

At $t = 2$,

$$s = 2^2 = 4$$

At $t = 3$,

$$s = 3^2 = 9$$

Hence,

$$\frac{9-4}{3-2} = 5$$

The average rate of change is now 5 m/s.

The average rate increases as the object moves through later intervals.

This suggests that the object's speed is changing continuously.

Differentiation gives us a way to investigate this changing rate.

8.5 Understanding Limits: Getting Closer and Closer

Before we calculate an instantaneous rate of change, we need to understand the idea of a limit.

A limit describes the value a mathematical expression approaches as its input approaches a particular value.

Consider the expression

$$f(x) = x + 2$$

As x approaches 3, the value of $f(x)$ approaches 5.

For example:

x	$f(x) = x + 2$
2	4
2.9	4.9
2.99	4.99
2.999	4.999

As x gets closer to 3, the expression gets closer to 5.

We write

$$\lim_{x \to 3}(x + 2) = 5$$

Read this as: "The limit of $x + 2$ as x approaches 3 is 5."

Why do limits matter?

Imagine trying to measure the speed of a moving object over an increasingly short period.

You could calculate its average speed over one second, then half a second, then one-tenth of a second, and then an even shorter interval.

As the interval becomes smaller, the average rate of change can approach the instantaneous rate.

The limiting value provides the mathematical foundation for the derivative.

A limit is not simply about substituting a number into an expression. It is about understanding what happens as the input approaches a particular value.

Example 1: A simple limit

Evaluate

$$\lim_{x \to 4} (2x + 3)$$

Step 1: Understand the expression.

The expression is

$$2x + 3$$

Step 2: Substitute the approaching value.

Because this expression is continuous at $x = 4$, direct substitution works.

$$2(4) + 3$$

Step 3: Calculate.

$$8 + 3 = 11$$

Therefore,

$$\boxed{\lim_{x \to 4} (2x + 3) = 11}$$

Example 2: A limit that requires simplification

Evaluate

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

Direct substitution gives

$$\frac{2^2-4}{2-2}=\frac{0}{0}$$

The expression $\frac{0}{0}$ is indeterminate. It does not tell us the limit.

We must simplify the expression.

Step 1: Factorise the numerator.

Using the difference of two squares,

$$x^2-4=(x-2)(x+2)$$

Therefore,

$$\frac{x^2-4}{x-2}=\frac{(x-2)(x+2)}{x-2}$$

Step 2: Cancel the common factor.

For $x\neq 2$,

$$\frac{(x-2)(x+2)}{x-2}=x+2$$

Step 3: Evaluate the limit.

Now let x approach 2:

$$\lim_{x\to 2}(x+2)=2+2$$

Thus,

$$\boxed{4}$$

The original expression is undefined at $x = 2$, but its limit as x approaches 2 exists and equals 4.

This distinction is important: the value of a function at a point and the limit of the function near that point are related, but they are not always the same.

8.6 Differentiation: Measuring the Rate of Change

Suppose the quantity y depends on another quantity x.

For example:

- distance may depend on time;
- cost may depend on output;
- revenue may depend on sales;
- population may depend on time.

If x changes, y may also change.

Differentiation helps us determine how quickly y changes in relation to x.

The derivative is often written as

$$\frac{dy}{dx}$$

This notation means "the rate of change of y with respect to x."

It is important to interpret the derivative in the context of the problem.

If y represents distance and x represents time, the derivative may represent velocity.

If y represents total cost and x represents output, the derivative may represent marginal cost.

If y represents the height of a growing plant and x represents time, the derivative represents the plant's rate of growth.

The mathematics may look similar, but the meaning depends on the quantities involved.

The derivative from first principles

Let us derive the derivative of

$$f(x) = x^2$$

rather than merely memorising the answer.

The derivative from first principles is defined by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Here:

- $f(x)$ is the original function value;
- $f(x+h)$ is the function value after a small change h;
- h represents the change in the input;
- the limit allows us to examine what happens as that change approaches zero.

The expression measures the average rate of change over a small interval. Taking the limit gives the instantaneous rate of change, when the limit exists.

Now apply the definition.

Step 1: Write the function.

$$f(x) = x^2$$

Step 2: Replace x with $x + h$.

$$f(x+h) = (x+h)^2$$

Expand:

$$f(x+h) = x^2 + 2xh + h^2$$

Step 3: Substitute into the derivative formula.

$$f'(x) = \lim_{h \to 0} \frac{(x^2 + 2xh + h^2) - x^2}{h}$$

Step 4: Simplify the numerator.

The x^2 terms cancel:

$$f'(x) = \lim_{h \to 0} \frac{2xh + h^2}{h}$$

Step 5: Factorise and cancel h.

$$f'(x) = \lim_{h \to 0} \frac{h(2x + h)}{h}$$

For $h \neq 0$, we obtain

$$f'(x) = \lim_{h \to 0} (2x + h)$$

Step 6: Evaluate the limit.

As h approaches zero,

$$2x + h \to 2x$$

Therefore,

$$\boxed{f'(x) = 2x}$$

We have discovered that the derivative of x^2 is $2x$.

The result is not an arbitrary rule. It follows from the definition of the derivative.

8.7 The Power Rule: Making Differentiation Easier

The first-principles method explains the foundation of differentiation. However, repeating the full process for every expression would be time-consuming.

Mathematicians developed rules that make differentiation easier.

One of the most useful is the power rule.

If

$$y = x^n$$

then

$$\boxed{\frac{dy}{dx} = nx^{n-1}}$$

In simple terms:

1. Bring the power down and multiply it by the coefficient.
2. Reduce the power by one.

Example 1: Differentiate x^3

Given

$$y = x^3$$

Bring down the power, 3:

$$3x^3$$

Reduce the power from 3 to 2:

$$\boxed{\frac{dy}{dx} = 3x^2}$$

Example 2: Differentiate $5x^4$

Given

$$y = 5x^4$$

Multiply the coefficient 5 by the power 4:

$$5 \times 4 = 20$$

Reduce the power from 4 to 3:

$$\boxed{\frac{dy}{dx} = 20x^3}$$

Example 3: Differentiate $3x^2 + 4x - 7$

Differentiate each term separately.

For the first term,

$$\frac{d}{dx}(3x^2) = 6x$$

For the second term,

$$\frac{d}{dx}(4x) = 4$$

The derivative of a constant is zero:

$$\frac{d}{dx}(-7) = 0$$

Combining the results gives

$$\boxed{\frac{dy}{dx} = 6x + 4}$$

Why does a constant have a derivative of zero?

Consider the function

$$y = 7$$

No matter what value x takes, y remains 7.

The quantity y does not change as x changes.

Therefore,

$$\boxed{\frac{d}{dx}(7) = 0}$$

Differentiation measures change. A constant has no change, so its derivative is zero.

Example 4: Differentiate a negative power

Given

$$y = \frac{4}{x^2}$$

Rewrite the expression using indices:

$$y = 4x^{-2}$$

Apply the power rule:

$$\frac{dy}{dx} = 4(-2)x^{-3}$$

Therefore,

$$\frac{dy}{dx} = -8x^{-3}$$

Rewriting with a positive power in the denominator,

$$\boxed{\frac{dy}{dx} = -\frac{8}{x^3}}$$

The ability to rewrite algebraic expressions using indices makes differentiation easier.

8.8 Differentiation as the Slope of a Graph

Differentiation has a geometric interpretation.

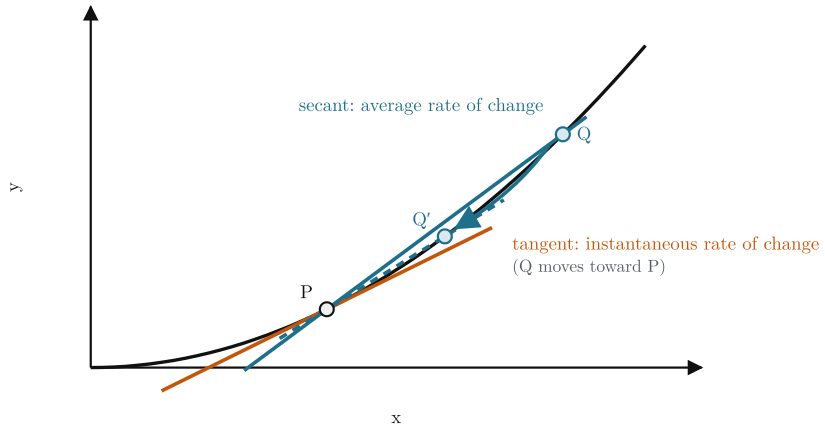
Imagine a curve drawn on a coordinate plane.

A straight line has a constant slope. A curved graph generally has a slope that changes from point to point.

Differentiation helps us determine the slope of a curve at a particular point, provided the derivative exists there.

FIGURE 10

Secant Line and Tangent Line



The blue secant through P and Q measures average rate of change. As Q slides toward P the secant turns into the orange tangent, the instantaneous rate of change.

Example: Finding the slope of $y = x^2$

We have already established that

$$\frac{dy}{dx} = 2x$$

Suppose we want the slope when $x = 3$.

Substitute $x = 3$:

$$\frac{dy}{dx} = 2(3)$$

Therefore,

$$\boxed{\text{Slope} = 6}$$

At $x = 3$, the curve has a tangent slope of 6.

Now consider $x = 1$:

$$\frac{dy}{dx} = 2(1) = 2$$

At $x = 1$, the slope is 2.

The slope is different at the two points. This demonstrates why a curved graph does not have one constant slope.

Where is the slope zero?

A point where the derivative equals zero may indicate a stationary point.

For example,

$$y = x^2$$

has derivative

$$\frac{dy}{dx} = 2x$$

Set the derivative equal to zero:

$$2x = 0$$

Therefore,

$$x = 0$$

Substitute into the original function:

$$y = 0^2 = 0$$

The stationary point is

$$\boxed{(0, 0)}$$

For this particular graph, the point is a minimum.

However, a zero derivative does not always mean a maximum or minimum. Some stationary points are neither. The shape of the graph or additional analysis determines the type of point.

8.9 Understanding Velocity and Acceleration

One of the most important applications of differentiation is the study of motion.

Suppose the position of an object is given by

$$s = t^2$$

where s is measured in metres and t in seconds.

The velocity is the rate at which position changes with time:

$$v = \frac{ds}{dt}$$

Differentiate:

$$\boxed{v = 2t}$$

The velocity depends on time.

At $t = 1$,

$$v = 2(1) = 2 \text{ m/s}$$

At $t = 3$,

$$v = 2(3) = 6 \text{ m/s}$$

At $t = 5$,

$$v = 2(5) = 10 \text{ m/s}$$

The object's velocity increases with time.

Now let us investigate how quickly the velocity changes.

Acceleration is the rate of change of velocity:

$$a = \frac{dv}{dt}$$

Since

$$v = 2t$$

differentiate again:

$$\boxed{a = 2 \text{ m/s}^2}$$

The acceleration is constant.

This example demonstrates an important chain of reasoning:

$$\text{Position} \longrightarrow \text{Velocity} \longrightarrow \text{Acceleration}$$

Differentiation connects these quantities.

The first derivative of position gives velocity, while the second derivative gives acceleration, when the position function is sufficiently differentiable.

A practical warning

The position function describes the mathematical model of the motion. Its predictions should be interpreted in the context of the actual situation.

For instance, the function $s = t^2$ predicts increasing velocity for all positive values of t. A real vehicle cannot accelerate indefinitely, so the model may only be useful over a limited period.

Mathematics becomes more powerful when we understand both what a model tells us and where its assumptions stop being realistic.

8.10 Think Like an Economist: Marginal Cost and Marginal Revenue

Calculus is useful in economics because many economic quantities change as production or sales change.

Imagine a small factory that produces notebooks.

Let its total cost of producing q notebooks be

$$C(q) = 1000 + 20q + q^2$$

where $C(q)$ is measured in naira.

This model contains three parts:

- 1000: a fixed cost;
- $20q$: a cost that increases in proportion to output;
- q^2: a cost that increases more rapidly as output grows.

The model is simplified, but it allows us to study how cost changes.

Finding marginal cost

Marginal cost describes the rate at which total cost changes as output changes.

In a continuous mathematical model, marginal cost is represented by the derivative of total cost:

$$MC = \frac{dC}{dq}$$

Differentiate the cost function:

$$C(q) = 1000 + 20q + q^2$$

The derivative of 1000 is zero.

The derivative of $20q$ is 20.

The derivative of q^2 is $2q$.

Therefore,

$$\boxed{MC = 20 + 2q}$$

Now suppose the factory is producing 30 notebooks.

Substitute $q = 30$:

$$MC = 20 + 2(30)$$
$$MC = 20 + 60$$

Thus,

$$\boxed{MC = 80}$$

The marginal cost is 80 naira per additional notebook at that output level, interpreted as the instantaneous rate of change of cost in the model.

It is not necessarily the exact cost of producing one extra notebook. For a discrete increase of one whole notebook, the actual additional cost can be calculated by comparing $C(31)$ with $C(30)$.

This distinction matters because real production usually occurs in whole units, while calculus treats output as a continuous variable for analysis.

Finding marginal revenue

Suppose revenue from selling q notebooks is modelled by

$$R(q) = 100q - q^2$$

Marginal revenue is

$$MR = \frac{dR}{dq}$$

Differentiate:

$$\boxed{MR = 100 - 2q}$$

At $q = 20$,

$$MR = 100 - 2(20)$$
$$MR = 100 - 40$$

Therefore,

$$\boxed{MR = 60}$$

The marginal revenue at that output is 60 naira per additional notebook in the model.

Finding marginal profit

Profit is revenue minus cost:

$$P(q) = R(q) - C(q)$$

Substitute the functions:

$$P(q) = (100q - q^2) - (1000 + 20q + q^2)$$

Expand and collect like terms:

$$P(q) = 100q - q^2 - 1000 - 20q - q^2$$

Therefore,

$$P(q) = 80q - 2q^2 - 1000$$

Differentiate:

$$\boxed{P'(q) = 80 - 4q}$$

If the factory wants to investigate the output level at which profit stops increasing and begins decreasing, it can examine where the derivative is zero.

Set

$$80 - 4q = 0$$

Rearrange:

$$80 = 4q$$

Divide both sides by 4:

$$\boxed{q = 20}$$

The second derivative is

$$P''(q) = -4$$

Because the second derivative is negative, the profit function is curved downward, so $q = 20$ gives a maximum for this model.

This result assumes the model applies at that output level and ignores other practical constraints.

The important lesson is not merely that calculus produces the answer 20. It is that calculus helps us investigate the relationship between output and profit.

8.11 Integration: Understanding Accumulation

Differentiation investigates rates of change.

Integration helps us calculate accumulated quantities.

Imagine water flowing into a tank.

The amount of water entering each second may change. If we know the flow rate over time, we may want to calculate the total amount of water collected.

Or imagine a car travelling at different speeds. If we know its velocity over time, we may want to calculate the distance travelled.

Integration provides methods for solving these kinds of problems.

One way to understand integration is to think of adding many small contributions together.

For example, the total distance travelled during a journey can be found by adding the distances travelled during small time intervals.

If the speed changes continuously, integration provides a mathematical way to calculate the accumulated distance.

8.12 Understanding Area Under a Curve

Consider a graph of velocity against time.

If an object moves at a constant velocity of 5 metres per second for 4 seconds, its distance travelled is

$$\text{Distance} = \text{Velocity} \times \text{Time}$$

Therefore,

$$s = 5 \times 4$$

$$\boxed{s = 20 \text{ m}}$$

On a velocity-time graph, this is the area of a rectangle with height 5 and width 4.

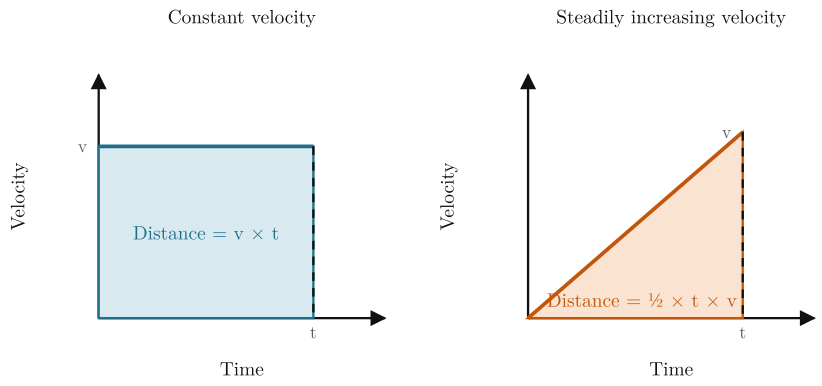
Now suppose velocity changes with time.

The graph may no longer form a rectangle. The area beneath the graph may have a curved boundary.

Integration allows us to calculate this accumulated area.

FIGURE 11

Area Under a Velocity-Time Graph



Area under a velocity-time graph is the distance travelled when velocity is non-negative: a rectangle for constant a triangle for steadily increasing velocity.

For a velocity-time graph, the signed area under the curve represents displacement. If velocity remains non-negative throughout the interval, displacement equals distance travelled. If velocity becomes negative, total distance requires accounting for the magnitude of motion in each direction.

8.13 The Power Rule for Integration

A useful integration rule reverses the power rule for differentiation.

For powers other than -1,

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

where $n \neq -1$.

The symbol $\int$ is the integral sign.

The constant C is called the constant of integration.

Why is it necessary?

Because the derivative of any constant is zero.

For example,

$$\frac{d}{dx}(x^2) = 2x$$

But also,

$$\frac{d}{dx}(x^2 + 5) = 2x$$

and

$$\frac{d}{dx}(x^2 - 9) = 2x$$

Differentiation cannot distinguish between these expressions using their derivatives alone. Integration therefore includes C when finding a general antiderivative.

Example 1: Integrate x^2

We want to evaluate

$$\int x^2\,dx$$

Step 1: Increase the power by one.

The power changes from 2 to 3.

Step 2: Divide by the new power.

$$\frac{x^3}{3}$$

Step 3: Add the constant of integration.

Therefore,

$$\boxed{\int x^2\,dx = \frac{x^3}{3} + C}$$

We can check the answer by differentiating:

$$\frac{d}{dx}\left(\frac{x^3}{3} + C\right) = \frac{3x^2}{3} + 0$$

Thus,

$$= x^2$$

The answer is correct.

Example 2: Integrate $6x^2$

Evaluate

$$\int 6x^2\,dx$$

Keep the coefficient 6 and apply the power rule:

$$6\left(\frac{x^3}{3}\right) + C$$

Simplify:

$$\boxed{2x^3 + C}$$

Example 3: Integrate a polynomial

Evaluate

$$\int (3x^2 + 4x + 2)\,dx$$

Integrate each term separately:

$$\int 3x^2\,dx + \int 4x\,dx + \int 2\,dx$$

The first term gives

$$x^3$$

The second gives

$$2x^2$$

The third gives

$$2x$$

Add the constant:

$$\boxed{x^3 + 2x^2 + 2x + C}$$

Checking by differentiation,

$$\frac{d}{dx}(x^3 + 2x^2 + 2x + C) = 3x^2 + 4x + 2$$

This matches the original integrand.

8.14 Definite Integrals: Calculating a Total

An indefinite integral gives a family of antiderivatives.

A definite integral calculates an accumulated quantity over a specified interval.

It is written as

$$\int_a^b f(x)\,dx$$

Here:

- a is the lower limit;
- b is the upper limit;
- $f(x)$ is the function being integrated.

The fundamental theorem of calculus connects a definite integral to an antiderivative.

If $F'(x) = f(x)$, then

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a)}$$

In simple terms:

1. Find an antiderivative.
2. Substitute the upper limit.
3. Substitute the lower limit.
4. Subtract the second result from the first.

Example: Find the area under $y = x$ from $x = 0$ to $x = 4$

We want to evaluate

$$\int_0^4 x\,dx$$

Step 1: Integrate the function.

$$\int x\,dx = \frac{x^2}{2} + C$$

For a definite integral, the constant cancels when the upper and lower values are subtracted.

Step 2: Substitute the upper limit.

$$F(4) = \frac{4^2}{2} = \frac{16}{2} = 8$$

Step 3: Substitute the lower limit.

$$F(0) = \frac{0^2}{2} = 0$$

Step 4: Subtract.

$$\int_0^4 x\,dx = 8 - 0$$

Therefore,

$$\boxed{8}$$

The integral equals 8 square units.

We can check the answer using ordinary geometry. The region beneath $y = x$, from $x = 0$ to $x = 4$, is a triangle with base 4 and height 4.

Its area is

$$A = \frac{1}{2} \times 4 \times 4 = 8$$

Both methods produce the same answer.

This illustrates an important principle: calculus can solve accumulation problems that ordinary geometry can sometimes solve more simply.

8.15 Using Integration to Recover Distance

Let us return to the moving object whose velocity is

$$v = 2t$$

Suppose we want to find its displacement between $t = 0$ and $t = 5$ seconds.

Because velocity is the rate of change of position,

$$v = \frac{ds}{dt}$$

Rearranging the relationship conceptually, we can recover position by integrating velocity:

$$s = \int 2t\, dt$$

Integrate:

$$s = t^2 + C$$

Suppose the object starts at position zero when $t = 0$.

Then

$$0 = 0^2 + C$$

Therefore,

$$C = 0$$

The position function is

$$s = t^2$$

Now calculate displacement from $t = 0$ to $t = 5$:

$$\int_0^5 2t\,dt$$

An antiderivative of $2t$ is t^2.

Therefore,

$$[t^2]_0^5 = 5^2 - 0^2$$

$$= 25$$

Thus,

$$\boxed{\text{Displacement} = 25 \text{ m}}$$

Since the velocity is non-negative throughout the interval, the distance travelled is also 25 metres.

Notice the connection:

- Differentiation changed the position function into a velocity function.
- Integration recovered the position change from the velocity function.

These processes are closely related.

8.16 The Fundamental Connection Between Differentiation and Integration

Imagine that you know how quickly water is entering a tank at every moment.

Differentiation can describe how an accumulated quantity changes.

Integration can combine the changing flow rates to calculate the total quantity accumulated.

The fundamental theorem of calculus formalises this connection.

If

$$F'(x) = f(x)$$

then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

This tells us that integration and differentiation are connected in a particularly powerful way.

Differentiation measures change. Integration accumulates change.

They are not exact opposites in every situation without qualification, but under the appropriate conditions, integration and differentiation undo one another in the sense described by the fundamental theorem.

This is one of the major reasons calculus is such an important branch of mathematics.

8.17 How Calculus Helps Solve Real Problems

Calculus is not limited to textbook questions. Its methods appear in many fields.

1. **Economics and business**

Economists use derivatives to study marginal cost, marginal revenue, and marginal profit.

Businesses can use these ideas to investigate how changes in production or prices affect financial outcomes.

Integrals can also help calculate accumulated costs or revenues when quantities are modelled continuously.

2. **Physics**

Physicists use differentiation to study velocity and acceleration.

Integration helps them calculate displacement and other accumulated quantities from known rates.

3. **Engineering**

Engineers use calculus in designing structures, analysing motion, studying electrical systems, and modelling heat transfer.

The precise application depends on the engineering problem.

4. **Medicine and biology**

Researchers use calculus to model changing populations of cells, the spread of some diseases, and the changing concentration of substances in the body.

These models simplify complex biological processes and must be tested against real observations.

5. **Environmental science**

Scientists use calculus to investigate changing temperatures, the movement of pollutants, population growth, and the accumulation of environmental substances.

6. **Technology**

Calculus supports many methods used in computer graphics, robotics, machine learning, and the analysis of continuously changing systems.

The common theme is change. Whenever a quantity changes in a way that can be described mathematically, calculus may provide a useful tool.

8.18 How to Make Calculus Easier

Calculus becomes easier when you understand the meaning of each operation instead of memorising rules without context.

Method 1: Identify the quantities

Before differentiating or integrating, ask what each variable represents.

For example, in

$$s = t^2$$

s is position and t is time.

That information helps you interpret the derivative.

Method 2: Ask what the question requires

If the question asks how quickly something changes, differentiation may be appropriate.

If it asks for a total accumulated amount from a changing rate, integration may be appropriate.

However, read the complete problem before deciding. The correct method depends on the information given and the quantity requested.

Method 3: Simplify before calculating

Consider

$$y = \frac{2x^3 + 4x}{2}$$

Instead of immediately applying complicated-looking rules, simplify:

$$y = x^3 + 2x$$

Now differentiate:

$$\boxed{\frac{dy}{dx} = 3x^2 + 2}$$

Simplifying the expression reduces unnecessary work.

Method 4: Learn the meaning of the symbols

Do not treat these symbols as decoration:

$$\frac{dy}{dx} \qquad \int f(x)\,dx \qquad \lim_{x \to a} f(x)$$

Each represents a different mathematical idea.

Understanding the notation helps you understand what the calculation is doing.

Method 5: Check your result

If you integrate a function, differentiate your answer to see whether you recover the original integrand.

For example,

$$\int 4x^3\,dx = x^4 + C$$

Check:

$$\frac{d}{dx}(x^4 + C) = 4x^3$$

The result matches the original function.

This is a powerful way to detect mistakes.

Method 6: Use units to check meaning

Suppose position is measured in metres and time in seconds.

Then

$$\frac{ds}{dt}$$

has units of metres per second.

If you claim that a velocity is measured in square metres, something is wrong with your interpretation.

Units help connect mathematical results to reality.

Method 7: Understand the assumptions

A mathematical model may not describe every aspect of a real situation.

A business cost function might assume that output changes continuously. A motion model might assume that acceleration remains constant.

Always ask whether those assumptions are reasonable for the problem being studied.

8.19 Common Mistakes in Calculus

Mistake 1: Confusing differentiation with integration

Differentiation measures rates of change. Integration can calculate accumulated quantities.

They are related, but they answer different questions.

Mistake 2: Forgetting to reduce the power when differentiating

For

$$y = x^5$$

the derivative is

$$\boxed{5x^4}$$

not $5x^5$.

The power is brought down and then reduced by one.

Mistake 3: Forgetting the constant of integration

For an indefinite integral,

$$\int 2x\,dx = x^2 + C$$

The constant is necessary because differentiating a constant gives zero.

Mistake 4: Treating every $\frac{0}{0}$ expression as zero

The expression $\frac{0}{0}$ is indeterminate.

It may be possible to simplify the expression and evaluate its limit, but further reasoning is required.

Mistake 5: Assuming a zero derivative always means a maximum

A zero derivative identifies a stationary point when the derivative exists.

That point may be a maximum, a minimum, or another type of stationary point.

Further analysis is necessary.

Mistake 6: Ignoring units and context

A correct numerical answer can still be misunderstood if the units are wrong or the result is interpreted incorrectly.

Always explain what your answer means in the original problem.

Mistake 7: Using a formula without understanding its purpose

Before applying a rule, ask what the question is trying to discover.

This habit makes it easier to choose the correct method and recognise when an answer is unreasonable.

8.20 Practical Workshop: Think Before the Formula

In this workshop, you will investigate real-life questions before deciding which calculus method to use.

Activity A: A moving object

An object's position is described by

$$s = t^2 + 3t$$

where s is measured in metres and t in seconds.

Think first:

1. What does the function describe?
2. Which quantity changes?
3. What would you need to calculate to find the object's instantaneous velocity?
4. What would you calculate to find its acceleration?

Now calculate.

Velocity is the derivative of position:

$$v = \frac{ds}{dt} = 2t + 3$$

At $t = 4$,

$$v = 2(4) + 3$$
$$\boxed{v = 11 \text{ m/s}}$$

Acceleration is the derivative of velocity:

$$a = \frac{dv}{dt} = 2$$

Therefore,

$$\boxed{a = 2 \text{ m/s}^2}$$

Activity B: A growing business

A business's total cost is modelled by

$$C(q) = 500 + 10q + 2q^2$$

where q is output.

Think first:

1. Which part of the cost remains fixed?
2. Which part increases with output?
3. How would you measure the rate at which cost changes?
4. Why might a business want to know that rate?

Now calculate.

Marginal cost is

$$MC = \frac{dC}{dq}$$

Differentiate:

$$MC = 10 + 4q$$

At $q = 15$,

$$\begin{aligned} MC &= 10 + 4(15) \\ &= 10 + 60 \end{aligned}$$

Therefore,

$$\boxed{MC = 70}$$

The marginal cost is 70 monetary units per unit of output at that production level, according to the model.

Activity C: Water flowing into a tank

Suppose water enters a tank at a rate described by

$$r(t) = 3t$$

litres per minute, where t is measured in minutes.

Think first:

1. Does the flow rate remain constant?
2. Is the question asking for a rate or a total?
3. Which operation could calculate the total water entering during the first four minutes?

Because the question asks for an accumulated quantity, integrate the rate:

$$\int_0^4 3t\,dt$$

An antiderivative is

$$\frac{3t^2}{2}$$

Evaluate the limits:

$$\left[\frac{3t^2}{2}\right]_0^4 = \frac{3(16)}{2} - 0$$
$$= \frac{48}{2}$$

Therefore,

$$\boxed{24 \text{ litres}}$$

The model predicts that 24 litres of water enter the tank during the first four minutes.

8.21 Exercises

Attempt these questions before looking for worked solutions elsewhere. Show your reasoning and explain what your results mean.

Section A: Understanding the ideas

1. Explain the difference between an average rate of change and an instantaneous rate of change.
2. What is a limit, and why is it useful in calculus?
3. Explain the difference between differentiation and integration.
4. Give three real-life situations in which differentiation may be useful.
5. Give three real-life situations in which integration may be useful.
6. Explain why the derivative of a constant is zero.
7. Why is the constant of integration included in an indefinite integral?

Section B: Limits

Evaluate the following limits.

8. $\lim_{x \to 2}(3x + 4)$

8. $\lim_{x \to 5}(x^2 - 2)$

$$x^2 - 9$$

8. $\lim_{x}$

$$\lim_{x \to 3} \frac{}{x - 3}$$
$$\lim \frac{x^2 - 1}{}$$

8. $\lim_{x}$

$$\lim_{x \to 1} \frac{}{x - 1}$$

Section C: Differentiation

Differentiate each function.

12. $y = x^4$

12. $y = 7x^3$

12. $y = 5x^2 + 3x - 8$

12. $y = 4x^5 - 2x^3 + x$

12. $y = \dfrac{6}{x^2}$

12. $y = 3x^4 + 2x^2 - 9$

Section D: Applications of differentiation

18. The position of an object is given by $s = t^2 + 2t$. Find its velocity function.

18. For the function in Question 18, calculate the velocity when $t = 3$ seconds.

18. A company's cost function is $C(q) = 200 + 15q + q^2$. Find its marginal cost function.

18. A profit function is $P(q) = 60q - q^2 - 100$. Find the output level at which the derivative of profit is zero, and use the second derivative to determine whether it is a maximum or minimum.

Section E: Integration

Evaluate the following indefinite integrals.

22. $\int x^3\,dx$

22. $\int 5x^2\,dx$

22. $\int (4x^3 + 2x + 6)\,dx$

22. $\int 8x^7\,dx$

Evaluate the following definite integrals.

$$.3$$

26. $\int 2x\,dx$

$$I_0 \; r^4$$

26. $\int 3\,dx$

$$I_1 \; r^2$$

26. $\int (x+1)\,dx$

I_0

Section F: Think before the formula

29. A car's position is described by $s = 3t^2$. Explain which calculus operation you would use to find its velocity and then calculate the velocity function.

29. A factory knows the rate at which its total cost changes with output. Explain how integration could help it investigate the change in total cost over a range of output levels. State what additional information may be needed to determine the actual total cost.

29. A tank receives water at a changing rate. Explain why multiplying one rate by the total time may give an incorrect result if the rate is not constant. Describe how integration can solve the problem.

29. In your own words, explain why understanding the situation is more important than memorising a calculus formula.

8.22 Chapter Review

Before moving to the next chapter, make sure you can explain the following ideas:

- Calculus studies quantities that change and the accumulation of changing quantities.
- Limits describe the values expressions approach.
- Differentiation measures instantaneous rates of change and can determine the slope of a curve.
- Integration can calculate accumulated quantities, including areas and displacement under appropriate conditions.
- The power rule makes differentiation easier.
- The power rule for integration helps us find antiderivatives.
- The fundamental theorem of calculus connects differentiation and definite integration.

- Calculus is used in economics, physics, engineering, biology, environmental science, and technology.
- Mathematical results must be interpreted using the correct units and assumptions.
- Understanding the problem helps you choose the appropriate mathematical method.

8.23 Conclusion: Calculus Is the Mathematics of Change

We began this chapter with a moving car and a simple question: how can we understand what happens when quantities change?

We then explored the historical development of calculus, the idea of limits, the meaning of differentiation, and the role of integration.

We saw how differentiation can help us understand velocity, acceleration, marginal cost, and marginal revenue. We also learned how integration can calculate areas, displacement, and other accumulated quantities.

But the most important lesson is not a particular formula.

It is the habit of asking what a calculation is meant to explain.

When you see a derivative, ask what quantity is changing and how quickly it is changing.

When you see an integral, ask what is accumulating and what total you need to find.

When you encounter a complicated expression, identify its meaning before deciding which operation to perform.

A formula is useful because it answers a question. Without the question, the formula becomes a collection of symbols.

> **Think before the formula. Understand the change before calculating it.**

In the next chapter, we will investigate another important field of mathematics: statistics and probability. We will learn how to understand data, describe patterns, measure uncertainty, and make decisions when the outcome is not known in advance.

CHAPTER 9

Statistics and Probability — Understanding Data and Uncertainty

09

9.1 Introduction: Numbers Tell Stories, but Do They Tell the Whole Truth?

Imagine that two students are applying for a scholarship.

Student A says, “I scored an average of 80 in my examinations.”

Student B says, “I also scored an average of 80.”

At first, their academic performance appears equal. But when you examine their individual scores, you discover something interesting.

Student A’s scores are:

$$70,\ 75,\ 80,\ 85,\ 90$$

Student B's scores are:

$$40,\ 60,\ 80,\ 100,\ 120$$

Both datasets have a mean of 80.

However, the students' results are not identical. Student A's scores are relatively close together, while Student B's scores are more widely spread.

The average alone does not tell the complete story.

Now imagine that a company announces that its employees earn an average monthly salary of ₦300,000. Does that mean every employee earns ₦300,000? Not necessarily. A few highly paid executives might raise the average even when many workers earn much less.

Or imagine that a weather forecaster says there is a 70% chance of rain tomorrow. Does this mean it will rain for 70% of the day? No. The statement concerns the probability of rain, not the proportion of the day

that will be rainy.

These examples introduce two important branches of mathematics:

- **Statistics**, which helps us collect, organise, summarise, analyse, and interpret data.
- **Probability**, which helps us study uncertainty and quantify how likely events are.

Before learning their formulas, we must understand the questions these fields were developed to answer.

Statistics helps us ask, "What does the available information tell us?"

Probability helps us ask, "What outcomes are possible, and how likely are they?"

Together, they help people make better decisions in education, business, economics, medicine, science, government, and everyday life.

9.2 Why Did People Need Statistics?

Imagine that you are responsible for managing a university with thousands of students.

You want to know:

- How many students are enrolled?
- Which courses have the highest failure rates?
- What percentage of students attend lectures regularly?
- How much money is needed to provide learning materials?
- Are students performing better this semester than last semester?

You cannot make good decisions without information.

However, collecting thousands of individual records is only the beginning. You also need a way to organise the information and discover what it means.

That is where statistics becomes useful.

Statistics in ancient societies

Long before modern statistical theory developed, governments and societies recorded information about people, land, agricultural production, and resources.

Such records helped rulers administer territories, collect taxes, organise labour, and plan public activities.

Ancient civilisations, including those in Egypt, Mesopotamia, China, and elsewhere, maintained different forms of administrative records. These were not all statistics in the modern scientific sense, but they demonstrate an early need to collect and organise information.

Over time, methods for summarising data and reasoning about uncertainty became more systematic.

The development of modern statistics

During the seventeenth century, European scholars investigated mortality records, population information, and games of chance. These studies contributed to the development of probability and statistical reasoning.

John Graunt analysed records of deaths in London in the seventeenth century. His work helped demonstrate how population records could reveal patterns that were difficult to see by examining individual cases.

Probability theory also developed through mathematical investigations of games of chance. Blaise Pascal and Pierre de Fermat exchanged ideas about gambling problems in the seventeenth century, contributing to the foundations of modern probability.

Later, statistical methods expanded into astronomy, biology, agriculture, economics, psychology, and many other fields.

Today, statistics helps researchers distinguish meaningful patterns from random variation, while probability provides mathematical models for uncertainty.

The historical lesson is simple: people needed to make decisions from incomplete or complicated information. Mathematics helped them organise that information and reason about it.

9.3 Understanding Data Before Calculating

Data is information collected for a purpose.

It may consist of numbers, categories, measurements, observations, or recorded events.

Examples include:

- students' examination scores;
- the prices of goods in a market;
- the ages of people in a community;
- the number of customers visiting a shop;
- the rainfall recorded each month;
- the opinions expressed in a survey.

Before analysing data, ask three questions.

First: What do I want to know?

A dataset is useful only in relation to a question.

For example, a school may collect examination scores to understand academic performance.

Second: Where did the data come from?

Information may come from a survey, experiment, official record, observation, or measurement.

The source affects how confidently we can interpret the results.

Third: Does the data represent the situation fairly?

If a school wants to understand the performance of all its students but collects results from only its highest-performing class, the information may give a misleading impression.

The quality of a statistical conclusion depends partly on the quality of the data.

Population and sample

A **population** is the entire group we want to study.

A **sample** is a smaller group selected from that population.

Suppose a university has 5,000 students and wants to understand their preferred method of learning.

The population consists of all 5,000 students.

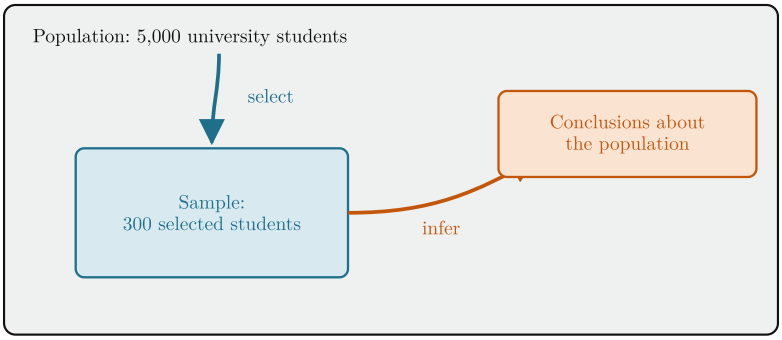
If the university surveys 300 students, those 300 form the sample.

Studying a sample can save time and money. However, the sample should be selected in a way that gives the population a fair chance of being represented.

If the survey includes only students from one department, the results may not accurately represent the whole university.

FIGURE 12

Population and Sample



A sample is drawn from the population, and what we learn from it is used to draw conclusions about the whole population.

Types of data

Data can be classified in different ways.

Qualitative data describes categories or qualities.

Examples include:

- department;
- type of transport;
- favourite subject;
- payment method.

Quantitative data represents numerical quantities.

Examples include:

- age;
- height;
- income;
- examination scores;

- number of customers.

Quantitative data can also be divided into two common types.

Discrete data usually involves countable values, such as the number of books sold.

Continuous data can take any value within a range, depending on the precision of measurement, such as time, distance, or temperature.

Understanding the type of data helps us decide how to summarise and analyse it.

9.4 Organising Data into Tables

Imagine that a teacher records the scores of ten students:

$$4,\ 6,\ 5,\ 7,\ 6,\ 4,\ 8,\ 6,\ 5,\ 9$$

Looking at the list, you can see the scores, but it may take time to determine which score occurs most often.

A frequency table makes the information easier to understand.

Score	Frequency
4	2
5	2
6	3
7	1
8	1
9	1
Total	**10**

Frequency means the number of times a value occurs.

For example, the score 6 occurs three times, so its frequency is 3.

The total frequency is

$$2 + 2 + 3 + 1 + 1 + 1 = 10$$

This agrees with the number of students.

That agreement is a useful check: the sum of the frequencies should equal the number of observations represented by the table.

Why frequency tables are useful

Frequency tables help us identify:

- the most common values;
- the least common values;
- the range of the data;
- patterns in repeated observations.

For larger datasets, tables can make information much easier to analyse than an unorganised list.

Grouped frequency tables

When a dataset contains many different values, it may be convenient to group the observations into intervals.

Suppose a teacher records the ages of 40 people.

Instead of listing every age separately, the teacher may use intervals such as:

Age group	Frequency
10–19	6
20–29	12
30–39	10
40–49	8
50–59	4
Total	**40**

The table summarises the distribution of ages.

However, grouping data removes some detail. From this table alone, we cannot identify the exact age of every person.

This illustrates a general principle: a summary makes information easier to understand, but it may hide details that are important for certain questions.

9.5 Measures of Central Tendency

Suppose a shopkeeper wants to know how much a typical customer spends.

The shopkeeper has hundreds of transactions. Examining every transaction individually may be inconvenient.

A measure of central tendency can provide a single value that summarises the centre of a dataset.

The three common measures are:

- mean;
- median;
- mode.

Each describes the data differently.

9.6 The Mean: Finding the Average

The arithmetic mean is commonly called the average.

It is calculated by adding all the observations and dividing by the number of observations.

The formula is

$$\boxed{\bar{x} = \frac{\sum x}{n}}$$

where:

- $\bar{x}$ represents the mean;
- $\sum x$ means the sum of all observations;
- n is the number of observations.

Example 1: Finding the mean

A student's scores are

$$60,\ 70,\ 80,\ 90,\ 100$$

Find the mean.

Step 1: Add the scores.

$$60 + 70 + 80 + 90 + 100 = 400$$

Step 2: Count the observations.

There are five scores.

Therefore,

$$n = 5$$

Step 3: Divide the total by the number of scores.

$$\bar{x} = \frac{400}{5}$$

Hence,

$$\boxed{\bar{x} = 80}$$

The mean score is 80.

Why do we divide by the number of observations?

Imagine sharing ₦400 equally among five people.

Each person receives

$$\frac{400}{5} = ₦80$$

The mean works similarly: it represents the value each observation would have if the total were distributed equally among all observations.

This interpretation is useful because it explains the purpose of the calculation rather than treating the formula as something to memorise.

Example 2: Finding an unknown score

Four students have a mean score of 15.

Their scores are

$$10,\ 12,\ 18,\ x$$

Find x.

The mean is 15, and there are four students.

Therefore, the total score must be

$$15 \times 4 = 60$$

The known scores add up to

$$10 + 12 + 18 = 40$$

Thus,

$$40 + x = 60$$

Subtract 40 from both sides:

$$\boxed{x = 20}$$

The missing score is 20.

Notice that we did not begin by searching for a complicated formula. We first understood that the mean and the number of observations determine the total.

9.7 The Median: Finding the Middle

The median is the middle value when observations are arranged in ascending or descending order.

Unlike the mean, the median is not calculated by adding all the values and dividing by the number of observations.

The order of the data matters.

Example 1: An odd number of observations

Consider

$$3,\ 5,\ 7,\ 9,\ 11$$

The observations are already arranged in ascending order.

The middle value is 7.

Therefore,

$$\boxed{\text{Median} = 7}$$

There are two values below 7 and two values above it.

Example 2: An even number of observations

Consider

$$2,\ 4,\ 6,\ 8,\ 10,\ 12$$

There is no single middle observation because the dataset contains six values.

The two middle values are 6 and 8.

Find their mean:

$$\frac{6+8}{2} = \frac{14}{2}$$

Therefore,

$$\boxed{\text{Median} = 7}$$

Why do we need the median?

Imagine that five workers earn these monthly salaries:

$$₦80{,}000,\ ₦90{,}000,\ ₦100{,}000,\ ₦110{,}000,\ ₦1{,}000{,}000$$

The mean is

$$\frac{80{,}000 + 90{,}000 + 100{,}000 + 110{,}000 + 1{,}000{,}000}{5}$$

The total is ₦1,380,000.

Therefore,

$$\text{Mean} = \frac{1{,}380{,}000}{5}$$

$$\boxed{\text{Mean} = ₦276{,}000}$$

But the median is ₦100,000.

The mean is much higher because the highest salary has a strong effect on the total.

The median provides another view of the typical salary because it is less sensitive to unusually high or low observations.

Neither measure is automatically correct for every purpose. The appropriate choice depends on the question and the shape of the data.

9.8 The Mode: Finding the Most Frequent Value

The mode is the value that occurs most frequently.

Consider the dataset

$$2,\ 3,\ 3,\ 4,\ 5,\ 3,\ 6,\ 7$$

The number 3 appears three times, more frequently than any other value. Therefore,

$$\boxed{\text{Mode} = 3}$$

The mode is particularly useful when we want to know the most common category or value.

For example, a shopkeeper may want to know which shoe size is sold most frequently.

A school may want to know which transport method students use most often.

A survey may identify the most popular response to a question.

Can a dataset have more than one mode?

Yes.

Consider

$$2,\ 2,\ 3,\ 4,\ 4,\ 5$$

The values 2 and 4 each occur twice.

Both share the highest frequency.

The dataset is therefore bimodal, with modes 2 and 4.

A dataset may also have no single mode if all observations occur equally often, depending on the convention being used.

The mode is not always unique.

9.9 Comparing Mean, Median, and Mode

Consider the dataset

$$2,\ 3,\ 3,\ 4,\ 8$$

The mean is

$$\frac{2+3+3+4+8}{5} = \frac{20}{5}$$
$$\boxed{\text{Mean} = 4}$$

The median is the middle observation:

$$\boxed{\text{Median} = 3}$$

The mode is the most frequent observation:

$$\boxed{\text{Mode} = 3}$$

Each measure tells us something different.

Measure	What it tells us
Mean	The total divided equally among all observations
Median	The middle position in ordered data
Mode	The most frequently occurring value

When choosing a measure, consider what you want to understand.

Use the mean when the overall numerical total is meaningful and the effect of extreme values is acceptable.

Use the median when you want a measure of the centre that is less affected by extreme observations.

Use the mode when the most common value or category is important.

A careful analyst does not automatically use the mean for every dataset.

9.10 Measures of Dispersion: How Spread Out Is the Data?

Central tendency describes the centre of a dataset.

But two datasets can have the same mean and very different levels of variation.

Consider two classes.

Class A:

$$48,\ 49,\ 50,\ 51,\ 52$$

Class B:

$$20,\ 35,\ 50,\ 65,\ 80$$

Both have a mean of 50.

For Class A,

$$\frac{48+49+50+51+52}{5} = \frac{250}{5} = 50$$

For Class B,

$$\frac{20+35+50+65+80}{5} = \frac{250}{5} = 50$$

Yet Class B's scores are more widely spread.

This is why statistics uses measures of dispersion, which describe how much observations differ from one another.

Common measures include:

- range;
- interquartile range;
- variance;
- standard deviation.

9.11 The Range

The range is the difference between the largest and smallest observations.

The formula is

$$\boxed{\text{Range} = \text{Maximum} - \text{Minimum}}$$

For Class A,

$$\text{Range} = 52 - 48$$
$$\boxed{\text{Range} = 4}$$

For Class B,

$$\text{Range} = 80 - 20$$
$$\boxed{\text{Range} = 60}$$

The range shows that Class B has a much wider spread between its smallest and largest scores.

However, the range uses only two observations. It does not describe how the remaining values are distributed.

For this reason, other measures of dispersion can provide additional information.

9.12 Variance: Measuring Squared Deviations

Variance measures the average squared distance of observations from their mean, with the exact formula depending on whether we are describing a population or estimating from a sample.

First, let us understand the idea.

Suppose the mean of a dataset is 10.

An observation of 11 is one unit above the mean.

An observation of 7 is three units below the mean.

The deviation of an observation is

$$\text{Deviation} = x - \bar{x}$$

where x is the observation and $\bar{x}$ is the mean.

If we simply add all deviations from the mean, their sum is zero. Positive and negative deviations cancel.

To measure spread, variance squares the deviations before averaging them.

Population variance

For a complete population, the variance is

$$\boxed{\sigma^2 = \frac{\sum(x-\mu)^2}{N}}$$

where:

- σ^2 is population variance;
- x represents each observation;
- μ is the population mean;
- N is the population size.

Example: Finding population variance

Consider the population

$$2,\ 4,\ 6$$

Step 1: Find the mean.

$$\mu = \frac{2+4+6}{3}$$

$$\mu = \frac{12}{3} = 4$$

Step 2: Calculate the deviations.

$$x \quad x-\mu$$

2 -2

4 0

6 2

Step 3: Square the deviations.

$x-\mu$	$(x-\mu)^2$
-2	4
0	0
2	4

Step 4: Add the squared deviations.

$$4+0+4=8$$

Step 5: Divide by the population size.

$$\sigma^2 = \frac{8}{3}$$

Therefore,

$$\boxed{\sigma^2 = \frac{8}{3} \approx 2.67}$$

The population variance is approximately 2.67 squared units.

Sample variance

If the observations are a sample used to estimate the variance of a larger population, the usual sample variance formula is

$$\boxed{s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}}$$

The denominator is $n - 1$, rather than n, because this formula provides an unbiased estimator of population variance under the usual independent random-sampling assumptions.

The distinction matters. Before calculating variance, determine whether the data represents the entire population or a sample used for estimation.

9.13 Standard Deviation

Variance is useful, but it is expressed in squared units.

For example, if the observations are measured in metres, variance is measured in square metres.

Standard deviation solves part of this interpretive problem by taking the square root of the variance.

For a population,

$$\boxed{\sigma = \sqrt{\sigma^2}}$$

For the population

$$2,\ 4,\ 6$$

we found that

$$\sigma^2 = \frac{8}{3}$$

Therefore,

$$\sigma = \sqrt{\frac{8}{3}}$$

$$\boxed{\sigma \approx 1.63}$$

The population standard deviation is approximately 1.63 units.

Standard deviation describes the typical scale of deviations from the mean, though it is not simply the average absolute distance from the mean.

A smaller standard deviation generally indicates that observations are more closely concentrated around the mean. A larger standard deviation indicates greater spread.

Why does standard deviation matter?

Imagine two investment options with similar average returns.

One has returns that vary only slightly from year to year.

The other has returns that fluctuate widely.

Their means alone may not capture the difference in uncertainty.

Measures of variation help analysts understand how stable or variable the observed results are.

However, standard deviation does not tell the whole story about risk. The nature of the outcomes, the possibility of extreme events, and the purpose of the analysis also matter.

9.14 Quartiles and the Interquartile Range

Sometimes we want to understand the spread of the middle portion of a dataset.

Quartiles divide ordered data into four parts.

The main quartiles are:

- Q_1: the first quartile;
- Q_2: the median;
- Q_3: the third quartile.

The interquartile range is

$$\boxed{IQR = Q_3 - Q_1}$$

It measures the spread of the middle 50% of the observations.

Different textbooks and statistical software may use slightly different methods to calculate quartiles, especially for small datasets. Therefore, when a particular method is required, use the stated convention consistently.

Example

Consider the ordered dataset

$$2,\ 4,\ 6,\ 8,\ 10,\ 12,\ 14$$

The median is 8.

Using the method that excludes the overall median when dividing an odd-sized dataset:

The lower half is

$$2,\ 4,\ 6$$

Its median is

$$Q_1 = 4$$

The upper half is

$$10,\ 12,\ 14$$

Its median is

$$Q_3 = 12$$

Therefore,

$$IQR = 12 - 4$$
$$\boxed{IQR = 8}$$

The interquartile range is 8.

This measure is less influenced by extreme minimum and maximum values than the range.

9.15 Presenting Data with Graphs

Tables organise data, but graphs can reveal patterns more quickly.

Different graphs serve different purposes.

Bar charts

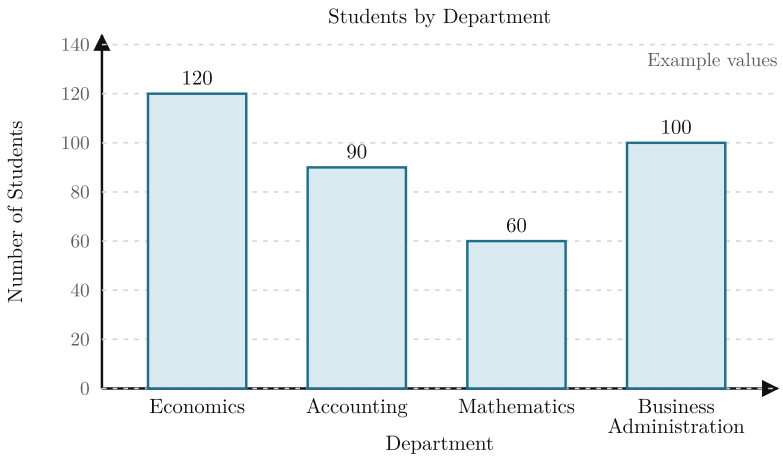
Bar charts compare categories.

For example, a university may compare the number of students in Economics, Accounting, Mathematics, and Business Administration.

The categories appear along one axis, while the frequencies or values appear along the other.

FIGURE 13

Bar Chart of Students by Department



Four separate bars with gaps, because the departments are distinct categories. The book gives no data, so the values are illustrative.

Histograms

Histograms display the distribution of numerical data grouped into intervals.

Unlike ordinary bar charts, histogram bars normally touch because the intervals represent adjacent ranges of a numerical variable.

For example, a histogram might show how many students scored between 0 and 9, 10 and 19, 20 and 29, and so on.

Pie charts

Pie charts show how a whole is divided into categories.

Each sector represents a proportion of the total.

They are most useful when the categories are mutually exclusive and collectively account for the whole being displayed.

Line graphs

Line graphs show how a quantity changes across an ordered sequence, often time.

For example, a line graph can show monthly rainfall, daily sales, or a company's revenue over several years.

Scatter plots

Scatter plots display the relationship between two numerical variables.

For example, a researcher may plot the number of hours students study against their examination scores.

A scatter plot can reveal whether the variables tend to increase together, decrease together, or show little apparent relationship.

However, a relationship between two variables does not automatically prove that one causes the other.

9.16 Correlation: Understanding Relationships

Imagine a researcher notices that students who spend more time studying often achieve higher examination scores.

This suggests that study time and examination performance may be associated.

Correlation is a statistical measure of the direction and strength of a particular type of relationship between variables.

The Pearson correlation coefficient is often represented by r.

Its value lies between -1 and 1:

$$-1 \leq r \leq 1$$

A value near 1 indicates a strong positive linear relationship.

A value near -1 indicates a strong negative linear relationship.

A value near zero indicates little or no linear association, although a nonlinear relationship may still exist.

Positive correlation

When one variable tends to increase as the other increases, the relationship is positive.

For example, there may be a positive association between the number of items sold and total revenue, depending on prices and other conditions.

Negative correlation

When one variable tends to increase as the other decreases, the relationship is negative.

For example, under suitable conditions, increasing the price of a product may be associated with lower demand.

Little or no linear correlation

Sometimes the data shows no clear linear pattern.

This does not prove that the variables are completely unrelated. They may have a nonlinear relationship or be influenced by other factors.

Correlation does not prove causation

Suppose ice cream sales and swimming activity both increase during hot weather.

It would be incorrect to conclude automatically that ice cream sales cause swimming activity.

Temperature may influence both.

A third variable can help explain why two other variables move together.

Before concluding that one factor causes another, researchers need stronger evidence, often involving carefully designed experiments or methods that address alternative explanations.

This is an important lesson in critical thinking: a statistical relationship is evidence of association, not automatic proof of cause.

9.17 Probability: Measuring Uncertainty

Statistics helps us understand collected data.

Probability helps us reason about uncertain events.

Imagine tossing a fair coin.

The possible outcomes are:

$$H,\ T$$

where H represents heads and T represents tails.

If the coin is fair, both outcomes have equal probability.

Therefore,

$$P(H) = \frac{1}{2}$$

and

$$P(T) = \frac{1}{2}$$

Probability measures how likely an event is, using values between 0 and 1.

$$\boxed{0 \leq P(A) \leq 1}$$

Here, $P(A)$ represents the probability of event A.

A probability of 0 means the event is impossible within the model.

A probability of 1 means the event is certain within the model.

A probability of 0.5 represents a probability of 50%.

Probability does not guarantee the outcome of an individual event. A fair coin can produce heads several times in succession. The probability describes the model, not a promise about the next result.

9.18 The Classical Probability Formula

When all outcomes are equally likely and the sample space is finite, probability can be calculated using

$$\boxed{P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}}$$

This formula applies under the equal-likelihood assumption.

Example 1: Rolling a fair die

A standard fair die has six possible outcomes:

$$1,\ 2,\ 3,\ 4,\ 5,\ 6$$

What is the probability of rolling a 4?

There is one favourable outcome.

There are six equally likely outcomes in total.

Therefore,

$$P(4) = \frac{1}{6}$$

Hence,

$$\boxed{P(4) = \frac{1}{6}}$$

Example 2: Rolling an even number

The even outcomes are

$$2,\ 4,\ 6$$

There are three favourable outcomes out of six.

Therefore,

$$P(\text{even}) = \frac{3}{6}$$

Simplify:

$$\boxed{P(\text{even}) = \frac{1}{2}}$$

Example 3: Selecting a ball

A bag contains 3 red balls, 2 blue balls, and 5 green balls. One ball is selected at random, with each ball equally likely to be chosen.

Find the probability of selecting a blue ball.

The total number of balls is

$$3 + 2 + 5 = 10$$

There are two blue balls.

Therefore,

$$P(\text{blue}) = \frac{2}{10}$$

Simplifying,

$$\boxed{P(\text{blue}) = \frac{1}{5}}$$

The probability is one-fifth, or 20%.

9.19 The Complement Rule

Sometimes it is easier to calculate the probability that an event does not happen.

The complement of event A is written as A^c.

The complement rule states

$$\boxed{P(A^c) = 1 - P(A)}$$

Why does this work?

An event either occurs or does not occur. These two possibilities cover the entire sample space and cannot both occur in the same trial.

Their probabilities therefore add up to 1.

Example

The probability that a student passes a particular test is

$$P(\text{pass}) = 0.8$$

Assuming the outcomes are classified as pass or not pass,

$$P(\text{not pass}) = 1 - 0.8$$

Therefore,

$$\boxed{P(\text{not pass}) = 0.2}$$

The probability of not passing is 20%.

The complement rule is especially useful when the probability of an event is easier to calculate than the probability of all the ways it could fail.

9.20 Independent Events

Two events are independent when the occurrence of one does not change the probability of the other.

For independent events A and B,

$$\boxed{P(A \cap B) = P(A)P(B)}$$

The symbol $\cap$ means that both events occur.

Example: Tossing a coin twice

Suppose a fair coin is tossed twice.

What is the probability of obtaining heads on both tosses?

The probability of heads on the first toss is

$$\frac{1}{2}$$

The probability of heads on the second toss is also

$$\frac{1}{2}$$

The tosses are independent.

Therefore,

$$P(HH) = \frac{1}{2} \times \frac{1}{2}$$

Hence,

$$\boxed{P(HH) = \frac{1}{4}}$$

The probability is one-quarter, or 25%.

Notice that independence is an assumption about how the events relate. It should not be assumed automatically in every problem.

9.21 Dependent Events

Two events are dependent when the outcome of one affects the probability of the other.

Imagine a bag containing three red balls and two blue balls.

You select one ball and do not replace it.

The first selection changes the contents of the bag, so the probability of the second selection changes.

Example: Selecting two red balls without replacement

There are five balls in total, three of which are red.

The probability that the first ball is red is

$$P(R_1) = \frac{3}{5}$$

If the first ball is red and is not replaced, two red balls remain among four balls.

Therefore, the probability that the second ball is red, given that the first was red, is

$$P(R_2 \mid R_1) = \frac{2}{4}$$

The probability of selecting two red balls in succession is

$$\frac{3}{5} \times \frac{2}{4}$$

Simplify:

$$\frac{3 \times 2}{5 \times 4} = \frac{6}{20}$$

Therefore,

$$\boxed{P(\text{two reds}) = \frac{3}{10}}$$

The key difference is that the probability of the second selection changed after the first ball was removed.

9.22 The Addition Rule

The addition rule helps us calculate the probability that event A or event B occurs.

The general rule is

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The symbol $\cup$ means that at least one of the events occurs.

Why do we subtract the intersection?

If we simply add $P(A)$ and $P(B)$, outcomes that belong to both events are counted twice. Subtracting the intersection corrects that double counting.

Example: Rolling a fair die

Find the probability of rolling a number that is either even or greater than 4.

The even outcomes are

$$A = 2, 4, 6$$

The outcomes greater than 4 are

$$B = 5, 6$$

The intersection is

$$A \cap B = 6$$

Therefore,

$$P(A) = \frac{3}{6}$$
$$P(B) = \frac{2}{6}$$
$$P(A \cap B) = \frac{1}{6}$$

Apply the addition rule:

$$P(A \cup B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6}$$

Thus,

$$P(A \cup B) = \frac{4}{6}$$

Simplify:

$$\boxed{P(A \cup B) = \frac{2}{3}}$$

We can check this by listing the favourable outcomes:

$$2, 4, 5, 6$$

There are four favourable outcomes out of six, confirming the result.

Mutually exclusive events

If two events cannot occur at the same time, they are mutually exclusive.

For mutually exclusive events,

$$P(A \cap B) = 0$$

The addition rule simplifies to

$$\boxed{P(A \cup B) = P(A) + P(B)}$$

For example, on one roll of a standard die, the events “roll a 2” and “roll a 5” cannot occur simultaneously.

9.23 Conditional Probability

Conditional probability describes the probability of an event when we know that another event has occurred.

It is written as

$$P(A \mid B)$$

This means the probability of A, given B.

When $P(B) > 0$, the formula is

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

Example: Selecting a student

Suppose a class contains 20 students:

- 12 study Economics;
- 8 study Accounting;
- 7 of the Economics students are members of a study group.

If a student is selected at random from the Economics students, what is the probability that the student belongs to the study group?

The question restricts our attention to the 12 Economics students.

Of those 12 students, 7 belong to the study group.

Therefore,

$$P(\text{study group} \mid \text{Economics}) = \frac{7}{12}$$

Hence,

$$\boxed{\frac{7}{12} \approx 0.5833}$$

The probability is approximately 58.33%.

The important step was recognising that the question changed the group being considered. We were not selecting from all 20 students; we were selecting from the 12 Economics students.

9.24 Expected Value: What Is the Average Outcome Over Time?

Probability can help us evaluate situations with different possible outcomes.

Expected value is the probability-weighted average of the possible numerical outcomes.

For a discrete random variable X,

$$\boxed{E(X) = \sum xP(X = x)}$$

This means that each possible value is multiplied by its probability, and the products are added.

Expected value does not predict exactly what will happen in one trial. It describes the long-

run average under the probability model.

Example: A simple prize game

Suppose a game has these possible outcomes:

Outcome	Prize	Probability
Win	₦1,000	0.2
Lose	₦0	0.8

The expected prize is

$$E(X) = 1000(0.2) + 0(0.8)$$

Calculate:

$$E(X) = 200 + 0$$

Therefore,

$$\boxed{E(X) = ₦200}$$

The expected prize is ₦200 per game.

This does not mean the player will receive ₦200 in a single game. The actual result is either ₦1,000 or ₦0.

If playing the game costs ₦250, the expected net outcome is

$$₦200 - ₦250 = -₦50$$

Under this model, the expected net result is a loss of ₦50 per play.

Expected value is useful in business, insurance, planning, and decision-making, but it should not be confused with certainty. Averages can hide substantial variation in possible outcomes.

9.25 Probability Distributions

A probability distribution describes the possible values of a random variable and the probabilities associated with those values.

Consider a fair coin tossed twice.

The possible outcomes are

$$HH,\ HT,\ TH,\ TT$$

Let X be the number of heads.

Then:

- HH produces $X = 2$;
- HT produces $X = 1$;
- TH produces $X = 1$;
- TT produces $X = 0$.

The probability distribution is:

Number of heads, X	Probability
0	$\frac{1}{4}$
1	$\frac{1}{2}$
2	$\frac{1}{4}$

The probabilities add up to 1:

$$\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$

This is a useful check for a complete probability distribution.

Probability distributions allow us to describe uncertainty systematically and calculate quantities such as expected value and variance.

9.26 The Binomial Distribution

The binomial distribution applies when we count the number of successes in a fixed number of independent trials, each with the same probability of success.

Typical examples include:

- the number of heads in repeated fair coin tosses;
- the number of defective items in a fixed sample, under suitable assumptions;
- the number of students who answer a question correctly when each response is modelled using the same success probability.

The probability of exactly r successes in n independent trials is

$$\boxed{P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}}$$

where:

- n is the number of trials;
- r is the number of successes;
- p is the probability of success on each trial;
- $1 - p$ is the probability of failure;
- $\binom{n}{r}$ counts the ways to choose which r trials are successes.

Example: Two coin tosses

A fair coin is tossed twice.

Find the probability of exactly one head.

Here,

$$n = 2, \qquad r = 1, \qquad p = \frac{1}{2}$$

Substitute into the binomial formula:

$$P(X=1)=\binom{2}{1}\left(\frac{1}{2}\right)^{1}\left(\frac{1}{2}\right)^{1}$$

Since

$$\binom{2}{1}=2$$

we have

$$P(X=1)=2\times\frac{1}{2}\times\frac{1}{2}$$

Therefore,

$$\boxed{P(X=1)=\frac{1}{2}}$$

The possible outcomes with exactly one head are HT and TH, so the answer agrees with the sample space.

9.27 Sampling and Bias

Suppose a university wants to discover whether students are satisfied with its library.

The university asks only students who are currently studying in the library.

The responses may be useful, but the sample may not represent all students. People who never visit the library have no chance of being included.

This is an example of possible selection bias.

Bias occurs when the data collection process systematically favours certain outcomes or groups.

Common sources of bias include:

- surveying only people who are easy to reach;
- asking leading questions;
- excluding important groups;
- collecting responses at an unrepresentative time;
- relying on voluntary responses that differ systematically from non-respondents.

How can bias be reduced?

Researchers can:

1. Define the population clearly.
2. Choose a suitable sampling method.
3. Give relevant members of the population a fair opportunity to be selected.
4. Use clear and neutral questions.
5. Record how the data was collected.
6. Consider who did not respond.

A large sample does not automatically guarantee reliable conclusions. A very large biased sample may still misrepresent the population.

Good statistics begins with good data collection.

9.28 Statistics, Probability, and Decision-Making

Imagine that a business owner wants to open a second shop.

The owner must consider:

- expected customer demand;
- rent and operating costs;
- likely sales;
- the possibility of unexpected expenses;
- uncertainty about future market conditions.

Statistics can summarise past sales and customer behaviour.

Probability can help model uncertain outcomes and compare possible risks.

Together, they support more informed decisions.

However, neither field guarantees perfect predictions.

A statistical model depends on the quality of its data and assumptions. Probability calculations depend on whether the chosen model reasonably represents the situation.

A good decision-maker uses mathematical evidence while remaining aware of uncertainty and limitations.

9.29 How to Make Statistics and Probability Easier

Method 1: Identify the question

Before calculating, ask what you need to know.

Are you trying to find a typical value, measure spread, describe a relationship, or calculate the likelihood of an event?

The question determines the method.

Method 2: Organise the information

Write down the observations, arrange them when necessary, and use a table if it makes the pattern clearer.

Organisation reduces errors and helps you understand the data.

Method 3: Choose the correct measure

Do not assume that the mean is always the best summary.

If extreme values strongly affect the average, the median may provide useful additional information.

If you want to identify the most common category, the mode may be more appropriate.

Method 4: Check the assumptions

The classical probability formula requires equally likely outcomes.

The binomial distribution requires a fixed number of independent trials with the same success probability.

Statistical conclusions require data that is suitable for the question.

Always check whether the assumptions fit the situation.

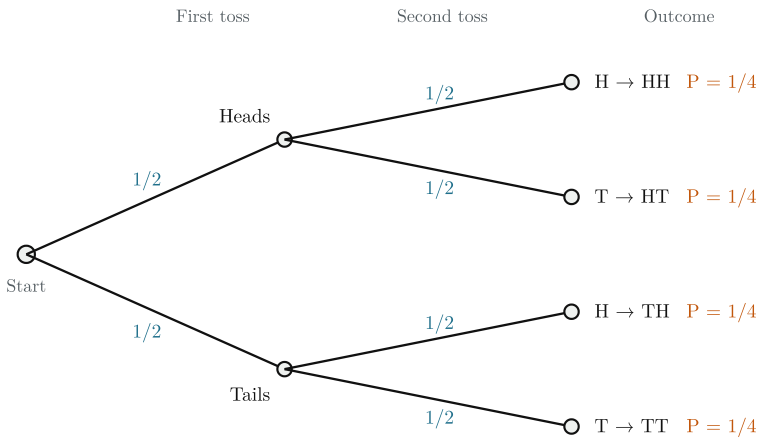
Method 5: Use a diagram or table

A probability tree, frequency table, or graph can make a complicated question easier to understand.

For example, a tree diagram can help organise a sequence of events.

FIGURE 14

Probability Tree for Two Coin Tosses



Each branch has probability 1/2. Multiplying along a path gives 1/4 for each of HH, HT, TH and TT.

Method 6: Check the result

For a complete probability distribution, check that all probabilities add up to 1.

For a frequency table, check that the frequencies add up to the total number of observations.

For a mean, multiply the mean by the number of observations to recover the total.

These checks can reveal errors.

Method 7: Explain the answer in ordinary language

Do not stop at a number.

If the mean salary is ₦150,000, explain what the figure represents and whether it gives a useful description of the workers' salaries.

If the probability of an event is 0.25, explain that the model assigns the event a probability of 25%, not that it must occur once in every four trials.

Interpretation is part of mathematical reasoning.

9.30 Common Mistakes

Mistake 1: Using the mean without examining the data

A few unusually high or low observations can strongly affect the mean.

Consider the median and the distribution when deciding how to describe a dataset.

Mistake 2: Finding the median before arranging the data

The median depends on the middle position in ordered data.

Arrange the observations before identifying the median.

Mistake 3: Confusing variance with standard deviation

Variance uses squared deviations.

Standard deviation is the square root of variance.

They are related, but they are not the same quantity.

Mistake 4: Assuming correlation proves causation

Two variables can move together because of a third factor or other circumstances.

Correlation alone does not prove that one variable causes the other.

Mistake 5: Forgetting that probability depends on assumptions

The formula for favourable outcomes divided by total outcomes works when the outcomes are equally likely.

If they are not equally likely, a different approach may be required.

Mistake 6: Multiplying probabilities without checking independence

For independent events, probabilities can be multiplied directly to find the probability that both occur.

For dependent events, the probability of the second event may change after the first event occurs.

Mistake 7: Treating probability as certainty

A probability describes uncertainty within a model. It does not guarantee a particular result in an individual trial.

Mistake 8: Ignoring sampling bias

A large sample is not necessarily representative.

Consider how participants were selected and whether important groups were excluded.

9.31 Practical Workshop: Investigate a Real Question

This workshop shows how statistics and probability can be applied to everyday situations.

Activity A: Study hours and examination results

Suppose you want to investigate whether students who study longer tend to achieve higher examination scores.

First, define your question:

“Is there an association between study time and examination performance among students in this group?”

Collect data on study hours and examination scores, with appropriate permission and respect for privacy.

Organise the observations in a table.

Create a scatter plot with study hours on the horizontal axis and examination scores on the vertical axis.

Then ask:

1. Do the points appear to form a pattern?
2. Is the association positive, negative, or unclear?
3. Are there unusual observations?
4. Could other factors influence examination performance?
5. Does the evidence establish causation, or only association?

The purpose is not to force a conclusion. It is to learn how to examine evidence carefully.

Activity B: Analyse a small dataset

Suppose the daily number of customers entering a shop over seven days is

$$12,\ 15,\ 10,\ 18,\ 15,\ 20,\ 14$$

Step 1: Find the mean.

Add the observations:

$$12 + 15 + 10 + 18 + 15 + 20 + 14 = 104$$

There are seven observations.

Therefore,

$$\bar{x} = \frac{104}{7}$$

$$\boxed{\bar{x} \approx 14.86}$$

The average is approximately 14.86 customers per day.

Step 2: Find the median.

Arrange the data in ascending order:

$$10,\ 12,\ 14,\ 15,\ 15,\ 18,\ 20$$

The middle value is 15.

Therefore,

$$\boxed{\text{Median} = 15}$$

Step 3: Find the mode.

The value 15 occurs twice, while every other value occurs once.

Therefore,

$$\boxed{\text{Mode} = 15}$$

Step 4: Find the range.

$$\text{Range} = 20 - 10$$
$$\boxed{\text{Range} = 10}$$

The dataset has a mean of approximately 14.86 customers, a median of 15, a mode of 15, and a range of 10.

These figures summarise different aspects of the same observations.

Activity C: Investigate uncertainty

A bag contains four red counters and six blue counters. One counter is selected at random, with every counter equally likely to be chosen.

Find the probability of selecting a red counter.

There are ten counters in total and four are red.

Therefore,

$$P(\text{red}) = \frac{4}{10}$$

Simplify:

$$\boxed{P(\text{red}) = \frac{2}{5}}$$

Now find the probability of not selecting a red counter.

Using the complement rule,

$$P(\text{not red}) = 1 - \frac{2}{5}$$

$$\boxed{P(\text{not red}) = \frac{3}{5}}$$

Check that the two probabilities add up to 1:

$$\frac{2}{5} + \frac{3}{5} = 1$$

The result is consistent with the model.

9.32 Exercises

Attempt the questions below. Show your working and explain your reasoning.

Section A: Understanding statistics

1. Explain the difference between a population and a sample.
2. Distinguish between qualitative and quantitative data.

3. Explain the difference between discrete and continuous data.
4. Why can two datasets have the same mean but different levels of variation?
5. Explain why a large sample may still produce misleading conclusions.
6. What is the difference between correlation and causation?

Section B: Central tendency

7. Find the mean of

$$5,\ 8,\ 10,\ 12,\ 15$$

8. Find the median of

$$9,\ 3,\ 7,\ 11,\ 5$$

9. Find the median of

$$4,\ 8,\ 10,\ 12,\ 16,\ 20$$

10. Find the mode of

$$2,\ 4,\ 4,\ 5,\ 6,\ 4,\ 7,\ 7$$

11. Five students have a mean score of 18. Four of their scores are 12, 16, 20, and 22. Find the fifth score.

Section C: Dispersion

12. Find the range of

$$11,\ 15,\ 18,\ 22,\ 27$$

13. Find the population variance of

$$2,\ 4,\ 6$$

14. Find the population standard deviation of

$$3,\ 3,\ 3,\ 3$$

15. Explain why the interquartile range may be more informative than the range when a dataset contains an extreme value.

Section D: Probability

16. A fair die is rolled once. Find the probability of obtaining a number greater than 4.

16. A bag contains 5 red balls, 3 blue balls, and 2 green balls. Find the probability of selecting a green ball at random.

16. A fair coin is tossed three times. Find the probability of obtaining heads on all three tosses.

16. The probability that a product passes inspection is 0.92. Assuming the only outcomes are pass and fail, find the probability that it fails inspection.

16. A bag contains 4 red balls and 6 blue balls. Two balls are selected without replacement. Find the probability that both are red.

16. A fair die is rolled once. Find the probability of rolling an even number or a number greater than 4.

Section E: Applied reasoning

22. Explain why a company's average salary might give a misleading impression of what a typical employee earns.

22. A weather forecast gives a 30% probability of rain. Explain what this means and what it does not mean.

22. A researcher finds that people who exercise more frequently tend to report better health. Give one reason why this association alone does

not prove causation.

22. A business owner wants to estimate future sales. Explain how historical data and probability could help, and identify one limitation of the resulting prediction.

9.33 Chapter Review

Before moving to the next chapter, make sure you understand the following ideas:

- Statistics helps us organise, summarise, analyse, and interpret data.
- Probability provides a mathematical way to describe uncertainty.
- A population is the complete group under study, while a sample is a subset.
- Frequency tables and graphs help reveal patterns in data.
- The mean, median, and mode describe central tendency in different ways.
- The range, variance, standard deviation, and interquartile range describe aspects of variation.
- Correlation measures association but does not automatically establish causation.
- Probability values range from 0 to 1.
- The complement rule calculates the probability that an event does not occur.
- The addition and multiplication rules help calculate probabilities for combined events.
- Conditional probability considers what is known about another event.
- Expected value describes a probability-weighted average, not a guaranteed outcome.
- Good conclusions require suitable data, reasonable assumptions, and careful interpretation.

9.34 Conclusion: Numbers Need Interpretation

We began this chapter with two students who had the same average score but different patterns of performance.

That example revealed an important truth: a single number rarely tells the whole story.

Statistics helps us understand the information we collect. It allows us to identify patterns, compare groups, describe typical values, and measure variation.

Probability helps us reason about uncertainty. It gives us tools for examining possible outcomes, evaluating risks, and making decisions when the future is not completely known.

But neither statistics nor probability removes the need for careful thinking.

A mean can hide an unusual observation. A graph can present a misleading picture if its scale is poorly chosen. A sample can fail to represent the population. A correlation can be mistaken for causation. A probability can be treated as a promise when it is only a measure of likelihood.

Mathematics becomes more valuable when we understand both what the numbers reveal and what they leave unanswered.

> **Think before the formula. Understand the data before drawing a conclusion, and understand the uncertainty before making a prediction.**

In the next chapter, we will think like scientists and engineers. We will explore how mathematical models help us represent real situations, test ideas, solve practical problems, and make decisions using evidence.

CHAPTER 10

Think Like a Scientist and Engineer — Using Mathematics to Solve Real Problems

10

10.1 Introduction: Mathematics Beyond the Classroom

Imagine that you are an engineer asked to design a bridge across a river.

You cannot simply choose some numbers, put them into a formula, and hope the bridge will stand. You must understand the river, the distance between the two sides, the materials available, the weight the bridge must support, and the forces acting on the structure.

You must ask questions before performing calculations.

How long must the bridge be? How much weight will it carry? What materials are suitable? How will wind, rain, and temperature affect it? What could happen if one of your assumptions is wrong?

Now imagine that you are a scientist investigating why a particular plant grows faster under certain conditions.

You need to identify the question, collect observations, control relevant conditions, record measurements, analyse the results, and decide whether the evidence supports your explanation.

These situations may appear different, but they share an important feature: both require people to understand a real problem and use evidence to develop a solution.

Mathematics provides tools for measuring, describing, predicting, and testing.

Scientists use mathematics to analyse observations and investigate relationships. Engineers use mathematics to design systems, estimate loads, calculate dimensions, and evaluate whether a design meets specified requirements.

The central lesson of this chapter is that **a calculation becomes useful when it answers a real question accurately enough for the purpose at hand**.

We will examine how scientists and engineers think, how mathematical models are developed, how formulas are selected, how errors are managed, and how calculations can be made easier without sacrificing accuracy.

10.2 The Difference Between Scientific and Engineering Thinking

Science and engineering often work together, but they do not always ask exactly the same questions.

Scientific thinking

Science seeks to understand the natural world through observation, measurement, experimentation, and reasoning.

A scientist might ask:

- Why does a metal expand when heated?

- How does light behave when it passes through different materials?
- What conditions influence the growth of a plant?
- How does the concentration of a substance change over time?

Scientific investigations aim to develop explanations that are consistent with evidence and can be tested against further observations.

Engineering thinking

Engineering applies scientific knowledge and mathematical reasoning to design, build, improve, and maintain systems or devices.

An engineer might ask:

- What material should be used for a particular structure?
- How much energy will a machine require?
- What dimensions will make a component fit?
- How can a system use less energy while meeting its performance requirements?

Engineering decisions must consider practical constraints such as cost, safety, reliability, available materials, and environmental effects.

Where does mathematics fit?

Mathematics helps both fields turn questions into quantities that can be analysed.

For example, a scientist investigating the motion of an object may use equations to describe its position and velocity.

An engineer designing a braking system may use motion equations to estimate stopping distances under specified conditions.

The mathematics may overlap, but the purpose of the calculation differs.

Science may focus on understanding the relationship between quantities. Engineering may use that relationship to decide how a system should be designed.

10.3 Begin with the Problem, Not the Formula

Suppose a school wants to install a water tank on a platform.

The tank must hold enough water for the school's needs, and the platform must support the tank safely.

A student might immediately search for a formula involving volume.

But volume is only one part of the problem.

Before calculating, ask:

1. How much water is needed?
2. What shape will the tank have?
3. What dimensions are available?
4. How much will the tank and its contents weigh?
5. What material will support the tank?
6. What safety requirements must the design satisfy?

These questions determine which calculations are necessary.

If the tank is cylindrical, its volume can be calculated using

$$V = \pi r^2 h$$

where r is the radius and h is the height.

But knowing the volume does not automatically prove that the platform is strong enough.

You would also need to consider the mass of the water, the mass of the tank, the distribution of the load, the strength of the platform, and relevant safety standards.

The formula for volume answers one question: how much space the tank contains.

It does not answer every question about the complete design.

This is why choosing the right formula begins with identifying the problem accurately.

10.4 The Scientific Method and Mathematical Reasoning

Scientists use systematic methods to investigate questions. The exact process varies by discipline, but many investigations share several stages.

Stage 1: Observe a situation

Suppose you notice that plants placed near a window appear to grow faster than plants kept in a darker room.

This is an observation, not yet a complete explanation.

Stage 2: Ask a question

You might ask:

“Does the amount of light received by a plant affect its growth under otherwise similar conditions?”

A clear question helps determine what information you need.

Stage 3: Develop a hypothesis

A hypothesis is a testable proposed explanation.

For example:

“Plants receiving more suitable light will grow taller over a specified period, provided other important conditions remain comparable.”

The hypothesis must be specific enough to investigate.

Stage 4: Design an investigation

You decide what to measure, how frequently to measure it, and which conditions to keep similar.

You might record the height of each plant every three days.

You would also need to consider factors such as water, soil, plant species, temperature, and the intensity and duration of light.

Stage 5: Collect data

Suppose you record the following plant heights:

Day	Plant A (cm)	Plant B (cm)
0	5.0	5.0
3	5.8	5.5
6	6.7	5.9
9	7.5	6.2
12	8.4	6.6

Plant A grows more in this particular set of observations.

However, before concluding that light caused the difference, you must consider whether the plants received different amounts of water, had different starting conditions, or experienced other relevant differences.

Stage 6: Analyse the data

You might calculate the change in height.

For Plant A,

$$\text{Growth} = 8.4 - 5.0$$

$$\boxed{\text{Growth} = 3.4 \text{ cm}}$$

For Plant B,

$$\text{Growth} = 6.6 - 5.0$$

$$\boxed{\text{Growth} = 1.6 \text{ cm}}$$

The observations show that Plant A grew 3.4 cm, while Plant B grew 1.6 cm.

The difference in growth is

$$3.4 - 1.6 = 1.8 \text{ cm}$$

This is a numerical description of the observed difference.

It is not, by itself, proof that light was the only cause.

Stage 7: Interpret the result

You must decide what the evidence supports and what remains uncertain.

A good scientist does not force the evidence to support a preferred conclusion.

If the data is limited, the conclusion should reflect that limitation.

Stage 8: Repeat and improve

Repeating an investigation helps researchers determine whether a pattern is consistent or might have resulted from chance, measurement error, or unusual conditions.

The scientific process is not simply a list of steps. It is a cycle of asking questions, testing ideas, evaluating evidence, and improving explanations.

Mathematics helps make that process more precise.

10.5 Engineering Design: From Need to Solution

Engineering often begins with a practical need.

Imagine that a community needs a footbridge across a small stream.

An engineering team must translate that need into design requirements.

Possible requirements include:

- the bridge must span a specified distance;
- it must safely carry the expected users;
- it must withstand relevant environmental conditions;
- it must use suitable materials;
- it must fit the available budget;
- it must be maintainable.

The team then investigates possible designs.

Step 1: Define the requirements

A vague instruction such as “build a strong bridge” is not enough.

The team needs measurable requirements, such as the required span, expected loading conditions, accessibility needs, and applicable safety standards.

Step 2: Develop possible designs

Different designs may meet the same requirements.

The team compares their advantages, limitations, costs, and risks.

Step 3: Model the design mathematically

Calculations help estimate dimensions, loads, forces, material quantities, and likely performance.

Step 4: Evaluate the design

The team checks whether the design meets the requirements and whether the assumptions are reasonable.

Depending on the structure, this may require specialist analysis, simulation, testing, and professional review.

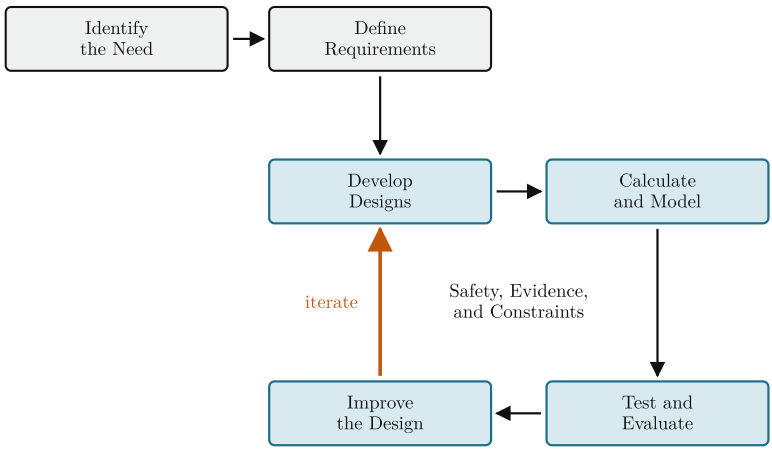
Step 5: Improve the design

If the calculations or tests reveal weaknesses, the design is revised.

This process may be repeated several times.

FIGURE 15

Engineering Design Cycle



Identify the need and define requirements, then loop through develop, calculate, test and improve. Improving sends the work back to developing designs.

The engineering design cycle shows why calculations are part of a larger decision-making process. A correct calculation cannot compensate for an incorrect problem definition or an unsafe design assumption.

10.6 Mathematical Models: Representing Reality

A mathematical model is a mathematical representation of some part of the real world.

Models may use equations, graphs, tables, diagrams, probability distributions, or computer simulations.

They simplify reality so that we can investigate a particular question.

For example, suppose a car travels at a constant speed of 60 kilometres per hour.

Its distance after t hours can be modelled by

$$d = 60t$$

where d is measured in kilometres.

After 2 hours,

$$d = 60(2)$$

Therefore,

$$\boxed{d = 120 \text{ km}}$$

This model is useful because it gives us a simple way to calculate distance.

But what assumptions does it make?

It assumes the car maintains a constant speed throughout the period. It does not account for traffic, stops, changing road conditions, or acceleration.

If the car stops for 30 minutes, the actual distance covered in two hours may be less than 120 km.

The model is not necessarily wrong. It may simply be unsuitable for a situation in which its assumptions do not hold.

Why models are useful

Models allow us to:

- investigate situations that are difficult to study directly;
- estimate quantities;
- compare alternative decisions;
- make predictions;
- understand relationships between variables;
- explore what might happen when conditions change.

A model is useful when its assumptions are appropriate for the question and its predictions are sufficiently reliable for the intended purpose.

Models are not the same as reality

A map represents a geographical area, but it does not contain every detail of the actual landscape.

Likewise, a mathematical model represents selected features of a real situation.

The person building the model must decide which features matter and which can reasonably be simplified.

This is an important responsibility. Leaving out an irrelevant detail may make a model easier to use. Leaving out an essential factor may make the result misleading.

10.7 Proportional Relationships in Science and Engineering

Many practical calculations involve proportional relationships.

Understanding the relationship helps you decide which formula is appropriate.

Direct proportion

Two quantities are directly proportional when one is a constant multiple of the other.

If y is directly proportional to x, then

$$y = kx$$

where k is the constant of proportionality.

For example, suppose one notebook costs ₦500 and there are no discounts.

The total cost C of buying n notebooks is

$$C = 500n$$

If 4 notebooks are purchased,

$$C = 500(4)$$

$$\boxed{C = ₦2{,}000}$$

The cost doubles when the number of notebooks doubles, provided the price per notebook remains constant.

Inverse proportion

Two quantities are inversely proportional when their product remains constant.

If y is inversely proportional to x, then

$$y = \frac{k}{x}$$

For example, suppose a fixed job requires 24 worker-hours.

If all workers have the same productivity and can work independently without interfering with one another, the time required may be modelled by

$$t = \frac{24}{n}$$

where n is the number of workers and t is the time in hours.

For 3 workers,

$$t = \frac{24}{3} = 8 \text{ hours}$$

For 6 workers,

$$t = \frac{24}{6} = 4 \text{ hours}$$

Under the model's assumptions, doubling the workforce halves the time.

Real projects may not behave so simply. Workers may need to coordinate, share equipment, or perform tasks that cannot be divided equally. The model is an approximation.

How to recognise the relationship

Before using a proportion formula, ask:

- Does one quantity increase when the other increases?
- Does one quantity decrease when the other increases?
- Is the relationship actually proportional, or merely increasing or decreasing?
- Are the conditions sufficiently stable for the relationship to hold?

Not every increasing relationship is a direct proportion, and not every decreasing relationship is an inverse proportion.

Understanding the relationship prevents incorrect formula selection.

10.8 Measurement: Every Number Needs a Unit

Suppose someone tells you that a table is 2 long.

The statement is incomplete.

Two metres? Two feet? Two centimetres?

A measurement requires both a numerical value and a unit.

Common units include:

Quantity	Common unit
Length	metre (m)
Mass	kilogram (kg)
Time	second (s)
Temperature	kelvin (K) or degree Celsius (°C)
Force	newton (N)
Energy	joule (J)
Electric current	ampere (A)

The International System of Units, commonly called SI, provides an internationally standardised system of measurement.

Using consistent units reduces confusion and helps people compare measurements from different sources.

Example: Converting kilometres to metres

A road is 3.5 kilometres long.

Since

$$1 \text{ km} = 1000 \text{ m}$$

we calculate

$$3.5 \times 1000 = 3500$$

Therefore,

$$\boxed{3.5 \text{ km} = 3500 \text{ m}}$$

Example: Converting hours to seconds

A machine operates for 2.5 hours.

Since

$$1 \text{ hour} = 3600 \text{ seconds}$$

we obtain

$$2.5 \times 3600 = 9000$$

Therefore,

$$\boxed{2.5 \text{ hours} = 9000 \text{ seconds}}$$

Why unit conversion matters

Suppose you calculate a speed using distance in kilometres and time in seconds, but then interpret the answer as kilometres per hour.

The numerical result will be incorrect unless you convert the units appropriately.

Always write the units during the calculation, especially when several different units appear in the same problem.

10.9 Dimensional Analysis: Checking Whether an Equation Makes Sense

Dimensional analysis uses the dimensions of physical quantities to check equations and calculations.

It is not a substitute for a complete scientific derivation, but it can reveal many errors.

For example, consider the equation

$$d = vt$$

where:

- d is distance;
- v is speed;
- t is time.

Speed has units of metres per second:

$$[v] = \frac{\mathrm{m}}{\mathrm{s}}$$

Time has units of seconds:

$$[t] = \mathrm{s}$$

Multiplying them gives

$$[v][t] = \frac{\mathrm{m}}{\mathrm{s}} \times \mathrm{s}$$

The seconds cancel:

$$[v][t] = \mathrm{m}$$

The result has units of distance, which matches the left-hand side.

The equation is dimensionally consistent.

Example of an inconsistent equation

Suppose someone proposes

$$d = v + t$$

where d is distance, v is speed, and t is time.

The right-hand side adds speed and time.

But speed has units of metres per second, while time has units of seconds.

These quantities have different dimensions and cannot be added meaningfully.

The equation is dimensionally inconsistent.

What dimensional analysis cannot prove

An equation can have consistent units and still be incorrect.

For example, an equation might use the wrong numerical constant or fail to describe the actual physical relationship.

Dimensional analysis is a useful check, not complete proof.

It helps answer one important question: do the units of the calculation make sense?

10.10 Significant Figures and Measurement Precision

Measurements are never perfectly precise.

A ruler may only allow measurements to the nearest millimetre. A digital scale may display mass to the nearest gram. A thermometer may have its own resolution and uncertainty.

A measurement written as 12 cm does not necessarily provide the same precision as a measurement written as 12.00 cm.

The second notation indicates that the value has been recorded to a finer decimal place, although the instrument and measurement method determine whether that precision is justified.

Significant figures help communicate the precision of numerical measurements.

Example: Multiplying measured quantities

Suppose a rectangle is measured as

$$4.2\ \text{m} \times 3.1\ \text{m}$$

Its calculated area is

$$A = 4.2 \times 3.1$$
$$A = 13.02\ \text{m}^2$$

However, both measurements are recorded to two significant figures.

If the measurements are approximate and their uncertainties are not otherwise specified, the area would ordinarily be reported to two significant figures:

$$\boxed{A \approx 13\ \text{m}^2}$$

The extra digits in 13.02 do not mean that the area is known to that level of precision.

This does not mean that every problem requires rounding immediately. During calculations, retaining extra digits can prevent accumulated rounding errors. Round the final result according to the measurement precision and the requirements of the problem.

Why precision matters

In a classroom exercise, a small rounding difference may be unimportant.

In engineering, medicine, manufacturing, or scientific experiments, the required precision may be critical.

The correct level of precision depends on the purpose, the quality of the measurements, and the consequences of an error.

10.11 Accuracy, Precision, and Error

These terms are related but not identical.

Accuracy describes how close a measurement or result is to the accepted reference value.

Precision can refer to how closely repeated measurements agree with one another, or to the fineness with which a value is reported. The intended meaning should be clear from the context.

Imagine that a reference length is known to be 10.0 cm.

Three measurements are recorded:

- 9.9 cm;
- 10.0 cm;
- 10.1 cm.

These measurements are close to the reference value.

Now imagine another instrument repeatedly gives 8.0 cm, 8.0 cm, and 8.1 cm.

Those measurements are closely grouped, but they are far from the reference value.

The second instrument may produce consistent measurements while still being inaccurate.

Absolute error

If a reference value is available, absolute error is the magnitude of the difference between the measured value and the reference value.

$$\boxed{\text{Absolute error} = |\text{Measured value} - \text{Reference value}|}$$

Suppose the reference value is 20 cm and a measurement gives 19.6 cm.

Then

$$\text{Absolute error} = |19.6 - 20|$$

$$\boxed{\text{Absolute error} = 0.4 \text{ cm}}$$

Percentage error

When a meaningful nonzero reference value is available, percentage error can be calculated as

$$\boxed{\text{Percentage error} = \frac{|\text{Measured value} - \text{Reference value}|}{|\text{Reference value}|} \times 100\%}$$

Using the same example,

$$\begin{aligned} \text{Percentage error} &= \frac{|19.6 - 20|}{20} \times 100\% \\ &= \frac{0.4}{20} \times 100\% \\ &\boxed{\text{Percentage error} = 2\%} \end{aligned}$$

Percentage error expresses the absolute error relative to the magnitude of the reference value.

However, error calculations depend on having an appropriate reference value. In some experiments, the true value is unknown, so scientists must estimate uncertainty using other methods.

10.12 Estimation: Knowing Whether an Answer Is Reasonable

Suppose you calculate the cost of buying 49 notebooks at ₦198 each.

A calculator can provide the exact product. But you can also estimate the answer before calculating.

Round 49 to 50 and ₦198 to ₦200.

Then,

$$50 \times 200 = 10{,}000$$

The exact cost is

$$49 \times 198$$

Calculate:

$$\begin{aligned} 198 \times 49 &= 198(50 - 1) \\ &= 9900 - 198 \\ &\boxed{₦9{,}702} \end{aligned}$$

The exact answer is close to the estimate of ₦10,000.

This agreement suggests that the answer is reasonable.

If your calculator instead displayed ₦97,020, estimation would help you notice that a decimal-

place or multiplication error may have occurred.

Why estimation matters in science and engineering

Estimation helps you:

- detect unreasonable answers;
- compare possible designs quickly;
- decide whether a more precise calculation is necessary;
- check the approximate size of a quantity;
- recognise unit-conversion errors.

Estimation does not replace exact calculations when precision is required. It provides an independent check on the result.

10.13 Forces and Motion: Thinking Like a Physicist

Imagine pushing a box across a floor.

The box may remain still, slide slowly, or accelerate, depending on the forces acting on it and other relevant conditions.

To investigate its motion, you need to understand force, mass, and acceleration.

Newton's second law of motion is commonly expressed as

$$\boxed{F = ma}$$

where:

- F is the net force;
- m is mass;
- a is acceleration.

The word *net* matters. The equation relates acceleration to the combined effect of the forces acting on an object, not necessarily to just one applied force.

Example: Calculating net force

Suppose an object of mass 4 kg accelerates at $3\ \mathrm{m/s^2}$.

Calculate the net force.

Given,

$$m = 4\ \mathrm{kg}$$

and

$$a = 3\ \mathrm{m/s^2}$$

Using

$$F = ma$$

we obtain

$$F = 4 \times 3$$

Therefore,

$$\boxed{F = 12\ \text{N}}$$

The net force is 12 newtons.

Think before calculating

Before applying the equation, ask:

- Is the mass known?
- Is the acceleration known?
- Is the question asking for net force or an individual force?
- Are the units consistent?
- Is the situation appropriate for the simplified model?

A box on a rough surface may experience friction. If several forces act on it, you must account for their directions and magnitudes when determining the net force.

A correct formula produces a meaningful result only when the quantities have been identified correctly.

10.14 Energy and Power

Energy is associated with the ability to do work or produce change.

Power describes the rate at which work is done or energy is transferred.

Average power can be calculated using

$$\boxed{P = \frac{E}{t}}$$

where:

- P is average power;
- E is energy transferred;

- t is the time taken.

Example

A machine transfers 6,000 joules of energy in 20 seconds.

Its average power is

$$P = \frac{6000}{20}$$

Therefore,

$$\boxed{P = 300 \text{ W}}$$

The machine transfers energy at an average rate of 300 joules per second during that interval.

Because

$$1 \text{ W} = 1 \text{ J/s}$$

the unit is watts.

Why power matters

Two machines may transfer the same amount of energy but take different amounts of time.

The machine that transfers the energy more quickly has the greater average power.

This distinction is useful when comparing motors, electrical appliances, pumps, and many other systems.

Again, the formula should be selected because it matches the question. If the question asks for energy, rearrange the relationship:

$$E = Pt$$

If it asks for the time taken,

$$t = \frac{E}{P}$$

The relationship is the same, but the unknown quantity changes.

10.15 Electricity: Calculating Electrical Power

Electrical systems provide another example of mathematical relationships being used in practical design.

For a direct-current circuit, electrical power can be calculated using

$$\boxed{P = VI}$$

where:

- P is electrical power in watts;
- V is voltage in volts;
- I is current in amperes.

Example

A device operates at 12 volts and draws a current of 2 amperes.

Then

$$P = 12 \times 2$$

Therefore,

$$\boxed{P = 24 \text{ W}}$$

The device uses electrical power at a rate of 24 watts under those operating conditions.

For alternating-current systems, the relationship between voltage, current, and real power can be more complicated because of waveform and power-factor effects. The simple formula should be applied only when its assumptions are appropriate.

Why context matters

A calculation involving electricity is not just an arithmetic exercise. It may have safety implications.

Electrical equipment must be selected and used according to its ratings and applicable safety requirements. Mathematical calculations alone do not make an electrical installation safe.

This illustrates a wider principle: when calculations influence real-world safety, they must be combined with appropriate standards, expertise, and verification.

10.16 Geometry in Engineering: Measuring Before Building

Engineers often use geometry to determine lengths, areas, volumes, and angles.

Imagine that a company wants to construct a rectangular storage room.

The room is 8 metres long and 5 metres wide.

To calculate the floor area,

$$A = lw$$

Substitute the dimensions:

$$A = 8 \times 5$$

Therefore,

$$\boxed{A = 40 \text{ m}^2}$$

If the room is 3 metres high, its volume is

$$V = lwh$$
$$V = 8 \times 5 \times 3$$

Thus,

$$\boxed{V = 120 \text{ m}^3}$$

The floor area may help estimate flooring materials. The volume may help describe the room's capacity or support ventilation calculations.

However, estimating the actual materials required may involve additional allowances for openings, waste, thickness, layout, and construction methods.

The geometric calculation gives the starting point, not necessarily the complete project estimate.

Using Pythagoras to find a diagonal

Suppose a rectangular room is 6 metres long and 8 metres wide.

The diagonal across the floor can be calculated using the Pythagorean theorem:

$$c^2 = a^2 + b^2$$

Substitute:

$$c^2 = 6^2 + 8^2$$
$$c^2 = 36 + 64$$
$$c^2 = 100$$

Take the square root:

$$\boxed{c = 10 \text{ m}}$$

The diagonal is 10 metres.

This calculation is useful when estimating diagonal distances or checking rectangular layouts.

In an actual construction project, the measurement must still be taken and checked against the design requirements.

10.17 Optimisation: Finding a Better Design

Engineering does not always involve finding one answer. Sometimes it involves comparing many possible answers and choosing the best one according to specific requirements.

Suppose a company wants to design a rectangular enclosure using 20 metres of fencing.

The enclosure must have the greatest possible area.

Let the length be l and the width be w.

The perimeter is

$$2l + 2w = 20$$

Divide both sides by 2:

$$l + w = 10$$

Therefore,

$$w = 10 - l$$

The area is

$$A = lw$$

Substitute $w = 10 - l$:

$$A = l(10 - l)$$

Expand:

$$A = 10l - l^2$$

We want to find the length that maximises the area.

Differentiate:

$$\frac{dA}{dl} = 10 - 2l$$

Set the derivative equal to zero:

$$10 - 2l = 0$$

Rearrange:

$$2l = 10$$

Therefore,

$$\boxed{l = 5 \text{ m}}$$

Since

$$w = 10 - l$$

we obtain

$$w = 10 - 5$$
$$\boxed{w = 5 \text{ m}}$$

The second derivative is

$$\frac{d^2A}{dl^2} = -2$$

Because it is negative, the area function is curved downward and the stationary point is a maximum.

The maximum area is

$$A = 5 \times 5$$

$$\boxed{A = 25 \text{ m}^2}$$

Under the stated assumptions, a square enclosure provides the greatest area for the available fencing.

What does this example teach us?

We did not begin by guessing a rectangle.

We described the constraint, expressed the area in terms of one variable, and used calculus to find the maximum.

This is called optimisation.

Scientists, engineers, and economists use optimisation methods to investigate questions such as:

- What design uses the least material?
- What route minimises travel time?
- What production level maximises profit under a model?
- What shape provides the greatest capacity for a given amount of material?

The best answer depends on what is being optimised and which constraints must be respected.

A design that maximises area may not be the best design if height, access, safety, cost, or land shape also matters.

10.18 Computer Models and Simulation

Some real-world problems are too complicated to solve conveniently with a simple formula.

For example, predicting the movement of air around a building may require a complex mathematical model.

Studying the behaviour of a large electrical network may involve thousands of connected components.

Investigating how a disease spreads may require a model that accounts for contact patterns, population structure, and changing conditions.

Computers can evaluate mathematical models repeatedly, allowing researchers and engineers to explore many scenarios.

This process is often called simulation.

A simple simulation example

Imagine that a business sells 100 products each week and expects sales to increase by 5% per week.

A simple model is

$$S_{n+1} = 1.05S_n$$

where S_n is the sales figure for week n.

Starting from

$$S_0 = 100$$

the next week is

$$S_1 = 1.05(100) = 105$$

The following week is

$$S_2 = 1.05(105) = 110.25$$

After three weeks,

$$S_3 = 1.05(110.25)$$

$$\boxed{S_3 \approx 115.76}$$

The model predicts a sales index of approximately 115.76 after three weekly increases.

If actual sales must be whole products, the business would need to interpret or round the prediction appropriately.

More importantly, the model assumes that the 5% growth rate continues each week. Actual sales may change because of prices, competitors, customer preferences, supply problems, or other conditions.

A computer can calculate the consequences of the assumption, but it cannot guarantee that the assumption is true.

Simulation is powerful because it allows us to explore possibilities. Its conclusions remain dependent on the quality of the model.

10.19 Scientific Notation: Handling Very Large and Very Small Numbers

Scientists and engineers often work with quantities that are extremely large or extremely small.

For example, the distance between astronomical objects may be enormous, while the size of certain microscopic structures may be tiny.

Scientific notation makes such numbers easier to write and compare.

A number in scientific notation is written as

$$\boxed{a \times 10^n}$$

where

$$1 \leq |a| < 10$$

and n is an integer.

Example 1: A large number

Write 450,000 in scientific notation.

Move the decimal point five places to the left:

$$450{,}000 = 4.5 \times 10^5$$

Therefore,

$$\boxed{450{,}000 = 4.5 \times 10^5}$$

Example 2: A small number

Write 0.00072 in scientific notation.

Move the decimal point four places to the right to obtain 7.2.

Therefore,

$$\boxed{0.00072 = 7.2 \times 10^{-4}}$$

The negative exponent indicates that the original number is smaller than 1.

Multiplying numbers in scientific notation

Calculate

$$(3 \times 10^4)(2 \times 10^3)$$

Multiply the coefficients:

$$3 \times 2 = 6$$

Add the exponents:

$$10^4 \times 10^3 = 10^{4+3} = 10^7$$

Therefore,

$$\boxed{6 \times 10^7}$$

Scientific notation makes calculations involving large and small quantities more manageable.

However, the notation does not remove the need to understand units, precision, and the physical meaning of the result.

10.20 Choosing the Right Tool

Suppose an engineer needs to calculate how long it will take a vehicle to travel a particular distance.

Several mathematical methods might be relevant, but the correct choice depends on what is known.

If the distance and constant speed are known, use

$$t = \frac{d}{v}$$

If acceleration is changing, a constant-speed calculation may not be suitable.

If the engineer needs to understand how velocity changes at a particular moment, differentiation may be useful.

If the engineer knows the velocity over a period and needs the displacement, integration may be appropriate.

The method depends on the question and the assumptions.

Here is a useful decision guide.

Question	Possible mathematical tool
How much space does the object occupy?	Geometry and volume
How fast is the quantity changing?	Rates of change or differentiation
What is the total accumulated amount?	Summation or integration
How are two quantities related?	Equations, graphs, or statistical analysis
How likely is an outcome?	Probability

What value gives the best result under constraints?

Optimisation

Is the answer reasonable? Estimation, units, and dimensional analysis

How does a system behave under different assumptions?

Mathematical modelling or simulation

This table is a guide rather than a set of rigid rules. A single problem may require several methods.

10.21 How to Make Scientific and Engineering Calculations Easier

Method 1: Translate words into quantities

Write down what is known and what must be found.

For example:

- known distance: 120 km;
- known constant speed: 60 km/h;
- unknown time.

The problem becomes easier once the information is organised.

Method 2: Draw a diagram

A diagram can reveal relationships that are difficult to see in a paragraph.

For a geometry problem, draw the shape and label its dimensions.

For a force problem, draw a diagram showing the forces and their directions.

For a process involving several stages, draw a flowchart.

Method 3: Write the units

Units help identify whether a formula has been used appropriately.

If the answer should be a time, check whether the final units represent time.

Method 4: Estimate first

Estimate the size of the answer before calculating precisely.

This helps detect arithmetic mistakes and unreasonable results.

Method 5: Simplify the model carefully

A simple model is often easier to analyse than a complicated one.

However, do not remove a factor that is essential to the question.

The goal is not to make every problem as simple as possible. It is to make the problem simple enough to solve while preserving the features that matter.

Method 6: Separate calculation from interpretation

First calculate the result correctly.

Then explain what it means in the real situation.

For example, a result of 300 W is a power measurement, not an energy measurement.

Method 7: Check assumptions and limitations

Ask whether the model's assumptions are reasonable.

If they are not, explain the limitation or use a more suitable model.

Method 8: Verify important results independently

For a critical calculation, use a second method, a measurement, a simulation, or an independent review when appropriate.

This is particularly important when errors could affect safety, cost, or health.

10.22 Common Mistakes in Scientific and Engineering Thinking

Mistake 1: Choosing a formula before understanding the question

A formula should match the relationship being investigated.

Using the wrong relationship can produce an incorrect answer even when the arithmetic is perfect.

Mistake 2: Ignoring units

A numerical value without a suitable unit may be incomplete or misleading.

Keep track of units throughout the calculation.

Mistake 3: Treating a model as an exact description of reality

Models simplify real situations.

Their usefulness depends on whether the assumptions are appropriate for the purpose.

Mistake 4: Reporting too many digits

A calculator may display many decimal places, but the measurements may not justify that precision.

Report results at a precision suitable for the data and task.

Mistake 5: Confusing accuracy with precision

Repeated measurements can be closely grouped and still be far from the reference value.

Consistency and closeness to the reference value are different ideas.

Mistake 6: Ignoring uncertainty

Measurements, assumptions, and predictions have limitations.

A good analysis recognises uncertainty rather than pretending it does not exist.

Mistake 7: Assuming that a mathematical answer guarantees a safe design

A design must satisfy relevant safety requirements and be checked by appropriately qualified professionals where necessary.

Mathematical calculations are essential, but they are not a substitute for complete engineering assessment.

Mistake 8: Accepting a result because it looks complicated

A complicated answer is not automatically correct.

Check the reasoning, units, assumptions, and arithmetic.

A simple result supported by sound reasoning is more valuable than a complicated result produced by an inappropriate method.

10.23 Practical Workshop: Solve a Problem Like a Scientist or Engineer

Activity A: Estimate the cost of flooring a room

A rectangular room measures 6 metres by 4 metres.

Floor tiles cost ₦8,000 per square metre. For this simplified estimate, ignore labour, cutting waste, transport, and additional materials.

Step 1: Identify the problem.

We need to estimate the tile cost.

Step 2: Identify the known information.

$$l = 6 \text{ m}$$
$$w = 4 \text{ m}$$

Cost per square metre:

₦8,000

Step 3: Calculate the floor area.

$$A = lw$$
$$A = 6 \times 4$$
$$\boxed{A = 24 \text{ m}^2}$$

Step 4: Calculate the tile cost.

$$\text{Cost} = 24 \times ₦8{,}000$$

Therefore,

$$\boxed{\text{Cost} = ₦192{,}000}$$

Step 5: Interpret the result.

The estimated tile cost is ₦192,000 under the stated assumptions.

A real project would need to allow for tile layout, cutting waste, the exact area to be covered, labour, and other expenses.

Activity B: Investigate the motion of a vehicle

Suppose a vehicle travels at a constant speed of 15 m/s for 12 seconds.

Find the distance travelled.

Step 1: Identify the relationship.

For constant speed,

$$d = vt$$

Step 2: Substitute the values.

$$d = 15 \times 12$$

Step 3: Calculate.

$$\boxed{d = 180 \text{ m}}$$

Step 4: Check the units.

$$\frac{\mathrm{m}}{\mathrm{s}} \times \mathrm{s} = \mathrm{m}$$

The units reduce to metres, as expected.

Step 5: Interpret the result.

Under the constant-speed assumption, the vehicle travels 180 metres in 12 seconds.

Activity C: Check a proposed equation

A student proposes the equation

$$d = vt^2$$

to calculate distance from speed and time.

Use dimensional analysis to investigate whether the equation is dimensionally consistent.

Speed has units

$$[v] = \frac{\mathrm{m}}{\mathrm{s}}$$

Time squared has units

$$[t^2] = \mathrm{s}^2$$

Multiplying gives

$$[v][t^2] = \frac{\mathrm{m}}{\mathrm{s}} \times \mathrm{s}^2$$

Therefore,

$$[v][t^2] = \mathrm{m} \cdot \mathrm{s}$$

The result has units of metre-seconds, not metres.

Therefore, the equation is dimensionally inconsistent for the stated quantities.

This check tells us that the equation is unsuitable as written. It does not require us to calculate a numerical answer.

Activity D: Optimise a design

A farmer has 40 metres of fencing and wants to enclose a rectangular plot with the greatest possible area.

Let the length be l and the width be w.

The perimeter condition is

$$2l + 2w = 40$$

Divide by 2:

$$l + w = 20$$

Therefore,

$$w = 20 - l$$

The area is

$$A = lw$$

Substitute:

$$A = l(20 - l)$$

Expand:

$$A = 20l - l^2$$

Differentiate:

$$\frac{dA}{dl} = 20 - 2l$$

Set the derivative to zero:

$$20 - 2l = 0$$

Therefore,

$$l = 10$$

Since

$$w = 20 - l$$

we obtain

$$w = 10$$

The maximum-area design is a square measuring 10 metres by 10 metres.

Its area is

$$A = 10 \times 10$$

$$\boxed{A = 100 \text{ m}^2}$$

This result assumes that the entire 40 metres of fencing is available for the boundary and that the plot can be shaped as required.

10.24 Exercises

Attempt the following questions. Explain your reasoning and include units where appropriate.

Section A: Scientific and engineering thinking

1. Explain the difference between scientific investigation and engineering design.
2. Why should a problem be clearly defined before a formula is selected?
3. What is a mathematical model?
4. Explain why a model may be useful even when it does not represent every detail of reality.
5. Describe four stages in a scientific investigation.
6. Explain why an engineering design may need to be tested and improved several times.

Section B: Measurement and estimation

7. Convert 4.6 km to metres.
7. Convert 3.5 hours to seconds.
7. A measurement is 48.5 cm, while the reference value is 50 cm. Calculate the absolute error.
7. Using the information in Question 9, calculate the percentage error.
7. Estimate 198×51 by rounding each number to a convenient value. Then calculate the exact product and compare the results.
7. Explain the difference between accuracy and precision.

Section C: Physical quantities

13. An object has a mass of 6 kg and a net acceleration of $4\ \text{m/s}^2$. Calculate the net force.
13. A machine transfers 9,000 J of energy in 30 seconds. Calculate its average power.
13. A direct-current device operates at 24 V and draws 3 A. Calculate

> its electrical power using $P = VI$.

13. Explain why the electrical power calculation in Question 15 does not, by itself, prove that the device is safe to use.
13. A rectangular room measures 9 m by 5 m. Calculate its floor area and volume if its height is 3 m.

13. A rectangular field measures 9 m by 12 m. Calculate the diagonal length.

Section D: Mathematical modelling

19. A car travels at a constant speed of 72 km/h for 2.5 hours. Calculate the distance travelled.

19. A worker can complete a job in 12 hours. Under a simplified model of equal productivity and perfectly divisible work, how long would four identical workers take to complete the job?

19. Explain one reason why adding more workers may not reduce the completion time in exact inverse proportion in a real project.

19. A business begins with 200 units of weekly sales. Sales increase by

> 10% each week according to the model $S_{n+1} = 1.10S_n$. Calculate the

predicted sales figure after two weekly increases.

19. Explain one limitation of assuming that a business will continue growing at the same percentage rate indefinitely.

Section E: Advanced reasoning

24. A proposed formula adds a length to a time. Explain why this is dimensionally inconsistent.

24. A scientist obtains a result that contradicts the original hypothesis. Explain what the scientist should do next.

24. A mathematical model predicts that a machine will operate successfully, but a physical test reveals a fault. Explain why the test result must be taken seriously.

24. An engineer has two designs that meet the same basic requirement. Identify four additional factors that could help determine which design is preferable.

24. Explain how estimation can help detect mistakes in a calculation.

24. A farmer wants to maximise the area enclosed by a fixed length of fencing. Explain why this is an optimisation problem.

24. In your own words, explain the difference between performing a calculation and solving a real-world problem.

10.25 Chapter Review

Before moving to the next chapter, make sure you understand these ideas:

- Scientists use observation, experimentation, and mathematical reasoning to investigate questions.
- Engineers use mathematical and scientific knowledge to design and improve practical systems.
- A mathematical model represents selected features of reality.
- Models depend on assumptions and must be evaluated for their intended purpose.
- Direct and inverse proportions describe different kinds of relationships.
- Measurements require appropriate units.
- Dimensional analysis can identify many equation and unit errors.
- Significant figures help communicate measurement precision.
- Accuracy and precision describe different aspects of measurement.
- Estimation provides a useful check on the size of a result.
- Geometry, calculus, statistics, and algebra can all support scientific and engineering work.
- Optimisation helps identify the best result under specified constraints.
- Computer simulations explore the consequences of a model's assumptions.
- Important calculations should be checked, especially when errors may affect safety.
- A numerical result must be interpreted in the context of the original problem.

10.26 Conclusion: Mathematics Is a Tool for Understanding and Building

We began this chapter with a bridge, a plant, and a water tank.

Each situation required more than memorising a formula. It required identifying a need, gathering information, understanding relationships, selecting suitable mathematical tools, and interpreting the result.

We explored how scientists investigate questions, how engineers develop designs, and how mathematical models simplify real situations.

We also examined measurement, dimensional analysis, uncertainty, forces, energy, electricity, geometry, optimisation, and simulation.

These examples demonstrate that mathematics is not isolated from the world. It is part of the process through which people understand natural phenomena, design technology, improve systems, and make decisions.

But calculations must be used responsibly.

A model may omit important details. A measurement may be uncertain. A formula may be inappropriate for the situation. A result may look precise while depending on weak assumptions.

The thoughtful problem-solver recognises these limitations and checks the result against evidence.

> **Think before the formula. Define the problem, understand the assumptions, calculate carefully, and test whether the answer makes sense.**

In the next chapter, we will focus on a skill that supports every mathematical topic in this book: how to make difficult calculations easier through better reasoning, careful organisation, estimation, simplification, and strategic problem-solving.

CHAPTER 11

How to Make Difficult Calculations Easy

11

11.1 Introduction: Difficulty Does Not Always Mean Complexity

Imagine that two students are given the same mathematical problem.

The first student immediately searches for a formula, writes down several steps, and becomes confused.

The second student reads the question carefully, identifies what is known, looks for a simpler way to represent the information, and solves the problem in a few clear steps.

Both students may have similar mathematical knowledge. However, the second student has learned something important: the way you approach a problem can be just as important as the formula you use.

Many calculations appear difficult because the information is poorly organised, the numbers are inconvenient, the wrong method is chosen, or too many steps are attempted at once.

Sometimes the calculation itself is genuinely advanced. In other cases, a small change in approach makes it much easier.

Consider the following problem:

$$25 \times 48$$

You could multiply 25 by 48 using long multiplication.

But you could also recognise that

$$25 = \frac{100}{4}$$

Therefore,

$$\begin{aligned} 25 \times 48 &= \frac{100}{4} \times 48 \\ &= 100 \times 12 \\ &\boxed{1200} \end{aligned}$$

The calculation becomes easy because we recognised a useful relationship.

This chapter explores how to simplify calculations without sacrificing accuracy, how to select efficient methods, how to avoid unnecessary work, and how to develop a disciplined approach to difficult mathematical problems.

The goal is not merely to calculate faster. It is to think more clearly.

11.2 Understand the Problem Before Solving It

One of the most common causes of mathematical difficulty is beginning the calculation before understanding what the question requires.

Consider this problem:

A trader buys 40 notebooks at ₦250 each. She sells every notebook for ₦320. What is her total profit?

A student may immediately calculate

$$40 \times 320$$

and obtain ₦12,800.

The calculation is correct, but it answers a different question: the total revenue from selling the notebooks.

The question asks for profit.

To calculate profit, we need both revenue and cost.

Step 1: Calculate the total cost

$$\text{Cost} = 40 \times 250$$

$$\boxed{\text{Cost} = ₦10{,}000}$$

Step 2: Calculate the total revenue

$$\text{Revenue} = 40 \times 320$$

$$\boxed{\text{Revenue} = ₦12{,}800}$$

Step 3: Calculate the profit

$$\begin{aligned}\text{Profit} &= \text{Revenue} - \text{Cost} \\ &= 12{,}800 - 10{,}000 \\ \boxed{\text{Profit} &= ₦2{,}800}\end{aligned}$$

The problem becomes easier when we separate it into smaller questions.

Before solving a problem, ask:

1. What exactly am I being asked to find?
2. What information has been provided?
3. Which information is relevant?
4. What relationship connects the known quantities to the unknown?
5. Can I divide the problem into smaller steps?
6. How will I check the final answer?

These questions help prevent unnecessary calculations and wrong answers.

11.3 Break Large Problems into Smaller Problems

A complicated problem often contains several simpler problems hidden inside it.

Suppose a company earns ₦850,000 in monthly revenue and spends ₦275,000 on materials, ₦180,000 on salaries, and ₦95,000 on transport.

Calculate the amount remaining after these expenses.

You could subtract all three expenses from the revenue in one long calculation.

Alternatively, organise the work.

First, add the expenses:

$$275{,}000 + 180{,}000 = 455{,}000$$

Then,

$$455{,}000 + 95{,}000 = 550{,}000$$

The total expenses are

$$\boxed{₦550{,}000}$$

Now subtract the total expenses from the revenue:

$$850{,}000 - 550{,}000$$

$$\boxed{₦300{,}000}$$

The amount remaining is ₦300,000.

Why this approach works

The problem has been divided into two stages:

- calculate total expenses;
- subtract the expenses from revenue.

This makes the reasoning easier to follow and allows each stage to be checked independently.

The same principle applies to algebra, geometry, statistics, calculus, and scientific calculations.

When a problem appears overwhelming, identify the smaller tasks that must be completed to solve it.

11.4 Use the Properties of Arithmetic

The basic properties of arithmetic are not merely rules to memorise. They are tools for making calculations easier.

Three especially useful properties are the commutative property, the associative property, and the distributive property.

11.4.1 The commutative property

The commutative property allows the order of numbers to be changed in addition and multiplication.

For addition,

$$a + b = b + a$$

For multiplication,

$$ab = ba$$

For example,

$$4 \times 25 = 25 \times 4$$

If you recognise that 4 groups of 25 make 100, the calculation becomes immediate.

$$\boxed{4 \times 25 = 100}$$

However, subtraction and division are not generally commutative.

For example,

$$10 - 4 = 6$$

but

$$4 - 10 = -6$$

Changing the order changes the answer.

11.4.2 The associative property

The associative property allows numbers to be regrouped in addition and multiplication.

For addition,

$$(a + b) + c = a + (b + c)$$

For multiplication,

$$(ab)c = a(bc)$$

Consider

$$25 \times 16 \times 4$$

A direct approach might multiply 25 by 16 first.

But we can regroup the calculation:

$$25 \times 16 \times 4 = 25 \times 4 \times 16$$

Now,

$$25 \times 4 = 100$$

Therefore,

$$100 \times 16 = 1600$$

Hence,

$$\boxed{25 \times 16 \times 4 = 1600}$$

The calculation becomes easier because we create a convenient intermediate result.

11.4.3 The distributive property

The distributive property allows multiplication to be expanded over addition or subtraction.

$$a(b + c) = ab + ac$$

Similarly,

$$a(b - c) = ab - ac$$

Consider

$$19 \times 36$$

Since

$$19 = 20 - 1$$

we can write

$$19 \times 36 = (20 - 1) \times 36$$

Expand:

$$= 20 \times 36 - 1 \times 36$$

$$= 720 - 36$$

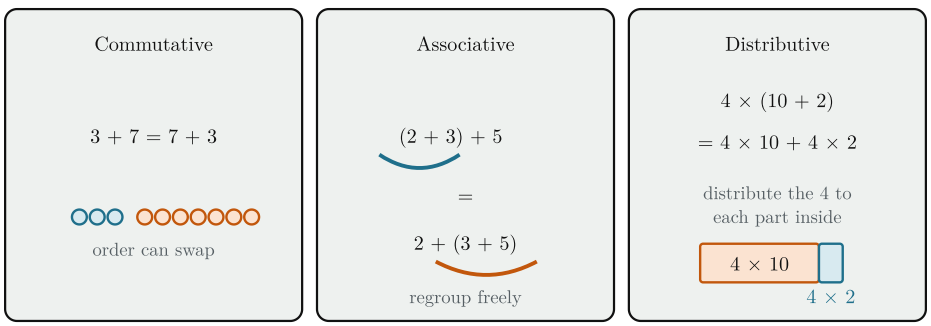
Therefore,

$$\boxed{19 \times 36 = 684}$$

This method is particularly useful when a number is close to 10, 20, 50, 100, or another convenient value.

FIGURE 16

Three Properties That Simplify Arithmetic



Commutative: 3 + 7 = 7 + 3. Associative: (2 + 3) + 5 = 2 + (3 + 5). Distributive: 4 × (10 + 2) = 4 × 10 + 4 × 2.

11.5 Make Multiplication Easier with Convenient Numbers

Some numbers have relationships that make multiplication especially simple.

Learning to recognise these relationships reduces unnecessary written work.

Example 1: Multiplying by 5

Calculate

$$48 \times 5$$

Since

$$5 = \frac{10}{2}$$

we obtain

$$\begin{aligned} 48 \times 5 &= \frac{48 \times 10}{2} \\ &= \frac{480}{2} \end{aligned}$$

Therefore,

$$\boxed{240}$$

Multiplying by 5 can often be done by multiplying by 10 and then dividing by 2.

Example 2: Multiplying by 25

Calculate

$$64 \times 25$$

Since

$$25 = \frac{100}{4}$$

we have

$$\begin{aligned} 64 \times 25 &= \frac{64 \times 100}{4} \\ &= \frac{6400}{4} \end{aligned}$$

Therefore,

$$\boxed{1600}$$

Example 3: Multiplying by 125

Calculate

$$32 \times 125$$

Since

$$125 = \frac{1000}{8}$$

we obtain

$$32 \times 125 = \frac{32 \times 1000}{8}$$

$$= 4 \times 1000$$

Therefore,

$$\boxed{4000}$$

Example 4: Multiplying by 99

Calculate

$$73 \times 99$$

Since

$$99 = 100 - 1$$

we obtain

$$\begin{aligned} 73 \times 99 &= 73(100 - 1) \\ &= 7300 - 73 \end{aligned}$$

Therefore,

$$\boxed{7227}$$

The general principle

Look for numbers that can be expressed using convenient values.

For example:

$$9 = 10 - 1$$
$$19 = 20 - 1$$
$$49 = 50 - 1$$
$$99 = 100 - 1$$
$$125 = \frac{1000}{8}$$

These relationships can turn an awkward multiplication into a simple calculation.

However, always verify that the transformation preserves the original expression.

11.6 Simplify Before You Calculate

One of the most useful mathematical habits is simplifying an expression before carrying out the main calculation.

Consider

$$\frac{18 \times 35}{6}$$

A direct approach would multiply 18 by 35 first and then divide by 6.

But 18 divides exactly by 6.

Therefore,

$$\begin{aligned}\frac{18 \times 35}{6} &= \frac{18}{6} \times 35 \\ &= 3 \times 35 \\ &\boxed{105}\end{aligned}$$

The result is the same, but the arithmetic is easier.

Example: Fractions

Simplify

$$\frac{24}{36}$$

Both numerator and denominator are divisible by 12.

Therefore,

$$\begin{aligned}\frac{24}{36} &= \frac{24 \div 12}{36 \div 12} \\ &\boxed{\frac{2}{3}}\end{aligned}$$

The fraction is now in its simplest form.

Example: Algebraic fractions

Simplify

$$\frac{6x}{9x}$$

provided that $x \neq 0$.

Divide the numerical coefficients by 3 and cancel the common factor x:

$$\frac{6x}{9x} = \frac{2}{3}$$

Therefore,

$$\boxed{\frac{2}{3}}, \qquad x \neq 0$$

The restriction matters because the original denominator is zero when $x = 0$, making the original expression undefined.

Simplification should preserve the meaning and domain of the original expression.

11.7 Use Estimation Before Exact Calculation

Estimation helps you predict the approximate size of an answer.

It is especially useful when you want to check whether a calculator result is reasonable.

Suppose you need to calculate

$$398 \times 51$$

Before calculating exactly, round the numbers:

$$398 \approx 400$$

and

$$51 \approx 50$$

Then,

$$400 \times 50 = 20{,}000$$

The answer should be near 20,000.

Now calculate exactly:

$$398 \times 51 = 398(50 + 1)$$

$$
\begin{aligned}
&= 398 \times 50 + 398 \\
&= 19{,}900 + 398 \\
&\boxed{20{,}298}
\end{aligned}
$$

The exact result is close to the estimate.

If your calculator displayed 202,980, you would have a reason to check your work.

Estimation is not guessing

A guess may have little reasoning behind it.

Estimation uses known quantities, convenient approximations, and mathematical relationships to establish a reasonable range or approximate value.

The purpose is to develop an independent sense of what the answer should look like.

11.8 Learn to Recognise Patterns

Patterns can make many calculations easier because they reveal a structure that repeats.

Example 1: Squaring a number ending in 5

Consider

$$35^2$$

The number 35 can be written as

$$35 = 30 + 5$$

Therefore,

$$35^2 = (30 + 5)^2$$

Using

$$(a+b)^2 = a^2 + 2ab + b^2$$

we obtain

$$35^2 = 30^2 + 2(30)(5) + 5^2$$

$$= 900 + 300 + 25$$

$$\boxed{1225}$$

There is also a convenient pattern for squaring whole numbers ending in 5.

For a number ending in 5, multiply the digits before the 5 by the next whole number, then append 25.

For 35:

- The digit before 5 is 3.
- Multiply $3 \times 4 = 12$.
- Append 25.

Therefore,

$$35^2 = 1225$$

Try the same method with 65:

$$6 \times 7 = 42$$

Append 25:

$$\boxed{65^2 = 4225}$$

This shortcut works for whole numbers ending in 5 because of the algebraic structure of their squares.

It is not a universal method for every number.

Example 2: Difference of two squares

The difference of two squares follows the identity

$$a^2 - b^2 = (a - b)(a + b)$$

Suppose we want to calculate

$$51^2 - 49^2$$

Directly calculating both squares would work, but the identity is simpler.

$$51^2 - 49^2 = (51 - 49)(51 + 49)$$

$$= 2 \times 100$$

Therefore,

$$\boxed{200}$$

The pattern eliminates the need to calculate the two large squares separately.

Example 3: Consecutive numbers

Consider

$$19 \times 21$$

These numbers lie one below and one above 20.

We can write

$$19 \times 21 = (20 - 1)(20 + 1)$$

Using the difference-of-

squares identity,

$$(a - b)(a + b) = a^2 - b^2$$

we obtain

$$\begin{aligned} 19 \times 21 &= 20^2 - 1^2 \\ &= 400 - 1 \end{aligned}$$

Therefore,

$$\boxed{399}$$

Pattern recognition becomes more powerful when you understand why the pattern works.

Memorising a shortcut without understanding its structure makes it harder to know when the shortcut is appropriate.

11.9 Change the Form of a Problem

Sometimes the best way to solve a problem is to represent it differently.

A word problem can become an equation. A long decimal can become a fraction. A table can become a graph. A complicated multiplication can become a difference of squares.

The underlying mathematical relationship remains the same, but the new representation may be easier to use.

Example: A percentage problem

A student scores 45 marks out of 60. Find the percentage score.

The formula is

$$\text{Percentage} = \frac{\text{Score}}{\text{Total}} \times 100\%$$

Substitute:

$$\frac{45}{60} \times 100\%$$

Simplify:

$$\frac{45}{60} = \frac{3}{4}$$

Therefore,

$$\frac{3}{4} \times 100\% = 75\%$$

Hence,

$$\boxed{75\%}$$

The fraction $\frac{3}{4}$ makes the calculation easier than working directly with the original numbers.

Example: A decimal problem

Calculate

$$0.25 \times 640$$

Since

$$0.25 = \frac{1}{4}$$

we have

$$\begin{aligned} 0.25 \times 640 &= \frac{1}{4} \times 640 \\ &= 640 \div 4 \end{aligned}$$

Therefore,

$$\boxed{160}$$

Changing the decimal into a fraction makes the operation more transparent.

11.10 Use Algebra to Reduce Repeated Work

Algebra allows us to describe general relationships rather than repeatedly calculating individual cases.

Suppose a taxi charges a fixed fee of ₦500 plus ₦200 per kilometre.

The total cost C for a journey of d kilometres is

$$C = 500 + 200d$$

If the journey is 5 km,

$$C = 500 + 200(5)$$
$$\boxed{C = ₦1{,}500}$$

For a 10 km journey,

$$C = 500 + 200(10)$$
$$\boxed{C = ₦2{,}500}$$

Once the relationship has been expressed algebraically, we do not need to invent a new method for each distance.

The same formula works for every journey for which the pricing rule applies.

Why this is easier

Without algebra, we might calculate each journey separately and repeatedly describe the pricing rule in words.

With algebra, the relationship is represented once.

This reduces repetition, helps us make predictions, and makes changes easier to analyse.

Suppose the price per kilometre increases to ₦250. The model becomes

$$C = 500 + 250d$$

Only one part of the expression changes.

Algebra is powerful because it separates the structure of a relationship from the particular numbers being used.

11.11 Cancel Common Factors Before Multiplying Fractions

Multiplying fractions becomes easier when common factors are cancelled before multiplication.

Consider

$$\frac{14}{15} \times \frac{25}{28}$$

A direct calculation gives

$$\frac{14 \times 25}{15 \times 28}$$

This is correct, but the numbers can be simplified first.

Since 14 and 28 have a common factor of 14,

$$\frac{14}{28} = \frac{1}{2}$$

Also, 25 and 15 have a common factor of 5:

$$\frac{25}{15} = \frac{5}{3}$$

The expression becomes

$$\frac{1}{2} \times \frac{5}{3}$$

Multiply the numerators and denominators:

$$\frac{1 \times 5}{2 \times 3}$$

Therefore,

$$\boxed{\frac{5}{6}}$$

This method reduces the size of the numbers before the main multiplication.

Remember that cancellation applies to common factors in multiplication, not to individual terms separated by addition or subtraction.

For example,

$$\frac{x+4}{x}$$

cannot be simplified by cancelling the x in the numerator against the x in the denominator. The numerator is a sum, not a product containing x as a common factor.

11.12 Use the Right Order of Operations

An expression can produce different results if its operations are performed in the wrong order.

Consider

$$8 + 4 \times 3$$

Multiplication is performed before addition.

Therefore,

$$8 + 4 \times 3 = 8 + 12$$
$$\boxed{20}$$

If you add first, you obtain

$$(8 + 4) \times 3 = 12 \times 3 = 36$$

But the brackets change the expression.

The two expressions are different:

$$8 + 4 \times 3 = 20$$

while

$$(8 + 4) \times 3 = 36$$

A reliable approach is to follow the standard order of operations:

1. Brackets and grouping symbols.
2. Powers and roots.
3. Multiplication and division, from left to right.
4. Addition and subtraction, from left to right.

Multiplication and division have equal precedence, as do addition and subtraction. When operations of the same precedence appear together, work from left to right unless brackets indicate otherwise.

Example

Evaluate

$$18 \div 3 \times 2$$

Work from left to right:

$$18 \div 3 = 6$$

Then,

$$6 \times 2 = 12$$

Therefore,

$$\boxed{12}$$

Do not multiply 3 by 2 first unless the expression contains brackets that require it.

Understanding the order of operations prevents errors that no amount of arithmetic speed can correct.

11.13 Use a Calculator Wisely

A calculator can perform arithmetic quickly and accurately when used correctly.

However, it cannot decide whether the formula is appropriate, whether the information has been interpreted correctly, or whether the result makes sense.

What calculators do well

Calculators are useful for:

- lengthy arithmetic;
- decimal calculations;
- repeated numerical evaluations;
- trigonometric functions;
- powers and roots;
- checking manual work.

What calculators do not automatically do

A calculator does not automatically:

- understand the real-world problem;
- identify irrelevant information;
- choose the correct mathematical model;
- detect every incorrect assumption;
- know whether a unit is appropriate;
- determine whether the final answer is reasonable.

Example: Parentheses matter

Suppose you want to calculate

$$\frac{20+10}{5}$$

The correct expression is

$$\begin{aligned}&(20+10)\div 5\\&=30\div 5\\&\boxed{6}\end{aligned}$$

If you enter

$$20+10\div 5$$

the calculator follows the order of operations:

$$20 + 2 = 22$$

That is a different expression.

When entering complicated calculations, use brackets deliberately and check the displayed expression before accepting the answer.

A useful calculator habit

Before pressing the equals button, predict the approximate answer.

After obtaining the result, compare it with your estimate.

This habit turns the calculator into a verification tool rather than a replacement for mathematical reasoning.

11.14 Work Backwards When It Helps

Some problems are easier to solve by starting with the final condition and reversing the operations.

Suppose a number is multiplied by 4 and then increased by 7 to produce 39.

Find the original number.

Let the original number be x.

The relationship is

$$4x + 7 = 39$$

Subtract 7 from both sides:

$$4x = 32$$

Divide by 4:

$$\boxed{x = 8}$$

We can also reason backwards from 39.

The last operation was adding 7, so reverse it by subtracting 7:

$$39 - 7 = 32$$

The previous operation was multiplying by 4, so reverse it by dividing by 4:

$$32 \div 4 = 8$$

Both methods produce the same result.

Why reversing operations works

Addition is reversed by subtraction.

Subtraction is reversed by addition.

Multiplication by a nonzero number is reversed by division by that number.

Division by a nonzero number is reversed by multiplication by that number.

The key is to reverse the operations in the opposite order from which they were performed.

This is useful in algebra, number puzzles, financial calculations, and many practical problems.

11.15 Use Graphs and Tables to Reveal Relationships

Not every mathematical problem is easiest to solve through algebra alone.

Tables organise numerical information. Graphs reveal patterns and relationships visually.

Suppose a student saves ₦1,000 each week, beginning with no savings.

The total savings after n weeks can be represented by

$$S = 1000n$$

A table gives:

Weeks (n)	Savings (S)
0	₦0
1	₦1,000
2	₦2,000
3	₦3,000
4	₦4,000
5	₦5,000

The pattern is clear: every additional week increases the savings by ₦1,000.

A graph of savings against weeks would be a straight line through the origin, with a gradient of ₦1,000 per week.

This representation makes the relationship easier to interpret.

When graphs are useful

Graphs can help you:

- identify increasing or decreasing trends;
- compare two quantities;
- detect unusual observations;
- estimate values;
- understand rates of change;
- see whether a relationship appears linear or nonlinear.

However, a graph should be constructed with appropriate scales, labelled axes, and accurate data.

A misleading graph can hide important differences or exaggerate small ones.

11.16 Learn the Meaning Behind Formulas

A formula becomes easier to remember when you understand what it represents.

Consider the formula for the area of a rectangle:

$$A = lw$$

This is not merely a rule to memorise.

Imagine covering a rectangular surface with identical square tiles, each measuring one unit by one unit.

If the rectangle is 5 units long and 3 units wide, there are 5 tiles in each row and 3 rows.

The total number of tiles is

$$5 \times 3 = 15$$

Therefore, the area is 15 square units.

The multiplication arises from counting equal rows of square units.

Understanding this reasoning helps explain why the formula works.

Another example: Average speed

The average speed over a journey is defined as total distance divided by total time:

$$\boxed{\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}}$$

If a cyclist travels 30 km in 2 hours, the average speed is

$$\frac{30}{2} = 15 \text{ km/h}$$

This does not mean that the cyclist travelled at exactly 15 km/h at every moment.

The cyclist may have travelled faster on some parts of the journey and slower on others.

The formula describes the average over the complete journey.

Knowing what the formula means prevents you from interpreting the result incorrectly.

11.17 Avoid Shortcuts That Destroy Understanding

Shortcuts are useful when they preserve the mathematical relationship.

They are dangerous when they are memorised without understanding the conditions under which they work.

For example, a student may learn to “cancel matching symbols” in algebra.

Consider

$$\frac{x+6}{x}$$

It may be tempting to cancel the x in the numerator with the x in the denominator.

But this is incorrect because the numerator contains a sum.

The expression can be rewritten as

$$\frac{x+6}{x} = \frac{x}{x} + \frac{6}{x}$$

provided that $x \neq 0$.

Thus,

$$\boxed{1 + \frac{6}{x}}$$

The x in the denominator does not cancel with the entire numerator.

Now consider

$$\frac{3x}{6x}$$

Here the numerator and denominator are products, so common factors can be cancelled:

$$\frac{3x}{6x} = \frac{1}{2}, \qquad x \neq 0$$

The difference lies in the structure of the expressions.

A good shortcut should make the reasoning shorter without changing the mathematics.

11.18 Develop Your Own Problem-Solving Routine

A reliable routine can help you approach unfamiliar questions with confidence.

Use the following sequence.

Step 1: Read carefully

Read the entire question before calculating.

Pay attention to what is being requested.

Step 2: Identify the unknown

Write down the quantity you need to find.

Step 3: Organise the information

List the known quantities, units, and relevant conditions.

Step 4: Represent the problem

Choose a suitable representation: a diagram, table, equation, graph, or written explanation.

Step 5: Choose the method

Select the mathematical relationship that matches the problem.

Do not select a formula merely because it contains numbers similar to those in the question.

Step 6: Simplify where possible

Look for common factors, convenient numbers, useful identities, or smaller steps.

Step 7: Calculate carefully

Show enough working to make your reasoning understandable and easy to check.

Step 8: Check the result

Use estimation, substitution, inverse operations, units, or a second method.

Step 9: Interpret the answer

Explain what the result means in the original situation.

Step 10: Learn from the process

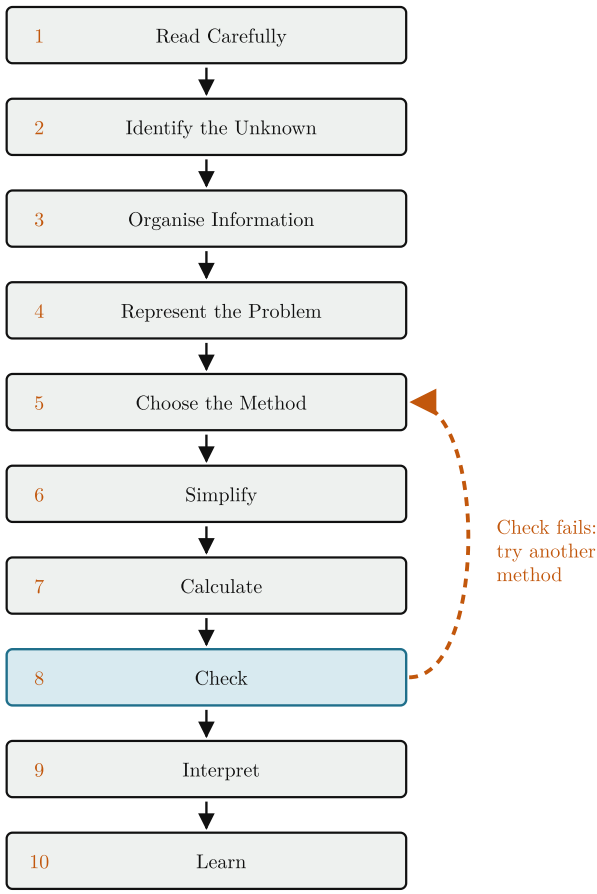
If the calculation was difficult, identify which part caused the difficulty.

Was it the arithmetic, the formula, the interpretation, or the organisation of information?

Understanding the source of the difficulty helps you improve.

FIGURE 17

The Ten-Step Problem-Solving Routine



Ten boxes, read top to bottom. If Check fails, the curved arrow sends you back to Choose the Method.

11.19 How to Practise Efficiently

Mathematical skill develops through practice, but not every form of practice is equally useful.

Repeating a calculation without understanding why it works may improve familiarity with that particular example while leaving you unprepared for a different problem.

A more effective approach combines understanding, variation, checking, and reflection.

Practise one idea at a time

If you are learning the distributive property, begin with examples such as

$$7(10+2)$$

Then practise

$$9(20-3)$$

After that, use less familiar numbers and expressions.

This allows you to recognise the same structure in different forms.

Change the numbers

Once you understand a method, change the numbers in the question.

For example, if you have learned to calculate

$$49 \times 18$$

using

$$49 = 50 - 1$$

try

$$99 \times 27$$

or

$$199 \times 14$$

The aim is to recognise the underlying relationship rather than memorise one answer.

Explain your method

After solving a problem, explain why you selected that method.

For example:

"I used $50-1$ because 49 is close to 50, and multiplying by 50 is convenient."

This explanation reveals whether you understand the method.

Compare different methods

Some problems can be solved in several ways.

Calculate

$$48 \times 25$$

Method 1:

$$\begin{aligned} 48 \times 25 &= 48 \times \frac{100}{4} \\ &= 4800 \div 4 = 1200 \end{aligned}$$

Method 2:

$$\begin{aligned} 48 \times 25 &= (50 - 2) \times 25 \\ &= 1250 - 50 = 1200 \end{aligned}$$

Both methods are correct.

Comparing them helps you understand that mathematical flexibility is often more valuable than memorising one preferred procedure.

Keep an error notebook

When you make a mistake, record:

- the original question;
- the incorrect step;
- the reason for the mistake;

- the corrected method;
- a similar question to practise.

For example, if you repeatedly forget to perform multiplication before addition, write down the rule and practise expressions that test it.

The goal is not to collect mistakes. It is to recognise and correct recurring patterns.

11.20 Practical Workshop: Make the Calculation Easier

In each activity, solve the problem using a method that reduces unnecessary work. Explain why your method is suitable.

Activity A: Multiplication

Calculate

$$48 \times 125$$

One approach is to recognise that

$$125 = \frac{1000}{8}$$

Therefore,

$$\begin{aligned} 48 \times 125 &= \frac{48 \times 1000}{8} \\ &= 6 \times 1000 \end{aligned}$$

Hence,

$$\boxed{6000}$$

Activity B: Percentage

A student obtains 72 marks out of 90. Find the percentage score.

Start with

$$\frac{72}{90} \times 100\%$$

Simplify:

$$\frac{72}{90} = \frac{4}{5}$$

Therefore,

$$\frac{4}{5} \times 100\% = 80\%$$

Hence,

$$\boxed{80\%}$$

Activity C: Difference of squares

Calculate

$$102^2 - 98^2$$

Use

$$a^2 - b^2 = (a - b)(a + b)$$

Then,

$$\begin{aligned} 102^2 - 98^2 &= (102 - 98)(102 + 98) \\ &= 4 \times 200 \end{aligned}$$

Therefore,

$$\boxed{800}$$

Activity D: Simplifying a fraction

Simplify

$$\frac{45}{60}$$

Both numbers are divisible by 15:

$$\frac{45}{60} = \frac{45 \div 15}{60 \div 15}$$

Therefore,

$$\boxed{\frac{3}{4}}$$

Activity E: Solve an equation

Find x if

$$5x + 12 = 47$$

Subtract 12 from both sides:

$$5x = 35$$

Divide by 5:

$$\boxed{x = 7}$$

Check:

$$5(7) + 12 = 35 + 12 = 47$$

The solution is correct.

Activity F: Estimate before calculating

Calculate

$$203 \times 49$$

First estimate:

$$200 \times 50 = 10{,}000$$

Now calculate exactly:

$$\begin{aligned} 203 \times 49 &= 203(50 - 1) \\ &= 10{,}150 - 203 \end{aligned}$$

Therefore,

$$\boxed{9947}$$

The exact answer is close to the estimate of 10,000.

Activity G: Identify the mistake

A student calculates

$$12 + 6 \times 3$$

and obtains

$$(12 + 6) \times 3 = 54$$

Explain the error and find the correct answer.

Multiplication must be performed before addition:

$$12 + 6 \times 3 = 12 + 18$$

Therefore,

$$\boxed{30}$$

The student's result of 54 corresponds to a different expression because the student added first.

Activity H: Choose a representation

A shop sells 8 identical bags for ₦12,000. Find the cost of one bag.

The total cost is divided equally among the 8 bags:

$$\text{Cost per bag} = \frac{12{,}000}{8}$$

Therefore,

$$\boxed{₦1{,}500}$$

The important step is recognising that equal sharing requires division.

11.21 Exercises

Section A: Understanding the problem

1. Explain why understanding the question is necessary before choosing a formula.
2. A trader buys 25 shirts at ₦4,000 each and sells them at ₦5,200 each. Calculate the total cost, total revenue, and total profit.
3. Describe three ways to check whether a mathematical answer is reasonable.
4. Explain the difference between estimation and guessing.
5. Give two examples of mathematical shortcuts that preserve the original relationship.

Section B: Arithmetic shortcuts

6. Calculate 25×72 using a convenient-number method.
7. Calculate 99×64 using the distributive property.

8. Calculate 125×48 using a fraction or another convenient representation.
9. Evaluate 35^2 using a pattern or algebraic expansion.
10. Evaluate $51^2 - 49^2$ without calculating the two squares separately.
11. Calculate 398×21 by expressing 21 as $20 + 1$.
12. Simplify $\frac{42}{56}$.
13. Calculate $\frac{15 \times 28}{5}$ by simplifying before multiplying.
14. Simplify $\frac{18}{24} \times \frac{16}{27}$.

Section C: Algebra and order of operations

15. Evaluate $7 + 4 \times 6$.
16. Evaluate $(7 + 4) \times 6$.
17. Evaluate $24 \div 4 \times 3$.
18. Solve $6x + 9 = 45$.
19. A number is multiplied by 5 and then reduced by 8 to produce 32. Find the original number.
20. Explain why $\frac{x+8}{x}$ cannot generally be simplified by cancelling the x in the numerator with the x in the denominator.

Section D: Estimation and reasoning

21. Estimate 497×19, then calculate the exact answer.
22. A calculator displays 0.0048 when you expected a result near 48. What should you check before accepting the result?
23. A student calculates the average speed of a journey by dividing the total distance by the time spent moving, but leaves out a 20-minute stop. Explain how this affects the result if the question asks for average speed over the entire journey, including the stop.
24. Explain why the same mathematical problem may have more than one correct solution method.
25. Choose one calculation you find difficult. Solve it in two different ways and explain which method you prefer and why.

11.22 Chapter Review

By the end of this chapter, you should understand that difficult calculations can often be made easier through better organisation and mathematical reasoning.

The most important ideas are:

- Understand the question before calculating.
- Separate complicated problems into smaller tasks.
- Use arithmetic properties to reorganise calculations.
- Recognise convenient numbers and useful patterns.
- Simplify fractions and expressions before performing lengthy calculations.
- Estimate the answer before calculating precisely.
- Change the representation of a problem when doing so makes the structure clearer.
- Use algebra to describe general relationships.
- Follow the correct order of operations.
- Use calculators carefully and verify the expression entered.
- Work backwards when reversing operations is helpful.
- Use tables and graphs to reveal relationships.
- Understand why a formula works instead of memorising it blindly.
- Avoid shortcuts that change the meaning of an expression.
- Practise with varied questions and learn from errors.
- Check the final answer and explain what it means.

11.23 Conclusion: Become a Better Thinker, Not Just a Faster Calculator

Mathematical ability is not measured only by how quickly someone produces an answer.

A person who calculates rapidly but repeatedly selects the wrong method will struggle with unfamiliar problems.

A person who understands the problem, recognises useful relationships, organises the information, and checks the result has developed a more dependable skill.

We have seen how arithmetic properties simplify multiplication, how algebra reduces repeated work, how estimation checks answers, and how

patterns reveal hidden structure.

We have also learned that calculators, graphs, and shortcuts are most useful when guided by understanding.

The goal is not to avoid every difficult calculation. Some problems require advanced mathematics, careful analysis, or substantial work.

The goal is to remove unnecessary difficulty while preserving the reasoning that makes the answer correct.

When you encounter a challenging problem, do not immediately conclude that you cannot solve it.

Pause.

Ask what the problem is really asking. Identify what you know. Look for relationships. Choose a suitable method. Simplify where possible. Calculate carefully. Check your result.

With practice, these habits become part of the way you think.

The best mathematical shortcut is often a clearer understanding of the problem itself.

In our final chapter, we will bring together the principles of this book. You will have the opportunity to solve practical problems independently, explain your reasoning, choose your own methods, and demonstrate that you can think before reaching for a formula.

CHAPTER 12

Now It is Your Turn

12

From Understanding Mathematics to Solving Real-Life Problems

12.1 Introduction

Throughout this book, we have explored an important idea: mathematics becomes easier when we understand the problem before searching for a formula.

We have examined how calculations help builders construct safe structures, how accountants manage money, how economists study costs and benefits, how algebra helps us find unknown quantities, and how calculus allows us to understand change.

We have also explored statistics, probability, scientific measurements, engineering calculations, and methods for simplifying difficult problems.

Now, it is time to bring everything together.

This chapter is different from the previous chapters. Instead of introducing many new mathematical ideas, it will challenge you to apply what you have learned.

You will encounter situations that do not immediately tell you which formula to use. You will need to identify the problem, select the relevant information, decide which mathematical tools are appropriate, perform the calculations, and explain what your answers mean.

That is what mathematical thinking looks like.

A person who memorizes formulas may solve familiar questions successfully but struggle when a problem is presented in an unfamiliar way. A person who

understands mathematical reasoning can often recognize the same underlying principle in different situations.

For example, the calculation used to determine the profit of a shopkeeper may also help an economist analyze a business. The same proportional reasoning used to calculate a discount can help a scientist interpret measurements. The geometry used by a builder can help an engineer determine

the dimensions of a structure.

The situations differ, but the mathematical principles connect them.

The purpose of this final chapter is to help you recognize those connections.

By the end of this chapter, you should be able to:

1. Identify the real problem hidden inside a word problem.
2. Separate relevant information from unnecessary information.
3. Choose suitable mathematical operations and formulas.
4. Break complicated problems into manageable stages.
5. Check calculations using estimation and alternative methods.
6. Interpret numerical answers in their practical context.
7. Explain why a mathematical method works instead of merely repeating its steps.
8. Approach unfamiliar problems with patience and confidence.

Remember the central principle of this book:

Do not begin by asking, “Which formula should I use?” Begin by asking, “What am I trying to understand or find?”

12.2 The Complete Mathematical Thinking Process

Before attempting the major exercises, let us establish a general method for approaching mathematical problems.

Consider a simple situation.

A student has ₦25,000. The student spends 30% of the money on books and ₦6,000 on transportation. How much money remains?

A person who rushes may begin multiplying numbers without deciding what each result represents.

A careful problem solver proceeds differently.

Step 1: Understand the situation

The student begins with ₦25,000.

Two expenses must be deducted:

- The cost of books, equal to 30% of the original money.
- Transportation, costing ₦6,000.

We need to find the amount remaining.

Step 2: Identify the unknown

Let R represent the remaining money.

Our goal is to determine R.

Step 3: Find the relevant information

The starting amount is ₦25,000.

The percentage spent on books is 30%.

Transportation costs ₦6,000.

Step 4: Decide what to calculate first

We cannot subtract the cost of books until we know that cost.

Therefore, we calculate 30% of ₦25,000 first.

$$\begin{aligned}\text{Cost of books} &= \frac{30}{100} \times 25{,}000 \\ &= 0.30 \times 25{,}000 \\ &= ₦7{,}500\end{aligned}$$

We multiply because 30% means 30 parts out of every 100 parts of the original amount.

Step 5: Calculate the total expenses

$$\begin{aligned}\text{Total expenses} &= 7{,}500 + 6{,}000 \\ &= ₦13{,}500\end{aligned}$$

We add the expenses because both amounts represent money spent.

Step 6: Calculate the remainder

$$\begin{aligned}R &= 25{,}000 - 13{,}500 \\ R &= ₦11{,}500\end{aligned}$$

We subtract because the expenses reduce the original amount.

Step 7: Check whether the answer makes sense

The student started with ₦25,000 and spent ₦13,500.

The remaining money must be less than ₦25,000.

Our answer satisfies that condition.

We can also check by adding the expenses to the remainder:

$$13{,}500 + 11{,}500 = 25{,}000$$

The calculation is consistent.

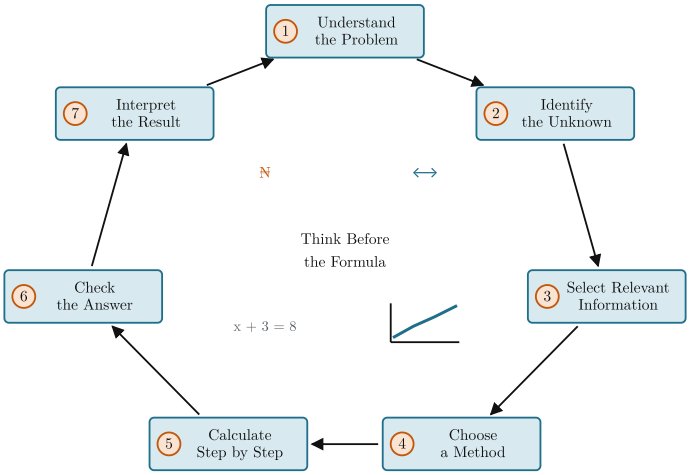
Answer: The student has ₦11,500 remaining.

Notice that we did not select a single formula and force the entire problem into it. We identified the separate questions and answered them in the correct order.

That is the method you will use throughout this chapter.

FIGURE 18

The Mathematical Thinking Process



Seven stages in a loop, with "Think Before the Formula" at the centre. Small symbols stand for money, measurement, equations and graphs.

12.3 Challenge One: The Student and the Budget

Imagine that you receive ₦80,000 to cover your educational expenses for a month.

You plan to spend:

- 25% on books and learning materials.
- ₦18,000 on transportation.
- ₦12,000 on food.
- ₦8,000 on internet access and other study resources.

You want to know how much money remains and what percentage of the original budget remains unspent.

Question 1: How much is spent on books?

The book allowance is 25% of ₦80,000.

$$\begin{aligned}\text{Books} &= \frac{25}{100} \times 80{,}000 \\ &= 0.25 \times 80{,}000 \\ &= ₦20{,}000\end{aligned}$$

Question 2: What is the total expenditure?

Add all four expenses:

$$20{,}000 + 18{,}000 + 12{,}000 + 8{,}000$$

First, add the book allowance and transportation:

$$20{,}000 + 18{,}000 = 38{,}000$$

Add food:

$$38{,}000 + 12{,}000 = 50{,}000$$

Add the remaining study expenses:

$$50{,}000 + 8{,}000 = 58{,}000$$

Therefore,

$$\text{Total expenditure} = ₦58{,}000$$

Question 3: How much money remains?

$$\begin{aligned}\text{Remaining money} &= 80{,}000 - 58{,}000 \\ &= ₦22{,}000\end{aligned}$$

Question 4: What percentage of the budget remains?

We divide the remaining amount by the original budget and multiply by 100%.

$$\text{Percentage remaining} = \frac{22{,}000}{80{,}000} \times 100\%$$
$$= 27.5\%$$

Therefore, 27.5% of the original budget remains unspent.

What have we learned?

This problem combines percentages, addition, subtraction, and interpretation.

Each operation answers a different question:

- Multiplication calculates a percentage of an amount.
- Addition combines expenses.
- Subtraction finds the balance.
- Division compares the remaining amount with the original budget.

The mathematics is manageable because each operation has a clear purpose.

Your turn

Suppose the transportation cost increases from ₦18,000 to ₦21,000.

1. What is the new total expenditure?
2. How much money remains?
3. What percentage of the budget remains?
4. By what percentage has transportation expenditure increased?

For Question 4, the increase is ₦3,000. The percentage increase is calculated using the original transportation cost:

$$\frac{3{,}000}{18{,}000} \times 100\% = 16.67\% \text{ approximately}$$

This last question teaches an important lesson: when calculating percentage change, compare the change with the original value, not the new value.

12.4 Challenge Two: The Builder and the Rectangular Hall

A builder is asked to estimate the dimensions and flooring requirements of a rectangular hall.

The hall is 12 metres long and 8 metres wide.

The builder needs to determine:

1. The area of the floor.
2. The perimeter of the hall.
3. The number of square tiles required if each tile covers $0.25\ \text{m}^2$.
4. The cost of the tiles if each tile costs ₦1,500.

Assume that the tiles cover the entire floor without gaps or wastage.

Question 1: What is the area?

The area of a rectangle is found by multiplying its length by its width.

$$A = l \times w$$

Substituting the measurements:

$$A = 12 \times 8$$
$$A = 96\ \text{m}^2$$

The floor covers 96 square metres.

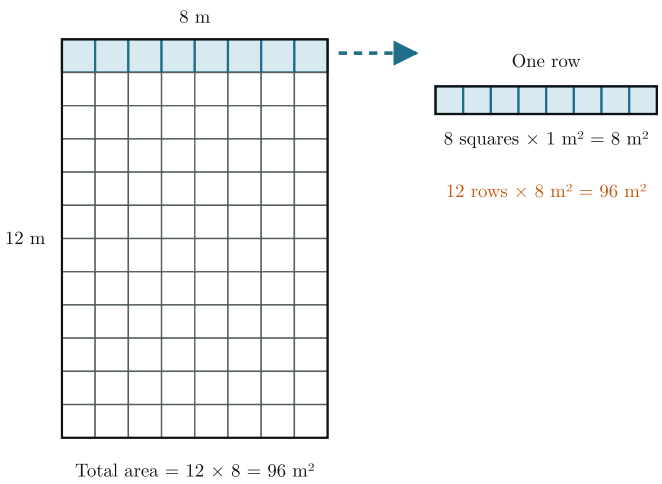
Why do we multiply?

Imagine dividing the floor into squares measuring one metre by one metre. Each row contains eight square metres of floor, and there are twelve such rows.

Therefore, the total number of square metres is $12 \times 8 = 96$.

FIGURE 19

Rectangular Hall Divided into Square Metres



The hall is a grid of 1 m2 squares. Each row holds 8 of them, and 12 rows give 96 m2.

Question 2: What is the perimeter?

Perimeter measures the total distance around a shape.

A rectangle has two lengths and two widths.

Therefore,

$$P = 2l + 2w$$

Substituting:

$$P = 2(12) + 2(8)$$
$$P = 24 + 16$$
$$P = 40 \text{ m}$$

The perimeter is 40 metres.

Notice the distinction between area and perimeter.

Area tells the builder how much floor surface must be covered. Perimeter may help estimate the length of material required around the edges.

The two quantities describe different properties of the same rectangle.

Question 3: How many tiles are required?

The floor area is 96 m^2.

Each tile covers 0.25 m^2.

The number of tiles is:

$$\begin{aligned}\text{Number of tiles} &= \frac{\text{Total floor area}}{\text{Area of one tile}} \\ &= \frac{96}{0.25}\end{aligned}$$

To simplify the division, multiply both the numerator and denominator by 100:

$$\begin{aligned}\frac{96}{0.25} &= \frac{9{,}600}{25} \\ &= 384\end{aligned}$$

Therefore, 384 tiles are required under the stated assumptions.

We divide because we are determining how many equal tile areas fit into the total floor area.

Question 4: What is the total cost?

$$\begin{aligned}\text{Total cost} &= 384 \times 1{,}500 \\ &= ₦576{,}000\end{aligned}$$

The estimated tile cost is ₦576,000.

This estimate excludes wastage, delivery, labour, adhesive, and any additional tiles needed for cutting or replacement.

A practical builder would normally allow for such factors when preparing a real quotation.

Your turn

A second rectangular room measures 10 m by 6 m.

Each tile covers 0.5 m^2, and each tile costs ₦2,000.

Calculate:

1. The floor area.
2. The perimeter.
3. The number of tiles required.
4. The total tile cost.

Then explain why the room's perimeter cannot be used directly to calculate the number of floor tiles.

12.5 Challenge Three: The Ladder and the Unknown Distance

A worker places a ladder against a vertical wall.

The foot of the ladder is 5 metres from the wall, and the ladder reaches a point 12 metres above the ground.

The worker wants to know the ladder's length.

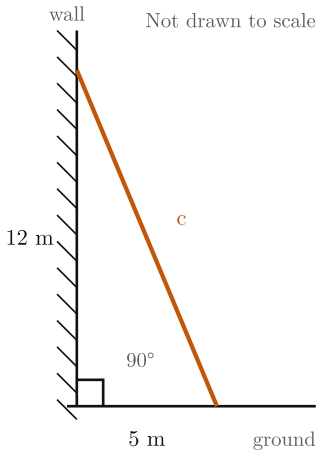
We can represent the situation using a right-angled triangle.

- The vertical wall provides one side.
- The ground provides another side.
- The ladder forms the sloping side.

The ladder is the hypotenuse because it lies opposite the right angle.

FIGURE 20

Ladder Against a Wall



A 5 m by 12 m right triangle with the ladder c as its hypotenuse. The sketch is not to scale.

Applying Pythagoras' theorem

For a right-angled triangle,

$$c^2 = a^2 + b^2$$

where c is the hypotenuse and a and b are the other two sides.

In this problem,

$$a = 5, \qquad b = 12$$

Therefore,

$$\begin{aligned} c^2 &= 5^2 + 12^2 \\ c^2 &= 25 + 144 \\ c^2 &= 169 \end{aligned}$$

Taking the positive square root,

$$c = \sqrt{169}$$
$$c = 13 \text{ m}$$

The ladder is 13 metres long.

Why does the theorem apply?

The wall and ground are assumed to meet at a right angle. This creates the right-angled triangle required by Pythagoras' theorem.

The ladder's length is not obtained by adding 5 and 12 because the ladder does not follow the ground and wall as a straight path. It forms the diagonal of the triangle.

The theorem provides the correct relationship between the three sides.

Your turn

A cable is stretched from the top of a 9-metre pole to a point on the ground 12 metres from the pole.

Find the cable's length, assuming the ground is level and the pole is vertical.

Then explain why adding 9 and 12 would give the wrong result.

12.6 Challenge Four: The Small Business

A student starts a small business selling notebooks.

The student purchases 100 notebooks at ₦800 each and sells every notebook at ₦1,100.

The business also incurs ₦5,000 in transportation expenses.

The student wants to know whether the business made a profit and how profitable the operation was.

Question 1: What is the cost of purchasing the notebooks?

$$\begin{aligned}\text{Purchase cost} &= 100 \times 800 \\ &= ₦80{,}000\end{aligned}$$

This is the total amount paid to acquire the notebooks.

Question 2: What is the total sales revenue?

$$\begin{aligned}\text{Revenue} &= 100 \times 1{,}100 \\ &= ₦110{,}000\end{aligned}$$

Revenue is the money received from selling the notebooks before deducting expenses.

Question 3: What is the total cost?

The business paid ₦80,000 for the notebooks and ₦5,000 for transportation.

$$\begin{aligned}\text{Total cost} &= 80{,}000 + 5{,}000 \\ &= ₦85{,}000\end{aligned}$$

Question 4: What is the profit?

$$\begin{aligned}\text{Profit} &= \text{Revenue} - \text{Total cost} \\ &= 110{,}000 - 85{,}000 \\ &= ₦25{,}000\end{aligned}$$

The business made a profit of ₦25,000.

Question 5: What is the profit margin?

Profit margin measures profit as a percentage of revenue.

$$\text{Profit margin} = \frac{\text{Profit}}{\text{Revenue}} \times 100\%$$

$$\begin{aligned}&= \frac{25{,}000}{110{,}000} \times 100\% \\ &\approx 22.73\%\end{aligned}$$

The profit margin is approximately 22.73%.

Question 6: What is the return on total cost?

This is a different measure.

$$\begin{aligned}\text{Return on cost} &= \frac{\text{Profit}}{\text{Total cost}} \times 100\% \\ &= \frac{25{,}000}{85{,}000} \times 100\% \\ &\approx 29.41\%\end{aligned}$$

The business earned approximately 29.41% of its total stated cost as profit.

Profit margin and return on cost are not identical because they use different denominators.

Question 7: What is the break-even selling price per notebook?

Break-even occurs when total revenue equals total cost.

The total cost is ₦85,000 for 100 notebooks.

Therefore, the break-even price per notebook is:

$$\frac{85{,}000}{100} = ₦850$$

The student must sell each notebook for an average of ₦850 to recover the stated total costs, assuming all 100 notebooks are sold and no other costs arise.

At ₦850 per notebook, the business neither makes a profit nor suffers a loss.

At ₦1,100 per notebook, the business makes a profit under the assumptions given.

Your turn

The student considers reducing the selling price to ₦1,000 per notebook.

1. What would the total revenue be?
2. What would the profit be?
3. What would the profit margin be?
4. Would the business still make a profit?
5. What additional information would you need before deciding whether reducing the price is a good business strategy?

The last question cannot be answered by arithmetic alone. You might need to consider customer demand, competitors' prices, the speed at which stock sells, and other business costs.

Mathematics helps you evaluate a decision, but good decisions also require an understanding of the situation.

12.7 Challenge Five: Algebra in Everyday Life

A transport company charges a fixed booking fee of ₦500 plus ₦200 for each kilometre travelled.

You want to understand how the total fare changes with distance.

Let x represent the number of kilometres travelled and C represent the total cost.

The total cost consists of two parts:

- A fixed charge of ₦500.
- A variable charge of ₦200 for every kilometre.

Therefore,

$$C = 500 + 200x$$

This is an algebraic model.

It describes the relationship between distance and cost.

Question 1: What is the fare for 10 kilometres?

Substitute $x = 10$:

$$C = 500 + 200(10)$$
$$C = 500 + 2{,}000$$
$$C = ₦2{,}500$$

Question 2: What is the fare for 25 kilometres?

Substitute $x = 25$:

$$C = 500 + 200(25)$$

$$C = 500 + 5{,}000$$
$$C = ₦5{,}500$$

Question 3: How far can a passenger travel with ₦4,500?

We know the total cost but do not know the distance.

Therefore,

$$4{,}500 = 500 + 200x$$

Subtract 500 from both sides:

$$4{,}000 = 200x$$

Divide both sides by 200:

$$x = \frac{4{,}000}{200}$$
$$x = 20$$

The passenger can travel 20 kilometres under the model.

Why did we use algebra?

Algebra allows us to describe a relationship once and then use it to answer many questions.

Without the model, we might calculate each situation separately and repeatedly explain the same relationship.

With the model, we can find the cost for any specified distance within the range where the pricing rule applies.

Question 4: What happens if the fixed fee changes?

Suppose the company increases its booking fee to ₦700 but keeps the distance charge at ₦200 per kilometre.

The new model becomes:

$$C = 700 + 200x$$

For a journey of 10 kilometres,

$$C = 700 + 200(10)$$
$$C = ₦2{,}700$$

The cost increases by ₦200 because the fixed fee increased by ₦200.

Your turn

A printing company charges ₦1,000 as a setup fee and ₦50 for every page printed.

1. Write an expression for the total cost C of printing n pages.
2. Calculate the cost of printing 40 pages.
3. How many pages can be printed for ₦6,000?
4. If the setup fee is removed, how does the model change?
5. Explain the difference between a fixed cost and a variable cost.

12.8 Challenge Six: Calculus and the Meaning of Change

Calculus can seem difficult when it is introduced as a collection of symbols and rules. However, its central ideas arise from questions about change.

Imagine a car travelling along a road.

Its position changes with time. Sometimes it moves quickly, sometimes slowly, and sometimes it stops.

A natural question is:

How quickly is the car's position changing at a particular moment?

This question leads to the idea of instantaneous velocity, which can be studied using derivatives.

A simple mathematical model

Suppose the position of a moving object, measured in metres, is given by

$$s(t) = t^2$$

where t is time measured in seconds.

This model is a mathematical example rather than a complete description of a real vehicle.

We can calculate the object's position at different times.

At $t = 1$,

$$s(1) = 1^2 = 1 \text{ m}$$

At $t = 2$,

$$s(2) = 2^2 = 4 \text{ m}$$

At $t = 3$,

$$s(3) = 3^2 = 9 \text{ m}$$

The position changes by different amounts over equal one-second intervals.

From $t = 1$ to $t = 2$, the object moves 3 metres.

From $t = 2$ to $t = 3$, it moves 5 metres.

This tells us that the position is not increasing at a constant rate.

Average velocity over an interval

Average velocity is the change in position divided by the change in time.

$$\text{Average velocity} = \frac{\text{Change in position}}{\text{Change in time}}$$

Between $t = 1$ and $t = 2$,

$$v_{\text{avg}} = \frac{4 - 1}{2 - 1}$$
$$v_{\text{avg}} = 3 \text{ m/s}$$

Between $t = 2$ and $t = 3$,

$$v_{\text{avg}} = \frac{9 - 4}{3 - 2}$$
$$v_{\text{avg}} = 5 \text{ m/s}$$

The average velocity is different in the two intervals.

Instantaneous velocity

To determine instantaneous velocity, calculus examines how the average rate of change behaves over smaller and smaller intervals around a particular time.

For the function

$$s(t) = t^2$$

the derivative is

$$\frac{ds}{dt} = 2t$$

Therefore, the instantaneous velocity at $t = 3$ is

$$v(3) = 2(3)$$
$$v(3) = 6 \text{ m/s}$$

The derivative tells us the instantaneous rate at which the position changes at that time.

Why is this useful?

The same idea of change appears in many fields.

- Economists study how costs and revenue change as output increases.
- Scientists study how temperature changes over time.
- Engineers study how moving systems behave.
- Businesses examine how production affects expenses and income.

The specific model may differ, but the question remains similar: how does one quantity change in response to another?

Your turn

Suppose an object's position is described by

$$s(t) = 3t^2$$

where s is measured in metres and t in seconds.

1. Find its position at $t = 2$.
2. Find its position at $t = 4$.
3. Calculate the average velocity between $t = 2$ and $t = 4$.
4. Differentiate the function to obtain the instantaneous velocity.
5. Find the instantaneous velocity at $t = 4$.

Explain the difference between average velocity and instantaneous velocity in your own words.

12.9 Challenge Seven: Statistics and Probability

A teacher records the following test scores for five students:

$$60,\ 70,\ 70,\ 80,\ 90$$

The teacher wants to describe the class's performance.

Question 1: What is the mean score?

The mean is the total of the observations divided by the number of observations.

$$\begin{aligned}\text{Mean} &= \frac{60+70+70+80+90}{5}\\ &= \frac{370}{5}\\ &= 74\end{aligned}$$

The mean score is 74.

Question 2: What is the median?

Arrange the scores in order:

$$60,\ 70,\ 70,\ 80,\ 90$$

There are five observations, so the middle observation is the third one.

$$\text{Median} = 70$$

Question 3: What is the mode?

The mode is the value that occurs most frequently.

The score 70 appears twice, while every other score appears once.

Therefore,

$$\text{Mode} = 70$$

Why are the three measures different?

The mean uses every observation and can be influenced by unusually high or low values.

The median identifies the middle observation in an ordered dataset.

The mode identifies the most frequently occurring value.

Each measure describes a different aspect of the data.

A teacher might use more than one measure to understand the class's performance.

Question 4: What is the probability of randomly selecting a student who scored at least 80?

The scores of at least 80 are:

$$80,\ 90$$

Two out of the five students satisfy the condition.

If each student is equally likely to be selected,

$$\begin{aligned} P(\text{score} \geq 80) &= \frac{2}{5} \\ &= 0.4 = 40\% \end{aligned}$$

The probability is 40%.

A warning about interpretation

This probability describes a random selection from these five students. It does not establish the probability that a different student in another class will score at least 80.

To make broader claims, we would need suitable data and a justified method of inference.

Your turn

Consider the dataset:

$$45,\ 55,\ 65,\ 65,\ 70,\ 90$$

Calculate:

1. The mean.
2. The median.
3. The mode.
4. The range.
5. The probability of selecting a score greater than 65 at random, assuming each observation is equally likely to be selected.

Then explain how the mean might change if the score of 90 were replaced by 100.

12.10 Challenge Eight: Science, Measurement and Estimation

A scientist measures the length of a small object three times:

$$12.1 \text{ cm}, \quad 12.3 \text{ cm}, \quad 12.2 \text{ cm}$$

The scientist wants to report a representative measurement.

A simple approach is to calculate the mean of the three readings.

$$\begin{aligned} \text{Mean} &= \frac{12.1 + 12.3 + 12.2}{3} \\ &= \frac{36.6}{3} \\ &= 12.2 \text{ cm} \end{aligned}$$

The mean measurement is 12.2 cm.

Why take repeated measurements?

Measurements may vary because of instrument limitations, the way the object is positioned, or other sources of uncertainty.

Repeated readings can help us assess consistency.

However, calculating the mean does not automatically eliminate every measurement error. The scientist must also consider the instrument's precision, the measurement procedure, and whether the readings are reliable.

Estimation as a check

Before calculating, we might estimate the result.

All three measurements are close to 12 cm.

Therefore, an average of 12.2 cm is reasonable.

If the calculation produced 122 cm, we would suspect an error in the arithmetic or in how the decimal point was handled.

Units matter

A numerical answer without a unit may be incomplete.

For example:

- 12.2 cm measures length.
- 12.2 cm^2 measures area.
- 12.2 cm^3 measures volume.

The number is the same, but the physical meaning is different.

Your turn

A student records three measurements of an object:

$$8.4 \text{ cm}, \quad 8.6 \text{ cm}, \quad 8.5 \text{ cm}$$

1. Calculate the mean measurement.
2. Estimate the answer before calculating and compare the estimate with the exact result.
3. Explain why the answer should include centimetres.
4. Give two reasons why repeated measurements might differ.

12.11 Challenge Nine: Comparing Two Investment Options

Imagine that a student has ₦100,000 and is comparing two hypothetical savings options.

- Option A: A simple annual return of 10% of the original amount.
- Option B: An annual return of 8%, compounded once per year.

Assume that the stated rates remain unchanged for the comparison period and ignore fees, taxes, inflation, and other risks. These are mathematical examples, not recommendations for a real financial product.

The student wants to compare the value of the two options after three years.

Option A: Simple interest

Simple interest is calculated on the original principal.

The formula is

$$I = Prt$$

where:

- P is the principal.
- r is the annual interest rate expressed as a decimal.
- t is the number of years.

Here,

$$P = 100{,}000, \quad r = 0.10, \quad t = 3$$

Therefore,

$$I = 100{,}000(0.10)(3)$$
$$I = ₦30{,}000$$

The final amount is

$$A = P + I$$
$$A = 100{,}000 + 30{,}000$$
$$A = ₦130{,}000$$

Option B: Compound interest

Compound interest is calculated on the accumulated amount, so each year's interest becomes part of the principal for the next year.

The formula is

$$A = P(1 + r)^t$$

Substituting:

$$A = 100{,}000(1.08)^3$$

First, calculate the powers step by step:

$$1.08^2 = 1.1664$$
$$1.08^3 = 1.259712$$

Therefore,

$$A = 100{,}000(1.259712)$$
$$A = ₦125{,}971.20$$

Compare the results

Option	Amount after three years
A: 10% simple interest	₦130,000
B: 8% annual compound interest	₦125,971.20

Under the assumptions given, Option A produces the larger final amount after three years.

However, we must not conclude that simple interest is always better than compound interest. The result depends on the rates, duration, fees, and terms of the options being compared.

The deeper lesson

A mathematical formula is only as useful as the assumptions behind it.

Before applying a financial formula, ask:

- Is the rate fixed or variable?

- Is the rate annual, monthly, or based on another period?
- Is interest simple or compound?
- Are fees or taxes included?
- What is the length of the investment?
- Are the options being compared under equivalent conditions?

Correct arithmetic cannot rescue an incorrect model.

Your turn

Suppose ₦200,000 earns 5% simple annual interest for four years.

1. Calculate the interest.
2. Calculate the final amount.
3. Explain why the interest is calculated on the original principal under simple interest.

Then calculate the final amount if the same principal earns 5% compounded annually for four years.

12.12 Challenge Ten: Finding the Best Use of Limited Resources

A school club has ₦120,000 available for an educational event.

The organizers want to spend the money on books, refreshments, and transportation.

Their proposed budget is:

- 40% for books.
- 25% for refreshments.
- ₦30,000 for transportation.
- The remainder kept as a reserve.

The organizers want to know whether their budget is consistent with the amount available.

Step 1: Calculate the book budget

$$\begin{aligned}\text{Books} &= 0.40 \times 120{,}000 \\ &= ₦48{,}000\end{aligned}$$

Step 2: Calculate the refreshment budget

$$\begin{aligned}\text{Refreshments} &= 0.25 \times 120{,}000 \\ &= ₦30{,}000\end{aligned}$$

Step 3: Calculate the total planned expenditure

$$48{,}000 + 30{,}000 + 30{,}000 = ₦108{,}000$$

Step 4: Find the reserve

$$\begin{aligned}\text{Reserve} &= 120{,}000 - 108{,}000 \\ &= ₦12{,}000\end{aligned}$$

The budget is consistent: the planned expenditure does not exceed the available money.

Step 5: Express the reserve as a percentage

$$\begin{aligned}\text{Reserve percentage} &= \frac{12{,}000}{120{,}000} \times 100\% \\ &= 10\%\end{aligned}$$

The reserve is 10% of the available budget.

Step 6: Consider a change

Suppose the transportation cost increases to ₦36,000.

The new expenditure becomes

$$48{,}000 + 30{,}000 + 36{,}000 = ₦114{,}000$$

The remaining reserve becomes

$$120{,}000 - 114{,}000 = ₦6{,}000$$

The reserve has fallen from ₦12,000 to ₦6,000.

That is a reduction of ₦6,000, or

$$\frac{6{,}000}{12{,}000} \times 100\% = 50\%$$

The reserve has decreased by 50%.

What does the result tell us?

The revised budget is still mathematically possible, but the smaller reserve leaves less room for unexpected expenses.

This is where calculation supports judgment.

Mathematics shows the effect of the transportation increase. The organizers must then decide whether the remaining reserve is adequate for the event.

Your turn

An event committee has ₦250,000.

It allocates 35% to equipment, 20% to refreshments, and ₦70,000 to transportation.

1. Calculate the equipment allocation.
2. Calculate the refreshment allocation.
3. Find the total planned expenditure.
4. Calculate the remaining reserve.
5. Express the reserve as a percentage of the original budget.
6. If transportation increases by ₦15,000, calculate the new reserve.

12.13 The Master Challenge: Plan a Small Business from Beginning to End

This is the major practical exercise of the book.

Imagine that you want to start a small printing business near a university. You have limited capital and want to decide whether the business is financially practical.

You do not need to solve everything in one step.

Your task is to use mathematical thinking to investigate the business.

The situation

You have ₦300,000 available.

You plan to use the money for:

- A printer costing ₦120,000.
- Initial paper and ink supplies costing ₦60,000.
- A table and other setup materials costing ₦25,000.
- Transportation and miscellaneous setup expenses costing ₦15,000.

You will charge ₦100 per printed page.

For the first month, you estimate that customers will request 1,500 pages.

Assume for this exercise that the paper and ink costs for the first month's printing are included in the ₦60,000 initial supply purchase. To estimate operating profit more realistically, you will need to identify the portion of those supplies consumed during the month.

For a simplified first calculation, assume that the 1,500 pages consume ₦30,000 worth of paper and ink. Also assume that rent, electricity, main-tenance, and other operating costs total ₦20,000 for the month.

These figures are hypothetical and are supplied only for mathematical practice.

Part A: Determine the startup expenditure

Add the initial costs:

$$120{,}000 + 60{,}000 + 25{,}000 + 15{,}000$$
$$= ₦220{,}000$$

The initial expenditure is ₦220,000.

Part B: Determine the remaining capital

$$300{,}000 - 220{,}000 = ₦80{,}000$$

The business retains ₦80,000 after the initial purchases.

This is important because a business may need cash to operate even after purchasing equipment and supplies.

Part C: Calculate expected monthly revenue

The business expects to print 1,500 pages at ₦100 per page.

$$\text{Revenue} = 1{,}500 \times 100$$
$$= ₦150{,}000$$

The expected monthly revenue is ₦150,000.

Part D: Estimate monthly operating profit

For this simplified model, the costs attributed to the month are:

- Paper and ink consumed: ₦30,000.
- Other monthly operating expenses: ₦20,000.

Total monthly operating costs are

$$30{,}000 + 20{,}000 = ₦50{,}000$$

Estimated operating profit is

$$150{,}000 - 50{,}000 = ₦100{,}000$$

The estimated monthly operating profit is ₦100,000 before any additional costs not included in the model.

Notice that we have not deducted the entire ₦220,000 startup expenditure from the month's revenue. Some of that expenditure purchased equipment and supplies that may be used over several months. A complete accounting analysis would distinguish equipment purchases, inventory, inventory consumed, depreciation, and other expenses.

Part E: Find the contribution per page

Under the simplified assumptions, ₦30,000 of supplies are consumed to print 1,500 pages.

The estimated supply cost per page is

$$\frac{30{,}000}{1{,}500} = ₦20$$

Each page sells for ₦100.

Therefore, the contribution per page after the stated paper and ink cost is

$$100 - 20 = ₦80$$

Each page contributes ₦80 toward the other operating costs and profit.

Part F: Calculate the monthly break-even volume

For this simplified model, assume the ₦20,000 in other monthly operating expenses is fixed and the ₦20 supply cost per page is variable.

The number of pages needed to cover those operating expenses is

$$\begin{aligned}\text{Break-even volume} &= \frac{\text{Fixed operating costs}}{\text{Selling price per page} - \text{Variable cost per page}} \\ &= \frac{20{,}000}{100 - 20} \\ &= \frac{20{,}000}{80} \\ &= 250 \text{ pages}\end{aligned}$$

The business would need to print 250 pages in the month to cover the stated ₦20,000 in fixed operating expenses, after accounting for the assumed ₦20 supply cost per page.

This is an operating break-even calculation. It does not recover the startup expenditure immediately, and it excludes any other costs that have not been specified.

Part G: Test a different demand level

Suppose the business prints only 200 pages in a month.

Revenue would be

$$200 \times 100 = ₦20{,}000$$

Variable supply cost would be

$$200 \times 20 = ₦4{,}000$$

The contribution after supply costs would be

$$20{,}000 - 4{,}000 = ₦16{,}000$$

After deducting the ₦20,000 fixed operating expenses,

$$16{,}000 - 20{,}000 = -₦4{,}000$$

The business would make an operating loss of ₦4,000 under these assumptions.

This demonstrates why revenue alone does not determine profitability.

Part H: Test a higher demand level

Suppose the business prints 2,000 pages.

Revenue:

$$2{,}000 \times 100 = ₦200{,}000$$

Variable supply cost:

$$2{,}000 \times 20 = ₦40{,}000$$

Contribution:

$$200{,}000 - 40{,}000 = ₦160{,}000$$

Estimated operating profit:

$$160{,}000 - 20{,}000 = ₦140{,}000$$

Under the simplified assumptions, printing 2,000 pages produces an estimated operating profit of ₦140,000.

Part I: Make a decision

You now have several pieces of information:

- Startup expenditure: ₦220,000.
- Capital remaining after initial purchases: ₦80,000.
- Expected monthly revenue at 1,500 pages: ₦150,000.
- Estimated operating profit at 1,500 pages: ₦100,000.
- Simplified operating break-even volume: 250 pages.
- Estimated operating loss at 200 pages: ₦4,000.
- Estimated operating profit at 2,000 pages: ₦140,000.

Does this mean the business is guaranteed to succeed?

No.

The calculations depend on assumptions. The printer may require repairs. Customer demand may be lower than expected. Competition may force the selling price down. Some supplies may cost more than anticipated. The business may need to account for rent, electricity, taxes, depreciation, and other expenses differently.

The correct conclusion is not that the business must succeed or fail. It is that the model gives you a starting point for investigating its feasibility.

Your master assignment

Prepare your own business assessment using the following questions:

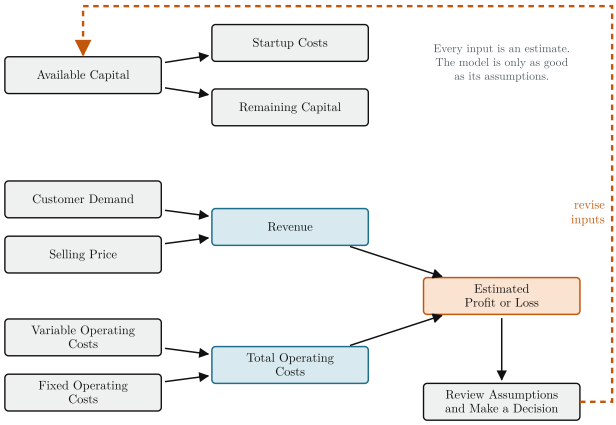
1. Which assumptions in the example would you investigate before starting the business?
2. What happens if the selling price falls to ₦80 per page?
3. What happens if the variable supply cost rises to ₦30 per page?
4. How many pages would be required to cover ₦30,000 in fixed monthly operating expenses if the selling price is ₦100 and variable cost is ₦20?
5. What information would you need to estimate the time required to recover the startup expenditure?
6. How could a table help you compare demand levels of 200, 500, 1,000, 1,500, and 2,000 pages?
7. Why is it dangerous to confuse revenue with profit?
8. What evidence would help you decide whether the expected demand of 1,500 pages is realistic?

Try to answer these questions without immediately searching for a formula. Describe the problem first, identify what you know, and then determine which calculations are necessary.

This is mathematical modelling: using mathematics to represent a real situation, examine possible outcomes, and support decisions.

FIGURE 21

Small Business Mathematical Model



Capital splits into startup costs and remaining capital. Demand and price feed revenue; variable and fixed costs f operating costs. Both lead to estimated profit or loss, which feeds a review of assumptions and a decision, and th back to the inputs.

12.14 How to Approach a Problem You Have Never Seen Before

At some point, you will encounter a mathematical problem that looks unfamiliar.

You may not immediately recognize its topic. You may not remember a relevant formula. You may even feel that the problem is beyond your ability.

Do not conclude too quickly that you cannot solve it.

Instead, return to the basic questions.

Question 1: What is happening?

Describe the situation in ordinary language.

If a question discusses a growing population, explain what is growing and over what period.

If it describes a business, identify the money received, the money spent, and what you need to find.

If it involves a geometric shape, identify the known dimensions and the unknown measurement.

Understanding the situation is the foundation of the solution.

Question 2: What do I know?

Write down the information provided.

For example:

$$\text{Initial amount} = ₦50{,}000$$
$$\text{Annual rate} = 5\%$$
$$\text{Time} = 3 \text{ years}$$

Organizing the information makes the relationship easier to see.

Question 3: What am I trying to find?

State the unknown explicitly.

It might be:

- A distance.
- A percentage.
- A total cost.
- An unknown number.
- A probability.
- A rate of change.
- A future value.

A problem becomes more manageable when its goal is clear.

Question 4: What relationship connects the known quantities to the unknown?

This is where mathematical knowledge becomes useful.

You might recognize that:

- Total cost is fixed cost plus variable cost.
- Profit is revenue minus cost.
- A percentage is a fraction with denominator 100.
- The area of a rectangle depends on length and width.
- A right-angled triangle satisfies Pythagoras' theorem.
- Average rate of change is change in quantity divided by change in the independent variable.
- The mean is the total of observations divided by the number of observations.

Do not memorize these relationships as disconnected statements. Understand what each one describes.

Question 5: Can I solve a simpler version first?

Suppose you are asked to calculate a 17% discount on ₦83,750.

If the numbers seem awkward, consider estimating first.

Seventeen per cent is close to 20%, and 20% of ₦80,000 is ₦16,000.

Therefore, a discount of approximately ₦14,000 to ₦17,000 would be plausible.

You can then perform the exact calculation:

$$0.17 \times 83{,}750 = ₦14{,}237.50$$

The estimate helps you judge whether the exact answer is reasonable.

Question 6: Can I check the result another way?

Possible checks include:

- Substituting a solution back into its equation.
- Adding expenses to a remaining balance.
- Estimating the size of the answer.
- Reversing an operation.
- Using a second method.

- Checking units.
- Comparing the result with the limits imposed by the situation.

Question 7: What does the answer mean?

Do not stop at the numerical result.

If the answer is 250 pages, explain that it represents the break-even monthly printing volume under the stated assumptions.

If the answer is 13 metres, explain that it is the calculated ladder length.

If the probability is 40%, explain which event has that probability and under what assumptions.

A number becomes useful when its meaning is clear.

12.15 Mathematical Habits That Make You Stronger

Mathematical ability develops through repeated practice, reflection, and correction.

The following habits can help you improve.

Habit 1: Explain every important step

After writing a line of calculation, ask yourself why it is valid.

For example,

$$3x + 5 = 20$$

Subtract 5 from both sides:

$$3x = 15$$

Why subtract 5?

Because the objective is to isolate the term containing x, and subtracting the same quantity from both sides preserves equality.

Divide both sides by 3:

$$x = 5$$

Why divide by 3?

Because $3x$ means three times x, and division by 3 reverses multiplication by 3.

The explanation is as important as the answer.

Habit 2: Keep an error notebook

When you make a mistake, do not simply cross it out and forget it.

Record:

1. The original question.
2. Your incorrect working.
3. The point where the reasoning failed.
4. The correct method.
5. A similar question to practise.

You may discover that your errors follow patterns.

Perhaps you frequently forget units, use the wrong denominator for a percentage change, or reverse an inequality incorrectly.

Identifying these patterns helps you practise more effectively.

Habit 3: Estimate before using a calculator

Suppose you calculate

$$498 \times 21$$

You can estimate:

$$500 \times 20 = 10{,}000$$

The exact answer should be close to 10,000.

Now calculate:

$$498 \times 21 = 10{,}458$$

The result is reasonably close to the estimate.

If your calculator instead showed 1,045,800, you would know to investigate the input or the placement of the digits.

Habit 4: Change the representation when necessary

A problem can become easier when expressed differently.

For example,

$$25\% = 0.25 = \frac{1}{4}$$

To calculate 25% of ₦48,000, you could multiply by 0.25:

$$0.25 \times 48{,}000 = 12{,}000$$

Or divide by 4:

$$48{,}000 \div 4 = 12{,}000$$

The second method may be easier mentally.

Likewise, a table may clarify a relationship that is difficult to understand from a paragraph. A graph may reveal a pattern that is not obvious from a list of numbers. An equation may summarize a relationship more clearly than repeated calculations.

Habit 5: Practise understanding, not only speed

Speed is useful, but it is not the only measure of mathematical ability.

A person who produces a quick answer without understanding the reasoning may struggle when the problem changes.

A person who understands the structure can often adapt the method to unfamiliar situations.

Aim to become accurate first, clear second, and faster through practice.

Habit 6: Learn when a model is insufficient

A mathematical model is a simplified representation of reality.

It may ignore some details to make a problem manageable.

For example, a model of transport cost may assume a fixed fee and a constant cost per kilometre. A real transport service may use different rates at different times or apply additional charges.

The model is useful only to the extent that its assumptions fit the situation.

Knowing when a model is insufficient is part of mathematical maturity.

Habit 7: Be patient with difficult ideas

Some topics require more time than others.

If you do not understand a concept immediately, return to its meaning, work through simpler examples, and gradually increase the difficulty.

You do not need to understand every topic in one sitting.

Progress comes from repeatedly connecting ideas, correcting errors, and practising thoughtfully.

12.16 Final Practice Examination

This examination combines the major ideas explored throughout the book.

Attempt the questions before checking the answer guide in Section 12.17.

Show your working, explain your reasoning where requested, and include units when appropriate.

Section A: Arithmetic and Percentages

Question 1

A student has ₦45,000 and spends 18% on books.

(a) Calculate the amount spent on books.

(b) How much money remains?

(c) What percentage of the original money remains?

Question 2

A trader purchases a product for ₦24,000 and sells it for ₦30,000.

(a) Calculate the profit.

(b) Calculate the profit as a percentage of cost.

(c) Calculate the profit margin as a percentage of revenue.

Question 3

A product originally costs ₦80,000. Its price is increased by 15% and then reduced by 15% of the increased price.

Calculate the final price and explain why it is not equal to the original price.

Section B: Geometry and Measurement

Question 4

A rectangular field measures 25 m by 16 m.

(a) Calculate its area.

(b) Calculate its perimeter.

(c) If fencing costs ₦2,500 per metre, calculate the cost of fencing the entire boundary, ignoring gates and other additional expenses.

Question 5

A right-angled triangle has shorter sides of 8 cm and 15 cm.

Calculate the hypotenuse.

Question 6

A box has length 10 cm, width 6 cm, and height 4 cm.

(a) Calculate its volume.

(b) Explain why the answer is expressed in cubic centimetres rather than square centimetres.

Section C: Algebra

Question 7

Solve:

$$4x + 7 = 31$$

Show why each operation is valid.

Question 8

A company charges a fixed fee of ₦1,200 and ₦150 per item processed.

(a) Write a formula for the total cost of processing n items.

(b) Calculate the cost of processing 20 items.

(c) How many items can be processed for ₦4,200 under this model?

Question 9

Two numbers have a sum of 42. The larger number is 8 greater than the smaller number.

Find both numbers.

Section D: Calculus and Change

Question 10

A moving object's position is given by

$$s(t) = t^2 + 2t$$

where s is in metres and t is in seconds.

(a) Calculate the position at $t = 3$.

(b) Find the derivative of the position function.

(c) Calculate the instantaneous velocity at $t = 3$.

Question 11

Explain the difference between average rate of change and instantaneous rate of change. Give one practical example of each.

Section E: Statistics and Probability

Question 12

The following scores are recorded:

$$50,\ 60,\ 60,\ 70,\ 80$$

Calculate the mean, median, and mode.

Question 13

A bag contains 4 red balls, 3 blue balls, and 5 green balls. One ball is selected at random, with every ball equally likely to be chosen.

Calculate the probability of selecting:

(a) A red ball.

(b) A blue ball.

(c) A ball that is not green.

Section F: Practical Reasoning

Question 14

A small business earns ₦180,000 in revenue and incurs ₦125,000 in total costs.

(a) Calculate its profit.

(b) Calculate its profit margin.

(c) Explain why the business's revenue alone is insufficient to determine whether it is profitable.

Question 15

A school has ₦200,000 for an event. It allocates 30% to equipment, 25% to refreshments, and ₦60,000 to transportation.

(a) Calculate the equipment allocation.

(b) Calculate the refreshment allocation.

(c) Calculate the total planned expenditure.

(d) Find the remaining reserve.

(e) Explain one reason why keeping a reserve may be useful.

Question 16

A student calculates the area of a rectangular room by adding its length and width.

Explain the error in the method and state what information and operation are required to calculate the area.

Question 17

A calculator displays an answer of 0.0036 for a calculation involving two measurements that are each approximately 100.

What should you do before accepting the answer? Explain at least three checks.

Question 18

A business model predicts a monthly profit of ₦500,000. Explain why this prediction should not automatically be treated as a guaranteed outcome.

Question 19

Choose one mathematical topic from this book and explain how it can be applied in two different fields.

Question 20

Create your own real-life mathematical problem. It must include at least three calculation steps, and your solution must explain why each operation is necessary.

12.17 Answer Guide and Worked Solutions

Use these solutions to check your work. If your answer differs, examine your method before assuming that your answer is wrong. Sometimes a problem allows different approaches, but the reasoning and final result must remain consistent with the stated conditions.

Answer 1: Arithmetic and Percentages

(a) Amount spent on books:

$$0.18 \times 45{,}000 = ₦8{,}100$$

(b) Remaining money:

$$45{,}000 - 8{,}100 = ₦36{,}900$$

(c) Percentage remaining:

$$100\% - 18\% = 82\%$$

The student has ₦36,900 remaining, which is 82% of the original amount.

Answer 2: Profit and Profitability

(a) Profit:

$$30{,}000 - 24{,}000 = ₦6{,}000$$

(b) Profit as a percentage of cost:

$$\frac{6{,}000}{24{,}000} \times 100\% = 25\%$$

(c) Profit margin:

$$\frac{6{,}000}{30{,}000} \times 100\% = 20\%$$

The answers differ because the first percentage uses cost as the denominator, while the second uses revenue.

Answer 3: Successive Percentage Changes

First, increase ₦80,000 by 15%:

$$80{,}000(1.15) = ₦92{,}000$$

Next, reduce ₦92,000 by 15%:

$$92{,}000(0.85) = ₦78{,}200$$

The final price is ₦78,200.

It is not equal to the original price because the 15% reduction is calculated on ₦92,000, not ₦80,000.

The increase was ₦12,000, while the reduction was ₦13,800.

Answer 4: Geometry and Fencing

(a) Area:

$$25 \times 16 = 400 \text{ m}^2$$

(b) Perimeter:

$$2(25) + 2(16) = 82 \text{ m}$$

(c) Fencing cost:

$$82 \times 2{,}500 = ₦205{,}000$$

The area measures the surface inside the field. The perimeter measures the boundary around it.

Answer 5: Pythagoras' Theorem

$$c^2 = 8^2 + 15^2$$
$$c^2 = 64 + 225 = 289$$

$$c = \sqrt{289} = 17 \text{ cm}$$

The hypotenuse is 17 cm.

Answer 6: Volume

a

$$V = 10 \times 6 \times 4$$
$$V = 240 \text{ cm}^3$$

(b) Volume measures three-dimensional space. The three dimensions are multiplied, producing cubic units. Square centimetres are units of area, not volume.

Answer 7: Solving an Equation

$$4x + 7 = 31$$

Subtract 7 from both sides:

$$4x = 24$$

Divide both sides by 4:

$$x = 6$$

Check:

$$4(6) + 7 = 24 + 7 = 31$$

The solution is correct.

Answer 8: A Cost Model

a

$$C = 1{,}200 + 150n$$

(b) For 20 items:

$$C = 1{,}200 + 150(20)$$
$$C = 1{,}200 + 3{,}000 = ₦4{,}200$$

(c) Set the cost equal to ₦4,200:

$$4{,}200 = 1{,}200 + 150n$$
$$3{,}000 = 150n$$
$$n = 20$$

The model allows 20 items to be processed for ₦4,200.

Answer 9: Two Unknown Numbers

Let the smaller number be x.

The larger number is $x + 8$.

Their sum is 42:

$$x + (x + 8) = 42$$
$$2x + 8 = 42$$
$$2x = 34$$
$$x = 17$$

The larger number is

$$17 + 8 = 25$$

The numbers are 17 and 25.

Answer 10: Calculus and Change

(a) Position at $t = 3$:

$$s(3) = 3^2 + 2(3)$$
$$s(3) = 9 + 6 = 15 \text{ m}$$

(b) Differentiate:

$$s(t) = t^2 + 2t$$
$$v(t) = \frac{ds}{dt} = 2t + 2$$

(c) Instantaneous velocity at $t = 3$:

$$v(3) = 2(3) + 2$$
$$v(3) = 8 \text{ m/s}$$

Answer 11: Rates of Change

Average rate of change describes the change in a quantity over an interval.

Instantaneous rate of change describes the rate at a particular point or moment, as defined by the relevant mathematical model.

For example, average speed describes distance travelled divided by elapsed time over a journey. Instantaneous velocity describes an object's velocity at a particular moment.

Answer 12: Statistics

Scores:

$$50,\ 60,\ 60,\ 70,\ 80$$

Mean:

$$\frac{50 + 60 + 60 + 70 + 80}{5} = \frac{320}{5} = 64$$

Median:

$$60$$

Mode:

$$60$$

Therefore, the mean is 64, while the median and mode are both 60.

Answer 13: Probability

Total balls:

$$4 + 3 + 5 = 12$$

(a) Red:

$$P(\text{red}) = \frac{4}{12} = \frac{1}{3}$$

(b) Blue:

$$P(\text{blue}) = \frac{3}{12} = \frac{1}{4}$$

(c) Not green:

$$P(\text{not green}) = \frac{4+3}{12} = \frac{7}{12}$$

These probabilities assume that every ball has an equal chance of being selected.

Answer 14: Business Profit

a

$$180{,}000 - 125{,}000 = ₦55{,}000$$

b

$$\frac{55{,}000}{180{,}000} \times 100\% \approx 30.56\%$$

(c) Revenue describes money received from sales. Profit depends on the relationship between revenue and costs. A business can generate substantial revenue and still make a loss if its costs are higher.

Answer 15: Event Budget

(a) Equipment:

$$0.30 \times 200{,}000 = ₦60{,}000$$

(b) Refreshments:

$$0.25 \times 200{,}000 = ₦50{,}000$$

(c) Total expenditure:

$$60{,}000 + 50{,}000 + 60{,}000 = ₦170{,}000$$

(d) Reserve:

$$200{,}000 - 170{,}000 = ₦30{,}000$$

(e) A reserve helps cover unexpected expenses without automatically exceeding the original budget.

Answer 16: Area of a Rectangle

Adding length and width does not calculate area. Area measures the surface covered by the rectangle.

The required operation is multiplication:

$$A = l \times w$$

The answer must be expressed in square units.

Answer 17: Checking a Calculator Result

Before accepting 0.0036, check:

1. Whether the numbers were entered correctly.
2. Whether the intended operation was entered correctly.
3. Whether parentheses were needed.
4. Whether the result is consistent with an estimate.
5. Whether the units and expected magnitude make sense.

Measurements of approximately 100 do not necessarily guarantee a particular result because the operation is unspecified. The correct response is to examine the original calculation rather than assume that the displayed answer is either correct or incorrect.

Answer 18: Interpreting a Business Model

A predicted profit depends on assumptions about customer demand, selling prices, expenses, and other conditions. Actual outcomes may differ. The prediction should therefore be treated as an estimate to be tested against evidence, not a guarantee.

Answer 19: Connecting Mathematical Topics

Answers will vary.

For example, percentages can be used by an accountant to calculate profit margins and by an economist to measure changes in prices.

The important point is to explain the relationship between the mathematical idea and the practical question being answered.

Answer 20: Your Own Problem

Answers will vary.

A strong solution should clearly state the situation, identify the unknown, organize the relevant information, justify the operations, show accurate working, include suitable units, and interpret the result.

12.18 The Final Lesson: Mathematics Is a Way of Thinking

We have reached the end of this book, but mathematical learning does not end here.

The real value of mathematics extends beyond examinations, formulas, and numerical answers. It lies in the ability to examine a situation, identify relationships, reason carefully, test conclusions, and make informed decisions.

When a builder calculates the dimensions of a structure, mathematics helps turn a plan into measurable quantities.

When an accountant calculates profit, mathematics helps reveal whether a business is earning more than it spends.

When an economist studies a change in price, mathematics helps examine relationships between costs, quantities, and incentives.

When a scientist measures an object, mathematics helps organize observations and assess patterns.

When an engineer models a system, mathematics helps predict how its parts may behave.

When you manage your personal budget, mathematics helps you understand what you can afford and how different choices affect your resources.

The applications are different, but the underlying discipline is similar: understand the situation, identify the relevant quantities, choose a suitable method, and examine the result.

Do not worship the formula

A formula is not a magical collection of symbols.

It is a compact expression of a relationship.

If you understand the relationship, you are better prepared to recognize when the formula applies, when it needs to be rearranged, and when a different method is required.

If you memorize a formula without understanding its meaning, you may apply it to the wrong problem.

That is why this book began with a question about thinking before calculation.

Do not fear difficult problems

A difficult problem may contain several simpler problems connected together.

Your responsibility is to identify those smaller problems and determine the order in which they should be solved.

You may need to calculate a percentage before finding a balance. You may need to determine an area before estimating material costs. You may need to construct an algebraic model before calculating a break-even point.

The complexity often becomes more manageable once the structure is clear.

Do not confuse an answer with an explanation

If you calculate that a business has made ₦50,000, ask what the amount represents.

Is it revenue, gross profit, operating profit, or cash remaining?

If you calculate 20%, ask what quantity forms the denominator.

If you calculate 15 metres, ask what distance has been measured and whether the assumptions support the result.

The explanation gives the answer its meaning.

Do not ignore the real world

Mathematical models simplify reality, but reality does not always follow our assumptions.

An estimated cost may differ from the final cost. A predicted demand may not occur. Measurements may contain uncertainty. A probability calculated from a small sample may not represent a wider population.

Good mathematical reasoning includes knowing what the calculations establish and what they do not establish.

Learn to ask better questions

As your mathematical understanding grows, your questions should become more precise.

Instead of asking only, “What is the answer?” ask:

- What information is missing?
- Which quantities are related?
- What assumptions are being made?
- Why is this operation appropriate?
- Is there another way to solve the problem?
- How can I check the result?
- What does the result mean?
- What would change if one of the assumptions changed?

These questions help turn calculation into reasoning.

12.19 A Personal Commitment to Mathematical Thinking

Before closing this book, make a commitment to the way you will approach mathematics.

You do not need to promise that you will never make a mistake. Mistakes are a normal part of learning.

Instead, promise yourself that you will investigate your mistakes, seek understanding, and practise with purpose.

Use the following principles as a personal guide.

1. I will try to understand a problem before selecting a formula.
2. I will identify what is known and what remains unknown.
3. I will explain why I choose a particular mathematical operation.
4. I will organize complicated problems into smaller steps.
5. I will use estimation to check whether an answer is reasonable.
6. I will pay attention to units, assumptions, and the meaning of results.
7. I will correct mistakes instead of merely hiding them.
8. I will practise applying mathematics to situations beyond the classroom.

9. I will remain patient when a topic requires more effort.
10. I will remember that understanding is more valuable than memorizing a procedure without knowing why it works.

Mathematics rewards careful thought, but it does not require you to understand everything immediately.

It requires you to keep asking questions, testing ideas, and improving your reasoning.

12.20 Conclusion

You began this book with a simple question: why should you search for a formula before understanding the problem?

Throughout the chapters, you have seen how this question changes the way mathematics can be learned.

You have explored numbers and percentages, geometry and Pythagoras, accounting and economics, algebra and calculus, statistics and probability, scientific measurement, and methods for simplifying difficult calculations.

You have seen that mathematical concepts are not isolated rules. They are connected ways of describing quantities, relationships, patterns, and change.

The same principle appears repeatedly:

Understand the situation.

Identify what you need to find.

Choose the appropriate relationship.

Calculate carefully.

Check the result.

Interpret what it means.

This process will not make every mathematical problem effortless. Some problems will still require advanced knowledge, long calculations, and repeated attempts.

But understanding the problem gives you a better starting point.

You will be less dependent on recognizing a familiar question and more capable of reasoning through an unfamiliar one.

That is the difference between merely performing calculations and thinking mathematically.

As you continue your studies, remember that every formula represents an idea, every equation expresses a relationship, and every mathematical model makes assumptions about the world it describes.

Your task is not simply to remember the symbols.

Your task is to understand what they mean.

Think before the formula. Understand before the calculation. Check before accepting the answer.

And whenever you meet a difficult problem, do not ask only whether you remember the formula.

Ask yourself:

What is happening, what do I know, what do I need to find, and how can mathematics help me understand it?

That is where meaningful mathematical learning begins.

— Santiano

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END OF BOOK